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TECHNICAL SPECIFICATION

Mission Critical (MC) services;
Mission Critical Push To Talk (MCPTT) Application Server (AS)
Protocol conformance specification for
server-to-client interface;
Part 1: Test structure, configurations,
conformance requirement and test purposes

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee TETRA and Critical Communications Evolution (TCCE).

The present document is based on the ETSI TS 136 579-3 [i.2] (3GPP TS 36.579-3) produced by the 3rd Generation Partnership Project (3GPP).

The present document is part 1 of a multi-part deliverable covering the Mission Critical (MC) services; Mission Critical Push To Talk (MCPTT) Application Server (AS) Protocol conformance specification for server-to-client interface, as identified below:

Part 1: "Test structure, configurations, conformance requirement and test purposes";

Part 2: "Test Applicability and Implementation Conformance Statement (ICS) pro forma specification".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The ETSI Technical Specification used as the reference document for the present document is part 3 of a multi-part deliverable covering conformance test specification for Mission Critical Services over LTE consisting of:

- ETSI TS 136 579-1 [1]: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)";
- ETSI TS 136 579-2 [i.6]: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)";
- **ETSI TS 136 579-3 [i.2]: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)";**
- ETSI TS 136 579-4 [2]: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)";
- ETSI TS 136 579-5 [i.7]: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)";
- ETSI TS 136 579-6 [i.8]: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)";
- ETSI TS 136 579-7 [i.9]: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)";
- 3GPP TS 36.579-8 [i.4]: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification";
- 3GPP TS 36.579-9 [i.3]: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

1 Scope

The present document specifies the protocol conformance testing for testing a MCPTT Server for compliance to the Mission Critical Push To Talk (MCPTT) over LTE protocol requirements defined by 3GPP. The present document addresses only MCPTT Server-Client communication scenarios. It does not cover e.g. MCPTT Server-EPS, MCPTT Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

In particular the present specification contains:

- the overall test structure;
- the test configurations;
- the conformance requirement and reference to the core specifications;
- the test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for MCPTT Servers implemented according to 3GPP releases starting from Release 14 up to the Release indicated on the cover page of the present document.

The following information relevant to testing specified in the present document could be found in accompanying specifications:

- default setting of the test parameters ETSI TS 136 579-1 [1] (3GPP TS 36.579-1);
- Implementation Conformance Statement (ICS) ETSI TS 136 579-4 [2] (3GPP TS 36.579-4) and Implementation extra Information for Testing (IXIT) ETSI TS 136 579-5 [i.7] (3GPP TS 36.579-5);
- the applicability of each test case ETSI TS 136 579-4 [2] (3GPP TS 36.579-4).

The present document does not specify the protocol conformance testing for the EPS (LTE) bearers which carry the MCPTT data sent or received by the MCPTT Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 136 579-1: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)".

[2] ETSI TS 136 579-4: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)".

[3] ETSI TS 123 179: "LTE; Functional architecture and information flows to support mission critical communication services; Stage 2 (3GPP TS 23.179)".

[4] ETSI TS 124 379: "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379)".

[5] ETSI TS 124 380: "LTE; Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification (3GPP TS 24.380)".

[6] ETSI TS 133 310: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Network Domain Security (NDS); Authentication Framework (AF) (3GPP TS 33.310)".

[7] IETF RFC 5245: "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols".

[8] IETF RFC 3556: "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[9] ETSI TS 124 282: "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282)".

[10] ETSI TS 124 281: "LTE; Mission Critical Video (MCVideo) signalling control; Protocol specification (3GPP TS 24.281)".

[11] ETSI TS 126 114: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction (3GPP TS 26.114)".

[12] IETF RFC 6335: "Internet Assigned Numbers Authority (IANA) Procedures for the Management of the Service Name and Transport Protocol Port Number Registry".

[13] IETF RFC 4867: "RTP Payload Format and File Storage Format for the Adaptive Multi-Rate (AMR) and Adaptive Multi-Rate Wideband (AMR-WB) Audio Codecs".

[14] IETF RFC 5621 (September 2009): "Message Body Handling in the Session Initiation Protocol (SIP)".

[15] IETF RFC 3261 (June 2002): "SIP: Session Initiation Protocol".

[16] IETF RFC 7134: "The Management Policy of the Resource Priority Header (RPH) Registry Changed to "IETF Review"".

[17] ETSI TS 136 509: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Special conformance testing functions for User Equipment (UE) (3GPP TS 36.509)".

[18] IETF RFC 8101: "IANA Registration of New Session Initiation Protocol (SIP) Resource-Priority Namespace for Mission Critical Push To Talk service".

[19] IETF RFC 3903 (October 2004): "Session Initiation Protocol (SIP) Extension for Event State Publication".

[20] IETF RFC 4412 (February 2006): "Communications Resource Priority for the Session Initiation Protocol (SIP)".

[21] IETF RFC 6665 (July 2012): "SIP-Specific Event Notification".

[22] IETF RFC 4028 (April 2005): "Session Timers in the Session Initiation Protocol (SIP)".

[23] ETSI TS 136 508: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing (3GPP TS 36.508)".

[24] IETF RFC 5366 (October 2008): "Conference Establishment Using Request-Contained Lists in the Session Initiation Protocol (SIP)".

[25] ETSI TS 123 283: "LTE; Mission Critical Communication Interworking with Land Mobile Radio Systems (3GPP TS 23.283)".

[26] ETSI TS 123 280: "LTE; Common functional architecture to support mission critical services; Stage 2 (3GPP TS 23.280)".

[27] ETSI TS 124 481: "LTE; Mission Critical Services (MCS) group management; Protocol specification (3GPP TS 24.481)".

[28] ETSI TS 124 482: "LTE; Mission Critical Services (MCS) identity management; Protocol specification (3GPP TS 24.482)".

[29] ETSI TS 124 484: "LTE; Mission Critical Services (MCS) configuration management; Protocol specification (3GPP TS 24.484)".

[30] ETSI TS 133 179: "LTE; Security of Mission Critical Push To Talk (MCPTT) over LTE (3GPP TS 33.179)".

[31] ETSI TS 133 180: "LTE; Security of the Mission Critical (MC) service (3GPP TS 33.180)".

[32] IETF RFC 4354: "A Session Initiation Protocol (SIP) Event Package and Data Format for Various Settings in Support for the Push-to-Talk over Cellular (PoC) Service".

[33] IETF RFC 2854: "The 'text/html' Media Type".

[34] W3C® Recommendation 24 December 1999: "HTML 4.01 Specification".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 121 905: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905)".

[i.2] ETSI TS 136 579-3: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)".

[i.3] 3GPP TS 36.579-9: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

[i.4] 3GPP TS 36.579-8: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification".

[i.5] ETSI TS 122 179: "LTE; 5G; Mission Critical Push to Talk (MCPTT); Stage 1 (3GPP TS 22.179)".

[i.6] ETSI TS 136 579-2: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)".

[i.7] ETSI TS 136 579-5: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)".

[i.8] ETSI TS 136 579-6: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)".

[i.9] ETSI TS 136 579-7: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply.

NOTE: A term defined in the present document takes precedence over the definition of the same term, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

For the purpose of the present document, the following terms and definitions given in ETSI TS 124 379 [4] (3GPP TS 24.379) apply:

- An MCPTT user is affiliated to an MCPTT group
- An MCPTT user is affiliated to an MCPTT group at an MCPTT client
- Affiliation status
- Group identity
- In-progress emergency private call state
- In-progress imminent peril group state
- MCPTT client ID
- MCPTT emergency alert state
- MCPTT emergency group state
- MCPTT emergency group call state
- MCPTT emergency private call state
- MCPTT emergency private priority state
- MCPTT imminent peril group call state
- MCPTT imminent peril group state
- MCPTT private emergency alert state
- MCPTT speech
- Media-floor control entity
- Temporary MCPTT group identity
- Trusted mutual aid
- Untrusted mutual aid

For the purposes of the present document, the following terms and definitions given in ETSI TS 122 179 [i.5] (3GPP TS 22.179) apply:

- In-progress emergency
- MCPTT emergency alert
- MCPTT emergency group call
- MCPTT emergency state

- Partner MCPTT system
- Primary MCPTT system

For the purpose of the present document, the following terms and definitions given in ETSI ETSI TS 124 380 [5] (3GPP TS 24.380) apply:

- MBMS subchannel

For the purpose of the present document, the following terms and definitions given in ETSI TS 123 179 [3] (3GPP TS 23.179) apply:

- Pre-selected MCPTT user profile

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply.

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

ECGI	E-UTRAN Cell Global Identification
GKTP	Group Key Transport Payload
ICS	Implementation Conformance Statement
IUT	Implementation Under Test
MBMS	Multimedia Broadcast and Multicast Service
MCPTT group ID	MCPTT group IDentity
MCPTT	Mission Critical Push To Talk
MEG	MCPTT Emergency Group
MEPC	MCPTT Emergency Private Call
MEPP	MCPTT Emergency Private Priority
MIG	MCPTT Imminent peril Group
MIME	Multipurpose Internet Mail Extensions
MPEA	MCPTT Private Emergency Alert
OIDC	OpenID Connect
PLMN	Public Land Mobile Network
QCI	QoS Class Identifier
RTP	Real-time Transport Protocol
SDP	Session Description Protocol
SIP	Session Initiation Protocol
SS	System Simulator
SSRC	Synchronization SouRCe
TMGI	Temporary Mobile Group Identity
TP	Test Purpose
TP	Transmission Point
URI	Uniform Resource Identifier
XCAP	eXtensible Markup Language (XML) Access Configuration Protocol

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure which is optional may be subject to a conformance test if it is implemented in the MCPTT Server.

A declaration by the Client supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 136 579-4 [2] (3GPP TS 36.579-4)) is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Detailed descriptions of the MCPTT Server to MCPTT Client test interfaces and special facilities for testing are provided in ETSI TS 136 509 [17] (3GPP TS 36.509). Descriptions of the MCPTT Server to MCPTT Server interface is specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1).

4.2 Implicit testing

For some 3GPP MCPTT protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a general rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of cores specifications

The conformance requirements which determine the scope of each test case are explicitly copy-pasted from relevant core specifications in the especially dedicated for this clause of each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present specification. This approach has been taken in order to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the cores specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the spec from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the spec with release = the TC Applicability release (which can be found in ETSI TS 136 579-4 [2] (3GPP TS 36.579-4), Table 4-2: Applicability of tests and additional information for testing, column 'Release'), or, as the Conformance requirements in the latest version of the spec of the release when the feature was introduced to the core specs.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). Where a test requires an environment that is different, this will be specified in the test itself.

4.6 Generic setup procedures

A set of basic generic procedures for MCPTT Client-Server and MCPTT Client-Client communication are described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). These procedures will be used in numerous test cases throughout the present document.

5 MCPTT Server - MCPTT Client Configuration

5.1 MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorisation / UE Configuration / User Profile / Key generation

5.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT client) activates an MCPTT application and requests MCPTT initialisation
}
    then { IUT (MCPTT Server) provides the initial UE configuration and performs MCPTT User
Authentication and provides id_token, access_token and refresh token to the successfully
authenticated user}
            }
```

(2)

```
with { IUT (MCPTT Server) having authenticated the SS-UE1 (MCPTT client) }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates key management authorization }
    then { IUT (MCPTT Server) provides identity management key material }
            }
```

(3)

```
with { IUT (MCPTT Server) having provided identity management key material }
ensure that {
  when { the SS-UE1 (MCPTT client) requests user service authorization }
    then { IUT (MCPTT Server) responds to the SS-UE1 (MCPTT Client) with SIP 200 (OK) messages }
            }
```

(4)

```
with { IUT (MCPTT Server) having provided service authorization }
ensure that {
  when { the SS-UE1 (MCPTT client) requests configuration management authorization}
    then { IUT (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message and
sends a SIP NOTIFY message containing the XCAP-URI of the documents and sends the MCPTT UE
Configuration Document and MCPTT User Profile Configuration Document MCPTT Service Configuration
Document via HTTP 200 (OK) messages in response to HTTP GET requests }
            }
```

(5)

```
with { IUT (MCPTT Server) having provided user configuration data }
ensure that {
  when { the SS-UE1 (MCPTT client) requests group management authorization }
    then { IUT (MCPTT Server) responds to the SS-UE1 (MCPTT Client) with SIP 200 (OK) messages and
sends the XCAP-URI of the Group documents via a SIP NOTIFY message and sends the Group Document
'MCPTT UE Configuration document' via a HTTP 200 (OK) message in response to a HTTP GET request and
sends the group key transport payloads (GKTP) document via a SIP NOTIFY message }
            }
```

(6)

```
with { IUT (MCPTT Server) having provided all required configuration data }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to refresh its service settings }
    then { IUT (MCPTT Server) responds to the SS-UE1 (MCPTT Client) with a SIP 200 (OK) message }
           }
```

5.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 482 [28] (3GPP TS 24.482) clause 6.3.1, clause A.2.1.3, clause A.2.3, ETSI TS 124 484 [29] (3GPP TS 24.484) clauses 4.3, 4.4, 6.2.3, 6.3.1.2, 6.3.2.3, 6.3.13.3.2.2, ETSI TS 124 481 [27] (3GPP TS 24.481) clauses 6.2.4, 6.3.3.3, 6.3.13.3.2.2, ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 7.3.2, 7.3.3, 7.3.4, ETSI TS 133 179 [30] (3GPP TS 33.179) clauses 5.6.1, 7.2.3, Annex D.1. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 482 [28] (3GPP TS 24.482), clause 6.3.1]

"Upon receipt of an OIDC Authentication Request message as specified in the OpenID Connect 1.0 [6] and IETF RFC 6749 [5] via a secure TLS tunnel between the identity management client and the authorisation endpoint of the IdM server, the IdM server:

1) shall validate the received OIDC Authentication Request message as specified in the OpenID Connect 1.0 [6] and IETF RFC 6749 [5];

2) shall generate an HTTP 200 (OK) response according to IETF RFC 2616 [4] including form data to prompt the MCPTT user for their username and password credentials; and

NOTE 1: The username will be the MCPTT user's MC ID.

3) shall send the HTTP 200 (OK) response towards the IdM client.

Upon receipt of an HTTP POST request method from the IdM client containing the MCPTT user's username and password, the IdM server authenticates the MCPTT user and:

NOTE 2: Other methods of authentication can be used by the MCPTT service provider and are not defined by the OIDC specifications. 3GPP TS 33.179 [2] has defined username and password as a mandatory authentication method to be supported for MCPTT, hence a procedure to realize that method is included here.

1) shall generate an OIDC Authentication Response message as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5] with the following clarifications:

a) shall generate an HTTP 302 (FOUND) response according to IETF RFC 2616 [4]; and

b) shall include the required parameters including the authorization_code as specified in 3GPP TS 33.179 [2] in the query component of the redirection URI contained in the Location header field of the HTTP FOUND request method using the "application/x-www-form-urlencoded" format as specified in W3C.REC-html401-19991224 [7]; and

2) shall send the HTTP 302 (FOUND) response towards the IdM client.

Upon receipt of an OIDC Token Request message via a secure TLS tunnel established between the identity management client and the token endpoint of the IdM server, the IdM server:

1) shall validate the OIDC Token Request message and if valid shall generate an OIDC Token Response message as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5] with the following clarifications:

a) shall generate an HTTP 200 (OK) response according to IETF RFC 2616 [4];

b) shall based on the received MC ID obtained from the received user authentication credentials, determine the MCPTT ID of the MCPTT user;

c) shall include an id_token, access_token and refresh_token and MCPTT ID as specified in 3GPP TS 33.179 [2]; and

d) shall include the other required parameters as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5]; and

2) shall send the HTTP 200 (OK) response towards the IdM client."

[ETSI TS 124 482 [28] (3GPP TS 24.482), clause A.2.1.3]

"The HTTP client in the network entity is configured with the following parameters:

1) a home HTTP proxy FQDN; and

2) a home HTTP proxy port.

The HTTP client in the network entity shall send and receive all HTTP messages via the home HTTP proxy.

The HTTP client in the network entity shall insert an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15] in the HTTP request and shall set X-3GPP-Asserted-Identity header field to the identity of the HTTP client in the network entity. The identity of the HTTP client in the network entity can be a public service identity, an MCPTT group ID, or an MCPTT ID."

[ETSI TS 124 482 [28] (3GPP TS 24.482), clause A.2.3]

"The HTTP server shall support the server role of IETF RFC 2616 [4].

Upon reception of an HTTP request:

1) if the received HTTP request does not contain an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14] and the received HTTP request does not contain an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15], the HTTP server shall reject the request with HTTP 403 (Forbidden) response;

2) if the received HTTP request contains an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14];

a) the HTTP server shall validate the bearer access token as specified in IETF RFC 6750 [14]; and

b) the HTTP server shall consider the MCPTT ID derived from the bearer access token as the identity of the sender of the HTTP request; and

3) if the received HTTP request does not contain an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14] and the received HTTP request contains an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15], the HTTP server shall consider the URI in the X-3GPP-Asserted-Identity header field as the identity of the sender of the HTTP request."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 4.3]

"The MCPTT server obtains the MCPTT service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCPTT service from the configuration management server.

The format of the MCPTT service configuration document downloaded to the MCPTT server is defined in subclause 7.5.

The MCPTT server obtains the MCPTT service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCPTT service from the configuration management server.

The MCPTT server subscribes to the MCPTT service configuration document for each mission critical organisation that is provisioned that is supported by the MCPTT server using the procedure specified in subclause 6.3.13.2.3. How the MCPTT server is provisioned with the identities of the mission critical organisations is out of scope of the present document.

If the MCPTT service configuration document has been updated since the current version stored at the MCPTT server, then the MCPTT server will receive a SIP NOTIFY request containing an HTTPS URI of the MCPTT service configuration document. Retrieval by the MCPTT server, using the notified HTTPS URI, of the MCPTT service configuration document is performed as specified in subclause 6.3.3.2.3.

NOTE: The MCPTT server can be notified of changes to the MCPTT service management configuration document at any time while operating the MCPTT service.

The format of the MCPTT service configuration document downloaded to the MCPTT server is defined in subclause 7.5."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 4.4]

"The following applies to the configuration management server used for online configuration.

The configuration management server needs to convert the MCPTT UE initial configuration document received from a MCPTT administrator into an appropriate format for configuration of the MCPTT UE initial configuration MO.

If the MCPTT UE initial configuration MO contains a <default-user-profile> element that identifies a MCPTT user profile configuration document, the configuration management server needs to convert the identified MCPTT user profile configuration document received from a MCPTT administrator into an appropriate format for configuration of the MCPTT user profile configuration MO.

Once an MCPTT User Profile configuration document has been created or updated by the MCPTT UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to store MCPTT user profile configuration document as the user profile in the MCPTT user database.

In order to download MCPTT the user profile configuration document to an MCPTT UE or to support an MCPTT UE updating the MCPTT user profile configuration document, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to obtain the MCPTT user profile from the MCPTT user database.

In order to be notified of changes to an MCPTT user profile configuration document that have been subscribed to by an MCPTT UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to be notified of changes to the MCPTT user profile stored in the MCPTT user database.

In order to delete the MCPTT user profile when requested by an MCPTT UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to delete the MCPTT user profile from the MCPTT user database.

NOTE: The configuration management server and group management server functionality for offline configuration is out of scope of the present document."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 6.2.3]

"The MCPTT server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.382 [6]."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 6.2.4]

"The CMS shall handle the HTTP request as specified for the HTTP server in annex A of 3GPP TS 24.382 [6].

The CMS shall be configured with an authorized MCPTT server list, containing public service identities of MCPTT servers of the MCPTT provider of the CMS.

When handling an HTTP request, the CMS shall determine the identity of the sender of the HTTP request as specified in 3GPP TS 24.382 [6], and shall use the identity of the sender of the HTTP request as an authenticated identity when performing the authorization."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 6.3.1.2]

"A CMS shall support subclause 6.2.1 "Document Management", and subclause 6.2.4 "Access Permissions" of OMA OMA-TS-XDM_Core-V2_1 [2] and subclause 6.3.13.3 for accepting subscriptions to configuration management documents."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 6.3.2.3]

"A CMS shall support receiving XML documents of the application usages specified in subclause 7.2.1, subclause 7.3.1, subclause 7.4.1 and subclause 7.5.1according to procedures specified in IETF RFC 4825 [14] "PUT Handling" where the Request-URI of the HTTP PUT request identifies an XML document and include the "auid" as per the appropriate application usage in clause 7."

[ETSI TS 124 484 [29] (3GPP TS 24.484), clause 6.3.13.3.2.2]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for performing subscription proxy function of the CMS;

c) with a P-Asserted-Identity header field not containing an identity listed in the authorized MCPTT server list specified in subclause 6.2.4;

d) with an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-access-token> element;

e) with an application/resource-lists+xml MIME body; and

f) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [23];

the CMS:

a) if an <EncryptedData> XML tag is included in the application/vnd.3gpp.mcptt-info+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/vnd.3gpp.mcptt-info+xml MIME body;

b) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body;

c) shall identify the originating MCPTT ID from <mcptt-access-token> element received in the application/vnd.3gpp.mcpttinfo+xml MIME body and shall use the originating MCPTT ID as an authenticated identity when performing the authorization;

d) if the authenticated identity is not authorized to subscribe to notification of changes of any resource in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;

e) act as a notifier according to IETF RFC 5875 [11]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the initial SIP SUBSCRIBE request contains an "auid" parameter set to an application usage identifying a configuration management document as described in clause 7;

shall return the XCAP URI identifying the configuration management document in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request.

Upon sending a SIP NOTIFY request associated with a subscription created as result of the received initial SIP SUBSCRIBE request, if the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, the CMS shall perform the confidentiality protection procedures and integrity protection procedures defined in 3GPP TS 24.379 [9] for MCPTT server.

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with an application/resource-lists+xml MIME body;

the CMS:

a) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body of the received SIP re-SUBSCRIBE request and the CSK was received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body; and

b) act as a notifier according to IETF RFC 5875 [11]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the SIP re-SUBSCRIBE request contains an "auid" parameter set to an application usage identifying a configuration management document as described in clause 7:

> *and for which there is no related subscription established according to the subclause 6.3.13.3.2.3, shall return the XCAP URI identifying the configuration management document in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request."*

[ETSI TS 124 481 [27] (3GPP TS 24.481), clause 6.2.4]

"The MCPTT server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.382 [10].

The MCPTT server shall perform the procedures in subclause 6.2.2 specified for GC."

[ETSI TS 124 481 [27] (3GPP TS 24.481), clause 6.3.3.3]

"A GMS shall support handling an HTTP GET request from a GMC according to procedures specified in IETF RFC 4825 [22] "GET Handling" where the Request-URI of the HTTP GET request identifies an XML document of the application usage specified in subclause 7.2."

[ETSI TS 124 481 [27] (3GPP TS 24.481), clause 6.3.13.3.2.2]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for performing subscription proxy function of the GMS;

c) with a P-Asserted-Identity header field not containing an identity listed in the authorized MCPTT server list specified in subclause 6.2.5.1 and not containing an identity listed in the authorized GMS list as specified in subclause 6.2.5.1;

d) with an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-access-token> element;

e) with an application/resource-lists+xml MIME body; and

f) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14];

the GMS:

a) if an <EncryptedData> XML tag is included in the application/vnd.3gpp.mcptt-info+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/vnd.3gpp.mcptt-info+xml MIME body;

b) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body;

c) shall identify the originating MCPTT ID from <mcptt-access-token> element received in the application/vnd.3gpp.mcptt-info+xml MIME body and shall use the originating MCPTT ID as an authenticated identity when performing the authorization;

d) if the authenticated identity is not authorized to subscribe to notification of changes of any resource in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;

e) act as a notifier according to IETF RFC 5875 [13]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the initial SIP SUBSCRIBE request identifies:

> *1) a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is an MCPTT group ID owned by an MCPTT provider other than the MCPTT provider of the GMS; or*

2) a element of an MCPTT GKTP document as described in subclause 7.7.10 where the group ID is an MCPTT group ID owned by an MCPTT provider other than the MCPTT provider of GMS;

shall perform the procedure in subclause 6.3.13.3.2.4 for each such MCPTT group ID and shall interwork information of received SIP NOTIFY requests in subclause 6.3.13.3.2.4 in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request.

Upon sending a SIP NOTIFY request associated with a subscription created as result of the received initial SIP SUBSCRIBE request, if the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, the GMS shall perform the confidentiality protection procedures and integrity protection procedures defined in 3GPP TS 24.379 [5] for MCPTT server.

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with an application/resource-lists+xml MIME body;

the GMS:

a) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body of the received SIP re-SUBSCRIBE request and the CSK was received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body; and

b) act as a notifier according to IETF RFC 5875 [13]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the SIP re-SUBSCRIBE request identifies:

1) a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is an MCPTT group ID owned by an MCPTT provider other than the MCPTT provider of the GMS; or

2) a element of an MCPTT GKTP document as described in subclause 7.7.10 where the group ID is an MCPTT group ID owned by an MCPTT provider other than the MCPTT provider of GMS;

and for which there is no related subscription established according to the subclause 6.3.13.3.2.4, shall perform the procedure in subclause 6.3.13.3.2.4 for each such MCPTT group ID and shall interwork information of received SIP NOTIFY requests in subclause 6.3.13.3.2.4 in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.2]

"The MCPTT server shall support obtaining service authorization specific information from the SIP REGISTER request sent from the MCPTT client and included in the body of a third-party SIP REGISTER request.

NOTE 1: 3GPP TS 24.229 [4] defines how based on initial filter criteria the SIP REGISTER request sent from the UE is included in the body of the third-party SIP REGISTER request.

Upon receiving a third party SIP REGISTER request with a message/sip MIME body containing the SIP REGISTER request sent from the MCPTT client containing an application/vnd.3gpp.mcptt-info+xml MIME body with an <mcptt-access-token> element and an <mcptt-client-id> element within a message/sip MIME body of the SIP REGISTER request sent from the MCPTT client, the MCPTT server:

1) shall identify the IMS public user identity from the third-party SIP REGISTER request;

2) shall identify the MCPTT ID from the SIP REGISTER request sent from the MCPTT client and included in the message/sip MIME body of the third-party SIP REGISTER request by following the procedures in subclause 7.3.1A;

3) shall perform service authorization for the identified MCPTT ID as described in 3GPP TS 33.179 [46];

4) if service authorization was successful, shall bind the MCPTT ID to the IMS public user identity; and

NOTE 2: The MCPTT server will store the binding MCPTT ID, IMS public user identity and an identifier addressing the MCPTT server in an external database.

5) if a Resource-Share header field with the value "supported" is contained in the "message/sip" MIME body of the third-party REGISTER request, shall bind the MCPTT ID to the identity of the MCPTT UE contained in the "+g.3gpp.registration-token" header field parameter in the Contact header field of the incoming third-party REGISTER request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.3]

"The MCPTT server shall support obtaining service authorization specific information from a SIP PUBLISH request for MCPTT server settings.

Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value;

2) an application/poc-settings+xml MIME body; and

3) an application/vnd.3gpp.mcptt-info+xml MIME body containing an <mcptt-access-token> element and an <mcptt-client-id> element;

the MCPTT server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall perform the procedures in subclause 7.3.1A;

3) if the procedures in subclause 7.3.1A were not successful shall send a SIP 403 (Forbidden) response towards the MCPTT server with the warning text set to: "140 unable to decrypt XML content " in a Warning header field as specified in subclause 4.4, and not continue with the rest of the steps in this subclause;

4) shall perform service authorization for the identified MCPTT ID as described in 3GPP TS 33.179 [46];

5) if service authorization was successful:

a) shall bind the MCPTT ID to the IMS public user identity;

b) if a Resource-Share header field with the value "supported" was included in the "message/sip" MIME body of the third-party REGISTER request, shall bind the MCPTT ID to the identity of the MCPTT UE contained in the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request that contained this IMS public user identity;

NOTE 1: The MCPTT server will store the binding MCPTT ID, IMS public user identity and an identifier addressing the MCPTT server in an external database.

c) shall download the MCPTT user profile from the MCPTT user database as defined in 3GPP TS 29.283 [73] if not already stored at the MCPTT server;

d) if multiple MCPTT user profiles are stored at the MCPTT server or downloaded for the MCPTT user from the MCPTT user database, shall determine the pre-selected MCPTT user profile by identifying the MCPTT user profile (see the MCPTT user profile document in 3GPP TS 24.384 [50]) in the collection of MCPTT user profiles that contains a <Pre-selected-indication> element; and

NOTE 2: If only one MCPTT user profile is stored at the MCPTT server or only one MCPTT user profile is downloaded from the MCPTT user database, then by default this MCPTT user profile is the pre-selected MCPTT user profile.

e) if an <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCPTT user profile document with one or more <entry> elements containing an MCPTT group ID (see the MCPTT user profile document in 3GPP TS 24.384 [50]) for the served MCPTT ID, shall perform implicit affiliation as specified in subclause 9.2.2.2.15;

6) if service authorization was not successful, shall send a SIP 403 (Forbidden) response towards the MCPTT server with the warning text set to: "101 service authorisation failed" in a Warning header field as specified in subclause 4.4, and not continue with the rest of the steps in this subclause;

7) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [37] and if processing of the SIP request was not successful, do not continue with the rest of the steps;

8) shall cache the received MCPTT service settings until the MCPTT service settings expiration timer expires;

9) shall send a SIP 200 (OK) response according 3GPP TS 24.229 [4]; and

10) shall use the Answer-Mode Indication setting in the <am-settings> element of the poc-settings event package as the current Answer-Mode Indication of the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.4]

"Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value;

2) an application/poc-settings+xml MIME body; and

3) an application/vnd.3gpp.mcptt-info+xml MIME body containing an <mcptt-request-uri> element and an <mcptt-client-id> element;

The MCPTT server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall perform the procedures in subclause 7.3.1A;

3) if the procedures in subclause 7.3.1A were not successful, shall send a SIP 403 (Forbidden) response towards the MCPTT server with the warning text set to: "140 unable to decrypt XML content" in a Warning header field as specified in subclause 4.4, and not continue with the rest of the steps in this subclause;

4) shall verify that a binding between the IMS public user identity in the Request-URI and the MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml exists at the MCPTT server;

5) if a binding exists between the IMS public user identity and the MCPTT ID in the request and the validity period of the binding has not expired:

a) shall download the MCPTT user profile from the MCPTT user database as defined in 3GPP TS 29.283 [73] if not already stored at the MCPTT server;

b) if multiple MCPTT user profiles are stored at the MCPTT server or downloaded for the MCPTT user from the MCPTT user database, shall determine the pre-selected MCPTT user profile by identifying the MCPTT user profile (see the MCPTT user profile document in 3GPP TS 24.384 [50]) in the collection of MCPTT user profiles that contains a <Pre-selected-indication> element; and

NOTE: If only one MCPTT user profile is stored at the MCPTT server or only one MCPTT user profile is downloaded from the MCPTT user database, then by default this MCPTT user profile is the pre-selected MCPTT user profile.

c) if an <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCPTT user profile document with one or more <entry> elements containing an MCPTT group ID (see the MCPTT user profile document in 3GPP TS 24.384 [50]) for the served MCPTT ID, shall perform implicit affiliation as specified in subclause 9.2.2.2.15.

6) if a binding does not exist between the IMS public user identity and the MCPTT ID in the request or the binding exists, but the validity period of the binding has expired, shall reject the SIP PUBLISH request with a SIP 404 (Not Found) response and not continue with any of the remaining steps;

7) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [37] and if processing of the SIP request was not successful, do not continue with the rest of the steps;

8) shall cache the received MCPTT service settings until the MCPTT service settings expiration timer expires;

9) shall send a SIP 200 (OK) response according 3GPP TS 24.229 [4]; and

10) shall use the Answer-Mode Indication setting in the <am-settings> element of the poc-settings event package as the current Answer-Mode Indication of the MCPTT client."

[ETSI TS 133 179 [30] (3GPP TS 33.179) clause 5.6.1]

"For key management authorization, the KM client in the UE presents an access token to the KMS over HTTP. The KMS validates the access token and if successful, provides user specific key material back to the UE KM client based on the MCPTT ID of the user. This includes identity based key information used for media and signalling protection.

For user service authorization, the MCPTT client in the UE presents an access token to the MCPTT server over SIP. The MCPTT server validates the access token and if successful, authorizes the user for full MCPTT services and sends an acknowledgement back to the MCPTT client. The MCPTT server then maps and maintains the IMPU to MCPTT ID association. The MCPTT ID to IMPU association shall only be known to the application layer. The SIP message used to convey the access token from the MCPTT client to the MCPTT server may be either a SIP REGISTER or SIP PUBLISH message.

The UE can now perform configuration management authorization and download the user profile. Following the flow described in subclause 10.1.4.2 of 3GPP TS 23.179 [2] "MCPTT user obtains the user profile (UE initiated)", the Configuration Management (CM) client in the UE sends an access token in the user profile query to the Configuration Management server over HTTP. The CM server receives the request and validates the access token, and if valid, the CM server uses the MCPTT ID to obtain the user profile from the MCPTT user database. The CM server then sends the user profile back to the CM client over HTTP.

Upon receiving the user's profile, the Group Management (GM) client in the UE can now perform group management authorization. The GM client obtains the user's group membership information from the user's profile, and following the flow shown in clause 10.1.5.2 of 3GPP TS 23.179 [2] "Retrieve group configurations at the group management client", the Group Management (GM) client in the UE sends an access token in the Get group configuration request to the host GM server of the group membership over HTTP. The GM server validates the access token, and if valid, completes the flow. As part of group management authorization, group key information is provided as per subclause 7.3.2 of the present document."

[ETSI TS 133 179 [30] (3GPP TS 33.179) clause 7.2.3]

"The procedure for the provision of identity-specific key material when the MCPTT proxy is supported between the KMS and the KMS client is described in figure 7.2.3-1. The procedure is the same whether the key management client in the MCPTT UE, MCPTT Server or group management server is making the request.

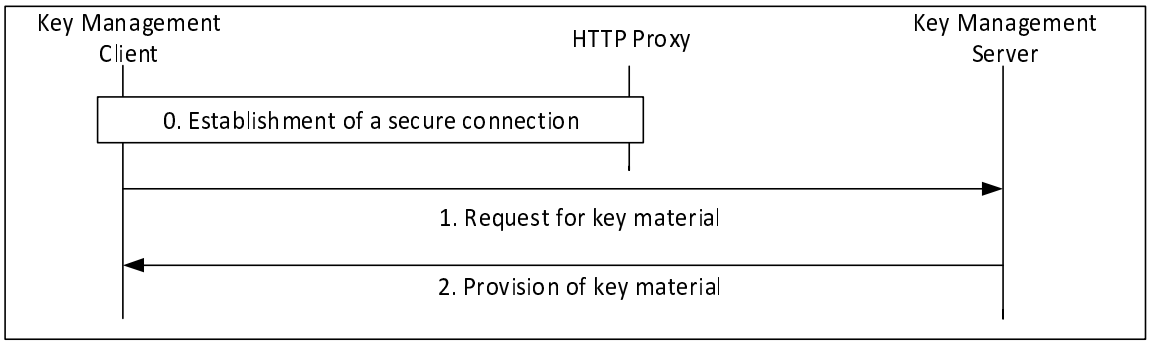


Figure 7.2.3-1: Provisioning of key material via the HTTP proxy

The procedure in figure 7.2.3-1 is now described step-by-step.

0) The key management client establishes a connection to the MCPTT KMS. As with other elements in the Common Services Core, the connection routed via, and secured by, the HTTP Proxy. The message flow below is within this secure connection.

NOTE: Additionally, the connection between the MCPTT KMS and the HTTP Proxy is secured according to clause 8.

1) The key management client makes a request for user key material from the MCPTT KMS. The request contains details of the identity (e.g. the MCPTT ID) requested for key management, and the time for which the key material is required.

2) The KMS provides a response containing key material. The response includes the type of key material, the period of use for the material and any domain-specific parameters required for its use. For public safety use, the key material itself shall be wrapped using a 256-bit transport key (TrK). The TrK is distributed via an out-of-band mechanism along with a 32-bit identifier, TrK-ID.

The procedure for the provisioning of identity-specific key material when the MCPTT proxy is not used between the KMS and the KMS client is as described in Figure 7.2.3-2.

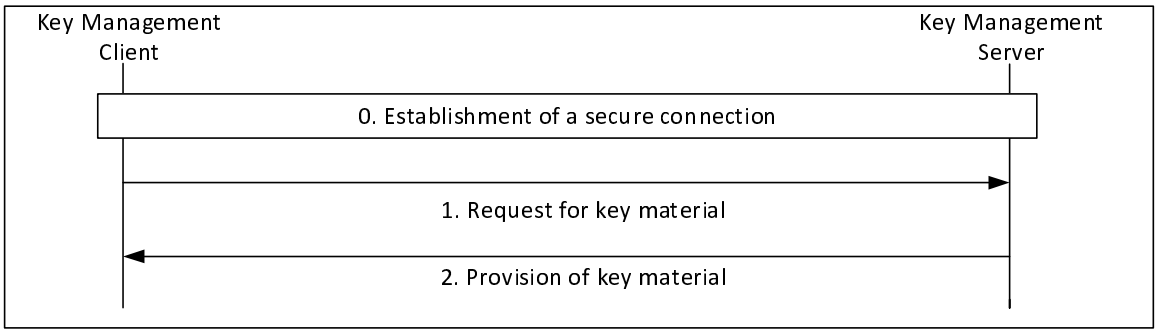


Figure 7.2.3-2: Provisioning of key material without a proxy

The procedure in Figure 7.2.3-2 is now described step-by-step:

0) The key management client establishes a direct HTTPS connection to the MCPTT KMS. The following message flow is within this secure connection.

1) The key management client makes a request for user key material from the MCPTT KMS. The request contains details of the identity requested for key management, and the time at which the key material is required.

2) The KMS provides a response containing key material. The response includes the type of key material, the period of use for the material and any domain-specific parameters required for its use. Optionally, the key material itself may also be wrapped using a 256-bit transport key (TrK), distributed via an out-of-band mechanism along with a 32-bit identifier (TrK-ID).

As a result of this procedure, the key management client has securely obtained key material for use within the MCPTT system."

[ETSI TS 133 179 [30] (3GPP TS 33.179) clause D.1]

"All KMS communications are made via HTTPS. The MCPTT key management client is provisioned via XML content in the KMS's response. The XML content is designed to be extendable to allow KMS/client providers to add further information in the XML. Where the interface is extended, a different XML namespace should be used (so that may be ignored by non-compatible clients).

It is assumed that transmissions between the KMS and the key management client are secure and that the KMS has authenticated the identity of the key management client.

Additionally, to allow the transmission of key material securely between a secure element within the KMS and a secure element within the key management client, a security extension is defined which allows messages to be signed and key material to be encrypted using a shared Transport Key (TrK)."

5.1.3 Test description

5.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1.
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.

5.1.3.2 Test procedure sequence

Table 5.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
0A	The SS-UE1 (MCPTT client) sends an HTTP GET (initial UE configuration) request to retrieve the initial UE configuration from the Server.	-->	HTTP GET	-	-
0B	Check: Does the IUT (MCPTT Server) respond by sending an HTTP 200 (OK) including the initial UE configuration document?	<--	HTTP 200 (OK)	1	P
0C	The SS-UE1 (MCPTT client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [6] (3GPP TS 33.310) to the authorisation endpoint of the IdM server as specified in ETSI TS 133 180 [31] (3GPP TS 33.180), using the configured URL of the authorisation endpoint of the IdM server as specified in the "<x>/OnNetwork/AppServerInfo/IDMSAuthEndpoint" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1).	-	-	-	-
1	The SS-UE1 (MCPTT client) sends an OpenID Connect Authentication Request using HTTP POST	-->	HTTP POST (Authorization)	-	-
2	Check: Does the IUT (MCPTT Server) respond by sending an HTTP 200 (OK) including the HTML form requesting username and password?	<--	HTTP 200 (OK)	1	P
3	The SS-UE1 (MCPTT client) sends an HTTP POST Request message to the SS containing user name and password	-->	HTTP POST	-	-
4	Check: Does the IUT (MCPTT Server) send an HTTP 302 (Found) as the OpenID Connect Authentication Response?	<--	HTTP 302 (Found)	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	The SS-UE1 (MCPTT client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [6] (3GPP TS 33.310) to the token endpoint of the IdM server as specified in ETSI TS 133 180 [31] (3GPP TS 33.180), using the configured URL of the token endpoint of the IdM server as specified in the "/<x>/OnNetwork/AppServerInfo/IDMSTokenEndpoint" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1)	-	-	-	-
6	The SS-UE1 (MCPTT client) sends an HTTP POST Request message to the IUT (MCPTT Server) over the TLS connection established to the IdM token endpoint (OIDC Token Request message) passing the authorization code sent in step 4.	-->	HTTP POST	-	-
7	Check: Does the IUT (MCPTT Server) send an HTTP 200 (OK) providing id_token, access_token and refresh token?	<--	HTTP 200 (OK)	1	P
7A	The SS-UE1 (MCPTT client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [6] (3GPP TS 33.310) to the HTTP Proxy as specified in ETSI TS 133 180 [31] (3GPP TS 33.180), using the configured URL of the HTTP Proxy as specified in the "/<x>/OnNetwork/AppServerInfo/HTTPproxy" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1)	-	-	-	-
8	The SS-UE1 (MCPTT client) sends an HTTP POST message presenting an access token sent in step 7 for Key Management Initialisation.	-->	HTTP POST	-	-
9	Check: Does the IUT (MCPTT Server) respond by sending identity-specific key information?	<--	HTTP 200 (OK)	2	P
10	The SS-UE1 (MCPTT client) sends an HTTP POST message presenting an access token for Key Material Request	-->	HTTP POST	-	-
11	Check: Does the IUT (MCPTT Server) respond by sending identity-specific key information?	<--	HTTP 200 (OK)	2	P
12	The SS-UE1 (MCPTT client) sends a SIP REGISTER request for service authorisation	-->	SIP REGISTER	-	-
13	Check: Does the IUT (MCPTT Server) respond by sending an SIP 200 (OK) message?	<--	SIP 200 (OK)	3	P
14	The SS-UE1 (MCPTT client) sends a SIP SUBSCRIBE - subscription to multiple documents simultaneously - containing the access token and a resource list mime body containing a list of the following documents: MCPTT UE Configuration document, MCPTT User Profile Configuration Document, and the MCPTT Service configuration document. The base URI of each list entry is set to the CMS XCAP-ROOT-URI	-->	SIP SUBSCRIBE	-	-
15	Check: Does the IUT (MCPTT Server) respond by sending an SIP 200 (OK) message?	<--	SIP 200 (OK)	4	P
16	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message containing the XCAP-URI of the documents?	<--	SIP NOTIFY	4	P
17	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
18	The SS-UE1 (MCPTT client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCPTT UE Configuration document	-->	HTTP GET	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
19	Check: Does the IUT (MCPTT Server) send the HTTP 200 (OK) message including the MCPTT UE Configuration Document?	<--	HTTP 200 (OK)	4	P
20	The SS-UE1 (MCPTT client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCPTT User Profile Configuration Document	-->	HTTP GET	-	-
21	Check: Does the IUT (MCPTT Server) send the HTTP 200 (OK) message including the MCPTT User Profile Configuration Document?	<--	HTTP 200 (OK)	4	P
22	The SS-UE1 (MCPTT client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCPTT Service Configuration Document	-->	HTTP GET	-	-
23	Check: Does the IUT (MCPTT Server) send the HTTP 200 (OK) message including the MCPTT Service Configuration Document?	<--	HTTP 200 (OK)	4	P
24	The SS-UE1 (MCPTT client) sends a SIP SUBSCRIBE containing the access token and a resource list mime body and a list of the Groups to be obtained. The base URI of each list entry is set to the GMS XCAP-ROOT-URI, and the MCPTT group ID identifies a group document	-->	SIP SUBSCRIBE	-	-
25	Check: Does the IUT (MCPTT Server) respond with a HTTP 200 (OK) message	<--	SIP 200 (OK)	5	P
26	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the UE that contains the XCAP-URI of the Group documents?	<--	SIP NOTIFY	5	P
27	The SS-UE1 (MCPTT client) sends a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
28	The SS-UE1 (MCPTT client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the Group Configuration document	-->	HTTP GET	-	-
29	Check: Does the IUT (MCPTT Server) send an HTTP 200 (OK) message including the Group Document 'MCPTT UE Configuration document'?	<--	HTTP 200 (OK)	5	P
29A	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the UE that contains the group key transport payloads (GKTP) document.	<--	SIP NOTIFY	5	P
29B	The SS-UE1 (MCPTT client) sends a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
30	The SS-UE1 (MCPTT client) sends a SIP PUBLISH request for update of PoC-settings. NOTE: The PoC-settings document contains the user profile index of the selected user profile.	-->	SIP PUBLISH	-	-
31	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK)?	<--	SIP 200 (OK)	6	P
32-33	Void	--	-	-	-

5.1.3.3 Specific message contents

Table 5.1.3.3-1: HTTP POST (Step 1, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition AUTH

Table 5.1.3.3-2: HTTP POST (Step 3, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition USERAUTH

Table 5.1.3.3-3: HTTP POST (Step 6, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TOKEN

Table 5.1.3.3-4: HTTP POST (Step 8, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition KMSINIT.

Table 5.1.3.3-5: HTTP POST (Step 10, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition KMSKEY.

Table 5.1.3.3-5A: HTTP 200 (OK) (Step 0B, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition UEINITIALCONFIG

Table 5.1.3.3-6: HTTP 200 (OK) (Step 2, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"text/html"		IETF RFC 2854 [33]	
Message-body				
HTML form	<!DOCTYPE html> <html> <body> <form action="/idms/userauth" method="post"> Username: <input type="text" name="user"> Password: <input type="password" name="password"><button type="submit">Login</button> </form> </body> </html>	"/idms/userauth" given by tsc_MCX_IdMS_userauth_UriPath is the URI to be used by the UE as request URI in the HTTP POST request for user authentication	HTML 4.01 Specification [34]	

Table 5.1.3.3-7: HTTP 200 (OK) (Step 7, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TOKEN

Table 5.1.3.3-8: HTTP 200 (OK) (Step 9, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition KMSINIT.

Table 5.1.3.3-9: HTTP 200 (OK) (Step 11, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition KMSKEY.

Table 5.1.3.3-10: HTTP 200 (OK) (Step 19, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition UECONFIG.

Table 5.1.3.3-11: HTTP 200 (OK) (Step 21, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition UEUSERPROF.

Table 5.1.3.3-12: HTTP 200 (OK) (Step 23, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition UESERVCONFIG.

Table 5.1.3.3-13: HTTP 200 (OK) (Step 29, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition GROUPCONFIG.

Table 5.1.3.3-14: HTTP 302 (Found) (Step 4, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.8-1, condition AUTH.

Table 5.1.3.3-14A: HTTP GET (Step 0A, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition UEINITIALCONFIG

Table 5.1.3.3-15: HTTP GET (Step 18, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition UECONFIG.

Table 5.1.3.3-16: HTTP GET (Step 20, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition UEUSERPROF.

Table 5.1.3.3-17: HTTP GET (Step 22, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition UESERVCONFIG.

Table 5.1.3.3-18: HTTP GET (Step 28, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition GROUPCONFIG

Table 5.1.3.3-19: SIP REGISTER (Step 12, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.13-1, condition SIP_REGISTER_INITIAL, CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	SIP URI of the home domain name		

Table 5.1.3.3-20: SIP SUBSCRIBE (Step 14, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition CONFIG

Table 5.1.3.3-20A: SIP SUBSCRIBE (Step 24, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3-20B			

Table 5.1.3.3-20B: Resource-lists in SIP SUBSCRIBE (Table 5.1.3.3-20A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition GROUPCONFIG

Table 5.1.3.3-21: SIP NOTIFY (Step 16, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			

Table 5.1.3.3-21A: SIP NOTIFY (Step 26, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			

Table 5.1.3.3-21B: SIP NOTIFY (Step 29A, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			
Message-body				
xcap-diff document	xcap-diff document as described in Table 5.1.3.3-21C			

Table 5.1.3.3-21C: Xcap-Diff Document (Table 5.1.3.3-21B)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.12-2, condition GROUPKEY

Table 5.1.3.3-22: SIP PUBLISH (Step 30, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		

Table 5.1.3.3-23: Void

Table 5.1.3.3-24: SIP 200 (OK) (Steps 13, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition REGISTER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**

Table 5.1.3.3-25: SIP 200 (OK) (Steps 15, 25, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			

Table 5.1.3.3-26: SIP 200 (OK) (Step 31, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition PUBLISH-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**

5.2 MCPTT Server - MCPTT Client / Configuration / Group Creation / Group Regroup Creation / Group Regroup Teardown

5.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and supports new MCPTT terminated group }
ensure that {
  when { the UE (MCPTT client) initiates creation of the terminated group by sending a HTTP PUT
request }
    then { IUT (MCPTT Server) responds to the UE (MCPTT Client) with a HTTP 201 (Created) }
            }
```

(2)

```
with { IUT (MCPTT Server) having establish terminated group }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates creation of the terminated temporary group by sending a
HTTP POST request }
    then { IUT (MCPTT Server) responds with a HTTP 200 (OK) containing the GMOP group-regroup-
creation-response }
            }
```

(3)

```
with { IUT (MCPTT Server) having establish a temporary group }
ensure that {
  when { the UE (MCPTT client) initiates tear down of the temporary group by sending a HTTP DELETE
request }
    then { IUT (MCPTT Server) responds to the UE (MCPTT Client) with HTTP 200 (OK) message }
            }
```

5.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 481 [27] (3GPP TS 24.481) clauses 6.3.2.3, 6.3.14.3.1 and 6.3.15.3.1; ETSI TS 133 180 [31] (3GPP TS 33.180), clauses 7.3.2 and 7.3.3.2. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 481 [27] (3GPP TS 24.481) clause 6.3.2.3]

"Upon reception of an HTTP PUT request where the Request-URI of the HTTP PUT request identifies an XML document of the application usage specified in subclause 7.2

then the GMS:

a) shall determine the identity of the sender of the received HTTP PUT request as specified in subclause 6.2.5;

b) if the identity of the sender of the received HTTP PUT request is not authorized to initiate group creation, shall respond with a HTTP 403 (Forbidden) response to the HTTP PUT request and shall not continue with rest of the steps;

c) if value of the "uri" attribute of the <list-service> element of the <group> element of the XML document specified in subclause 7.2.1 of the received HTTP PUT request does not conform to local policy, shall respond with an HTTP 409 (Conflict) response to the HTTP PUT request. The <uniqueness-failure> error element shall identify the error condition. The GMS shall include at least one <alt-value> element in the <uniqueness-failure> error element, whereby each <alt-value> element contains an MCS Group ID acceptable for the GMS. The GMS shall not continue with rest of the steps; and

d) shall support receiving an XML document of the application usage specified in subclause 7.2.1 according to procedures specified in IETF RFC 4825 [22] "PUT Handling" where the Request-URI of the HTTP PUT request identifies an XML document of the application usage specified in subclause 7.2."

[ETSI TS 124 481 [27] (3GPP TS 24.481) clause 6.3.14.3.1]

"Upon reception of an HTTP POST request:

a) with a Request-URI with an XCAP URI identifying a non-existing group document; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup creation specified in subclause 7.3.4.3;

then the GMS:

a) shall determine the identity of the sender of the received HTTP POST request as specified in subclause 6.2.5;

b) if the identity of the sender of the received HTTP POST request is not authorized to initiate temporary MCS group formation, shall respond with HTTP 403 (Forbidden) response to the HTTP POST request and shall not continue with rest of the steps;

c) if value of the "uri" attribute of the <list-service> element of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request does not conform to local policy, shall respond with an HTTP 409 (Conflict) response to the HTTP POST request. The <uniqueness-failure> error element shall identify the error condition. The GMS shall include at least one <alt-value> element in the <uniqueness-failure> error element, whereby each <alt-value> element contains a MCS Group ID acceptable for the GMS. The GMS shall not continue with rest of the steps; and

d) for each MCS group ID of an MCS group to be combined indicated in content of a <constituent-MCPTT-group-ID> element of the <constituent-MCPTT-group-IDs> element of the <on-network-temporary> element of the group document of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request:

1) shall send a HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.5. In the HTTP POST request, the GMS:

…

Upon reception of a HTTP 3xx, 4xx, or 5xx responses to a sent HTTP POST request or upon timeout, the GMS shall send a HTTP 403 (Forbidden) response to the received HTTP request and shall not continue with rest of the steps.

…

Upon reception of HTTP 2xx responses to all sent HTTP POST requests, the GMS shall create the group document of the temporary MCS group at the location specified by the Request-URI of the received HTTP POST request and shall send an HTTP 2xx response to the received HTTP request. In the HTTP 2xx response, the GMS shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document with group regroup creation response specified in subclause 7.3.4.6 with a <temporary-group-document-ETag> element set to the current value of the entity tag for the created group document of the temporary MCS group."

[ETSI TS 124 481 [27] (3GPP TS 24.481) clause 6.3.15.3.1]

"Upon reception of an HTTP DELETE request with Request-URI with an XCAP URI identifying a group document of a temporary MCS group, the GMS:

a) for each constituent MCS group indicated in the group document indicated by Request-URI:

1) shall send an HTTP DELETE request. In the HTTP DELETE request, the GMS:

…

Upon reception of an HTTP response to all sent HTTP DELETE requests, the GMS shall remove the group document of the temporary MCS group and shall send an HTTP 2xx response to the received HTTP request."

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 7.3.2]

"The group creation procedure is described in clause 10.2.3 of 3GPP TS 23.280 [36] and applies to the MCPTT scenario of normal group creation by an MC administrator and user regrouping operations by an authorized user/dispatcher. To establish the security context for the group, the GMS follows the procedures in clause 5.7 to create a new GMK and GMK-ID.

The encapsulated GMK and GUK-ID is sent to group members by the GMS within a notification message (Step 4 in clause 10.2.3 of 3GPP TS 23.280 [36]). The procedure is equivalent to that described in clause 5.7 of this specification."

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 7.3.3.2]

"Group Regroup procedures for the MC system are described in clause 10.2.4.1 of 3GPP TS 23.280 [36]. To create the security context for the temporary group, the GMS follows the procedures in clause 5.7, creating a new GMK and GMK-ID for the temporary group.

An encapsulated GMK and GUK-ID is sent to the temporary group members by the GMS within a notification message (Step 5 in clause 10.2.4.1 of 3GPP TS 23.280 [36]). The procedure is equivalent to that described in clause 5.7."

5.2.3 Test description

5.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Group Management Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) is member of group A.

5.2.3.2 Test procedure sequence

Table 5.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCX Server Terminated Group Creation' as described in Annex A, Table A.2.4.3-1 to create group B?	-	-	1	P
2	Check: Does the MCPTT Server correctly perform procedure 'MCX Server Terminated Temporary Group Creation' as described in Annex A, Table A.2.5.3-1 to create temporary group T?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document of group A and retrieve the GMK for the temporary group T	-	-	2	P
4	Check: Does the MCPTT Server correctly perform procedure 'MCX Server Terminated Temporary Group Tear Down ' as described in Annex A, Table A.2.6.3-1 to delete temporary group T?	-	-	3	P
5	Check: Does the MCPTT Server correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document and the GKTP document of group A?	-	-	3	P

5.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications:

- Subclauses referring to messages from UE relate to SS-UE1 (MCX Client).

5.3 MCPTT Server - MCPTT Client / Configuration / Group Affiliation / Implicit Affiliation / Remote change / De-affiliation / Home MCPTT system

5.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group}
ensure that {
  when { the SS-UE1 (MCPTT client ) requests for own affiliation status and to subscribe to
affiliation status changes for the MCPTT User with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(2)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to de-affiliate user A from an MCPTT group with SIP
PUBLISH }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
            }
```

(3)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to affiliate user A to an MCPTT group with SIP PUBLISH
message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
            }
```

(4)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group }
ensure that {
  when { the SS-UE2 (MCPTT client) requests to de-affiliate user B from an MCPTT group with SIP
PUBLISH message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
            }
```

(5)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is allowed to make affiliation
changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to affiliate status changes of targeted user with SIP
SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(6)

with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a pointer to the group information, indicating that the MCPTT Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to affiliate targeted user B to an MCPTT group via mandatory mode with SIP PUBLISH message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(7)

with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a pointer to the group information, indicating that the MCPTT Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to de-affiliate targeted user from an MCPTT group via mandatory mode with SIP PUBLISH message }
    then { IUT (MCPTT Server) responds to the SS-UE (MCPTT client) with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(8)

with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a pointer to the group information, indicating that the MCPTT Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { SS-UE1 (MCPTT Client) requests that a target user B be affiliated to an MCPTT group via negotiated mode by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }

(9)

with { MCPTT Client already affiliated to the MCPTT group }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to affiliate targeted user to a group via negotiated mode }
    then { IUT (MCPTT Server) informs user B about a remote request by sending a SIP MESSAGE message to the SS-UE2 (MCPTT Client) }
            }

(10)

with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with the specified group }
ensure that {
  when { the SS-UE2 (MCPTT client) requests to affiliate user B to an MCPTT group with SIP PUBLISH message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(11)

with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a pointer to the group information, indicating that the MCPTT Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { SS-UE1 (MCPTT Client) requests that a target user B be de-affiliated to an MCPTT group via negotiated mode by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }

(12)

with { MCPTT Client already de-affiliated from the MCPTT group }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to de-affiliate targeted user to a group via negotiated mode }
    then { IUT (MCPTT Server) informs user B about a remote request by sending a SIP MESSAGE message to the SS-UE2 (MCPTT Client) }
            }

(13)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group}
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to de-subscribe from affiliation status changes for the
MCPTT User with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
          }
```

5.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.2.2.3, 9.2.2.2.4, 9.2.2.2.5, 9.2.2.2.8 and 9.2.2.2.15. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user affiliation information according to clause 9.3.1;*

then the MCPTT server:

1) *shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;*

2) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;*

3) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;*

4) *if the originating MCPTT ID is different than the served MCPTT ID and the originating MCPTT ID is not authorized to modify affiliation status of the served MCPTT ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps;*

5) *if the Expires header field of the SIP PUBLISH request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP PUBLISH request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;*

6) *if the Expires header field of the SIP PUBLISH request has nonzero value, shall determine the candidate expiration interval to according to IETF RFC 3903 [37];*

7) *if the Expires header field of the SIP PUBLISH request has zero value, shall set the candidate expiration interval to zero;*

8) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37]. In the SIP 200 (OK) response, the MCPTT server:

a) shall set the Expires header field according to IETF RFC 3903 [37], to the candidate expiration time;

9) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCPTT ID, shall not continue with the rest of the steps;

10) shall identify the served MCPTT client ID in the "id" attribute of the <tuple> element of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request;

11) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

12) shall consider an MCPTT client information entry such that:

a) the MCPTT client information entry is in the list of MCPTT client information entries of the served MCPTT user information entry; and

b) the MCPTT client ID of the MCPTT client information entry is equal to the served MCPTT client ID;

as the served MCPTT client information entry;

13) shall consider a copy of the list of the MCPTT group information entries of the served MCPTT client information entry as the served list of the MCPTT group information entries;

14) if the candidate expiration interval is nonzero:

a) shall construct the candidate list of the MCPTT group information entries as follows:

i) for each MCPTT group ID which has an MCPTT group information entry in the served list of the MCPTT group information entries, such that the expiration time of the MCPTT group information entry has not expired yet, and which is indicated in a "group" attribute of an <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall copy the MCPTT group information entry into a new MCPTT group information entry of the candidate list of the MCPTT group information entries;

B) if the affiliation status of the MCPTT group information entry is "deaffiliating" or "deaffiliated", shall set the affiliation status of the new MCPTT group information entry to the "affiliating" state and shall set the affiliating p-id of the new MCPTT group information entry to the value of the <p-id> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

C) shall set the expiration time of the new MCPTT group information entry to the current time increased with the candidate expiration interval;

ii) for each MCPTT group ID which has an MCPTT group information entry in the served list of the MCPTT group information entries, such that the expiration time of the MCPTT group information entry has not expired yet, and which is not indicated in any "group" attribute of the <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall copy the MCPTT group information entry into a new MCPTT group information entry of the candidate list of the MCPTT group information entries; and

B) if the affiliation status of the MCPTT group information entry is "affiliated" or "affiliating":

- *shall set the affiliation status of the new MCPTT group information entry to the "de-affiliating" state; and*
- *shall set the expiration time of the new MCPTT group information entry to the current time increased with twice the value of timer F; and*

iii) for each MCPTT group ID:

A) which does not have an MCPTT group information entry in the served list of the MCPTT group information entries; or

B) which has an MCPTT group information entry in the served list of the MCPTT group information entries, such that the expiration time of the MCPTT group information entry has already expired;

and which is indicated in a "group" element of the <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall add a new MCPTT group information entry in the candidate list of the MCPTT group information list for the MCPTT group ID;

B) shall set the affiliation status of the new MCPTT group information entry to the "affiliating" state;

C) shall set the expiration time of the new MCPTT group information entry to the current time increased with the candidate expiration interval; and

D) shall set the affiliating p-id of the new MCPTT group information entry to the value of the <p-id> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request;

b) determine the candidate number of MCPTT group IDs as number of different MCPTT group IDs which have an MCPTT group information entry:

i) in the candidate list of the MCPTT group information entries; or

ii) in the list of the MCPTT group information entries of an MCPTT client information entry such that:

A) the MCPTT client information entry is in the list of the MCPTT client information entries of the served MCPTT user information entry; and

B) the MCPTT client ID of the MCPTT client information entry is not equal to the served MCPTT client ID;

with the affiliation status set to the "affiliating" state or the "affiliated" state and with the expiration time which has not expired yet; and

c) if the candidate number of MCPTT group IDs is bigger than N2 value of the served MCPTT ID, shall based on MCPTT service provider policy reduce the candidate MCPTT group IDs to that equal to N2;

NOTE: The MCPTT service provider policy can determine to remove an MCPTT group ID based on the order it appeared in the PUBLISH request or based on the importance or priority of the MCPTT group or some other policy to determine which MCPTT groups are preferred.

15) if the candidate expiration interval is zero, constructs the candidate list of the MCPTT group information entries as follows:

a) for each MCPTT group ID which has an entry in the served list of the MCPTT group information entries:

i) shall copy the MCPTT group entry of the served list of the MCPTT group information into a new MCPTT group information entry of the candidate list of the MCPTT group information entries;

ii) shall set the affiliation status of the new MCPTT group information entry to the "de-affiliating" state; and

iii) shall set the expiration time of the new MCPTT group information entry to the current time increased with twice the value of timer F;

16) shall replace the list of the MCPTT group information entries stored in the served MCPTT client information entry with the candidate list of the MCPTT group information entries;

…

18) shall identify the handled p-id in the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

19) shall perform the procedures specified in clause 9.2.2.2.5 for the served MCPTT ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.4]

"*Upon receiving a SIP SUBSCRIBE request such that:*

1) Request-URI of the SIP SUBSCRIBE request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

the MCPTT server:

1) shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

4) if the originating MCPTT ID is different than the served MCPTT ID and the originating MCPTT ID is not authorized to modify affiliation status of the served MCPTT ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps; and

5) shall generate a 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify the subscriber about changes of the information of the served MCPTT ID, as described in clause 9.2.2.2.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.5]

"In order to notify the subscriber about changes of the served MCPTT ID, the MCPTT server:

1) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

2) *shall consider the list of the MCPTT client information entries of the served MCPTT user information entry as the served list of the MCPTT client information entries;*

3) *shall generate an application/pidf+xml MIME body indicating per-user affiliation information according to clause 9.3.1 and the served list of the MCPTT client information entries with the following clarifications:*

   a) *the MCPTT server shall not include information from an MCPTT group information entry with the expiration time already expired;*

   b) *the MCPTT server shall not include information from an MCPTT group information entry with the affiliation status set to the "deaffiliated" state;*

   c) *if the SIP SUBSCRIBE request creating the subscription of this notification contains an application/simple-filter+xml MIME body indicating per-client restrictions of presence event package notification information according to clause 9.3.2, the MCPTT server shall restrict the application/pidf+xml MIME body according to the application/simple-filter+xml MIME body; and*

   d) *if this procedures is invoked by procedure in clause 9.2.2.2.3 where the handled p-id value was identified, the MCPTT server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and*

4) *send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9.2.2.2.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-user affiliation information."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.8]

"Upon receiving a SIP MESSAGE request such that:

1) *Request-URI of the SIP MESSAGE request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;*

2) *the SIP MESSAGE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element and the <mcptt-calling-user-id> element;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and*

4) *the SIP MESSAGE request contains an application/vnd.3gpp.mcptt-affiliation-command+xml MIME body;*

then the MCPTT server:

1) *shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP MESSAGE request;*

2) *shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP MESSAGE request;*

3) *if the originating MCPTT ID is not authorized to send an affiliation status change request in negotiated mode to the served MCPTT ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps;*

4) *shall set the Request-URI of the SIP MESSAGE request to the public user identity bound to the served MCPTT ID in the MCPTT server; and*

5) *shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

before forwarding the SIP MESSAGE request further."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.15]

"If the participating MCPTT function has successfully performed service authorization for the MCPTT ID identified in the service authorisation procedure as described in 3GPP TS 33.180 [78], the participating MCPTT function:

1) shall identify the MCPTT ID included in the SIP request received for service authorisation procedure as the served MCPTT ID;

2) shall identify the MCPTT client ID from the <mcptt-client-id> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body included in the SIP request received for service authorisation as the served MCPTT client ID;

3) shall download the MCPTT user profile from the MCPTT user database as defined in 3GPP TS 29.283 [73] if not already stored at the participating MCPTT function;

4) if no <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) for the served MCPTT ID or the <ImplicitAffiliations> element contains no <entry> elements containing an MCPTT group ID, shall skip the remaining steps;

5) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

6) shall consider an MCPTT client information entry such that:

a) the MCPTT client information entry is in the list of MCPTT client information entries of the served MCPTT user information entry; and

b) the MCPTT client ID of the MCPTT client information entry is equal to the served MCPTT client ID;

as the served MCPTT client information entry;

7) shall consider a copy of the list of the MCPTT group information entries of the served MCPTT client information entry as the served list of the MCPTT group information entries;

8) shall construct the candidate list of the MCPTT group information entries as follows:

a) for each MCPTT group ID which has an MCPTT group information entry in the served list of the MCPTT group information entries shall copy the MCPTT group information entry into a new MCPTT group information entry of the candidate list of the MCPTT group information entries;

b) for each MCPTT group ID contained in an <entry> element of the <ImplicitAffiliations> element in the <OnNetwork> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) for the served MCPTT ID that does not have an MCPTT group information entry in the served list of the MCPTT group information entries or has an MCPTT group information entry in the served list of the MCPTT group information entries such that the expiration time of the MCPTT group information entry has already expired:

i) shall add a new MCPTT group information entry in the candidate list of the MCPTT group information list for the MCPTT group ID;

ii) shall set the affiliation status of the new MCPTT group information entry to the "affiliating" state; and

iii) shall set the expiration time of the new MCPTT group information entry to the current time increased with the candidate expiration interval;

…

9) determine the candidate number of MCPTT group IDs as the number of different MCPTT group IDs which have an MCPTT group information entry:

a) in the candidate list of the MCPTT group information entries; or

b) in the list of the MCPTT group information entries of an MCPTT client information entry such that:

i) the MCPTT client information entry is in the list of the MCPTT client information entries of the served MCPTT user information entry; and

ii) the MCPTT client ID of the MCPTT client information entry is not equal to the served MCPTT client ID;

with the affiliation status set to the "affiliating" state or the "affiliated" state and with the expiration time which has not expired yet; and

…

10) shall replace the list of the MCPTT group information entries stored in the served MCPTT client information entry with the candidate list of the MCPTT group information entries; and

11) for each MCPTT group ID contained in an <entry> element of the <ImplicitAffiliations> element in the <OnNetwork> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) for the served MCPTT ID and which has an MCPTT group information entry in the candidate list of the MCPTT group information entries with an affiliation status of "affiliating", shall perform the procedures specified in clause 9.2.2.2.6 for the served MCPTT ID and each MCPTT group ID."

5.3.3 Test description

5.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

5.3.3.2 Test procedure sequence

Table 5.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.3.7.3-1 to subscribe the SS-UE1 (MCPTT client) to own affiliation status changes for user A? (see note)	-	-	1	P
2	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to de-affiliate user A from group A using the SS-UE1 (MCPTT client)?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to affiliate user A to group A using the SS-UE1 (MCPTT client)?	-	-	3	P
4	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.3.7.3-1 to subscribe the SS-UE2 (MCPTT client) to own affiliation status changes for user B? (see note)	-	-	1	P
5	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to de-affiliate user B from group A using the SS-UE2 (MCPTT client)?	-	-	4	P
6	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.3.7.3-1 to subscribe the SS-UE1 (MCPTT client) to affiliation status changes of targeted user B?	-	-	5	P
7	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to request from the SS-UE1 (MCPTT Client) to affiliate targeted user B to group A (mandatory mode)?	-	-	6	P
8	Check: Does the MCPTT Server correctly perform steps 3-6 of procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to inform user B in the SS-UE2 (MCPTT client) that is being remotely affiliated to group A?	-	-	6	P
9	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to request from the SS-UE1 (MCPTT Client) to de-affiliate targeted user B from group A (mandatory mode)?	-	-	7	P
10	Check: Does the MCPTT Server correctly perform steps 3-6 of procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to inform user B in the SS-UE2 (MCPTT Client) that is being remotely de-affiliated from group A?	-	-	7	P
11	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating	-	-	8	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to request from the SS-UE1 (MCPTT Client) to affiliate targeted user B to group A via negotiated mode?				
12	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to inform user B in the SS-UE2 (MCPTT Client) about a remote request to affiliate to group A via negotiated mode?	-	-	9	P
13	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to affiliate user B to group A using the SS-UE2 (MCPTT client)?	-	-	10	P
14	Check: Does the MCPTT Server correctly perform steps 3-6 of procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to inform user A in the SS-UE1 (MCPTT Client), that user B is being affiliated to group A?	-	-	10	P
15	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to request from the SS-UE1 (MCPTT Client) to de-affiliate targeted user B to group A via negotiated mode?	-	-	11	P
16	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to inform user B in the SS-UE2 (MCPTT Client) about a remote request to de-affiliate to group A via negotiated mode	-	-	11	P
17	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to de-affiliate user B to group A using the SS-UE2 (MCPTT client)?	-	-	4	P
18	Check: Does the MCPTT Server correctly perform steps 3-6 of procedure 'MCPTT Group Affiliation status change' as described in Annex A, Table A.3.8.3-1 to inform user A in the SS-UE1 (MCPTT Client), that user B is being de-affiliated from group A?	-	-	4	P
19	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.3.7.3-1 to de-subscribe the SS-UE1 (MCPTT Client) from affiliation status changes for targeted user B?	-	-	13	P
NOTE: Affiliation status change triggered by implicit affiliation.					

5.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.3.3.3-1: SIP SUBSCRIBE from the SS-UE1 (Step 1, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Annex A, Table A.2.7.4-2			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.3.3.3-2			

Table 5.3.3.3-2: Simple-Filter in SIP SUBSCRIBE (Table 5.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.6-1, condition PER-CLIENT

Table 5.3.3.3-3: SIP SUBSCRIBE from the SS-UE2 (Step 4, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [15]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-4			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.3.3.3-2			

Table 5.3.3.3-4: MCPTT-Info in SIP SUBSCRIBE (Table 5.3.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.3.3.3-5: SIP SUBSCRIBE from the SS-UE1 (Step 6, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-4			

Table 5.3.3.3-6: SIP SUBSCRIBE from the SS-UE1 (Step 19, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) , Table A.3.7.4-1, condition re-SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-4			

Table 5.3.3.3-7: SIP PUBLISH from the SS-UE1 (Step 2, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			

Table 5.3.3.3-8: SIP PUBLISH from the SS-UE1 (Step 3, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1

Table 5.3.3.3-9: SIP PUBLISH from the SS-UE2 (Steps 5 and 17, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [15]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-10			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-11			

Table 5.3.3.3-10: MCPTT-Info in SIP PUBLISH (Table 5.3.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.3.3.3-11: PIDF in SIP PUBLISH (Table 5.3.3.3-9)

Derivation Path: Annex A, Table A.3.8.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status	not present			
..p-id	any allowed value	p-id shall be present acc. to ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5 and clause 9.2.2.3.5		
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.3.3.3-12: SIP PUBLISH from the SS-UE1 (Step 7, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-13			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-14			

Table 5.3.3.3-13: MCPTT-Info in SIP PUBLISH (Table 5.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.3.3.3-14: PIDF in SIP PUBLISH (Table 5.3.3.3-12)

Derivation Path: Annex A, Table A.3.8.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status	not present			

Derivation Path: Annex A, Table A.3.8.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
..p-id	any allowed value	p-id shall be present acc. to ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5 and clause 9.2.2.3.5		
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.3.3.3-15: SIP PUBLISH from the SS-UE1
(Step 9, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-13			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-14			

Table 5.3.3.3-16: SIP PUBLISH from the SS-UE2
(Step 13, Table 5.3.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-10			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-11			

Table 5.3.3.3-17: SIP NOTIFY from the MCPTT Server
(Step 1, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-18			

Table 5.3.3.3-18: PIDF in SIP NOTIFY (Table 5.3.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			

Table 5.3.3.3-19: SIP NOTIFY from the MCPTT Server (Step 3, Table 5.3.3.2-1; Step 5, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-20			

Table 5.3.3.3-20: PIDF in SIP NOTIFY (Table 5.3.3.3-19)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 3)			

Table 5.3.3.3-21: SIP NOTIFY from the MCPTT Server (Step 3, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-22			

Table 5.3.3.3-22: PIDF in SIP NOTIFY (Table 5.3.3.3-21)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 3)			

Table 5.3.3.3-23: SIP NOTIFY from the MCPTT Server (Step 2, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-24			

Table 5.3.3.3-24: PIDF in SIP NOTIFY (Table 5.3.3.3-23)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"deaffiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 2)			

Table 5.3.3.3-25: SIP NOTIFY from the MCPTT Server (Step 2, Table 5.3.3.2-1; Step 5, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-26			

Table 5.3.3.3-26: PIDF in SIP NOTIFY (Table 5.3.3.3-25)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	not present			

Table 5.3.3.3-27: SIP NOTIFY from the MCPTT Server (Step 4, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-28			

Table 5.3.3.3-28: PIDF in SIP NOTIFY (Table 5.3.3.3-27)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliated"			
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.13.3-1.				

Table 5.3.3.3-29: SIP NOTIFY from the MCPTT Server (Steps 7, 8, 13, 14, Table 5.3.3.2-1; Step 5, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-30			

Table 5.3.3.3-30: PIDF in SIP NOTIFY (Table 5.3.3.3-29)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note1) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliated"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 7 and 13)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-31: SIP NOTIFY from the MCPTT Server (Steps 7, 8, 13, 14, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-32			

Table 5.3.3.3-32: PIDF in SIP NOTIFY (Table 5.3.3.3-31)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 7 and 13)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-33: SIP NOTIFY from the MCPTT Server (Steps 5, 9, 10, 17, 18, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-34			

Table 5.3.3.3-34: PIDF in SIP NOTIFY (Table 5.3.3.3-33)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"deaffiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 5, 9 and 17)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-35: SIP NOTIFY from the MCPTT Server (Steps 5, 6, 9, 10, 17, 18, Table 5.3.3.2-1; Step 5, Annex A, Table A.3.8.3-1; Step 3, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-36			

Table 5.3.3.3-35: PIDF in SIP NOTIFY (Table 5.3.3.3-36)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
status	not present			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-36: SIP NOTIFY from the MCPTT Server (Step 19, Table 5.3.3.2-1; Step 3, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [21]	
substate-value	"terminated"			
expires	not present			
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-35			

Table 5.3.3.3-37: SIP MESSAGE from the SS-UE1 (Step 11, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-38			

Table 5.3.3.3-38: MCPTT-Info in SIP MESSAGE (Table 5.3.3.3-37)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.3.3.3-39: SIP MESSAGE from the MCPTT Server (Step 12, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-40			

Table 5.3.3.3-40: MCPTT-Info in SIP MESSAGE (Table 5.3.3.3-39)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1A.				

Table 5.3.3.3-41: SIP MESSAGE from the SS-UE1 (Step 15, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Affiliation-Command**		
MIME-part-body	Affiliation-Command as described in Table 5.3.3.3-42			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-43			

Table 5.3.3.3-42: MCPTT-Info in SIP MESSAGE (Table 5.3.3.3-41)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.3.3.3-43: Affiliation-Command in SIP MESSAGE (Table 5.3.3.3-41)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
command-list				
affiliate	not present			
de-affiliate				
group[1]	px_MCPTT_Group_A_ID	MCPTT group name		

Table 5.3.3.3-44: SIP MESSAGE from the MCPTT Server (Step 16, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Affiliation-Command**		
MIME-part-body	Affiliation-Command as described in Table 5.3.3.3-45			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.3.3.3-46			

Table 5.3.3.3-45: MCPTT-Info in SIP MESSAGE (Table 5.3.3.3-44)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B			
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1A.				

Table 5.3.3.3-46: Affiliation-Command in SIP MESSAGE (Table 5.3.3.3-44)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
command-list				
affiliate	not present			
de-affiliate				
group[1]	px_MCPTT_Group_A_ID	MCPTT group name		

5.4 MCPTT Server - MCPTT Client / Configuration / Pre-established Session Establishment / Pre-established Session Modification / Pre-established Session Release

5.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) is able to create pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests creation of pre-established session with sending SIP
INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
          }
```

(2)

```
with { IUT (MCPTT Server) is able to create/modify Terminated session without provisional responses
except 100 Trying }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests creation modification session by usage of implicit
control with sending SIP re-INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
          }
```

(3)

```
with { IUT (MCPTT Server) is able to release pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to terminate MCX session with sending SIP BYE message }
    then { IUT (MCPTT Server) responds with 200 (OK) message }
          }
```

(4)

```
with { IUT (MCPTT Server) is able to modify pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests modification session by usage of implicit control with
sending SIP UPDATE message }
    then { IUT (MCPTT Server) responds with 200 (OK) message }
          }
```

(5)

```
with { IUT (MCPTT Server) is able to modify pre-established Originating session }
ensure that {
  when { the IUT (MCPTT Server) triggers pre-established session modification }
    then { IUT (MCPTT Server) requests modification session with sending SIP re-INVITE message }
           }
```

(6)

```
with { IUT (MCPTT Server) is able to modify pre-established Originating session }
ensure that {
  when { the IUT (MCPTT Server) triggers pre-established session modification }
    then { the IUT (MCPTT Server) requests modification session by sending SIP UPDATE message }
           }
```

(7)

```
with { IUT (MCPTT Server) is able to release pre-established Originating session }
ensure that {
  when { the IUT (MCPTT Server) triggers release session }
    then { the IUT (MCPTT Server) requests release session sending SIP BYE message }
```

5.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 8.2.2, 8.3.2.1, 8.3.2.2, 8.4.2.1 and 8.4.2.2. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 8.2.2]

"Upon receipt of a "SIP INVITE request for establishing a pre-established session" the participating MCPTT function:

1) shall check whether the public service identity is allocated and perform the actions specified in clause 6.3.7.1 if it is not allocated. Otherwise, continue with the rest of the steps;

2) shall determine the MCPTT ID of the MCPTT user establishing the pre-established session and perform actions to verify the MCPTT ID of the MCPTT client and authorise the request according to local policy, and if not authorised, the participating MCPTT function shall return a SIP 403 (Forbidden) response with the warning text set to "100 function not allowed due to <detailed reason>" as specified in clause 4.4. Otherwise, continue with the rest of the steps;

3) shall determine whether resource sharing is supported (see clause 8.1A);

4) if resource sharing is supported by the SIP core, determine that there is a binding between the MCPTT ID of the MCPTT user establishing the pre-established session and the MCPTT UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request (see clause 8.1A) and that this UE identity matches the identity in the "+g.3gpp.registration-token" header field parameter in the Feature-Caps header field in the "SIP INVITE request for establishing a pre-established session";

5) if resource sharing is not supported or if there is no binding between the MCPTT ID of the MCPTT user and the identity of the MCPTT UE identified by the "+g.3gpp.registration-token" header field parameter in the Feature-Caps header field, then the participating MCPTT function shall return a SIP 403 (Forbidden) response with the warning text set to "100 function not allowed due to pre-established session not supported" as specified in clause 4.4 and not continue with the rest of the steps;

6) shall validate the media parameters and if the MCPTT speech codec is not offered in the SIP INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

7) shall verify that the media resources are available to support the media parameters and if not shall reject the request with a SIP 500 (Server Internal Error) response, and shall not continue with the rest of the steps;

8) shall allocate a URI to be used to identify the pre-established session;

9) shall generate a SIP 200 (OK) response to the SIP INVITE request according to clause 6.3.2.1.5.2; and

a) shall include a Contact header field containing the URI that identifies the pre-established session;

b) shall include the public service identity in the P-Asserted-Identity header field;

c) shall include a Supported header field containing the "norefersub" option tag;

d) shall if the SIP core supports resource sharing, include a Resource-Share header field answer as specified in 3GPP TS 24.229 [4] with:

A) the value "media-sharing";

B) an "origin" header field parameter set to "session-initiator";

C) a "timestamp" header field parameter; and

D) a "rules" header field parameter with one resource sharing rule per media stream in the same order the corresponding m-line appears in the SDP. Each resource sharing rule is constructed as follows:

- *a "new-sharing-key" part; and*

- *a "directionality" part indicating the direction of the pre-established media stream; and*

e) shall include an SDP answer as specified in 3GPP TS 24.229 [4] with the clarifications in clause 6.3.2.1.2.2 and include ICE candidates in the SDP answer as per IETF RFC 8839 [90];

10) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 1: Resulting media plane processing is completed before the next step is performed.

11) shall send the SIP 200 (OK) response towards the MCPTT client according to the rules and procedures of the 3GPP TS 24.229 [4]; and

12) shall evaluate the ICE candidates according to IETF RFC 8445 [89].

NOTE 2: If ICE candidate evaluation results in candidate pairs other than the default candidate pair being selected a further offer answer exchange using the procedures in clause 8.3 will be needed."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 8.3.2.1]

"Upon receiving a SIP UPDATE request or a SIP re-INVITE request to modify an existing pre-established session without associated MCPTT call, the participating MCPTT function:

1) shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec is acceptable by the participating MCPTT function and if not reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps; and

2) shall generate a SIP 200 (OK) response as follows:

a) include an SDP answer according to 3GPP TS 24.229 [4] based on the received SDP offer with the clarifications given in the 6.3.2.1.2.2, and include ICE candidates in the SDP answer as per IETF RFC 8839 [90], if required; and

b) include a Contact header field containing the URI that identifies the pre-established session and send a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 8.3.2.2]

"When the participating MCPTT function needs to modify the pre-established session outside of an MCPTT call, the participating MCPTT function:

1) shall generate a SIP UPDATE request or a SIP re-INVITE request according to 3GPP TS 24.229 [4];

2) shall include an SDP offer according to 3GPP TS 24.229 [4], and include ICE candidates in the SDP offer as per IETF RFC 8839 [90], if required;

3) shall interact with the media plane as specified in 3GPP TS 24.380 [5], if removing a media-floor control entity; and

4) shall send the SIP request towards the MCPTT client according to rules and procedures of 3GPP TS 24.229 [4].

On receipt of the SIP 200 (OK) response the participating MCPTT function:

1) shall interact with media plane as specified in 3GPP TS 24.380 [5], if there is change in media parameters or codecs in the received SDP answer, compared to those in the previously agreed SDP;

2) shall interact with media plane as specified in 3GPP TS 24.380 [5], if there is a media stream, that is currently used in the pre-established session, marked as rejected in the received SDP answer; and

3) shall interact with media plane as specified in 3GPP TS 24.380 [5], if there is a media stream accepted in the received SDP answer, that is not currently used by the participant in the pre-established session.

NOTE: The participating MCPTT function keeps resources for previously agreed media stream, media-floor control entities, media parameters and codecs until it receives a SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 8.4.2.1]

"Upon receiving a SIP BYE request from the MCPTT client within a pre-established session the participating MCPTT function:

1) shall check whether there is a MCPTT call using the pre-established session, and:

a) if there is an established MCPTT session then the participating MCPTT function shall remove the MCPTT client from the MCPTT session by performing the procedures as specified in clause 6.3.2.1.6; and

b) if there is a MCPTT session in the process of being established, then the participating MCPTT function:

i) shall send a SIP CANCEL request to cancel the MCPTT session in the process of being established as specified in 3GPP TS 24.229 [4]; and

ii) shall release the MCPTT session as specified in the clause clause 6.3.2.1.6, if a SIP 200 (OK) response for the SIP INVITE request is received from the remote side; and

c) if there is no MCPTT call using the pre-established session, then the participating MCPTT function shall:

i)_ interact with the media plane as specified in 3GPP TS 24.380 [5] for disconnecting the media plane resources towards the MCPTT client; and

ii) shall generate and send a SIP 200 (OK) response to the SIP BYE request according to rules and procedures of 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request from the remote side, the participating MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources towards the remote side;

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources towards the MCPTT client; and

3) shall send a SIP 200 (OK) response to the SIP BYE request to the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 8.4.2.2]

"When a participating MCPTT function needs to release a pre-established session as created in clause 8.2.2, the participating MCPTT function:

1) *shall first release any participants of all MCPTT calls that are using the pre-established session using the procedures as follows. The participating MCPTT function:*

   a) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];*

   b) *shall generate a SIP BYE request as specified in 3GPP TS 24.229 [4];*

   c) *shall set the Request-URI to the MCPTT session identity;*

   d) *shall set the contents of the P-Asserted-Identity header field to the P-Asserted-Identity header field of the MCPTT client that's pre-established session is being release;*

   e) *shall send the SIP BYE request toward the controlling MCPTT function, according to 3GPP TS 24.229 [4]; and*

   f) *shall, upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCPTT function shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];*

2) *shall generate a SIP BYE request according to rules and procedures of 3GPP TS 24.229 [4];*

3) *shall set the Request-URI of the SIP BYE request to the URI that identifies the pre-established session;*

4) *shall send the SIP BYE request towards the MCPTT client within the SIP dialog of the pre-established session according to rules and procedures of the 3GPP TS 24.229 [4]; and*

5) *shall, upon receiving a SIP 200 (OK) response to the SIP BYE request interact with the media plane as specified in 3GPP TS 24.380 [5]."*

5.4.3 Test description

5.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

5.4.3.2 Test procedure sequence

Table 5.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure 'MCX Participating Terminated pre-established session establishment' as described in Annex A, Table A.2.13.3-1 to create a pre-established session without implicit floor request?	-	-	1	P
2	Check: Does the MCPTT server correctly perform test procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.2.14.3-1 to modify the pre-established session by requesting usage of implicit floor control?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session?	-	-	3	P
4	Check: Does the MCPTT server correctly perform test procedure 'MCX Participating Terminated pre-established session establishment' as described in Annex A, Table A.2.13.3-1 to create a pre-established session without implicit floor request?	-	-	1	P
5	The SS-UE1 (MCPTT client) sends a SIP UPDATE message to modify the pre-established session by requesting usage of implicit floor control	-->	SIP UPDATE	4	-
6	Check: Does the MCPTT Server respond to the SIP UPDATE message with a SIP 200 (OK) message	<--	SIP 200 (OK)	4	P
-	EXCEPTION: Steps 7a1-8a1 describe a behaviour that depends on the MCPTT Server implementation. The "lower case letter" identifies a step sequence that takes place if the server is capable to provide pre-established session modification, by using SIP UPDATE or SIP re-INVITE messages.	-	-	-	-
7a1	Check: Does the MCPTT server correctly perform test procedure 'MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.2.16.3-1 to modify the floor priority of the pre-established session?	-	-	5	P
8a1	Check: Does the MCPTT Server send a SIP UPDATE message to modify the pre-established session?	<--	SIP UPDATE	6	P
8a2	The SS-UE1 responds to the SIP UPDATE message with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
9	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Originated call release' as described in Annex A, Table A.2.17.3-1 to release the pre-established session?	-	-	7	P

5.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.4.3.3-1: SIP INVITE from the SS-UE1
(Steps 1 and 4, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.13.3-1)

Derivation Path: Annex A, Table A.2.13.4-1, condition MCPTT

Table 5.4.3.3-2: SIP 200 (OK) from the MCPTT Server (Steps 1 and 4, Table 5.4.3.2-1; Step 3, Annex A, Table A.2.13.3-1)

Derivation Path: Annex A, Table A.2.13.4-2, condition MCPTT

Table 5.4.3.3-3: SIP re-INVITE from the SS-UE1 (Step 2, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.2.14.4-1, condition MCPTT and re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact	Contact header with the same Contact URI and the same mandatory feature parameters as in the INVITE creating the dialog			
Message-body				
SDP Message	As described in Table 5.4.3.3-4			

Table 5.4.3.3-4: SDP in SIP re-INVITE (Table 5.4.3.3-3) or SIP UPDATE (Table 5.4.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER, PRE_ESTABLISHED_SESSION, IMPLICIT_GRANT_REQUESTED

Table 5.4.3.3-5: SIP 200 (OK) from the MCPTT Server (Step 2, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.2.13.4-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 5.4.3.3-6			

Table 5.4.3.3-6: SDP in 200 (OK) (Table 5.4.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER, PRE_ESTABLISHED_SESSION, IMPLICIT_GRANT_REQUESTED

Table 5.4.3.3-7: SIP UPDATE from the SS-UE1 (Step 5, Table 5.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.15.1-1, condition MO_CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 5.4.3.3-4			

Table 5.4.3.3-8: SIP 200 (OK) from the MCPTT Server (Step 6, Table 5.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 5.4.3.3-6			

Table 5.4.3.3-9: SIP re-INVITE from the MCPTT Server (Step 7a1, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.16.3-1)

Derivation Path: Annex A, Table A.2.16.4-1, condition MCPTT and re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact	same as in the response for the INVITE creating the dialog			
Message-body				
SDP Message	As described in Table 5.4.3.3-10			

Table 5.4.3.3-10: SDP in SIP INVITE (Table 5.4.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_OFFER, PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description [2]		Media description for media control		
media attribute		a= line attribute = fmtp		
fmtp				
format specific parameters				
mc_priority	"2"			

Table 5.4.3.3-11: SIP 200 (OK) from the SS-UE1 (Steps 7a1 and 8a2, Table 5.4.3.2-1; Step 3, Annex A, Table A.2.16.3-1)

Derivation Path: Annex A, Table A.2.16.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact	Contact header with the same Contact URI and the same mandatory feature parameters as in the INVITE creating the dialog			
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP Message	As described in Table 5.4.3.3-12			

Table 5.4.3.3-12: SDP in SIP 200 (OK) (Table 5.4.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER, PRE_ESTABLISHED_SESSION

Table 5.4.3.3-13: SIP UPDATE from the MCPTT Server (Step 8a1, Table 5.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.15.2-1, condition MO_CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 5.4.3.3-14			

Table 5.4.3.3-14: SDP in SIP UPDATE (Table 5.4.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_OFFER, PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description [2]		**Media description for media control**		
media attribute		a= line attribute = fmtp		
fmtp				
format specific parameters				
mc_priority	"3"			

Table 5.4.3.3-9: SIP BYE from the SS-UE1
(Step 3, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

Table 5.4.3.3-10: SIP BYE from the MCPTT-Server
(Step 9, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1, condition MO_CALL

5.5 MCPTT Server - MCPTT Client / Configuration / Determination of MCPTT Service Settings / Current Active MCPTT Settings / De-subscribe

5.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to determine the currently active MCPTT service
settings with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(2)

```
with { IUT (MCPTT Server) having determined MCPTT service settings}
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to re-subscribe to MCPTT service settings with SIP
SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(3)

```
with { IUT (MCPTT Server) having determined MCPTT service settings }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to de-subscribe from MCPTT service settings with SIP
SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

5.5.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 7.3.6.1 and 7.3.6.2. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.6.1]

"Upon receiving a SIP SUBSCRIBE request such that:

1) *Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCPTT function of the served MCPTT user;*

2) *the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and*

3) *the Event header field of the SIP SUBSCRIBE request contains the 'poc-settings' event type.*

the MCPTT server:

1) *shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;*

2) *if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;*

3) *if the originating MCPTT ID is different than the served MCPTT ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps; and*

4) *shall generate a 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26] and IETF RFC 4354 [55].*

For the duration of the subscription, the MCPTT server shall notify subscriber about changes of the MCPTT service settings of the subscribed MCPTT user, as described in clause 7.3.6.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.6.2]

"In order to notify the subscriber about changes of the MCPTT service settings of the subscribed MCPTT client of the subscribed MCPTT user, the MCPTT server:

1) *shall generate an application/poc-settings+xml MIME body containing:*

   a) *the <am-settings> element of the poc-settings event package set to the current answer mode setting of the MCPTT client according to IETF RFC 4354 [55]; and*

   b) *the <selected-user-profile-index> element as defined in clause 7.4.1.2.2 identifying the active MCPTT user profile; and*

2) *send a SIP NOTIFY request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26] and IETF RFC 4354 [55] with the constructed application/poc-settings+xml MIME body."*

5.5.3 Test description

5.5.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

5.5.3.2 Test procedure sequence

Table 5.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to determine the currently active MCPTT service settings of the UE (MCPTT client)?	-	-	1	P
2	Check: Does the MCPTT Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to re-subscribe to MCPTT service settings and to receive later notifications?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to de-subscribe from MCPTT service settings?	-	-	3	P

5.5.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

**Table 5.5.3.3-1: SIP SUBSCRIBE from the SS-UE1
(Step 1, Table 5.5.3.2-1; Step 1, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 7.2.4	
value	"4294967295"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.5.3.3-2			

Table 5.5.3.3-2: MCPTT-Info in SIP SUBSCRIBE (Table 5.5.3.3-1, Table 5.5.3.3-4 and Table 5.5.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition POC-SETTINGS-EVENT

**Table 5.5.3.3-3: SIP 200 (OK) from the MCPTT Server
(Steps 1, 2, 3, Table 5.5.3.2-1; Step 2, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions SUBSCRIBE-RSP

**Table 5.5.3.3-4: SIP SUBSCRIBE from the SS-UE1
(Step 2, Table 5.5.3.2-1; Step 1, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT, re_SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 7.2.4	
value	"4294967295"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.5.3.3-2			

**Table 5.5.3.3-5: SIP SUBSCRIBE from the SS-UE1 (Step 3, Table 5.5.3.2-1;
Step 1, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT, re_SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 7.2.4	
value	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.5.3.3-2			

Table 5.5.3.3-6: SIP NOTIFY from the MCPTT Server (Steps 1, 2, Table 5.5.3.2-1; Step 3, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PoC-Settings**	IETF RFC 4354 [32]	
MIME-part-body	PoC-Settings as described in Table 5.5.3.3-7			

Table 5.5.3.3-7: PoC-Settings in SIP NOTIFY (Table 5.5.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.11.2-1, condition AUTOMATIC

5.6 MCPTT Server - MCPTT Client / Configuration / Download CSK

5.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { IUT (MCPTT Server) requests to update the existing CSK }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE1 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having registered the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an E2E group call establishment with manual
commencement mode and with implicit floor }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
registered users of the group and enforces floor control (Floor Taken, Floor Granted) }
            }
```

(3)

```
with { IUT (MCPTT Server) having registered the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an MCPTT floor release }
    then { IUT (MCPTT Server) accepts the floor release and sends floor control messages (Floor Ack,
Floor Idle) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT E2E group call with Manual Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the E2E group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

5.6.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clause 7.3.7; ETSI TS 133 180 [31] (3GPP TS 33.180), clause 9.2.1.4; and ETSI TS 124 380 [5] (3GPP TS 24.380) 13.3.2. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 7.3.7]

"If confidentiality protection is enabled as specified in clause 6.6.2.3.1, and if the participating MCPTT function received a Client Server Key (CSK) within a SIP REGISTER request for service authorisation or SIP PUBLISH request for service authorisation, the participating MCPTT function may decide to update the CSK. In this case, the participating MCPTT function shall perform a key download procedure for the CSK. The participating MCPTT function:

1) shall generate an SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

2) shall set the Request-URI to the URI received in the To header field in a third-party SIP REGISTER request;

3) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media-feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

4) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";

5) shall include an application/mikey MIME body containing the CSK-ID and the CSK encrypted within a MIKEY message to the MC client as specified in clause 9.2.1 of 3GPP TS 33.180 [78] in the body of the SIP MESSAGE request;

6) shall send the SIP MESSAGE request towards the MCPTT client according to 3GPP TS 24.229 [4]."

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 9.2.1.4]

"The MCX Server may decide to update an existing CSK at any time. This may be due to CSK revocation or expiry.

The CSK shall be updated by the MCX Server using the 'key download' procedure, defined in clause 5.8. Upon receipt of a CSK via a 'key download' procedure, the MC client shall identify the type of key as a CSK via the 4 most significant bits of the CSK-ID. The MC client shall:

- *discard any previous CSKs associated with the MC Server FQDN, and*
- *use the new CSK for uplink signaling with the MC Server."*

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 5.8.1]

"The 'key download' procedure is used to send keys from the MCX server to the MC client. It is used to distribute Multicast Signalling Keys (MuSiKs) to the MC clients, and it is used to update both the CSKs and MuSiKs.

Within the 'key download' procedure, keys (CSK or MuSiKs) are encrypted specifically to the MC user and signed using an identity representing the MC Server. Prior to group key distribution, each MC client shall be provisioned by the KMS with time-limited key material associated with the MC User as described clause 5.3. The MC Server shall also be provisioned by the MC KMS with key material for an identity which is authorised to act as an MCX Server.

The key (CSK or MuSiK) is distributed from the MCX Server to a MC client using the security mechanism described in clause 5.2.2, transported over the SIP bearer. End-point diversity is not required as end-points do not encrypt data, hence the extension in clause 5.2.3 is not applied. Additional parameters may be included as defined in clause 5.2.4. The SAKKE-to-self extension may be included as defined in clause 5.2.5. Identity hiding may be supported as defined in clause 5.2.6.

The initiating entity shall be the initiating MCX Server and the receiving entity shall be the terminating MC user. The initiating entity URI shall be the FQDN of the MCX Server (e.g. MDSI of the MC Domain) and the receiving entity URI shall be the MC Service ID of the terminating user. The distributed key, K, shall be the CSK or MuSiK and the key identifier K-ID shall be the CSK-ID or MuSiK-ID (respectively).

As a result of this 'key download' mechanism, the MC clients receive a new signalling key, CSK or MuSiK, identified by the 4 most significant bits of the key ID.

The MCData Service server may use the Key Download procedure to indicate or modify the algorithm used to protect the MCData signalling fields (i.e. MCData signaling parameters, Data signaling payload and End to end security parameters) by including a 'signalling algorithm' parameter. The 'signalling algorithm' parameter is described in clause 8.5.4.1. The available algorithms shall be as defined in clause 8.5.4.2."

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 5.8.2]

"The procedure for key download is described in figure 5.8.2-1:

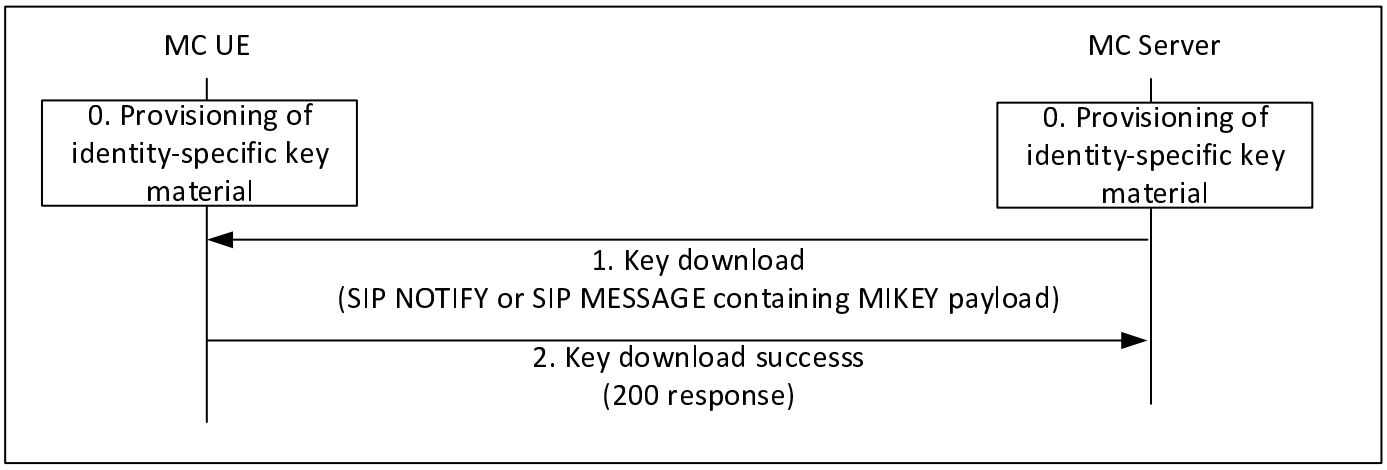


Figure 5.8.2-1: Procedures for key download

0. The MCX UE has been provisioned by a KMS with key material associated with the MC user. The MC UE has also registered with an MCX Server. As a consequence of this registration, the MC UE is subscribed to key download notifications from the MCX Server.

1. The MCX Server sends a key download message (SIP NOTIFY or SIP MESSAGE) to the MC UE. The MC UE extracts the signalling key from the key download message.

2. Upon successful extraction of the signalling key, the MC UE returns a key download success message (200 OK response) to the MCX Server. Upon receipt of a notification of success, the MCX Server is able to begin to use the key for protection of signalling traffic."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 13.3.2]

"The participating MCPTT function:

1. if protection of media is negotiated, shall be transparent to RTP media streams and shall forward encrypted RTP media streams without decrypting the payload;

2. if protection of floor control messages sent using unicast between the participating MCPTT function and the MCPTT client is negotiated and the CSK and the CSK-ID were received from the MCPTT client using SIP signalling according to 3GPP TS 24.379 [2]:

A) shall encrypt floor control messages sent using unicast to the served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2; and

B) shall decrypt floor control messages received using unicast from the served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2;

…

6. if protection of pre-established session call control messages between the participating MCPTT function and the MCPTT client is negotiated and the CSK and the CSK-ID were received from the MCPTT client using SIP signalling according to 3GPP TS 24.379 [2]:

A) shall encrypt pre-established session call control messages sent to the served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2; and

B) shall decrypt pre-established session call control messages received from served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2;

7. if protection of media control messages sent using unicast between the participating MCPTT function and the MCPTT client is negotiated between the participating MCPTT function and the MCPTT client and the CSK and the CSK-ID were received from the MCPTT client using SIP signalling according to 3GPP TS 24.379 [2];

A) shall encrypt media control messages sent using unicast to the served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2; and

B) shall decrypt media control messages received using unicast from the served MCPTT client according to IETF RFC 3711 [16] and 3GPP TS 33.180 [18] using SRTP-MK, SRTP-MS and SRTP-MKI generated using the CSK and CSK-ID as specified in clause 13.2;"

5.6.3 Test description

5.6.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT Client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT Client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT Client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1, with manual commencement mode as the default answer mode within the poc-settings.

5.6.3.2 Test procedure sequence

Table 5.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to update the existing CSK between the MCPTT Server and SS-UE1 (MCPTT Client)?	-	-	1	P
2a1	Check: Does the MCPTT Server correctly perform procedure 'MCPTT E2E group call establishment, manual commencement' as described in Annex A, Table A.3.2.3-1 to establish a group call with manual commencement mode, and with implicit floor but without implicit granted; and using the updated CSK to unprotect the SIP INVITE message received from SS-UE1 (MCPTT Client), with Option b.ii of note on Table A.3.2.3-1?	-	-	2	P
2a2	Check: Does the MCPTT server correctly protect the Floor Granted message described in step 2a1 of procedure 'MCPTT E2E group call establishment, manual commencement' as described in Annex A, Table A.3.2.3-1, by using the updated CSK?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Floor release - Floor Idle' as described in Annex A, Table A.3.15.3-1, to release the floor without acknowledgement from SS-UE1 (MCPTT Client), and using the updated CSK to protect the Floor Release and Floor Idle messages?	-	-	3	P
4	Check: Does the MCPTT Server send a Floor Idle message to the SS-UE2 (MCPTT client)?	<--	Floor Idle	3	P
5	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1, to release the call?	-	-	4	P

5.6.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.6.3.3-1: SIP MESSAGE from the MCPTT Server (Step 1, Table 5.6.3.2-1; Step 1, Annex A, Table A.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 7.3.7	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
Content-Type				
media-type	"application/mikey"			
Message-body				
MIKEY message	base64 encoded MIKEY message as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-1A	MIKEY message, containing the updated CSK		

Table 5.6.3.3-2: Floor Idle (Step 4, Table 5.6.3.2-1)

Derivation Path: Table 5.5.6.6-1

5.7 MCPTT Server - MCPTT Client / Configuration / Subscription to group dynamic data / De-subscribe

5.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) having registered the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an E2E group call with manual commencement mode and
without implicit floor }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
registered users of the group }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an E2E group call with manual commencement mode }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to subscribe to group dynamic data for group call
ongoing information with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(3)

```
with { IUT (MCPTT Server) having determined MCPTT subscription to group dynamic data }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to de-subscribe from MCPTT group dynamic data for group
call ongoing information with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT E2E group call with Manual Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the E2E group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

5.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.2.2.16, 9.2.2.3.9 and 9.2.2.3.10. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.16]

"Upon receiving a SIP SUBSCRIBE request such that:

1) *Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user;*

2) *the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info MIME body containing the<mcptt-request-uri> element which identifies an MCPTT group ID not served by the MCPTT server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and*

4) *the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;*

then the MCPTT server:

1) *shall identify the target MCPTT group ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP SUBSCRIBE request;*

2) *shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;*

3) *shall if the originating MCPTT ID is not authorized for this subscription reject the SIP SUBSCRIBE request with SIP 403 (Forbidden) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;*

4) *shall generate a SIP SUBSCRIBE request from the received SIP SUBSCRIBE request. In the generated SIP SUBSCRIBE request, the MCPTT server:*

   a) *shall set the Request-URI to the public service identity identifying the controlling MCPTT function serving the target MCPTT group ID;*

NOTE 1: The public service identity can identify the controlling MCPTT function in the primary MCPTT system or in a partner MCPTT system.

NOTE 2: If the controlling MCPTT function is in a partner MCPTT system in a different trust domain, then the public service identity can identify the MCPTT gateway server that acts as an entry point in the partner MCPTT system from the primary MCPTT system.

NOTE 3: If the controlling MCPTT function is in a partner MCPTT system in a different trust domain, then the primary MCPTT system can route the SIP request through an MCPTT gateway server that acts as an exit point from the primary MCPTT system to the partner MCPTT system

NOTE 4: How the participating MCPTT function determines the public service identity of the controlling MCPTT function serving he target MCPTT group ID or of the MCPTT gateway server in the partner MCPTT system is out of the scope of the present document.

NOTE 5: How the primary MCPTT system routes the SIP request through an exit MCPTT gateway server is out of the scope of the present document.

   b) *shall include a P-Asserted-Identity header field containing the public service identity identifying the originating participating MCPTT function serving the MCPTT user;*

   c) *shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:*

      A) *shall include the <mcptt-request-uri> element set to the target MCPTT group ID; and*

      B) *shall include the <mcptt-calling-user-id> element set to the originating MCPTT ID; and*

   d) *shall include other signalling elements from the received SIP SUBSCRIBE request; and*

4) *shall send the generated SIP SUBSCRIBE request according to 3GPP TS 24.229 [4].*

The MCPTT server shall forward any received SIP responses to the SIP SUBSCRIBE request, any received SIP NOTIFY requests and any received SIP responses to the SIP NOTIFY request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.9]

"Upon receiving a SIP SUBSCRIBE request such that:

1) *Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCPTT function associated with the served MCPTT group;*

2) *the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element and the <mcptt-calling-user-id> element;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

4) *the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and*

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-group dynamic data of presence event package notification information according to clause 9.3.2 indicating the same MCPTT group ID as in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCPTT server:

1) shall identify the served MCPTT group ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

…

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify subscriber about changes of the information of the served MCPTT ID, as described in clause 9.2.2.3.10."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.10]

"In order to notify the subscriber identified by the handled MCPTT ID about changes of the per-group dynamic data of the served MCPTT group ID, the MCPTT server:

1) shall consider an MCPTT group information entry such that:

a) the MCPTT group information entry is in the list of MCPTT group information entries described in clause 9.2.2.3.2; and

b) the MCPTT group ID of the MCPTT group information entry is equal to the served MCPTT group ID;

2) shall generate an application/pidf+xml MIME body indicating per-group dynamic data according to clause 9.3.1 with the following clarifications:

a) the MCPTT server shall include the "expires" attribute in the <affiliation> element; and

3) shall send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9.2.2.3.8. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-group dynamic data."

5.7.3 Test description

5.7.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT Client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)

- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1

- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT Client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT Client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

5.7.3.2 Test procedure sequence

Table 5.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCPTT E2E group call establishment, manual commencement' as described in Annex A, Table A.3.2.3-1 to establish a group call with manual commencement mode, and without implicit floor, with Option b.ii of note on Table A.3.2.3-1?	-	-	1	P
2	Check: Does the MCPTT Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to subscribe to group dynamic data for group call ongoing information?	-	-	2	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to de-subscribe to group dynamic data for group call ongoing information?	-	-	3	P
4	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1, to release the call?	-	-	4	P

5.7.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.7.3.3-1: SIP SUBSCRIBE from the SS-UE1 (Step 2, Table 5.7.3.2-1; Step 1, Annex A, Table A.2.7.3-1)

Derivation Path: Annex A, Table A.2.7.4-1, condition PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-headers				
MIME-part-body	MCPTT-Info as described in Table 5.7.3.3-2			
MIME body part		**SIMPLE-FILTER**		
MIME-part-headers				
MIME-part-body	SIMPLE-FILTER as described in Table 5.7.3.3-3			

Table 5.7.3.3-2: MCPTT-Info in SIP SUBSCRIBE (Table 5.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A		

Table 5.7.3.3-3: SIMPLE-FILTER in SIP SUBSCRIBE (Table 5.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.6-1, condition PER-GROUP

Table 5.7.3.3-4: SIP NOTIFY from the MCPTT Server (Step 2, Table 5.7.3.2-1; Step 3, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.7.3.3-5			

Table 5.7.3.3-5: PIDF in SIP NOTIFY (Table 5.7.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_Group_A_ID		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
tuple			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
id attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
status			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
affiliation			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
group	not present			
client	Encrypted URI (see note) with value set to the mcptt-client-id as provided by the UE at registration		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
status	not present			
expires	Current time plus one hour		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.10	
tuple			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
id attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
status			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
affiliation			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
group	not present			
client	Encrypted URI (seen ote) with value set to px_MCPTT_Client_B_ID		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
status	not present			
expires	Current time plus one hour		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.10	
additionalData			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
groupCallOngoing	"true"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.13.3-1.				

Table 5.7.3.3-6: SIP SUBSCRIBE from the SS-UE1 (Step 3, Table 5.7.3.2-1; Step 1, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition PRESENCE-EVENT, re_SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-headers				
MIME-part-body	MCPTT-Info as described in Table 5.7.3.3-2			
MIME body part		**SIMPLE-FILTER**		
MIME-part-headers				
MIME-part-body	SIMPLE-FILTER as described in Table 5.7.3.3-3			

Table 5.7.3.3-7: SIP NOTIFY from the MCPTT Server (Step 2, Table 5.7.3.2-1; Step 3, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [21]	
substate-value	"terminated"			
expires	not present			
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.7.3.3-5			

5.8 MCPTT Server - MCPTT Client / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias

5.8.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to subscribe to functional alias status change
notifications with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
           }
```

(2)

```
with { IUT (MCPTT Server) having received subscription to determine the status of a functional alias
}
ensure that {
  when { the SS-UE1 (MCPTT client) requests to activate a functional alias with SIP PUBLISH }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
           }
```

(3)

```
with { IUT (MCPTT Server) having received subscription to determine the status of a functional alias
}
ensure that {
  when { the SS-UE1 (MCPTT client) requests to deactivate a functional alias with SIP PUBLISH }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
           }
```

(4)

```
with { IUT (MCPTT Server) having received subscription to determine the status of a functional alias
}
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to de-subscribe from functional alias status change
notifications with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
           }
```

5.8.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9A.2.2.2.3, 9A.2.2.2.4, and 9A.2.2.2.5. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user functional alias information according to clause 9A.3.1;*

then the MCPTT server:

1) *shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;*

2) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;*

3) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;*

4) *if the originating MCPTT ID is different than the served MCPTT ID or the originating MCPTT ID is not authorized to modify functional alias status of the served MCPTT ID, shall send a SIP 403 (Forbidden) response and shall not continue with the rest of the steps;*

5) *if the Expires header field of the SIP PUBLISH request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP PUBLISH request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;*

6) *if the Expires header field of the SIP PUBLISH request has nonzero value, shall determine the candidate expiration interval to according to IETF RFC 3903 [37];*

7) *if the Expires header field of the SIP PUBLISH request has zero value, shall set the candidate expiration interval to zero;*

8) *shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37]. In the SIP 200 (OK) response, the MCPTT server:*

   a) *shall set the Expires header field according to IETF RFC 3903 [37], to the candidate expiration time;*

9) *if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCPTT ID, shall not continue with the rest of the steps;*

10) *shall consider an MCPTT user information entry such that:*

    a) *the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and*

    b) *the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;*

    as the served MCPTT user information entry;

11) *shall consider a copy of the list of the MCPTT functional alias entries of the served MCPTT user information entry as the served list of the MCPTT functional alias information entries;*

12) *if the candidate expiration interval is nonzero, shall construct the candidate list of the MCPTT functional alias entries as follows:*

    a) *for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is indicated in a "functionalAliasID" attribute of a <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:*

       i) *shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries;*

       ii) *if the functional alias status of the functional alias information entry is "deactivating" or "deactivated", shall set the functional alias status of the new functional alias information entry to the "activated" state and shall reset the activating p-id-fa of the new functional alias information entry; and*

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval;

b) for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is not indicated in any "functionalAliasID" attribute of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries; and

ii) if the functional alias status of the functional alias information entry is "activated" or "activating":

- *shall set the functional alias status of the new functional alias entry to the "deactivating" state; and*
- *shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F; and*

c) for each functional alias ID:

i) which does not have a functional alias information entry in the served list of the functional alias entries; or

ii) which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has already expired;

and which is indicated in a "functionalAliasID" element of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall add a new functional alias information entry in the candidate list of the functional alias information list for the functional alias ID;

ii) shall set the functional alias status of the new functional alias information entry to the "activating" state;

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval; and

iv) shall reset the activating p-id-fa of the new functional alias information entry;

13) if the candidate expiration interval is zero, constructs the candidate list of the functional alias information entries as follows:

a) for each functional alias ID which has an entry in the served list of the functional alias information entries:

i) shall copy the functional alias entry of the served list of the functional alias information into a new functional alias information entry of the candidate list of the functional alias information entries;

ii) shall set the functional alias status of the new functional alias information entry to the "deactivating" state; and

iii) shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F;

14) shall replace the list of the functional alias information entries stored in the served MCPTT user information entry with the candidate list of the functional alias information entries;

…

16) shall identify the handled p-id-fa in the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

17) shall perform the procedures specified in clause 9A.2.2.2.5 for the served MCPTT ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.4]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing:

a) the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server; and

b) the <mcpttinfo> element with the <mcptt-Params> element containing an <anyExt> element with the <request-type> element set to a value of "functional-alias-status-determination";

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

the MCPTT server:

1) shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

4) if the originating MCPTT ID is different than the served MCPTT ID and the originating MCPTT ID is not authorized to modify functional alias status of the served MCPTT ID, shall send a SIP 403 (Forbidden) response and shall not continue with the rest of the steps; and

5) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify the subscriber about changes of the information of the served MCPTT ID, as described in clause 9A.2.2.2.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.5]

"In order to notify the subscriber about changes of functional aliases of the served MCPTT ID, the MCPTT server:

1) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

2) shall generate an application/pidf+xml MIME body indicating per-user functional alias information according to clause 9A.3.1 and the served list of the MCPTT user information entries with the following clarifications:

a) the MCPTT server shall not include information from functional alias information entry with the expiration time already expired;

b) the MCPTT server shall not include information from a functional alias information entry with the functional alias status set to the "deactivated" state;

c) if this procedures is invoked by procedure in clause 9A.2.2.2.3 where the handled p-id-fa value was identified, the MCPTT server shall set the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id-fa value; and

3) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9A.2.2.2.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-user functional alias information."

5.8.3 Test description

5.8.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

5.8.3.2 Test procedure sequence

Table 5.8.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Functional Alias Status Determination and Subscription' as described in Annex A, Table A.3.9.3-1 to determine the current status of a functional alias and to get later notification of status changes of a functional alias?	-	-	1	P
2	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Functional Alias Status Change' as described in Annex A, Table A.3.10.3-1 to activate a functional alias?	-	-	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Functional Alias Status Change' as described in Annex A, Table A.3.10.3-1 to deactivate a functional alias?	-	-	3	P
4	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Functional Alias Status Determination and Subscription' as described in Annex A, Table A.3.9.3-1 to de-subscribe from functional alias status change notifications?	-	-	4	P

5.8.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.8.3.3-1: SIP PUBLISH from the SS-UE1
(Step 3, Table 5.8.3.2-1; Step 1, Annex A, Table A.3.10.3-1)

Derivation Path: Annex A, Table A.3.10.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			

Table 5.8.3.3-2: SIP NOTIFY from the MCPTT Server
(Step 3, Table 5.8.3.2-1; Step 3, Annex A, Table A.3.10.3-1)

Derivation Path: Annex A, Table A.3.10.4-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table 5.8.3.3-3			

Table 5.8.3.3-3: PIDF in SIP NOTIFY (Table 5.8.3.3-2)

Derivation Path: Annex A, Table A.3.10.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"deactivating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	

Table 5.8.3.3-4: SIP NOTIFY from the MCPTT Server (Step 3, Table 5.8.3.2-1; Step 5, Annex A, Table A.3.10.3-1)

Derivation Path: Annex A, Table A.3.10.4-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table 5.8.3.3-5			

Table 5.8.3.3-5: PIDF in SIP NOTIFY (Table 5.8.3.3-4)

Derivation Path: Annex A, Table A.3.10.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status	not present			

Table 5.8.3.3-6: SIP NOTIFY from the MCPTT Server (Step 4, Table 5.8.3.2-1; Step 3, Annex A, Table A.3.9.3-1)

Derivation Path: Annex A, Table A.3.9.4-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [21]	
substate-value	"terminated"			
expires	not present			
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.8.3.3-5			

6 MCPTT Server - MCPTT Client operation

6.1 Group Calls

6.1.1 Pre-arranged Group Calls

6.1.1.1 MCPTT Server - MCPTT Client / Pre-arranged Group Call / On-demand / Automatic Commencement Mode / Floor Control / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group Call / Cancel Imminent Peril State

6.1.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having registered the clients}
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates a pre-arranged group call with automatic commencement
mode and implicit floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
affiliated users of the group }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) releases the floor }
    then { IUT (MCPTT Server) sends floor idle to the SS-UE1 (MCPTT client) and sends floor idle to
the SS-UE2 (MCPTT client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode}
ensure that {
  when { the SS-UE1 (MCPTT Client) upgrades the call to emergency group call with implicit floor
control }
    then { IUT (MCPTT Server) accepts the upgrade from the initiator and updates the call for all
affiliated users of the group }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) requests the floor }
    then { IUT (MCPTT Server) sends floor granted to the SS-UE1 (MCPTT client) and sends floor taken
to the SS-UE2 (MCPTT client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) requests downgrade of the call to a normal group call }
    then { IUT (MCPTT Server) accepts the downgrade from the initiator and updates the call for all
affiliated users of the group without implicit floor request }
            }
```

(6)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode}
ensure that {
  when { the SS-UE1 (MCPTT Client) upgrades the call to imminent peril group call with implicit
floor request }
    then { IUT (MCPTT Server) accepts the upgrade from the initiator and updates the call for all
affiliated users of the group }
            }
```

(7)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the Pre-arranged group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

6.1.1.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 10.1.1.2.1.1, 10.1.1.2.1.3, 10.1.1.2.1.4, 10.1.1.2.1.5, 10.1.1.3.1.1, 10.1.1.3.1.3, 10.1.1.3.2, 10.1.1.3.3.1, 10.1.1.3.4.1, 10.1.1.3.6, 10.1.1.4.1.1, 6.3.2.1.1.1, 6.3.2.1.2.1, 6.3.2.1.5.2, 6.3.2.1.6, 6.3.2.2.1, 6.3.2.2.2.1, 6.3.2.2.3, 6.3.2.2.4.2, 6.3.2.2.5.2, 6.3.2.2.8.1, 6.3.2.2.10, 6.4, and ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 5.4, 6.3.2.2, 6.3.4.2.2, 6.3.5.2.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.2.1.1]

"Upon receiving a request from an MCPTT user to establish an MCPTT prearranged group session the MCPTT client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

…

4) shall include the g.3gpp.mcptt media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [16];

5) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP INVITE request;

7) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

8) should include the "timer" option tag in the Supported header field;

9) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

10) shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCPTT function serving the MCPTT user;

NOTE 1: The MCPTT client is configured with public service identity identifying the participating MCPTT function serving the MCPTT user.

11) may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [4];

…

14) shall contain in an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged";

b) the <mcptt-request-uri> element set to the group identity;

c) the <mcptt-client-id> element set to the MCPTT client ID of the originating MCPTT client;

…

15) shall include an SDP offer according to 3GPP TS 24.229 [4] with the clarifications given in clause 6.2.1;

16) if an implicit floor request is required, shall indicate this as specified in clause 6.4; and

17) shall send the SIP INVITE request towards the MCPTT server according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.2.1.3]

"Upon receiving a request from an MCPTT user to upgrade the MCPTT group session to an emergency condition or an imminent peril condition on an MCPTT prearranged group, the MCPTT client shall generate a SIP re-INVITE request as specified in 3GPP TS 24.229 [4], with the clarifications given below.

…

3) if the MCPTT user has requested to upgrade the MCPTT group session to an MCPTT emergency call, the MCPTT client:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body by following the procedures in clause 6.2.8.1.1; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2.

4) if the MCPTT user has requested to upgrade the MCPTT group session to an MCPTT imminent peril call, the MCPTT client:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body by following the procedures in clause 6.2.8.1.9; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

5) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.2.1.4]

"Upon receiving a request from an MCPTT user to cancel the in-progress emergency condition on a prearranged MCPTT group, the MCPTT client shall generate a SIP re-INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

…

2) shall, if the MCPTT user is cancelling an in-progress emergency condition and optionally an MCPTT emergency alert originated by the MCPTT user, include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.3;

3) shall, if the MCPTT user is cancelling an in-progress emergency condition and an MCPTT emergency alert originated by another MCPTT user, include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.14;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged"; and

b) the <mcptt-request-uri> element set to the group identity;

NOTE 1: The MCPTT ID of the originating MCPTT user is not included in the body, as this will be inserted into the body of the SIP INVITE request that is sent by the originating participating MCPTT function.

5) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP re-INVITE request according to IETF RFC 3840 [16];

6) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…

8) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2; and

9) shall send the SIP re-INVITE request according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.2.1.5]

"Upon receiving a request from an MCPTT user to cancel the in-progress imminent peril condition on a prearranged MCPTT group, the MCPTT client shall generate a SIP re-INVITE request by following the procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

…

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.11; and

3) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged"; and

b) the <mcptt-request-uri> element set to the group identity;

NOTE 1: The MCPTT ID of the originating MCPTT user is not included in the body, as this will be inserted into the body of the SIP re-INVITE request that is sent by the originating participating MCPTT function.

5) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP re-INVITE request according to IETF RFC 3840 [16];

6) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [18]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [18];

12) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCPTT client as specified in clause 6.3.2.1.1.1;

13a) if the received SIP request contains a <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body, then check if the status of the functional alias is activated for the MCPTT ID. If the functional alias status is activated, then set the <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request to the received value, if the status is unequal activated then do not include a <functional-alias-URI> element;

NOTE 8: The participating MCPTT server learns the functional alias state for an MCPTT ID from procedures specified in clause 9A.2.2.2.7.

…

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCPTT client;

NOTE 9: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

17) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

…

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request, the participating MCPTT function:

1) void;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2 with the clarification that if an <MKFC-GKTPs> element was contained in the received SIP 2xx response it is not included in the generated SIP 200 (OK) response;

NOTE 10: If an <MKFC-GKTPs> element is received in a SIP 2xx response, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the originating client.

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

6A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

7) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

9) shall interact with Media Plane as specified in 3GPP TS 24.380 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [7]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.3]

"Upon receipt of a SIP re-INVITE request for an MCPTT session identifying an on-demand prearranged MCPTT group session or pre-established session, the participating MCPTT function:

…

3) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.1.9;

4) shall include the MCPTT ID of the originating user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request;

5) shall, if the SIP re-INVITE request was received within an on-demand session, include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.1.1;

…

7) if the received SIP re-INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

NOTE 3: The controlling MCPTT function will determine the validity of the Resource-Priority header field.

8) shall forward the SIP re-INVITE request according to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the above SIP re-INVITE request in step 7) the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

4) shall copy the contents received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

5) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [4]; and

6) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

This clause covers both on-demand session and pre-established sessions.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.3.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.4.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.6]

"In the procedures in this clause:

1) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and*

2) *imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.*

Upon receipt of a SIP re-INVITE request for a terminating MCPTT client of a MCPTT group containing an emergency indication or imminent peril indication, the participating MCPTT function:

1) *shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may perform further actions outside the scope of this specification to act upon an included Resource-Priority header field as specified in 3GPP TS 24.229 [4];*

2) *shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.2.10;*

3) *shall include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.2.1; and*

4) *shall send the SIP re-INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP 200 (OK) response to the above SIP re-INVITE request sent to the MCPTT client, the participating MCPTT function:

1) *shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;*

2) *shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;*

3) *shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and*

4) *shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4]."*

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCPTT function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

…

4) *shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:*

   a) *the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and*

   b) *the <mcptt-calling-group-id> element set to the group identity;*

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;*

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCPTT user and MCPTT group are authorised for the initiation of MCPTT emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) void

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCPTT function:

1) shall contain only one SDP media-level section for MCPTT speech as contained in the received SDP offer; and

2) shall contain one SDP media-level section for media plane control messages, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCPTT function, if required;

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the participating MCPTT function; and

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media plane control channel, if present in the received SDP offer.

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.5.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCPTT function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [4] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". If the "refresher" parameter is not included in the received request, the "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

5) shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];

6) may include a Resource-Share header field in accordance with clause 5.7.1.20.2 in 3GPP TS 24.229 [4]; and

7) if the incoming SIP 200 (OK) response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.6]

"Upon receiving a SIP BYE request from the MCPTT client, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [4];

3) shall set the Request-URI to the MCPTT session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCPTT function, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCPTT function shall forward a SIP 200 (OK) response to the MCPTT client and shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the controlling MCPTT function."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.1]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.2.1]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.3]

"The participating MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4] and:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

c) the isfocus media feature tag;

d) an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

5) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

6) shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];

7) may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4]; and

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.4.2]

"When sending SIP 200 (OK) responses, the participating MCPTT function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [4] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCPTT function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.5.2]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" for an on-demand session that requires automatic commencement mode the participating MCPTT function:

1) if:

a) the incoming SIP INVITE request contained a Priv-Answer-Mode header field set to the value of "Auto";

b) no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) the incoming SIP INVITE request contained an Answer-Mode header field set to "Auto" and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer";

then:

a) shall generate a SIP 183 (Session Progress) response to the "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.4.1;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <ambient-listening-type> element set to a value of "remote-init" shall include in the SIP 183 (Session Progress) response an alert-info header field set to a value of "<file:///dev/null>" according to IETF RFC 3261 [24] and as updated by IETF RFC 7462 [77]; and

NOTE: The SIP 183 (session Progress) response can be sent reliably or unreliably depending on the content of the received SIP INVITE request. Regardless of if the SIP 183 (Session Progress) response is sent reliably or unreliably, SDP is not included in the SIP 183 (Session Progress) response.

c) shall set the P-Answer-State header field to "Unconfirmed" in the SIP 183 (Session Progress) response;

2) shall copy the public user identity contained in the Request-URI of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the SIP 183 (Session Progress) response;

3) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

4) shall set the Request-URI to the public user identity associated to the MCPTT ID of the MCPTT user to be invited;

5) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request, into the outgoing SIP INVITE request;

6) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

7) if the Priv-Answer-Mode header field is present in the incoming SIP INVITE request with a value of "Auto", shall include a Priv-Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request. Otherwise, if the Answer-Mode header field is present in the incoming SIP INVITE request, the participating MCPTT function shall include an Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request;

8) if no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer", shall set the Answer-Mode header field to "Auto" in the outgoing SIP INVITE request;

9) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.1;

10) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

11) shall send the SIP INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].

If the SIP 183 (Session Progress) response was sent reliably, then upon receiving a SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the above SIP INVITE request sent to the MCPTT client, the participating MCPTT function:

1) if the SIP 183 (Session Progress) was sent unreliably, shall send the SIP 200 (OK) response immediately; and

2) if the SIP 183 (Session Progress) was sent reliably and,

a) if the SIP PRACK request to the SIP 183 (Session Progress) response has been received by the participating MCPTT function and the SIP 200 (OK) response to the SIP PRACK request has been sent, shall send the SIP 200 (OK) response immediately;

b) if the SIP PRACK request to the SIP 183 (Session Progress) response has not yet been received, then upon receipt of the SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4], before sending the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function".

When the participating MCPTT function sends the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

3A) if the SIP 183 (Session Progress) was sent with the P-Answer-State header field set to "Unconfirmed", shall set the P-Answer-State header field to "Confirmed" in the outgoing SIP 200 (OK) response;

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4].

The participating MCPTT function shall forward any other SIP response that does not contain SDP along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resource associated with the SIP session with the controlling MCPTT function;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [4];*

3) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

4) *if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

5) *shall send the SIP BYE request to the MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCPTT function:

1) *shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCPTT function according to 3GPP TS 24.229 [4]; and*

2) *shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT client."*

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.10]

"The participating MCPTT function shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [4] and:

1) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];*

2) *may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4];*

3) *shall perform the procedures specified in clause 6.3.2.2.9 to copy any MIME bodies in the received SIP (re-)INVITE request to the outgoing SIP re-INVITE request; and*

4) *if the received SIP (re-)INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field."*

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.4]

"An initial SIP INVITE request fulfilling the following criteria shall be regarded by the MCPTT server as an implicit floor request by the originating MCPTT client when the MCPTT client:

1) *initiates an MCPTT speech session or initiates a pre-established session that is not used for a remotely initiated MCPTT ambient listening call; and*

2) *includes the "mc_implicit_request" 'fmtp' attribute in the associated UDP stream for the floor control in the SDP offer/answer as specified in 3GPP TS 24.380 [5] clause 12.*

An initial SIP INVITE request fulfilling the following criteria shall be regarded by the MCPTT server as an implicit request to grant the floor to the terminating MCPTT client when the originating MCPTT client:

1) *initiates a remotely initiated MCPTT ambient listening call; and*

2) *includes the "mc_implicit_request" 'fmtp' attribute in the associated UDP stream for the floor control in the SDP offer/answer as specified in 3GPP TS 24.380 [5] clause 12.*

A SIP re-INVITE request fulfilling the following criteria shall be regarded by the MCPTT server as an implicit floor request when the MCPTT client:

1) *performs an upgrade of:*

   a) *an MCPTT group call to an emergency MCPTT group call;*

   b) *an MCPTT private call to an emergency MCPTT private call; or*

c) an MCPTT group call to an imminent peril MCPTT group call; and

2) includes the "mc_implicit_request" 'fmtp' attribute in the associated UDP stream for the floor control in the SDP offer/answer as specified in 3GPP TS 24.380 [5] clause 12.

In all other cases the SIP (re-)INVITE request shall be regarded as received without an implicit floor request."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 5.4]

"An MCPTT server providing the participating MCPTT function shall:

1. *support the role of a participating MCPTT function as specified in 3GPP TS 23.379 [5];*
2. *distribute floor control messages as described in subclause 6.3 and support associated procedures as specified in subclause 6.4;*
3. *support call setup control over a pre-established session as specified in subclause 9.3;*
4. *when operating in systems that support and employ MBMS functionality, act as a GCS AS as specified in 3GPP TS 29.468 [6] and execute the related MBMS subchannel control procedures as specified in subclause 10.2; and*
5. *support media plane security as specified in clause 13."*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.2.2]

"When an MCPTT call is established a new instance of the floor control server state machine for 'general floor control operation' is created.

For each MCPTT client added to the MCPTT call, a new instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the floor control server could queue the implicit floor control request for the media-floor control entity.

The initial SIP INVITE request or SIP REFER request to establish an MCPTT chat group call or to rejoin an ongoing MCPTT call is not handled as an implicit floor control request message by the floor control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

…

Before the floor control server sends the first floor control message in the MCPTT call, the floor control server has to assign itself a SSRC identifier to be included in media floor control messages and quality feedback messages if the MCPTT server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The floor participant and the floor control server can negotiate the maximum priority level that the floor participant is permitted to request. The floor control server can pre-empt the current sender based on the negotiated maximum priority level that the floor participant is permitted to request and the priority level included in the Floor Request message.

NOTE: The maximum priority level that a floor participant can use is negotiated as specified in subclause 14.3.3 and is based on group configuration data retrieved by the controlling MCPTT function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCPTT function from the configuration management server as described in 3GPP TS 24.484 [13].

…

The floor participant and the floor control server can negotiate queueing of floor requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the floor control server queues the floor control request if a Floor Request message is received when another floor participant has the floor and the priority of the current speaker is the same or higher. Queueing is not permitted for groups configured for audio cut-in floor control."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.4.2.2]

"When an MCPTT call is initiated as specified in 3GPP TS 24.379 [2] and

1. *if a confirmed indication is required and at least one invited MCPTT client has accepted the invitation;*
2. *if a confirmed indication is not required; or*
3. *if the initialised MCPTT call is a temporary group session;*

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

then the floor control arbitration logic in the floor control server:

1. *shall create an instance of the 'general floor control operation' state machine;*
2. *shall wait for the 'basic floor control operation towards the floor participant' to be initialized before continuing the following steps;*
3. *when the 'basic floor control operation towards the floor participant' state machine is initialized and the initialised session is not a temporary group session:*
    a. *if the "mc_granted" fmtp attribute is not negotiated as specified in clause 14 or this is an ambient listening call:*
        i. *if the floor control server is granting an implicit floor request at MCPTT call establishment, shall act as if a Floor Request message was received and perform the actions specified in subclause 6.3.4.3.3; or*
        ii. *if the floor control server is not granting an implicit floor request at MCPTT call establishment, shall enter the'G: Floor Idle' state as specified in subclause 6.3.4.3.2; or*
    b. *if the "mc_granted" fmtp attribute is negotiated as specified in clause 14 and this is not an ambient listening call, shall enter the 'G: Floor Taken' state as specified in subclause 6.3.4.4.2; and"*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.5.2.2]

"When a SIP Session is established and if:

1. *the session is not a temporary group call session;*
2. *the session is a temporary group call session and the associated floor participant is an invited MCPTT client (i.e. not a constituent MCPTT group); or*
3. *the session is not an ambient listening call;*

then:

NOTE 1: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

1. *if an MCPTT client initiates an MCPTT call with an implicit floor request, and the MCPTT call does not exist yet, the floor control interface towards the MCPTT client in the floor control server:*
    a. *shall initialize a general state machine as specified in subclause 6.3.4.2.2; and*

NOTE 2: In the subclause 6.3.4.2.2 the 'general floor control operation' state machine will continue with the initialization of the 'general floor control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the subclause 6.3.5.5.2;*

…

4. if the MCPTT client is invited to the MCPTT call and

a. if another MCPTT client has permission to send a media, the floor control interface towards the MCPTT client in the floor control server:

i. should send a Floor Taken message to the MCPTT client. The Floor Taken message:

A. shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested;

B. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

C. if the session is a broadcast group call, shall include the Permission to Request the floor field set to '0';

D. if the session is not a broadcast group call, may include the Permission to Request the floor field set to '1'; and

E. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Floor Taken' state as specified in the subclause 6.3.5.4.2; and

b. if no other MCPTT client has the permission to send a media; the floor control interface towards the MCPTT client in the floor control server:

i. should send a Floor Idle message to the MCPTT client. The Floor Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Floor Idle' state as specified in the subclause 6.3.5.3.2.

…"

6.1.1.1.3 Test description

6.1.1.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)

- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1

- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.

- The IUT (MCPTT Server) is the controlling server of the Prearranged group.

6.1.1.1.3.2 Test procedure sequence

Table 6.1.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the SS-UE1 (MCPTT client) request the establishment of a group call to SS-UE2 with automatic commencement mode and implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a pre-arranged group call with automatic commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
5	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
6	Make the SS-UE1 (MCPTT client) request upgrade of the call to an emergency group call with implicit floor request.	-	-	-	-
7	Check: Does the MCPTT server correctly perform procedure 'MCX E2E session establishment/modification without provisional responses other than 100 Trying' with MCPTT as MCX, as described in Annex A, Table A.3.1.3-1 to upgrade the call to emergency group call with implicit floor request according to option b.ii of note in Annex A, Table A.3.1.3-1?	-	-	3	P
8	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
9	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
10	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
11	Make the SS-UE1 (MCPTT client) request the floor.	-	-	-	-
12	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	4	P
13	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	4	P
14	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-
15	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
16	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
17	Make the SS-UE1 (MCPTT client) request downgrade of the call to a normal group call.	-	-	-	-
18	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to downgrade the call without implicit floor request, with option a of note on Table A.3.1.3-1?	-	-	5	P
19	Make the SS-UE1 (MCPTT client) request the floor.	-	-	-	-
20	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	4	P
21	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	4	P
22	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-
23	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
24	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
25	Make the SS-UE1 (MCPTT client) request upgrade of the call to an imminent peril group call with implicit floor request.	-	-	-	-
26	Check: Does the MCPTT server correctly perform procedure 'MCX E2E session establishment/modification without provisional responses other than 100 Trying' with MCPTT as MCX, as described in Annex A, Table A.3.1.3-1 to upgrade the call to imminent peril group call with implicit floor request according to option b.ii of note in Annex A, Table A.3.1.3-1?	-	-	6	P
27	void	-	-	-	-
28	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
29	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
30	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
31	Make the SS-UE1 (MCPTT client) request the floor.	-	-	-	-
32	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	4	P
33	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	4	P
34	Make the SS-UE1 (MCPTT client) release the floor.	-	-	-	-
35	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
36	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	2	P
37	Make the SS-UE1 (MCPTT client) request downgrade of the call to a normal group call.	-	-	-	-
38	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to downgrade the call without implicit floor request, with option a of note on Table A.3.1.3-1?	-	-	5	P
39	Make the SS-UE1 (MCPTT client) release the call.	-	-	-	-
40	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	7	P

6.1.1.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.1.1.3.3-1: SIP INVITE from the SS-UE1 (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-1B			

Table 6.1.1.1.3.3-1A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions GROUP-CALL, IMPLICIT_GRANT_REQUESTED

Table 6.1.1.1.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL

Table 6.1.1.1.3.3-2: SIP INVITE from the MCPTT Server (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-2B			

Table 6.1.1.1.3.3-2A: SDP in SIP INVITE from the MCPTT Server (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions GROUP-CALL

Table 6.1.1.1.3.3-2B: MCPTT-Info in SIP INVITE from the MCPTT Server (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL

Table 6.1.1.1.3.3-3: SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-3A			

Table 6.1.1.1.3.3-3A: SDP in SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.1.1.1.3.3-4: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-4A			

Table 6.1.1.1.3.3-4A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.1.1.1.3.3-5: SIP re-INVITE from the SS-UE1 (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions GROUP-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-5B			

Table 6.1.1.1.3.3-5A: SDP in SIP re-INVITE from the SS-UE1 (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), GROUP-CALL, IMPLICIT_GRANT_REQUESTED
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-5B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL, EMERGENCY-CALL

Table 6.1.1.1.3.3-6: SIP re-INVITE from the MCPTT Server (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions GROUP-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-6B			

Table 6.1.1.1.3.3-6A: SDP in SIP re-INVITE from the MCPTT Server (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-6B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL, EMERGENCY-CALL

Table 6.1.1.1.3.3-7: SIP 200 (OK) from the SS-UE2 (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-7A			

Table 6.1.1.1.3.3-7A: SDP in SIP 200 (OK) from the SS-UE2 (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.1.1.1.3.3-8: SIP 200 (OK) from the MCPTT Server (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-8A			

Table 6.1.1.1.3.3-8A: SDP in SIP 200 (OK) from the MCPTT Server (Step 7, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.1.1.1.3.3-9: SIP re-INVITE from the SS-UE1 (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions GROUP-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-9A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-9B			

Table 6.1.1.1.3.3-9A: SDP in SIP re-INVITE from the SS-UE1 (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-9B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL

Table 6.1.1.1.3.3-10: SIP re-INVITE from the MCPTT Server (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions GROUP-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-10B			

Table 6.1.1.1.3.3-10A: SDP in SIP re-INVITE from the MCPTT Server (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-10B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL

Table 6.1.1.1.3.3-11: SIP 200 (OK) from the SS-UE2 (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-11A			

Table 6.1.1.1.3.3-11A: SDP in SIP 200 (OK) from the SS-UE2 (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.1.1.1.3.3-12: SIP 200 (OK) from the MCPTT Server (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-12A			

Table 6.1.1.1.3.3-12A: SDP in SIP 200 (OK) from the MCPTT Server (Steps 18 and 38, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.1.1.1.3.3-13: SIP re-INVITE from the SS-UE1 (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions GROUP-CALL, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-13A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-13B			

Table 6.1.1.1.3.3-13A: SDP in SIP re-INVITE from the SS-UE1 (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), GROUP-CALL, IMPLICIT_GRANT_REQUESTED
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-13B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL, IMMPERIL-CALL

Table 6.1.1.1.3.3-14: SIP re-INVITE from the MCPTT Server (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions GROUP-CALL, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-14A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-14B			

Table 6.1.1.1.3.3-14A: SDP in SIP re-INVITE from the MCPTT Server (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.1.1.3.3-14B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL, IMMPERIL-CALL

Table 6.1.1.1.3.3-15: SIP 200 (OK) from the SS-UE2 (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-15A			

Table 6.1.1.1.3.3-15A: SDP in SIP 200 (OK) from the SS-UE2 (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.1.1.1.3.3-16: SIP 200 (OK) from the MCPTT Server (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.1.1.3.3-16A			

Table 6.1.1.1.3.3-16A: SDP in SIP 200 (OK) from the MCPTT Server (Step 26, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.1.1.1.3.3-17: Floor Granted (Step 7, Table 6.1.1.1.3.2-1; Step 2a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-18: Floor Taken (Step 7, Table 6.1.1.1.3.2-1; Step 3a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-19: Floor Release (Steps 9 and 15, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-20: Floor Idle (Steps 9 and 15, Table 6.1.1.1.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-21: Floor Idle (Steps 10 and 16, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-22: Floor Request (Step 12, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-23: Floor Granted (Step 12, Table 6.1.1.1.3.2-1; Steps 2a1 and 2b1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-24: Floor Taken (Step 13, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-25: Floor Granted (Step 26, Table 6.1.1.1.3.2-1; Step 2a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-26: Floor Taken (Step 26, Table 6.1.1.1.3.2-1; Step 3a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-27: Floor Release (Steps 29 and 35, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-28: Floor Idle (Steps 29 and 35, Table 6.1.1.1.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-29: Floor Idle (Steps 30 and 36, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-30: Floor Request (Step 32, Table 6.1.1.1.3.2-1; Step 1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-31: Floor Granted (Step 32, Table 6.1.1.1.3.2-1; Steps 2a1 and 2b1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions IMMPERIL-CALL

Table 6.1.1.1.3.3-32: Floor Taken (Step 33, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions IMMPERIL-CALL

6.1.1.2 MCPTT Server - MCPTT Client / Pre-arranged Group Call / Pre-established Session / Automatic Commencement Mode

6.1.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having established a pre-established session with
the clients}
ensure that {
  when { the SS-UE1 (MCPTT Client) requests the call establishment using a pre-established group
call with automatic commencement mode and implicit floor request with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and establishes the call with all users
of the group }
            }
```

(2)

```
with { IUT (MCPTT Server) having established a pre-established session with the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) establishes a pre-arranged group call with automatic commencement
mode }
    then { IUT (MCPTT Server) enforces floor granted to the SS-UE1 (MCPTT client) and floor taken to
SS-UE2 (MCPTT client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established a call using  pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests the call release keeping the pre-established group call
with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and releases the SS-UE2 (MCPTT Client) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established a call using  pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
            }
```

(5)

```
with { IUT (MCPTT Server) having established a call using  pre-established session }
ensure that {
  when { the SS-UE2 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
            }
```

6.1.1.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 10.1.1.3.1.2, 10.1.1.3.2, 10.1.1.3.3.2, 10.1.1.3.4.2, 6.3.2.2.5.3 and ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 9.3.2.2.2, 9.3.2.3.2, 9.3.2.3.3, 9.3.2.4.5, 9.3.2.4.7, 9.3.2.4.10. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.2]

"Upon receipt of a "SIP REFER request for a pre-established session", with:

1) *the Refer-To header field containing a Content-ID ("cid") Uniform Resource Locator (URL) as specified in IETF RFC 2392 [62] that points to an application/resource-lists MIME body as specified in IETF RFC 5366 [20] containing an <entry> element with a "uri" attribute containing a SIP URI set to a pre-arranged group identity;*

2) *an hname "body" parameter in the headers portion of the SIP URI specified above containing an application/vnd.3gpp.mcptt-info MIME body with the <session-type> element set to "prearranged"; and*

3) a Content-ID header field set to the "cid" URL;

the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP REFER request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

6) if the "SIP REFER request for a pre-established session" contained a Refer-Sub header field containing "false" value and a Supported header field containing "norefersub" value, shall handle the SIP REFER request as specified in 3GPP TS 24.229 [4], IETF RFC 3515 [25] as updated by IETF RFC 6665 [26], and IETF RFC 4488 [22] without establishing an implicit subscription;

7) if received SIP REFER request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.2.1.8.3;

…

9) shall retrieve the group identity within the <entry> element of the application/resource-lists MIME body, referenced by the "cid" URL contained in the Refer-To header field of the SIP REFER request;

10) shall determine the public service identity of the controlling MCPTT function associated with the group identity in the application/resource-lists MIME body referenced by the Refer-To header of the SIP REFER request. If the participating MCPTT function is unable to identify the controlling MCPTT function associated with the group identity, it shall reject the REFER request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

NOTE 4: The public service identity can identify the controlling function in the primary MCPTT system or a partner MCPTT system.

NOTE 5: How the participating MCPTT server discovers the public service identity of the controlling MCPTT function associated with the group identity is out of scope of the current document.

11) if the user identified by the MCPTT ID is not affiliated to the group identified in the SIP REFER request as determined by clause 9.2.2.2.11 and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, shall perform the actions specified in clause 9.2.2.2.12 for implicit affiliation;

…

13) shall generate a final SIP 200 (OK) response to the "SIP REFER request for a pre-established session" according to 3GPP TS 24.229 [4];

NOTE 8: In accordance with IETF RFC 4488 [22], the participating MCPTT function inserts the Refer-Sub header field containing the value "false" in the SIP 200 (OK) response to the SIP REFER request to indicate that it has not created an implicit subscription.

14) shall send the response to the "SIP REFER request for a pre-established session" towards the MCPTT client according to 3GPP TS 24.229 [4];

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.4;

16) shall set the Request-URI of the SIP INVITE request to the public service identity of the controlling MCPTT function associated with the group identity;

17) shall copy the group identity from the "uri" attribute of the <entry> element of the application/resource-lists MIME body pointed to by the "cid" URL in the Refer-to header field of the SIP REFER request, to the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request;

18) if the received SIP REFER request contained a Resource-Priority header field, shall include in the outgoing SIP INVITE request a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the received SIP REFER request from the MCPTT client;

NOTE 9: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

19) shall forward the SIP INVITE request according to 3GPP TS 24.229 [4].

Upon receiving SIP provisional responses for the SIP INVITE request the participating MCPTT function:

1) shall discard the received SIP responses without forwarding them.

…

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the participating MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

2) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13; and

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

This clause covers both on-demand session and pre-established sessions.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.3.2]

"Upon receiving from the MCPTT client a SIP REFER request when using a pre-established session with the "method" SIP URI parameter set to value "BYE" in the URI in the Refer-To header field the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.7."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.4.2]

"Upon receiving a SIP BYE request from the controlling MCPTT function and if the MCPTT session id refers to an MCPTT user that has a pre-established session with the participating MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.5.3]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" for a pre-established session that requires automatic commencement mode the participating MCPTT function:

…

2) shall generate a SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function" as described in the clause 6.3.2.2.4.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.2.2.2 based on the SDP negotiated during the pre-established session establishment and SDP offer received in the "SIP INVITE request for terminating participating MCPTT function";

4) shall set the P-Answer-State header field to "Unconfirmed" in the SIP 200 (OK) response;

5) shall set the P-Asserted-Identity header field to the same value as included in the SIP INVITE request that was sent by the MCPTT client when the pre-established session was established, in the SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [4]; and

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 9.

NOTE 2: Received SIP INVITE requests handled by the present clause in steps 2) through 7) will result in the terminating client not receiving data normally contained in an application/ vnd.3gpp.mcptt-info+xml MIME body. This includes but is not limited to call type indicators such as XML elements <broadcast-ind> or <associated-group>."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.2.2]

"When a pre-established session is created between the participating MCPTT function and an MCPTT client, the participating MCPTT function:

1. shall initiate any needed user plane resources for the pre-established session; and

2. shall enter the 'G: Pre-established session not in use' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.3.2]

"Upon receiving a SIP REFER request from the MCPTT client the participating MCPTT function:

1. shall reserve user plane resources for the call session, if not yet reserved; and,

2. shall enter the 'G: Pre-established session in use' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.3.3]

"Upon receiving a SIP INVITE request from the controlling MCPTT function, if in automatic answer mode, the participating MCPTT function:

1. shall send the Connect message to the invited MCPTT client using the media plane control transport channel defined for this call. The Connect message:

a. shall include the SIP URI received in the Contact header field of the SIP INVITE request in the <MCPTT Session Identity> value in the MCPTT Session Identity field;

b. if the SIP INVITE request included an "application/vnd.3gpp.mcptt-info+xml" MIME body with the <session-type> element set to a value of "prearranged" or "chat":

i. shall include the content of the <mcptt-calling-group-id> element in the MCPTT Group Identity field;

ii. shall include the value 'prearranged' or 'chat' in the <Session Type> value in the MCPTT Session Identity field;

iii. shall include the content of the <mcptt-calling-user-id> element in the Inviting MCPTT User Identity field if available from the SIP INVITE message and privacy of the inviting MCPTT ID is not required; and

iv. shall include 'anonymous@anonymous.invalid' in the Inviting MCPTT User Identity field if the MCPTT ID of the inviting MCPTT user is not available from the SIP INVITE message or privacy of the inviting MCPTT ID is required;

…

d. shall include a Media Streams field, if the negotiated SDP in the pre-arranged session contains more media streams than required by the SDP in the SIP INVITE request; and

NOTE 1: For example if the call is a private call without floor control the Media Streams field needs to be included with the <Control Channel> value set to '0'.

e. shall set the first bit in the subtype of the Connect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;

2. shall start timer T55 (Connect);

3. shall initialize counter C55 (Connect) to 1;

4. shall reserve the media channel resources if not yet reserved; and

5. shall enter the 'G: Pre-established session in use' state.

NOTE 2: The case of manual answer is described in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.5]

"Upon receiving an indication from the controlling MCPTT function that the call is released (see 3GPP TS 24.379 [2]), the participating MCPTT function:

1. shall stop timer T55 (Connect)if running;

2. shall send Disconnect message to the MCPTT client. The Disconnect message:

a. shall include the MCPTT Session Identity field with the same content as sent in the Connect message when the MCPTT call was established; and

b. should set the first bit in the subtype of the Disconnect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;

3. shall start timer T56 (Disconnect);

4. shall initialise counter C56 (Disconnect) to 1; and

5. shall enter the 'G: Call releasing' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.7]

"Upon receiving an Acknowledge message from the MCPTT client, the participating MCPTT function:

1. shall stop timer T55 (Connect);

2. if the Reason Code is not 'Accepted' the participating MCPTT function:

a. shall send Disconnect message to the MCPTT client;

i. shall include the MCPTT Session Identity field with the same content as sent in the Connect message when the MCPTT call was established; and

ii. should set the first bit in the subtype of the Disconnect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;

b. shall start timer T56 (Disconnect);

c. shall initialise counter C56 (Disconnect) to 1;

d. shall send a call release indication to the controlling MCPTT function (see 3GPP TS 24.379 [2]); and

e. shall enter the 'G: Call releasing' state; and

3. otherwise the participating MCPTT function:

a. shall send a SIP 200 (OK) response to the SIP INVITE request as specified in 3GPP TS 24.379 [2] if the SIP 200 (OK) response to the SIP INVITE request is not already sent; and

b. shall remain in the 'G: Pre-established session in use' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.10]

"Upon receiving a SIP 200 (OK) response from the controlling MCPTT function, the participating MCPTT function:

1. shall send the Connect message to the MCPTT client which initiated this call using the media plane control transport channel defined for this call. The Connect message:

a. shall include the SIP URI received in the Contact header field of the SIP 200 (OK) response in the <MCPTT Session Identity> value in the MCPTT Session Identity field;

b. if an "application/vnd.3gpp.mcptt-info+xml" MIME body with an <session-type> element is included in the SIP 200 (OK) response, shall include the <Session Type> value in the MCPTT Session Identity field with a value corresponding to the <session-type> element;

c. if an "application/vnd.3gpp.mcptt-info+xml" MIME body with an <session-type> element is not included in the SIP 200 (OK) response, shall include the <Session Type> value set to 'no session type' in the MCPTT Session Identity field;

d. if a Warning header field with a SIP "warn-code" set to the value "399" is received in the SIP 200 (OK) response, shall include the content of the SIP "warn-text" parameter from the received Warning header field in the <Warning Text> value in the Warning Text field;

e. if a P-Answer-State header field is included in the SIP 200 (OK) response, shall include an Answer State field with an <Answer State> value corresponding to the "answer-type" parameter; and

f should set the first bit in the subtype of the Connect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;

2. shall start timer T55 (Connect);

3. shall increment counter C55 (Connect) by 1; and,

4. shall remain in the 'G: Pre-established session in use' state."

6.1.1.2.3 Test description

6.1.1.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Pre-established Session Establishment as specified in step 1 on Table 5.4.3.2-1 in sclause 5.4.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.

6.1.1.2.3.2 Test procedure sequence

Table 6.1.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) request the establishment of a pre-arranged group call to SS-UE2 for Group A, with automatic commencement mode and implicit floor request, reusing the pre-established session.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E call establishment using a pre-established session, automatic commencement' as described in Annex A, Table A.3.5.3-1 to establish a pre-arranged group call with automatic commencement mode?	-	-	1	P
3	Check: Does the MCPTT server send a Floor Granted message to SS-UE1 (MCPTT client)?	<--	Floor Granted	2	P
4	Check: Does the MCPTT server send a Floor Taken message to SS-UE2 (MCPTT client)?	<--	Floor Taken	2	P
5	The SS-UE1 (MCPTT client) releases the call keeping the pre-established session.	-	-	-	-
6	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E call release keeping the pre-established session' as described in Annex A, Table A.3.6.3-1?	-	-	3	P
7	The SS-UE1 (MCPTT client) releases the pre-established session.	-	-	-	-
8	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE1?	-	-	4	P
9	The SS-UE2 (MCPTT client) release the pre-established session.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE2?	-	-	5	P

6.1.1.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.1.2.3.3-1: SIP REFER from SS-UE1
(Step 2, Table 6.1.1.2.3.2-1; Step 1, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-1, with conditions GROUP-CALL and IMPLICIT_GRAN_REQUESTED for all clauses and subclauses referenced on that Table.

Table 6.1.1.2.3.3-2: SIP 200 (OK)
(Step 2, Table 6.1.1.2.3.2-1; Step 2, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-3, with conditions GROUP-CALL and IMPLICIT_GRAN_REQUESTED for all clauses and subclauses referenced on that Table.

Table 6.1.1.2.3.3-3: Connect
(Step 2, Table 6.1.1.2.3.2-1; Steps 3 and 5, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-4, with condition GROUP-CALL.

Table 6.1.1.2.3.3-4: Floor Granted (Step 3, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 6.1.1.2.3.3-5: Floor Taken (Step 4, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 6.1.1.2.3.3-6: SIP BYE from SS-UE1 and SS-UE2
(Steps 8 and 10, Table 6.1.1.2.3.2-1; Step 1, Annex A, Table A.2.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

6.1.1.3 MCPTT Server - MCPTT Client / Broadcast Group Call / On-demand / Manual Commencement Mode

6.1.1.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having registered the clients}
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates an On-demand Broadcast group call with manual
commencement mode and implicit floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
affiliated users of the group }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Broadcast Group Call with Manual
Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) release the floor }
    then { IUT (MCPTT Server) sends floor idle to the SS-UE1 (MCPTT client) and sends floor idle to
the SS-UE2 (MCPTT client) }
           }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Broadcast Group Call with
Manual Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the Broadcast group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
           }
```

6.1.1.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 4.12, 6.2.8.2, 6.3.8.1, 10.1.1.2.1.1, 10.1.1.3.1.1, 10.1.1.3.2, and ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 6.3.4.3.2, 6.3.4.4.2, 6.3.5.2.2, 6.3.5.4.5. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 4.12]

"A broadcast group call is a group call where the initiating MCPTT user expects no response from the other MCPTT users, so that when the user's transmission is complete, so is the call. The functionality in the present release of the specification for broadcast group calls is not compliant to the requirements for user-broadcast group and group-broadcast group calls as specified in 3GPP TS 22.179 [2], 3GPP TS 22.280 [76] and 3GPP TS 23.379 [3]. In the present release of the specification, a broadcast group call can be initiated by an MCPTT user on any MCPTT group that the MCPTT user is part of.

NOTE 1: Configuration related to the authorisation to create a user-broadcast group or a group-broadcast exists in the user profile document as specified in 3GPP TS 24.484 [50], but is not used by any procedures in 3GPP TS 24.481 [31] in the current release, as the ability for an authorised user to create user-broadcast groups and group-broadcast groups is not provided in the current release.

NOTE 2: Configuration related to broadcast group hierarchies can be found in the group document as specified in 3GPP TS 24.481 [31] and in the service configuration document as specified in 3GPP TS 24.484 [50]. However, this configuration is not used by any procedures in 3GPP TS 24.380 [5] in the current release."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.2.8.2]

"When the MCPTT user initiates a broadcast group call, the MCPTT client:

1) in the case of the prearranged group call is initiated on-demand, shall include in the application/vnd.3gpp.mcptt-info+xml MIME body the <broadcast-ind> element set to "true" as defined in clause F.1; and

2) in the case the prearranged group call is initiated using a pre-established session, shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the hname "body" parameter in the headers portion of the SIP URI in the Refer-To header field the <broadcast-ind> element set to "true" as defined in clause F.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.8.1]

"If:

1) the call is a pre-arranged group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

2) there are only one or no participants in the MCPTT session;

3) the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the MCPTT session;

4) the minimum number of affiliated MCPTT group members is not present;

5) timer TNG3 (group call timer) expires; or

6) the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

the controlling MCPTT function shall release the MCPTT session for the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.2.1.1]

"Upon receiving a request from an MCPTT user to establish an MCPTT prearranged group session the MCPTT client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

…

3) if the MCPTT user has requested the origination of a broadcast group call, the MCPTT client shall comply with the procedures in clause 6.2.8.2;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [18]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [18];

12) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCPTT client as specified in clause 6.3.2.1.1.1;

13a) if the received SIP request contains a <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body, then check if the status of the functional alias is activated for the MCPTT ID. If the functional alias status is activated, then set the <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request to the received value, if the status is unequal activated then do not include a <functional-alias-URI> element;

NOTE 8: The participating MCPTT server learns the functional alias state for an MCPTT ID from procedures specified in clause 9A.2.2.2.7.

…

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCPTT client;

NOTE 9: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

17) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

…

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request, the participating MCPTT function:

1) void;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2 with the clarification that if an <MKFC-GKTPs> element was contained in the received SIP 2xx response it is not included in the generated SIP 200 (OK) response;

NOTE 10: If an <MKFC-GKTPs> element is received in a SIP 2xx response, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the originating client.

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

6A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

7) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

9) shall interact with Media Plane as specified in 3GPP TS 24.380 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [7]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

This clause covers both on-demand session and pre-established sessions.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCPTT client negotiated support of queueing floor requests as described in clause 14, and the state machine specified in subclause 6.3.6 does not exist, the floor control arbitration logic in the floor control server:

…

2. if the active floor request queue is empty the floor control server:

a. shall send Floor Idle message to all floor participants. The Floor Idle message:

…

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

…

When entering this state from any state except the 'Start-stop' state and the state machine specified in subclause 6.3.6 exists, the floor control arbitration logic in the floor control server:

…

2. shall send Floor Idle message to all floor participants which are configured to listen to the overridden participant. The Floor Idle message:

…

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

3. shall send Floor Taken message to floor participants which are configured to listen only to the overridden participant. The Floor Taken message:

…

c. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. *shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:*

   …

   d. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Floor Indicator field with appropriate indications; and*

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.5.2.2]

"When a SIP Session is established and if:

1. *the session is not a temporary group call session;*
2. *the session is a temporary group call session and the associated floor participant is an invited MCPTT client (i.e. not a constituent MCPTT group); or*
3. *the session is not an ambient listening call;*

then:

…

2. *if the associated MCPTT client rejoins an ongoing MCPTT call without an implicit floor request or initiates or joins a chat group call without an implicit floor request or attempts to initiate an already existing MCPTT call without an implicit floor request, and*

   a. *if an MCPTT call already exists but no MCPTT client has the permission to send a media, the floor control interface towards the MCPTT client in the floor control server:*

      i. *should send a Floor Idle message to the MCPTT client. The Floor Idle message:*

         A. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*

         B. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and*

…

4. *if the MCPTT client is invited to the MCPTT call and*

   a. *if another MCPTT client has permission to send a media, the floor control interface towards the MCPTT client in the floor control server:*

      i. *should send a Floor Taken message to the MCPTT client. The Floor Taken message:*

         A. *shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested;*

         B. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

         C. *if the session is a broadcast group call, shall include the Permission to Request the floor field set to '0';*

         D. *if the session is not a broadcast group call, may include the Permission to Request the floor field set to '1'; and*

E. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Floor Taken' state as specified in the subclause 6.3.5.4.2; and

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.3.5.4.5]

"Upon receiving a Floor Release message from the associated floor participant and if the MCPTT client did not negotiate support of queueing of floor requests or included a floor priority in the "mc_priority" fmtp attribute as specified in clause 14, the floor control interface towards the MCPTT client in the floor control server:

…

2. shall send a Floor Taken message to the associated floor participant. The Floor Taken message:

…

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

3. shall remain in the 'U: not permitted and Floor Taken' state.

Upon receiving a Floor Release message from the associated floor participant and if the MCPTT client negotiated support of queueing of floor requests as specified in clause 14, the floor control interface towards the MCPTT client in the floor control server:

1. if the first bit in the subtype of the Floor Release message is set to '1' (Acknowledgment is required) as described in subclause 8.2.2, shall send a Floor Ack message. The Floor Ack message:

a. shall include the Message Type field set to '4' (Floor Release); and

b. shall include the Source field set to '2' (the controlling MCPTT function is the source);

…

4. shall send a Floor Taken message to the associated floor participant. The Floor Taken message:

a. shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested;

b. if the session is a broadcast group call, shall include the Permission to Request the floor field set to '0';

c. if the session is not a broadcast group call, may include the Permission to Request the floor field set to '1';

..

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

5. may set the first bit in the subtype of the Floor Taken message is set to '1' (Acknowledgment is required) as described in subclause 8.2.2; and

NOTE 2: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

6. shall remain in the 'U: not permitted and Floor Taken' state."

6.1.1.3.3 Test description

6.1.1.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated with Broadcast Group.
- The IUT (MCPTT Server) is the controlling server for Broadcast Group.

6.1.1.3.3.2 Test procedure sequence

Table 6.1.1.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) request the establishment of a broadcast group with manual commencement mode and implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.2.3-1 to establish a broadcast group call with manual commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.2.3-1?	-	-	1	P
3	The SS-UE1 (MCPTT client) release the floor.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	2	P
5	Check: Does the MCPTT server correctly perform 'MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the group call for SS-UE1?	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6	Check: Does the MCPTT server correctly perform 'MCX Participating Originated call release' as described in Annex A, Table A.2.17.3-1 to release the group call for SS-UE2?	-	-	3	P

6.1.1.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.1.3.3.3-1: MCPTT-Info in SIP INVITE from the SS-UE1 (Step2, Table 6.1.1.3.3.2-1; step 1, Table A.3.2.3-1; step 1, Table A.2.20.3-1)

Derivation Path: Annex A, Table A.3.2.4-3, with condition BROADCAST-CALL

Table 6.1.1.3.3.3-2: Floor Granted (Step2, Table 6.1.1.3.3.2-1; step 2a1, Table A.3.2.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with condition BROADCAST-CALL

Table 6.1.1.3.3.3-3: Floor Taken (Step2, Table 6.1.1.3.3.2-1; step 3a1, Table A.3.2.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with condition BROADCAST-CALL

Table 6.1.1.3.3.3-4: Floor Release (Step4, Table 6.1.1.3.3.2-1; step 1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with condition BROADCAST-CALL

Table 6.1.1.3.3.3-5: Floor Idle (Step4, Table 6.1.1.3.3.2-1; steps 3a1 and 3b1, Table A.3.15.3-1)

Derivation Path: Table 5.5.6.6-1, with condition BROADCAST-CALL

6.1.1.4 MCPTT Server - MCPTT Client / Emergency Alert / Cancel Emergency Alert

6.1.1.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having to support emergency alert}
ensure that {
  when { SS-UE1 (MCPTT Client) requests to send an emergency alert by sending a SIP MESSAGE message
}
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and sends SIP MESSAGE message to request
an emergency alert for SS-UE1 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having to support emergency alert}
ensure that {
  when { SS-UE1 (MCPTT Client) requests to cancel an emergency alert by sending a SIP MESSAGE
message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and sends SIP MESSAGE message to cancel
an emergency alert for SS-UE1 (MCPTT Client) }
            }
```

6.1.1.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.1, 12.1.1.2, 12.1.2.1, 12.1.2.2, 12.1.2.3, 12.1.3.1 and 12.1.3.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.1]

"…

The MCPTT client:

…

4) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <mcptt-request-uri> element set to the group identity;

b) the <alert-ind> element set to a value of "true";

c) the <mcptt-client-id> element set to the MCPTT client ID of the originating MCPTT client; and

d) if the MCPTT client needs to include an active functional alias in the SIP MESSAGE request, the <functional-alias-URI> set to the URI of the used functional alias;

NOTE 2: The MCPTT client learns the functional aliases that are activated for an MCPTT ID from procedures specified in clause 9A.2.1.3.

5) shall include in the SIP MESSAGE request the specific location information for MCPTT emergency alert as specified in clause 6.2.9.1;

…

8) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the group identity; and

9) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.2]

"…

The MCPTT client:

…

4) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <mcptt-request-uri> element set to the MCPTT group identity;

b) the <alert-ind> element set to a value of "false"; and

c) if the MCPTT user is cancelling an MCPTT emergency alert originated by another MCPTT user, include the <originated-by> element set to the MCPTT ID of the MCPTT user who originated the MCPTT emergency alert;

5) if the MCPTT user has additionally requested the cancellation of the in-progress emergency state of the MCPTT group and this is an authorised request for an in-progress emergency group state cancellation as determined by clause 6.2.8.1.7, shall include an <emergency-ind> element set to a value of "false" in the <mcpttinfo> element containing the <mcptt-Params> element;

6) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the group identity;

…

8) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.2.1]

"Upon receipt of a "SIP MESSAGE request for emergency notification for originating participating MCPTT function", the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

3) if the MCPTT user is not affiliated with the MCPTT group as determined by clause 9.2.2.2.11, shall perform the actions specified in clause 9.2.2.2.12 for implicit affiliation;

…

5) shall determine the public service identity of the controlling MCPTT function associated with the group identity in the received SIP MESSAGE request;

6) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

7) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCPTT function associated with the group identified by the <mcptt-request-uri> element contained in the <mcpttinfo> element containing the <mcptt-Params> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the incoming SIP MESSAGE request;

8) shall copy the contents of the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 included in the outgoing SIP MESSAGE request;

8a) if the received SIP MESSAGE request contains a <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body, shall check the status of the functional alias for the MCPTT ID. If the functional alias status is activated, then the participating MCPTT function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP MESSAGE request to the received value, otherwise it shall not include a <functional-alias-URI> element;

9) shall set the <mcptt-calling-user-id> element of the <mcpttinfo> element containing the <mcptt-Params> element to the MCPTT ID determined in step 2) above;

10) if the received SIP MESSAGE request contains an application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause F.3 shall copy the contents of the application/vnd.3gpp.mcptt-location-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-location-info+xml MIME body included in the outgoing SIP MESSAGE request;

11) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

12) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response in response to the SIP MESSAGE request sent in step 10):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [4] with the follow clarifications:

a) shall include the public user identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

2) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13; and

3) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.2.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP MESSAGE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) alert indication in an incoming SIP MESSAGE request refers to the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP MESSAGE requests for emergency notification for terminating participating MCPTT function", the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

3) if the binding between the MCPTT ID and public user identity does not exist, then the participating MCPTT function shall reject the SIP MESSAGE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

4) shall generate an outgoing SIP MESSAGE request as specified in clause 6.3.2.2.11; and

5) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall follow the procedures specified in 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.2.3]

"Upon receipt of a SIP MESSAGE request routed to the terminating participating MCPTT function as a result of initial filter criteria with the Request-URI set to the public service identity of the terminating participating MCPTT function and the SIP MESSAGE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with an <alert-ind-rcvd> element present, the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

…

4) shall generate an outgoing SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] and:

a) shall include in the SIP MESSAGE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] that were received (if any) in the incoming SIP MESSAGE request;

b) shall set the Request-URI of the outgoing SIP MESSAGE request to the public user identity associated to the MCPTT ID of the MCPTT user that was in the Request-URI of the incoming SIP MESSAGE request;

c) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body received in the incoming SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP MESSAGE request; and

d) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP MESSAGE request to the P-Asserted-Identity header field of the outgoing SIP MESSAGE request; and

5) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall follow the procedures specified in 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1]

"Upon receipt of a "SIP MESSAGE request for emergency notification for controlling MCPTT function", the controlling MCPTT function:

…

3) if the received SIP MESSAGE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false", shall perform the procedures specified in clause 12.1.3.2 and skip the rest of the steps;

4) if the received SIP MESSAGE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "true":

…

b) if the received SIP MESSAGE request is an authorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1:

i) if the sending MCPTT user identified by the <mcptt-calling-user-id> element included in the application/vnd.3gpp.mcptt-info+xml MIME body is not affiliated with the MCPTT group identified by the <mcptt-request-uri> element of the MIME body as determined by the procedures of clause 6.3.6:

I) shall check if the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined by clause 9.2.2.3.6;

…

III) if the procedures of clause 9.2.2.3.6 determined the MCPTT user to be eligible to be implicitly affiliated to the MCPTT group shall, perform the implicit affiliation as specified in clause 9.2.2.3.7;

ii) for each of the other affiliated members of the group:

A) generate an outgoing SIP MESSAGE request notification of the MCPTT user's emergency alert indication as specified in clause 6.3.3.1.11 with the clarifications of clause 6.3.3.1.12;

B) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <mcptt-calling-user-id> element set to the value of the <mcptt-calling-user-id> element in the received SIP MESSAGE request; and

C) send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [4];

iii) shall generate a SIP 200 (OK) response to the received SIP MESSAGE request as specified in 3GPP TS 24.229 [4] with the following clarifications:

A) shall cache the information that the MCPTT user has initiated an MCPTT emergency alert;

iv) shall send the SIP 200 (OK) response to the received SIP MESSAGE according to rules and procedures of 3GPP TS 24.229 [4].

v) shall generate a SIP MESSAGE request as described in clause 6.3.3.1.20 to indicate successful receipt of an emergency alert, and shall include in the application/vnd.3gpp.mcptt-info+xml MIME body:

A) the <alert-ind> element set to a value of "true";

B) the <alert-ind-rcvd> element set to a value of true; and

C) the <mcptt-client-id> element with the MCPTT client ID that was included in the incoming SIP MESSAGE request; and

vi) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the controlling MCPTT function shall follow the procedures specified in 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2]

"Upon receipt of a "SIP MESSAGE request for emergency notification for controlling MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false", the controlling MCPTT function:

...

2) if the received SIP MESSAGE request is an authorised request for an MCPTT emergency alert cancellation as specified in clause 6.3.3.1.13.1:

a) if the received SIP MESSAGE request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, shall clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency alert;

b) if the received SIP MESSAGE request does not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the sender of the SIP MESSAGE request as having an outstanding MCPTT emergency alert;

c) if the received SIP MESSAGE request does not contain an <emergency-ind> element or is an unauthorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.13.4, for each of the affiliated but not joined members of the group shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's emergency alert as specified in clause 6.3.3.1.11;

ii) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <mcptt-calling-user-id> element set to the value of the <mcptt-calling-user-id> element in the received SIP MESSAGE request;

iii) if the received SIP MESSAGE request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, copy the contents of the received <originated-by> element to an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP MESSAGE request;

iv) shall include an <alert-ind> element set to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP MESSAGE request; and

v) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4];

…

e) shall generate a SIP 200 (OK) response to the received SIP MESSAGE request as specified in 3GPP TS 24.229 [4];

f) shall send the SIP 200 (OK) response to the received SIP MESSAGE as specified in 3GPP TS 24.229 [4].

g) shall generate a SIP MESSAGE request as described in clause 6.3.3.1.20 to indicate successful receipt of the request for emergency alert cancellation;

h) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body, the <alert-ind> element set to a value of "false" and the <alert-ind-rcvd> set to "true";

i) shall populate the <mcptt-client-id> element with the MCPTT client ID that was included in the incoming SIP MESSAGE request;

j) if the received SIP MESSAGE request contains an <emergency-ind> element of the <mcpttinfo> element set to a value of "false":

i) if this is an authorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.13.4, shall include an <emergency-ind> element set to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP MESSAGE request; and

B) otherwise, if this is an unauthorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.13.4, and the in-progress emergency state of the group is set to a value of "true", shall include an <emergency-ind> element set to a value of "true" in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP MESSAGE request; and

k) shall send the SIP MESSAGE request according to according to the rules and procedures of 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the controlling MCPTT function shall follow the procedures specified in 3GPP TS 24.229 [4]."

6.1.1.4.3 Test description

6.1.1.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.

6.1.1.4.3.2 Test procedure sequence

Table 6.1.1.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the SS-UE1 (MCPTT client) send an emergency alert.	-	-	-	-
2	Check: Does the MCPTT Server correctly perform 'MCX E2E SIP MESSAGE Request - Accept' as described in Annex A, Table A.2.11.3-1 to send an emergency alert for SS-UE1?	-	-	1	P
3	Make the SS-UE1 (MCPTT client) cancel an emergency alert.	-	-	-	-
4	Check: Does the MCPTT Server correctly perform 'MCX E2E SIP MESSAGE Request - Accept' as described in Annex A, Table A.2.11.3-1 to cancel an emergency alert for SS-UE1?	-	-	2	P

6.1.1.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.1.4.3.3-1: SIP MESSAGE from the SS-UE1 (Step 2, Table 6.1.1.4.3.2-1; Step 1, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, with conditions MCPTT and LOCATION_REPORT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-2			
MIME body part		**Location-info**		
MIME-part-body	Location-info as described in Table 6.1.1.4.3.3-3			

Table 6.1.1.4.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.1	
mcptt-client-id	Encrypted <mcptt-client-id> with mcpttString set to the mcptt-client-id as provided by the UE at registration	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.1	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to true	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.1	

Table 6.1.1.4.3.3-3: Location-Info in SIP MESSAGE (Table 6.1.1.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
location-info				
Report				
CurrentLocation				
CurrentCoordinate				
longitude	Encrypted <longitude> with px_MCX_CoordinateLongitude_Client_A value	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.4.1-1		
latitude	Encrypted <latitude> with px_MCX_CoordinateLatitude_Client_A value	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.4.1-1		

Table 6.1.1.4.3.3-4: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Step 2, Table 6.1.1.4.3.2-1; Step 3, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with conditions MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-5			

Table 6.1.1.4.3.3-5: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	not present		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to true	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1	
mcptt-client-id	Encrypted <mcptt-client-id> with mcpttString set to the client-id as received from the UE	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1	
alert-ind-rcvd	true		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1	

Table 6.1.1.4.3.3-6: SIP MESSAGE from the MCPTTServer to SS-UE2 (Step 2, Table 6.1.1.4.3.2-1; Step 5, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and LOCATION-INFO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-7			
MIME body part		**Location-info**		
MIME-part-body	Location-info as described in Table 6.1.1.4.3.3-3			

Table 6.1.1.4.3.3-7: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.11	
mcptt-calling-user-id	Encrypted <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.1	
mcptt-calling-group-id	Encrypted < mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.11	
mc-org	px_MCX_DomainName_Organization_A	Indicates the organization an MCPTT user belongs to	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.12	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to true	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1		

Table 6.1.1.4.3.3-8: SIP MESSAGE from the SS-UE1 (Step 4, Table 6.1.1.4.3.2-1; Step 1, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-9			

Table 6.1.1.4.3.3-9: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, with conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.2	
mcptt-client-id	Encrypted <mcptt-client-id> with mcpttString set to the mcptt-client-id as provided by the UE at registration	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.2	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to false	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.1.2	

Table 6.1.1.4.3.3-10: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Step 4, Table 6.1.1.4.3.2-1; Step 3, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with conditions MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-11			

Table 6.1.1.4.3.3-11: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	not present		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
emergency-ind	Encrypted <emergency-ind> with mcpttBoolean set to false	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to false	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
mcptt-client-id	Encrypted <mcptt-client-id> with mcpttString set to the client-id as received from the UE	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
alert-ind-rcvd	true		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	

Table 6.1.1.4.3.3-12: SIP MESSAGE from the MCPTT Server to SS-UE2 (Step 4, Table 6.1.1.4.3.2-1; Step 5, Annex A, Table A.2.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.4.3.3-13			

Table 6.1.1.4.3.3-13: MCPTT-Info in SIP MESSAGE (Table 6.1.1.4.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	Encryption according to NOTE 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.11	
mcptt-calling-user-id	Encrypted <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	Encryption according to NOTE 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
mcptt-calling-group-id	Encrypted < mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID	Encryption according to NOTE 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.11	
alert-ind	Encrypted <alert-ind> with mcpttBoolean set to false	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	
emergency-ind	Encrypted <emergency-ind> with mcpttBoolean set to false	Encryption according to note 1 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.3.2	

6.1.2 Chat Group Calls

6.1.2.1 MCPTT Server - MCPTT Client / Chat Group Call / Pre-established Session

6.1.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having established a chat group call using pre-
established session }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to establish a chat group call with automatic
commencement mode and without implicit floor control with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and establishes the call with all users
of the group }
            }
```

(2)

```
with { IUT (MCPTT Server) having established a chat group call using pre-established session }
ensure that {
  when { the SS-UE2 (MCPTT Client) requests to establish a chat group call with automatic
commencement mode and without implicit floor control with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and establishes the call with all users
of the group }
            }
```

(3)

```
with { IUT (MCPTT Server) having established a chat group call with the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) requests the floor }
    then { IUT (MCPTT Server) sends floor granted to the SS-UE1 (MCPTT client) and sends floor taken
to the SS-UE2 (MCPTT client) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established a chat group call using pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client) release the floor }
    then { IUT (MCPTT Server) sends floor idle to the SS-UE1 (MCPTT client) and sends floor idle to
the SS-UE2 (MCPTT client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established a chat group call using pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests the call release keeping the pre-established session
with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and releases the SS-UE2 (MCPTT Client) }
            }
```

(6)

```
with { IUT (MCPTT Server) having established a chat group using pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
            }
```

(7)

```
with { IUT (MCPTT Server) having established a chat group using pre-established session }
ensure that {
  when { the SS-UE2 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
```

6.1.2.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.2.1, 10.1.2.3.2.1, 10.1.2.3.3.2, 10.1.2.3.4.2 and 10.1.2.4.3.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.2.1]

"Upon receiving a request from an MCPTT user to initiate or join an MCPTT group session using an MCPTT group identity identifying a chat MCPTT group within the pre-established session, the MCPTT client shall generate a SIP REFER request outside a dialog as specified in IETF RFC 3515 [25] as updated by IETF RFC 6665 [26] and IETF RFC 7647 [27], and in accordance with the UE procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

1) shall set the Request URI of the SIP REFER request to the session identity of the pre-established session;

2) shall set the Refer-To header field of the SIP REFER request as specified in IETF RFC 3515 [25] with a Content-ID ("cid") Uniform Resource Locator (URL) as specified in IETF RFC 2392 [62] that points to an application/resource-lists MIME body as specified in IETF RFC 5366 [20], and with the Content-ID header field set to this "cid" URL;

3) shall include in the application/resource-lists MIME body a single <entry> element containing a "uri" attribute set to the chat group identity, extended with the following parameters in the headers portion of the SIP URI:

NOTE 1: Characters that are not formatted as ASCII characters are escaped in the following parameters in the headers portion of the SIP URI.

a) the Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

b) an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6]; and

c) an hname "body" parameter populated with:

i) an application/sdp MIME body containing an SDP offer, if the session parameters of the pre-established session require modification or if implicit floor control is required, according to the conditions specified in clause 6.4;

ii) an application/vnd.3gpp.mcptt-info MIME body with:

A) the <session-type> element set to a value of "chat";

B) the <mcptt-client-id> element set to the MCPTT client ID of the originating MCPTT client; and

C) if the MCPTT client is aware of active functional aliases, and an active functional alias is to be included in the SIP REFER request, the <functional-alias-URI> set to the URI of the used functional alias; and

iii) if:

A) implicit floor control is required; and

B) an application/vnd.3gpp.mcptt-info MIME body with the <allow-location-info-when-talking> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true";

then shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element;

NOTE 2: The MCPTT client learns the functional aliases that are activated for an MCPTT ID from procedures specified in clause 9A.2.1.3.

…

8) shall include a P-Preferred-Service header field set to the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), according to IETF RFC 6050 [9];

9) shall include the following according to IETF RFC 4488 [22]:

a) the option tag "norefersub" in the Supported header field; and

b) the value "false" in the Refer-Sub header field.

10) shall include a Target-Dialog header field as specified in IETF RFC 4538 [23] identifying the pre-established session;

11) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP REFER request according to IETF RFC 3840 [16]; and

12) shall send the SIP REFER request according to 3GPP TS 24.229 [4].

On receiving a final SIP 2xx response to the SIP REFER request, the MCPTT client:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.2.1]

"Upon receipt of a "SIP REFER request for a pre-established session", with:

1) the Refer-To header containing a Content-ID ("cid") Uniform Resource Locator (URL) as specified in IETF RFC 2392 [62] that points to an application/resource-lists MIME body as specified in IETF RFC 5366 [20] containing an <entry> element with a "uri" attribute containing a SIP URI set to a chat group identity;

2) an hname "body" parameter in the headers portion of the SIP URI specified above containing an application/vnd.3gpp.mcptt-info MIME body with the <session-type> element set to "chat"; and

3) a Content-ID header field set to the "cid" URL;

the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP REFER request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

5) shall retrieve the group identity within the <entry> element of the application/resource-lists MIME body, referenced by the "cid" URL contained in the Refer-To header field of the SIP REFER request;

…

9) if the user identified by the MCPTT ID is not affiliated to the group identified in the SIP REFER request as determined by clause 9.2.2.2.11, shall perform the actions specified in clause 9.2.2.2.12 for implicit affiliation;

…

11) shall generate a final SIP 200 (OK) response to the "SIP REFER request for a pre-established session" according to 3GPP TS 24.229 [4];

12) if the "SIP REFER request for a pre-established session" contained a Refer-Sub header field containing the value "false" and a Supported header field containing "norefersub" value, shall handle the SIP REFER request as specified in 3GPP TS 24.229 [4], IETF RFC 3515 [25] as updated by IETF RFC 6665 [26], and IETF RFC 4488 [22] without establishing an implicit subscription;

13) shall determine the public service identity of the controlling MCPTT function associated with the group identity in the application/resource-lists MIME body pointed to by the "cid" URL in the Refer-To header field of the SIP REFER request. If the participating MCPTT function is unable to identify the controlling MCPTT function associated with the group identity, it shall reject the REFER request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with the remaining steps;

NOTE 6: The public service identity can identify the controlling function in the primary MCPTT system or a partner MCPTT system.

NOTE 7: How the participating MCPTT server discovers the public service identity of the controlling MCPTT function associated with the group identity is out of scope of the current document.

14) shall send the SIP 200 (OK) response to the SIP REFER request towards the MCPTT client according to 3GPP TS 24.229 [4];

NOTE 8: In accordance with IETF RFC 4488 [22], the participating MCPTT function inserts the Refer-Sub header field containing the value "false" in the SIP 200 (OK) response to the SIP REFER request to indicate that it has not created an implicit subscription.

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.4;

16) shall set the Request-URI of the SIP INVITE request to the public service identity of the controlling MCPTT function associated with the group identity;

17) shall copy the group identity from the "uri" attribute of the <entry> element of the application/resource-lists MIME body pointed to by the "cid" URL in the Refer-to header field of the SIP REFER request, to the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request;

18) if the received SIP REFER request contained a Resource-Priority header field, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the received SIP REFER request from the MCPTT client; and

…

19) shall forward the SIP INVITE request according to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the above SIP INVITE request in step 19) the participating MCPTT function shall follow procedures specified in 3GPP TS 24.229 [4], with the clarifications given below:

1) shall cache the contact received in the Contact header field;

2) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

3…

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.3.2]

"Upon receiving from the MCPTT client a SIP REFER request when using a pre-established session with the "method" SIP URI parameter set to value "BYE" in the URI in the Refer-To header field the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.7."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.4.2]

"Upon receiving a SIP BYE request from the controlling MCPTT function and if the MCPTT session id refers to an MCPTT user that has a pre-established session with the participating MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.3.2]

"When the MCPTT session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.5."

6.1.2.1.3 Test description

6.1.2.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Pre-established Session Establishment as specified in step 1 on Table 5.4.3.2-1 in clause 5.4.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Chat group.
- The IUT (MCPTT Server) is the controlling server of the Chat group.

6.1.2.1.3.2 Test procedure sequence

Table 6.1.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a chat group call for Group A, without implicit floor request using pre-established session.	-	-	-	-
2	Check: Does the MCPTT server correctly perform steps 1-4 of procedure 'MCPTT E2E call establishment using a pre-established session, automatic commencement' as described in Annex A, Table A.3.5.3-1 to establish a chat group call without implicit floor control for SS-UE1?	-	-	1	P
3	The SS-UE2 (MCPTT client) requests the establishment of a chat group call for Group A, without implicit floor request using pre-established session.	-	-	-	-
4	Check: Does the MCPTT server correctly perform steps 1-4 of procedure 'MCPTT E2E call establishment using a pre-established session, automatic commencement' as described in Annex A, Table A.3.5.3-1 to establish a chat group call without implicit floor control for SS-UE2?	-	-	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	The SS-UE1 (MCPTT client) requests the floor.	-	-	-	-
6	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	3	P
7	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	3	P
8	The SS-UE1 (MCPTT client) release the floor.	-	-	-	-
9	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	4	P
10	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	4	P
11	The SS-UE1 (MCPTT client) release the call keeping the pre-established session.	-	-	-	-
12	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E call release keeping the pre-established session' as described in Annex A, Table A.3.6.3-1? (see note)	-	-	5	P
13	The SS-UE1 (MCPTT client) release the pre-established session.	-	-	-	-
14	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE1?	-	-	6	P
15	The SS-UE2 (MCPTT client) release the pre-established session.	-	-	-	-
16	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE2?	-	-	7	P
NOTE: The MCPTT server initiates the call release towards SS-UE2 as there is just one participant in the call.					

6.1.2.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.2.1.3.3-1: SIP REFER from SS-UE1
(Step 2, Table 6.1.2.1.3.2-1; Step 1, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-1, with condition CHAT-GROUP-CALL for all clauses and subclauses referenced on that Table.

Table 6. 1.2.1.3.3-2: SIP 200 (OK)
(Step 2, Table 6.1.2.1.3.2-1; Step 2, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-3, with conditions CHAT-GROUP-CALL for all clauses and subclauses referenced on that Table.

Table 6.1.2.1.3.3-3: Connect
(Steps 2 and 4, Table 6.1.2.1.3.2-1; Step 3, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-4, with condition CHAT-GROUP-CALL.

Table 6.1.2.1.3.3-4: SIP REFER from SS-UE2
(Step 4, Table 6.1.2.1.3.2-1; Step 1, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-1, with condition CHAT-GROUP-CALL for all clauses and subclauses referenced on that Table.

Table 6. 1.2.1.3.3-5: SIP 200 (OK)
(Step 4, Table 6.1.2.1.3.2-1; Step 2, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-3, with conditions CHAT-GROUP-CALL for all clauses and subclauses referenced on that Table.

Table 6.1.2.1.3.3-6: Floor Taken (Step 7, Table 6.1.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 6.1.2.1.3.3-7: Floor Idle (Step 10, Table 6.1.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 6.1.2.1.3.3-8: SIP BYE from SS-UE1 and SS-UE2
(Steps 14 and 16, Table 6.1.2.1.3.2-1; Step 1, Annex A, Table A.2.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

6.1.2.2 MCPTT Server - MCPTT Client / Chat Group Call / On-demand / Join Chat Group Session / Upgrade to Emergency / Cancel Emergency / Upgrade to Imminent Peril / Cancel Imminent Peril

6.1.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests session establishment of chat group call without
implicit control with sending SIP INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an ongoing On-demand Chat Group Call }
ensure that {
  when { the SS-UE2 (MCPTT client ) requests session establishment by usage of implicit control with
sending SIP INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an ongoing On-demand Chat Group Call }
ensure that {
  when { the SS-UE1 (MCPTT Client) upgrades the call to emergency group call with implicit floor
control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
affiliated users of the group }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an ongoing Emergency Chat Group Call }
ensure that {
  when { the SS-UE1 (MCPTT client) release the floor }
    then { IUT (MCPTT Server) sends floor idle to the SS-UE1 (MCPTT client) and sends floor idle to
the SS-UE2 (MCPTT client) }
           }
```

(5)

```
with { IUT (MCPTT Server) having established an ongoing On-demand Chat Group Call }
ensure that {
  when { the SS-UE1 (MCPTT client) requests downgrade of the call to a normal group call }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the downgraded
call with all affiliated users of the group without implicit floor request }
           }
```

(6)

```
with { IUT (MCPTT Server) having established an ongoing On-demand Chat Group Call }
ensure that {
  when { the SS-UE1 (MCPTT Client) upgrades the call to imminent peril group call with implicit
floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
affiliated users of the group and sends floor taken to the SS-UE2 (MCPTT client) }
           }
```

(7)

```
with { IUT (MCPTT Server) having established an ongoing On-demand Pre-aranged Chat Group Call }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
```

6.1.2.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.1.1, 10.1.2.2.1.3, 10.1.2.2.1.4, 10.1.2.2.1.5, 10.1.2.3.1.1, 10.1.2.3.1.2, 10.1.2.3.1.4, 10.1.2.3.3.1, 10.1.2.3.4.1, 10.1.2.4.1.2, and 10.1.2.4.3.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.1.1]

"Upon receiving a request from an MCPTT user to initiate or join an MCPTT group session using an MCPTT group identity, identifying a chat MCPTT group, the MCPTT client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

1) *if the MCPTT user has requested the origination of an MCPTT emergency group call or is originating an MCPTT chat group call and the MCPTT emergency state is already set, the MCPTT client shall comply with the procedures in clause 6.2.8.1.1;*

2) *if the MCPTT user has requested the origination of an MCPTT imminent peril group call, shall comply with the procedures in clause 6.2.8.1.9;*

3) *shall include the g.3gpp.mcptt media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [16];*

4) *shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

5) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP INVITE request;*

6) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

7) should include the "timer" option tag in the Supported header field;

8) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the refresher parameter is omitted. If included, the refresher parameter shall be set to "uac";

9) shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCPTT function serving the MCPTT user;

NOTE 1: The MCPTT client is configured with public service identity identifying the participating MCPTT function serving the MCPTT user.

10) may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [4];

11) if the MCPTT client emergency group state for this group is set to "MEG 2: in-progress" or "MEG 4: confirm-pending", the MCPTT client shall comply with the procedures in clause 6.2.8.1.2;

12) if the MCPTT client imminent peril group state for this group is set to "MIG 2: in-progress" or "MIG 4: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

13) shall contain an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "chat";

b) the <mcptt-request-uri> element set to the group identity;

c) the <mcptt-client-id> element set to the MCPTT client ID of the originating MCPTT client; and

d) if the MCPTT client is aware of active functional aliases, and an active functional alias is to be included in the initial SIP INVITE request, the <functional-alias-URI> set to the URI of the used functional alias;

NOTE 2: The MCPTT ID of the originating MCPTT user is not included in the body, as this will be inserted into the body of the SIP INVITE request that is sent by the originating participating MCPTT function.

14) shall include in the SIP INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

15) if an implicit floor request is required, shall indicate this as specified in clause 6.4; and

16) shall send the SIP INVITE request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.1.3]

"This clause covers both on-demand session and pre-established sessions.

Upon receiving a request from an MCPTT user to cancel the in-progress emergency condition on a chat MCPTT group, the MCPTT client shall generate a SIP re-INVITE request as specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

…

2) shall, if the MCPTT user is cancelling an in-progress emergency condition and optionally an MCPTT emergency alert originated by the MCPTT user, include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.3;

3) shall, if the MCPTT user is cancelling an in-progress emergency condition and optionally an MCPTT emergency alert originated by another MCPTT user, include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.14;

4) shall, if the SIP re-INVITE request is to be sent within an on-demand session, include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…

6) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2; and

7) shall send the SIP re-INVITE request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.1.4]

"This clause covers both on-demand session and pre-established sessions.

Upon receiving a request from an MCPTT user to upgrade the MCPTT group session to an emergency condition or an imminent peril condition on a chat MCPTT group, the MCPTT client shall generate a SIP re-INVITE request as specified in 3GPP TS 24.229 [4], with the clarifications given below.

…

3) if the MCPTT user has requested to upgrade the MCPTT group session to an MCPTT emergency call, the MCPTT client:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body by following the procedures in clause 6.2.8.1.1; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2.

4) if the MCPTT user has requested to upgrade the MCPTT group session to an MCPTT imminent peril call, the MCPTT client:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body by following the procedures in clause 6.2.8.1.9; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

5) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…

8) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2; and

9) shall send the SIP re-INVITE request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.2.1.5]

"This clause covers both on-demand session and pre-established sessions.

Upon receiving a request from an MCPTT user to cancel the in-progress imminent peril condition on a chat MCPTT group, the MCPTT client shall generate a SIP re-INVITE request by following the procedures specified in 3GPP TS 24.229 [4], with the clarifications given below.

The MCPTT client:

…

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.2.8.1.11;

3) shall include a Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "chat"; and

b) the <mcptt-request-uri> element set to the group identity;

NOTE 1: The MCPTT ID of the originating MCPTT user is not included in the body, as this will be inserted into the body of the SIP re-INVITE request that is sent by the originating participating MCPTT function.

5) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP re-INVITE request according to IETF RFC 3840 [16];

6) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.2.1;

…

8) shall send the SIP re-INVITE request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body;

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" for a group identity identifying a chat MCPTT group containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "chat", the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

6) if the user identified by the MCPTT ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCPTT function" as determined by clause 9.2.2.2.11, shall perform the actions specified in clause 9.2.2.2.12 for implicit affiliation;

…

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity present in the incoming SIP INVITE request;

11) shall include the MCPTT ID of the calling user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request;

11a) if the received SIP request contains a <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body, then check if the status of the functional alias is activated for the MCPTT ID. If the functional alias status is activated, then set the <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request to the received value, if the status is unequal activated then do not include a <functional-alias-URI> element;

NOTE 8: The participating MCPTT server learns the functional alias state for an MCPTT ID from procedures specified in clause 9A.2.2.2.7.

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request as specified in clause 6.3.2.1.1.1;

…

14) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of IETF RFC 4412 [29] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCPTT client;

NOTE 10: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

16) shall forward the SIP INVITE request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.1.2]

"This clause covers both on-demand session and pre-established sessions.

Upon receipt of a SIP re-INVITE request for a served MCPTT client of a chat MCPTT group, the participating MCPTT function:

…

3) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.1.9;

4) shall include the MCPTT ID of the originating user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request;

5) shall, if the SIP re-INVITE request was received within an on-demand session, include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.1.1.1;

…

7) if the received SIP re-INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

NOTE 3: The controlling MCPTT function will determine the validity of the Resource-Priority header field.

8) shall forward the SIP re-INVITE request according to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the above SIP re-INVITE request in step 7) the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

…

3) if the SIP 200 (OK) response is to be sent within a pre-established session shall include in the SIP 200 (OK) response an SDP answer based upon the previously negotiated SDP for the pre-established session;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response; and

5) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.1.4]

"This clause covers both on-demand session and pre-established sessions.

Upon receipt of a SIP re-INVITE request for a terminating MCPTT client of a chat MCPTT group, the participating MCPTT function:

1) shall check if a Resource-Priority header field is included in the incoming SIP re-INVITE request and may perform further actions outside the scope of this specification to act upon an included Resource-Priority header field as specified in 3GPP TS 24.229 [4];

…

3) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.2.10;

4) shall include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.2.1; and

5) shall send the SIP re-INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the above SIP re-INVITE request sent to the MCPTT client, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1; and

3) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.3.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.4.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.1.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP re-INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCPTT session identity identifying a chat MCPTT group session, the controlling MCPTT function:

…

4) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCPTT function shall:

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency call;

…

v) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other affiliated and joined participants of the chat MCPTT group as specified in clause 6.3.3.1.6. The MCPTT controlling function:

A) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

B) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

vi) shall generate SIP INVITE requests for the MCPTT emergency group call to the affiliated but not joined members of the chat MCPTT group as specified in clause 6.3.3.1.7. The controlling MCPTT function:

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

vii) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";

6) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.13.4 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCPTT function shall:

b) shall set the in-progress emergency group state of the group to a value of "false";

c) shall clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency group call;

…

e) shall generate SIP re-INVITE requests to the other affiliated and joined members of the MCPTT group as specified in clause 6.3.3.1.6. The MCPTT controlling function:

i) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

…

8) if the received SIP re-INVITE request contains an imminent peril indication, shall perform the procedures specified in clause 10.1.2.4.1.3 and skip the rest of the steps;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.3.2]

"When the MCPTT session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.5."

6.1.2.2.3 Test description

6.1.2.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Chat group.
- The IUT (MCPTT Server) is the controlling server of the Chat group.

6.1.2.2.3.2 Test procedure sequence

Table 6.1.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a chat group call to Group A without implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' as described in Table A.2.14.3-1 with MCPTT to establish a chat group call without implicit floor request for SS-UE1?	-	-	1	P
3	The SS-UE2 (MCPTT client) requests the establishment of a chat group call to Group A, to join chat group session without implicit floor request.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' as described in Table A.2.14.3-1 with MCPTT to establish a chat group call without implicit floor request for SS-UE2?	-	-	2	P
5	The SS-UE1 (MCPTT client) requests upgrade of the call to an emergency group call with implicit floor request.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to upgrade the call to emergency group call with implicit floor request, with option b.ii of note on Table A.3.1.3-1?	-	-	3	P
7	The SS-UE1 (MCPTT client) releases the floor.	-	-	-	-
8	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	4	P
9	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	4	P
10	The SS-UE1 (MCPTT client) requests downgrade of the call to a normal group call.	-	-	-	-
11	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to downgrade the call without implicit floor request, with option a of note on Table A.3.1.3-1?	-	-	5	P
12	The SS-UE1 (MCPTT client) requests upgrade of the call to an imminent peril group call with implicit floor request.	-	-	-	-
13	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to upgrade the call to imminent peril group call with implicit floor request, with option b.ii of note on Table A.3.1.3-1?	-	-	6	P
14	The SS-UE1 (MCPTT client) releases the floor.	-	-	-	-
15	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	4	P
16	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	4	P
17	The SS-UE1 (MCPTT client) requests downgrade of the call to a normal group call.	-	-	-	-
18	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to downgrade the call without implicit floor request, with option a of note on Table A.3.1.3-1?	-	-	5	P
19	The SS-UE1 (MCPTT client) releases the call.	-	-	-	-
20	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1? (see note)	-	-	7	P
NOTE: The MCPTT server initiates the call release towards SS-UE2 as there is just one participant in the call.					

6.1.2.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.2.2.3.3-1: SIP INVITE from the SS-UE1
(Step 2, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-1B			

Table 6.1.2.2.3.3-1A: SDP in SIP INVITE from the SS-UE1
(Step 2, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1
(Step 2, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-2: SIP 200 (OK) from the MCPTT Server
(Step 2, Table 6.1.2.2.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-2A			

Table 6.1.2.2.3.3-2A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 2, Table 6.1.2.2.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-3: SIP INVITE from the SS-UE2
(Step 4, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-3A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-3B			

**Table 6.1.2.2.3.3-3A: SDP in SIP INVITE from the SS-UE2
(Step 4, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)**

Derivation Path: Annex A, Table A.3.1.4-2, with conditions CHAT-GROUP-CALL

**Table 6.1.2.2.3.3-3B: MCPTT-Info in SIP INVITE from the SS-UE2
(Step 4, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.2.14.3-1)**

Derivation Path: Annex A, Table A.3.1.4-3, with conditions CHAT-GROUP-CALL

**Table 6.1.2.2.3.3-4: SIP 200 (OK) from the MCPTT Server
(Step 4, Table 6.1.2.2.3.2-1; Step 3, Annex A, Table A.2.14.3-1)**

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-4A			

**Table 6.1.2.2.3.3-4A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 4, Table 6.1.2.2.3.2-1; Step 3, Annex A, Table A.2.14.3-1)**

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

**Table 6.1.2.2.3.3-5: SIP re-INVITE from the SS-UE1 (Step 6, Table 6.1.2.2.3.2-1;
Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)**

Derivation Path: Annex A, Table A.3.1.4-1, with conditions CHAT-GROUP-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-5B			

**Table 6.1.2.2.3.3-5A: SDP in SIP re-INVITE from the SS-UE1 (Step 6, Table 6.1.2.2.3.2-1;
Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)**

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

**Table 6.1.2.2.3.3-5B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Step 6, Table 6.1.2.2.3.2-1;
Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)**

Derivation Path: Annex A, Table A.3.2.4-3, with conditions CHAT-GROUP-CALL, EMERGENCY-CALL

Table 6.1.2.2.3.3-6: SIP re-INVITE from the MCPTT Server (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions CHAT-GROUP-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-6B			

Table 6.1.2.2.3.3-6A: SDP in SIP re-INVITE from the MCPTT Server (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.2.2.3.3-6B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.2.4-6, with conditions CHAT-GROUP-CALL, EMERGENCY-CALL

Table 6.1.2.2.3.3-7: SIP 200 (OK) from the SS-UE2 (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-7A			

Table 6.1.2.2.3.3-7A: SDP in SIP 200 (OK) from the SS-UE2 (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-8: SIP 200 (OK) from the MCPTT Server (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-8A			

Table 6.1.2.2.3.3-8A: SDP in SIP 200 (OK) from the MCPTT Server (Step 6, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-9: SIP re-INVITE from the SS-UE1 (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions CHAT-GROUP-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-9A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-9B			

Table 6.1.2.2.3.3-9A: SDP in SIP re-INVITE from the SS-UE1 (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note1), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.2.2.3.3-9B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.2.4-3, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-10: SIP re-INVITE from the MCPTT Server (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions CHAT-GROUP-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-10B			

Table 6.1.2.2.3.3-10A: SDP in SIP re-INVITE from the MCPTT Server (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.2.2.3.3-10B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.2.4-6, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-11: SIP 200 (OK) from the SS-UE2 (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-11A			

Table 6.1.2.2.3.3-11A: SDP in SIP 200 (OK) from the SS-UE2 (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-12: SIP 200 (OK) from the MCPTT Server (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-12A			

Table 6.1.2.2.3.3-12A: SDP in SIP 200 (OK) from the MCPTT Server (Steps 11 and 18, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-13: SIP re-INVITE from the SS-UE1 (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions CHAT-GROUP-CALL, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-13A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-13B			

Table 6.1.2.2.3.3-13A: SDP in SIP re-INVITE from the SS-UE1 (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.2.2.3.3-13B: MCPTT-Info in SIP re-INVITE from the SS-UE1 (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.2.4-3, with conditions CHAT-GROUP-CALL, IMMPERIL-CALL

Table 6.1.2.2.3.3-14: SIP re-INVITE from the MCPTT Server (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions CHAT-GROUP-CALL, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.2.3.3-14A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.2.2.3.3-14B			

Table 6.1.2.2.3.3-14A: SDP in SIP re-INVITE from the MCPTT Server (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), CHAT-GROUP-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.1.2.2.3.3-14B: MCPTT-Info in SIP re-INVITE from the MCPTT Server (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.2.4-6, with conditions CHAT-GROUP-CALL, IMMPERIL-CALL

Table 6.1.2.2.3.3-15: SIP 200 (OK) from the SS-UE2 (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-15A			

Table 6.1.2.2.3.3-15A: SDP in SIP 200 (OK) from the SS-UE2 (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-16: SIP 200 (OK) from the MCPTT Server (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.2.2.3.3-16A			

Table 6.1.2.2.3.3-16A: SDP in SIP 200 (OK) from the MCPTT Server (Step 13, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

Table 6.1.2.2.3.3-17: Floor Granted
(Step 6, Table 6.1.2.2.3.2-1; Step 2a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions EMERGENCY-CALL

Table 6.1.2.2.3.3-18: Floor Taken
(Step 6, Table 6.1.2.2.3.2-1; Step 3a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions EMERGENCY-CALL

Table 6.1.2.2.3.3-19: Floor Release
(Step 8, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with conditions EMERGENCY-CALL

Table 6.1.2.2.3.3-20: Floor Idle
(Step 8, Table 6.1.2.2.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions EMERGENCY-CALL

Table 6.1.2.2.3.3-21: Floor Granted
(Step 13, Table 6.1.2.2.3.2-1; Step 2a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions IMMPERIL-CALL

Table 6.1.2.2.3.3-22: Floor Taken
(Step 13, Table 6.1.2.2.3.2-1; Step 3a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions IMMPERIL-CALL

Table 6.1.2.2.3.3-23: Floor Release
(Step 15, Table 6.1.2.2.3.2-1; Step 1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with conditions IMMPERIL-CALL

Table 6.1.2.2.3.3-24: Floor Idle
(Step 15, Table 6.1.2.2.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions IMMPERIL-CALL

6.1.3 Conference Event Package

6.1.3.1 MCPTT Server - MCPTT Client / Conference Event Package / Subscription to Conference Event Package / Termination of subscription

6.1.3.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests session establishment of chat group call without
implicit control with sending SIP INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
            }
```

(2)

```
with { IUT (MCPTT Server) having established a Chat Group Call with automatic commencement mode and
without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to the conference event package for the ongoing group
call with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
           }
```

(3)

```
with { IUT (MCPTT Server) having established a Chat Group Call with automatic commencement mode and
without implicit floor request }
ensure that {
  when { the SS-UE2 (MCPTT client ) requests session establishment of chat group call without
implicit control with sending SIP INVITE message }
    then { IUT (MCPTT Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message }
           }
```

(4)

```
with { IUT (MCPTT Server) having established a Chat Group Call with the clients }
ensure that {
  when { the IUT (MCPTT Server) triggers notification to indicate status joint client to the group }
    then { the IUT (MCPTT Server) requests notification by sending SIP NOTIFY message }
           }
```

(5)

```
with { IUT (MCPTT Server) having established a Chat Group Call with the clients }
ensure that {
  when { the SS-UE1 (MCPTT client ) requests to de-subscribe from MCPTT service settings with SIP
SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
           }
```

(6)

```
with { IUT (MCPTT Server) having established a Chat Group Call with the clients }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
```

6.1.3.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 10.1.3.3 and 10.1.3.4.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.3.3]

"Upon receipt of a SIP SUBSCRIBE request for conference event status subscription from a MCPTT user served by the participating MCPTT function and if the SIP SUBSCRIBE request contains:

1) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

2) *an Accept header field containing the application/conference-info+xml"MIME type; and*

3) *an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-request-uri> set to a MCPTT group ID;*

then the participating MCPTT function:

1) *shall attempt to resolve the received Request-URI to an existing MCPTT session identity;*

2) *if the participating MCPTT function could not resolve the received Request-URI to an existing MCPTT session identity, shall reject the SIP SUBSCRIBE response with a SIP 404 (Not Found) response with a warning text set to "137 the indicated group call does not exists" as specified in clause 4.4 and shall skip the rest of the steps*

3) shall generate a SUBSCRIBE request as specified in TS 24.229 [4]

4) shall set the SIP URI in the Request-URI with the MCPTT session identity that is mapped to the MCPTT session identity in the received Request-URI;

5) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body the <mcptt-calling-user-id> element set to the MCPTT ID of the served user: and

6) shall insert a Record-Route header containing a URI identifying its own address; and

7) shall send the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4].

Upon receiving a SIP response to the SIP SUBSCRIBE request the participating MCPTT function:

1) shall copy the content of the incoming SIP response to an outgoing SIP response;

2) if a SIP 200 (OK) response, shall include in the Contact header field of the outgoing SIP response an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response in the outgoing SIP response; and

3) shall forward the SIP response according to 3GPP TS 24.229 [4].

Upon receiving a SIP NOTIFY request within the dialog created by the SIP SUBSCRIBE request destined to a served MCPTT client, the participating MCPTT function:

1) shall include the public service identity of the MCPTT user in the Request-URI;

2) shall copy the content of the incoming SIP NOTIFY request to the outgoing SIP NOTIFY request; and

3) shall send the SIP NOTIFY request according to 3GPP TS 24.229 [4].

Upon receiving a SIP response to the SIP NOTIFY request the participating MCPTT function:

1) shall copy the content of the incoming SIP response to an outgoing SIP response;

2) if a SIP 200 (OK) response, shall include an MCPTT session identity constructed from the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response in the outgoing SIP response; and

3) shall forward the SIP response according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.3.4.2]

"The procedures in this clause is triggered by:

1) the receipt of a SIP SUBSCRIBE request as specified in clause 10.1.3.4.1;

2) the receipt of a SIP BYE request from one of the participants in a pre-arranged or a chat group session; or

3) when a new participant is added in a pre-arranged or chat group session.

When sending a conference event notification, the controlling MCPTT function:

1) shall generate a notification package as specified in clause 6.3.3.4 to all MCPTT clients which have subscribed to the conference event package; and

NOTE: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

2) shall send a SIP NOTIFY request to all participants which have subscribed to the conference event package as specified in 3GPP TS 24.229 [4]."

6.1.3.1.3 Test description

6.1.3.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Chat group.
- The IUT (MCPTT Server) is the controlling server of the Chat group.

6.1.3.1.3.2 Test procedure sequence

Table 6.1.3.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the SS-UE1 (MCPTT client) request the establishment of a chat group call to Group A with automatic commencement mode and without implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' as described in Table A.2.14.3-1 with MCPTT to establish a chat group call without implicit floor request for SS-UE1?	-	-	1	P
3	Make the SS-UE1 (MCPTT client) subscribe to the conference event package for the ongoing group call to receive the current status and later notifications.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the MCPTT server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to subscribe to the conference event package for the ongoing group call to receive the current status and later notifications?	-	-	2	P
5	Make the SS-UE2 (MCPTT client) request the establishment of a chat group call to Group A, to join chat group session, without implicit floor request.	-	-	-	-
6	Check: Does the MCPTT server correctly perform procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' as described in Table A.2.14.3-1 with MCPTT to establish a chat group call without implicit floor request for SS-UE2?	-	-	3	P
7	Check: Does the MCTT Server send a SIP NOTIFY message indicating status with SS-UE2 joint to the chat group call?	<--	SIP NOTIFY	4	P
8	The SS-UE1 (MCPTT Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
9	Make the SS-UE1 (MCPTT client) de-subscribe to the conference event package for the ongoing group call to stop receiving status and notifications.	-	-	-	-
10	Check: Does the MCPTT server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to de-subscribe to the conference event package for the ongoing group call to receive the current status and later notifications?	-	-	5	P
11	Make the SS-UE1 (MCPTT client) release the call.	-	-	-	-
12	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	6	P

6.1.3.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.3.1.3.3-1: SIP SUBSCRIBE from the SS-UE1 (Step 4, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.7.3-1)

Derivation Path: Annex A, Table A.2.7.3-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	MCPTT session identity of the group session as provided by the SS in the Contact header field of the SIP 200 OK at call establishment		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	
Accept-Contact				
ac-value				
feature-param	"+g.3gpp.icsi-ref=urn:urn-7:3gpp-service.ims.icsi.mcptt"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	
req-param	"require"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	

Derivation Path: Annex A, Table A.2.7.3-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
explicit-param	"explicit"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	
Expires				
value	"4294967295"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	
Accept				
media-range	"application/conference-info+xml"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.3.1.3.3-2			

Table 6.1.3.1.3.3-2: MCPTT-Info in SIP SUBSCRIBE (Table 6.1.3.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	The URI of the group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.3.1.3.3-3: SIP NOTIFY from the MCPTT Server (Step 4, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.4	
Event				
event-type	"conference"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.4	
Expires				
value	"3600"	3600 seconds	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.4	
Subscription-State				
expires	"3600"			
P-Preferred-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.4	
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.3.1.3.3-4			
MIME body part		**Conference-Info**		
MIME-part-body	Conference-Info as described in Table 6.1.3.1.3.3-5			

Table 6.1.3.1.3.3-4: MCPTT-Info in SIP NOTIFY (Table 6.1.3.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	not present			
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-1A.				

Table 6.1.3.1.3.3-5: Conference-Info in SIP NOTIFY (Table 6.1.3.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.15-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
conference-info				
users				
user [2]	not present			
user [3]	not present			

Table 6.1.3.1.3.3-6: SIP NOTIFY from the MCPTT Server (Step 7, Table 6.1.3.1.3.2-1)

Derivation Path: Table 6.1.3.1.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.3.1.3.3-4			
MIME body part		**Conference-Info**		
MIME-part-body	Conference-Info as described in Table 6.1.3.1.3.3-7			

Table 6.1.3.1.3.3-7: Conference-Info in SIP NOTIFY (Table 6.1.3.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.15-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
conference-info				
users				
user [2]				
endpoint				
status	connected			
user [3]	not present			

Table 6.1.3.1.3.3-8: SIP SUBSCRIBE from the SS-UE1 (Step 10, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.7.3-1)

Derivation Path: Table 6.1.3.1.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"0"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.3.2	

Table 6.1.3.1.3.3-9: SIP NOTIFY from the MCPTT Server (Step 10, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.7.3-1)

Derivation Path: Table 6.1.3.1.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [21]	
substate-value	"terminated"			
expires	not present			

Table 6.1.3.1.3.3-10: SIP INVITE from the SS-UE1 (Step 2, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.1.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.3.1.3.3-10B			

Table 6.1.3.1.3.3-10A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions CHAT-GROUP-CALL

Table 6.1.3.1.3.3-10B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions CHAT-GROUP-CALL

Table 6.1.3.1.3.3-11: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.3.1.3.3-11A			

Table 6.1.3.1.3.3-11A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

Table 6.1.3.1.3.3-12: SIP INVITE from the SS-UE2
(Step 6, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.1.3.3-12A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.3.1.3.3-12B			

Table 6.1.3.1.3.3-12A: SDP in SIP INVITE from the SS-UE2
(Step 6, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions CHAT-GROUP-CALL

Table 6.1.3.1.3.3-12B: MCPTT-Info in SIP INVITE from the SS-UE2
(Step 6, Table 6.1.3.1.3.2-1; Step 1, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	not present or encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.1.3.1.3.3-13: SIP 200 (OK) from the MCPTT Server
(Step 6, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.1.3.1.3.3-13A			

Table 6.1.3.1.3.3-13A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 6, Table 6.1.3.1.3.2-1; Step 3, Annex A, Table A.2.14.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions CHAT-GROUP-CALL

6.1.4 Remote Change of Selected Group

6.1.4.1 MCPTT Server - MCPTT Client / Remote Change of Selected Group / Selected Group Change of Targeted User

6.1.4.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCPTT Client) requests to change the selected group of targeted user B, in the SS-
UE2 (MCPTT Client), to group B by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to UE2 (MCPTT Client) and responds with a
SIP 200 (OK) to UE1 (MCPTT Client) }
            }
```

6.1.4.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 10.1.4.2.1, 10.1.4.2.2, 10.1.4.3.1, 10.1.4.3.2 and 10.1.4.4. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.4.2.1]

"Upon receiving a request from the MCPTT user to send a group selection change request to change the selected group of a targeted MCPTT user to a specific MCPTT group, the MCPTT client:

…

2) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4]; and

d) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-request-uri> set to the MCPTT group identity to be selected by the targeted MCPTT user; and

ii) the <anyExt> element containing the <request-type> element set to a value of "group-selection-change-request"; and

e) shall insert in the SIP MESSAGE request a MIME resource-lists body with the MCPTT ID of the targeted MCPTT user, according to rules and procedures of IETF RFC 5366 [20];

3) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the MCPTT user; and

4) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.4.2.2]

"Upon receiving a "SIP MESSAGE request for group selection change request for terminating client", the MCPTT client:

1) if the received SIP MESSAGE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing an <affiliation-required> element set to a value of "true":

a) shall invoke the procedures of clause 9.2.1.2 to affiliate to the MCPTT group identified by the contents of the <mcptt-calling-group-id> included in the application/vnd.3gpp.mcptt-info+xml MIME body;

b) if the MCPTT client has not already invoked the procedures of clause 9.2.1.3, shall invoke the procedures of clause 9.2.1.3; and

c) upon receiving a SIP NOTIFY request including a <p-id> element set to a value matching the <p-id> value included in the SIP PUBLISH request sent in step 1) a) above as specified in clause 9.2.1.3, shall determine if the affiliation procedure to the MCPTT group identified by the contents of the <mcptt-calling-group-id> in the received SIP MESSAGE request was successful;

2) if the received SIP MESSAGE request contained an application/vnd.3gpp.mcptt-info+xml MIME body containing an <affiliation-required> element set to a value of "true" and the affiliation was successful as determined in step 1) c) above, or if the <affiliation-required> element was not present in the received SIP MESSAGE request:

a) shall change the MCPTT client's selected group to the MCPTT group identified by the contents of the <mcptt-calling-group-id> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body included in the received SIP MESSAGE request; and

b) shall determine the success or failure of the change of selected group action;

3) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

d) shall include in an application/resource-lists+xml MIME body, the MCPTT ID contained in the <mcptt-calling-user-id> element in the application/ vnd.3gpp.mcptt-info+xml MIME body of the received SIP MESSAGE request; and

e) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-request-uri> set to the MCPTT group identity to be selected by the MCPTT user; and

ii) the <anyExt> element containing:

A) the <response-type> element set to a value of "group-selection-change-response";

B) if the MCPTT client was able to successfully change the selected group as determined in step 2) b) above, then a <selected-group-change-outcome> element set to a value of "success"; and

C) if the MCPTT client:

I) was required to affiliate to the MCPTT group identified by the contents of the <mcptt-calling-group-id> in the received SIP MESSAGE request and the affiliation failed as determined in step 1) c); or

II) failed to change the selected group as determined in step 2) b);

then a <selected-group-change-outcome> element set to a value of "fail";

4) should indicate to the MCPTT user the success or failure of the requested change of selected group action;

5) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the MCPTT user; and

6) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.4.3.1]

"Upon receiving a "SIP MESSAGE request for group-selection-change for originating participating MCPTT function" the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;

NOTE: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

3) if the "SIP MESSAGE request for group selection change for originating participating MCPTT function" contains the <request-type> element set to a value of "group-selection-change-request":

a) if:

i) the <RemoteGroupSelectionURIList> element does not exist in the MCPTT user profile document with one or more <entry> elements (see the MCPTT user profile document in 3GPP TS 24.484 [50]); or

ii) if the MCPTT ID contained in the <mcptt-request-uri> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body included in the received "SIP MESSAGE request for group selection change for originating participating MCPTT function" does not match with one of the <entry> elements of the <RemoteGroupSelectionURIList> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]);

then:

i) shall reject the SIP MESSAGE request with a SIP 403 (Forbidden) response including warning text set to "155 user not authorised to change user's selected group" in a Warning header field as specified in clause 4.4, and not continue with the rest of the steps in this clause;

4) shall determine the public service identity of the controlling MCPTT function associated with the group identity contained in the <mcptt-request-uri> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body;

5) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

6) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCPTT function associated with the group identity contained in the received SIP MESSAGE request;

7) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 included in the outgoing SIP MESSAGE request;

8) shall copy the contents of the application/resource-lists MIME body into the outgoing SIP MESSAGE request;

9) shall set the <mcptt-calling-user-id> element of the <mcpttinfo> element containing the <mcptt-Params> element to the MCPTT ID determined in step 2) above;

10) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request;

11) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

12) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

14) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response in response to the sent SIP MESSAGE request, the participating MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the MCPTT client.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, shall forward the error response to the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.4.3.2]

"Upon receiving a "SIP MESSAGE request for group-selection-change for terminating participating MCPTT function" the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

…

4) shall generate an outgoing SIP MESSAGE request as specified in clause 6.3.2.2.11;

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

6) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall forward the SIP 2xx response to the controlling MCPTT function.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, shall forward the response to the controlling MCPTT function."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.4.4]

"Upon receiving:

- *a "SIP MESSAGE request for group selection change request for controlling MCPTT function"; or*
- *a "SIP MESSAGE request for group selection change response for controlling MCPTT function";*

the controlling MCPTT function:

…

3) if there is a <request-type> element set to a value of "group-selection-change-request" contained in the <anyExt> element in the <mcptt-Params> element contained in the <mcpttinfo> root element contained in the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request:

a) if the MCPTT user identified by the MCPTT ID in the application/resource-lists MIME body contained in the SIP MESSAGE request is not affiliated with the MCPTT group identified by the <mcptt-request-uri> in the application/vnd.3gpp. mcptt-info+xml MIME body as determined by the procedures of clause 6.3.6:

i) shall determine if the MCPTT user is eligible to be affiliated with the MCPTT group as determined by clause 9.2.2.3.8; and

ii) if the MCPTT user is not eligible for affiliation, shall reject the SIP MESSAGE request with a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

4) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

5) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

6) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

7) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP MESSAGE request with the following clarifications:

a) shall set the <mcptt-calling-group-id> to the MCPTT group identity contained in the <mcptt-request-uri> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body included in the received SIP MESSAGE request; and

b) shall set the <mcptt-request-uri> element of the application/vnd.3gpp. mcptt-info+xml MIME body in the outgoing SIP MESSAGE request to the MCPTT ID of the targeted MCPTT user contained in the application/resource-lists MIME body contained in the received SIP MESSAGE request;

8) if the received SIP MESSAGE request is a "SIP MESSAGE request for group selection change request for controlling MCPTT function":

a) if the targeted MCPTT user is not affiliated to the identified MCPTT group and was determined to be eligible to be affiliated with the MCPTT group in step 3) a) i) above, shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <anyExt> element an <affiliation-required> element set to a value of "true";

9) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated with the targeted MCPTT user;

NOTE: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

10) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";

11) shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request; and

12) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the controlling MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the originating participating MCPTT function.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, controlling MCPTT function shall forward the error response to the originating participating MCPTT function."

6.1.4.1.3 Test description

6.1.4.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE2 (MCPTT client) is affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.

6.1.4.1.3.2 Test procedure sequence

Table 6.1.4.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the SS-UE1 (MCPTT client) request to change the selected group of user B to group A for SS-UE2.	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to change the selected group of targeted user B, in the SS-UE2, to group A?	-	-	1	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to notify SS-UE1 about the successful change of the selected group A in the SS-UE2? (see note)	-	-	1	P
NOTE: During this procedure the SS-UE1 performs the role of SS-UE2 and viceversa.					

6.1.4.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.4.1.3.3-1: SIP MESSAGE from the SS-UE1 (Step 2, Table 6.1.4.1.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.4.1.3.3-2			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.1.4.1.3.3-3			

Table 6.1.4.1.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.1.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	The MCPTT group identity to be selected by the targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.1	
anyExt				
request-type	"group-selection-change-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.4.1.3.3-3: Resource lists in SIP MESSAGE (Table 6.1.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_B	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.1	

Table 6.1.4.1.3.3-4: SIP MESSAGE from the MCPTT-Server (Step 2, Table 6.1.4.1.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.4.1.3.3-5			

Table 6.1.4.1.3.3-5: MCPTT-Info in SIP MESSAGE (Table 6.1.4.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
anyExt				
request-type	"group-selection-change-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.4.1.3.3-6: SIP MESSAGE from the SS-UE2 (Step 3, Table 6.1.4.1.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.4.1.3.3-7			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.1.4.1.3.3-8			

Table 6.1.4.1.3.3-7: MCPTT-Info in SIP MESSAGE (Table 6.1.4.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	The MCPTT group identity to be selected by the targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.2	
anyExt				
response-type	"group-selection-change-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.4	
selected-group-change-outcome	"success"			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.4.1.3.3-8: Resource lists in SIP MESSAGE (Table 6.1.4.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_A	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.2.2	

Table 6.1.4.1.3.3-9: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Step 3, Table 6.1.4.1.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity				
name-addr	px_MCX_SIP_PublicUserId_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.4	
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.4.1.3.3-5			

Table 6.1.4.1.3.3-5: MCPTT-Info in SIP MESSAGE (Table 6.1.4.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
anyExt				
response-type	"group-selection-change-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.4.4	
selected-group-change-outcome	"success"			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

6.1.5 Remotely initiated group call

6.1.5.1 MCPTT Server - MCPTT Client / Remotely initiated group call

6.1.5.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) having received the provisioned MCPTT Client's group information or a
pointer to the group information, indicating that the MCPTT Client is authorized to affiliate with
the specified group}
ensure that {
  when { the SS-UE2 (MCPTT client ) requests to subscribe from affiliation status changes for the
SS-UE 1 MCPTT User with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(2)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { SS-UE2 (MCPTT Client) requests the remote initiation of a group call for Group A by SS-UE1
(MCPTT Client) by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to SS-UE1 (MCPTT Client) and responds with
a SIP 200 (OK) to SS-UE2 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having requested the remote initiation of a group call }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of group call with automatic commencement
mode and implicit floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with all
affiliated users of the group }
            }
```

(4)

```
with { IUT (MCPTT Server) having requested the remote initiation of a group call }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

6.1.5.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 10.1.5.2.1, 10.1.5.2.2, 10.1.5.3.1, 10.1.5.3.2, 10.1.5.4. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.5.2.1]

"Upon receiving a request from the requesting MCPTT user to send a remotely initiated group call request to the remote MCPTT user for a targeted MCPTT group, the MCPTT client:

…

3) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

d) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-request-uri> set to the MCPTT group identity of the targeted MCPTT group for the remotely initiated call; and

ii) an <anyExt> element containing:

A) the <request-type> element set to a value of "remotely-initiated-group-call-request";

B) the <notify-remote-user> element set to a value of "true" if the requesting MCPTT user has indicated that the remote MCPTT user be notified of the remotely initiated group call request; and

C) the <notify-remote-user> element set to a value of "false" if the requesting MCPTT user has indicated that the remote MCPTT user not be notified of the remotely initiated group call request;

e) shall insert in the SIP MESSAGE request a MIME resource-lists body with the MCPTT ID of the remote MCPTT user, according to rules and procedures of IETF RFC 5366 [20]; and

f) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the remote MCPTT user; and

4) shall send the SIP MESSAGE request towards the MCPTT server according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.5.2.2]

"Upon receiving a "SIP MESSAGE request for remotely initiated group call request for terminating client", the MCPTT client:

1) if the <notify-remote-user> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body contained in the received SIP MESSAGE request is set to a value of "true", may indicate to the remote MCPTT user that a remotely initiated call request to call the targeted MCPTT group has been received; and

2) shall invoke the procedures of clause 10.1.1.2.1.1 to originate an MCPTT group call to the targeted MCPTT group with the following clarifications:

a) if the <notify-remote-user> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body contained in the received SIP MESSAGE request is set to a value of "false":

i) shall not indicate to the remote MCPTT user that a remotely initiated call request to call the targeted MCPTT group has been received; and

ii) shall not give any indication to the remote MCPTT user that the remotely initiated call origination is in progress.

Upon completion of the procedures of clause 10.1.1.2.1.1, the MCPTT client:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4]; and

d) shall include in an application/resource-lists+xml MIME body the MCPTT ID contained in the <mcptt-calling-user-id> element in the application/ vnd.3gpp.mcptt-info+xml MIME body of the received SIP MESSAGE request; and

e) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-request-uri> set to the MCPTT group identity called by the remote MCPTT user; and

ii) an <anyExt> element containing:

A) the <response-type> element set to a value of "remotely-initiated-group-call-response";

B) if the procedures of clause 10.1.1.2.1.1 were successful in originating an MCPTT group call to the targeted MCPTT group, a <remotely-initiated-call-outcome> element set to a value of "success"; and

C) if the procedures of clause 10.1.1.2.1.1 were not successful in originating an MCPTT group call to the targeted MCPTT group, a <remotely-initiated-call-outcome> element set to a value of "fail";

2) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the requesting MCPTT user; and

3) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.5.3.1]

"In the procedures of this clause:

- *the term "requesting MCPTT user" is used to refer to the user who sent a request to initiate a remotely initiated group call; and*
- *the term "remote MCPTT user" is used to refer to the user who is sent a request to initiate a remotely initiated group call.*

Upon receiving a "SIP MESSAGE request for remotely initiated group call for originating participating MCPTT function" the participating MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP MESSAGE request with a SIP 500 (Server Internal Error) response. The participating MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24] and skip the rest of the steps;

2) shall determine the MCPTT ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;

NOTE: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

2a) if the participating MCPTT function cannot find a binding between the public user identity and an MCPTT ID or if the validity period of an existing binding has expired, then the participating MCPTT function shall reject the SIP MESSAGE request with a SIP 404 (Not Found) response with the warning text set to "141 user unknown to the participating function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

3) if the "SIP MESSAGE request for remotely initiated group call for originating participating MCPTT function" contains the <request-type> element set to a value of "remotely-initiated-group-call-request":

a) if the <allow-request-remote-init-group-call> element of the <ruleset> element is not present in the requesting MCPTT user's MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) or is set to a value of "false", shall reject the SIP MESSAGE request with a SIP 403 (Forbidden) response including warning text set to "1xx user not authorised to request a remotely initiated group call" in a Warning header field as specified in clause 4.4, and not continue with the rest of the steps in this clause;

4) shall determine the public service identity of the controlling MCPTT function associated with the group identity contained in the <mcptt-request-uri> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body;

5) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

6) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCPTT function associated with the group identity contained in the received SIP MESSAGE request;

7) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 included in the outgoing SIP MESSAGE request;

8) shall copy the contents of the application/resource-lists MIME body into the outgoing SIP MESSAGE request;

9) shall set the <mcptt-calling-user-id> element of the <mcptt-Params> element of the <mcpttinfo> element to the MCPTT ID determined in step 2) above;

10) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request;

11) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

12) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

14) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the sent SIP MESSAGE request, the participating MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the MCPTT client.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, shall forward the error response to the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.5.3.2]

"Upon receiving a "SIP MESSAGE request for remotely initiated group call for terminating participating MCPTT function" the participating MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP MESSAGE request with a SIP 500 (Server Internal Error) response. The participating MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24] and skip the rest of the steps;

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

3) *if the binding between the MCPTT ID and public user identity does not exist, then the participating MCPTT function shall reject the SIP MESSAGE request with a SIP 404 (Not Found) response and skip the rest of the steps;*

4) *shall generate an outgoing SIP MESSAGE request as specified in clause 6.3.2.2.11;*

5) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request; and*

6) *shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].*

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall forward the SIP 2xx response to the controlling MCPTT function.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, shall forward the response to the controlling MCPTT function."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.5.4]

"Upon receiving:

- *a "SIP MESSAGE request for remotely initiated group call request for controlling MCPTT function"; or*
- *a "SIP MESSAGE request for remotely initiated group call response for controlling MCPTT function";*

the controlling MCPTT function:

1) *if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP MESSAGE request with a SIP 500 (Server Internal Error) response. The controlling MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24] and skip the rest of the steps;*

2) *shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";*

3) *if:*

   a) *there is a <request-type> element set to a value of "remotely-initiated-group-call-request" contained in the <anyExt> element in the <mcptt-Params> element contained in the <mcpttinfo> root element contained in the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request; and*

   b) *if the MCPTT user identified by the MCPTT ID in the application/resource-lists MIME body contained in the SIP MESSAGE request is not affiliated with the MCPTT group identified by the <mcptt-request-uri> in the application/vnd.3gpp. mcptt-info+xml MIME body as determined by the procedures of clause 6.3.6;*

   then:

   a) *shall reject the SIP MESSAGE request with a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;*

4) *shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];*

5) *shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

6) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];*

7) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP MESSAGE request with the following clarifications:

a) shall set the <mcptt-calling-group-id> to the MCPTT group identity contained in the <mcptt-request-uri> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body included in the received SIP MESSAGE request; and

b) shall set the <mcptt-request-uri> element of the application/vnd.3gpp. mcptt-info+xml MIME body in the outgoing SIP MESSAGE request to the MCPTT ID contained in the application/resource-lists MIME body contained in the received SIP MESSAGE request;

8) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated with the MCPTT user identified by the MCPTT ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp. mcptt-info+xml MIME body in the outgoing SIP MESSAGE request;

NOTE: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

9) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";

10) shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request; and

11) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the controlling MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the originating participating MCPTT function.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request, the controlling MCPTT function shall forward the error response to the originating participating MCPTT function."

6.1.5.1.3 Test description

6.1.5.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1

- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.

6.1.5.1.3.2 Test procedure sequence

Table 6.1.5.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the SS-UE2 (MCPTT client) subscribe to affiliation status changes of targeted user A, in the SS-UE1.	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCPTT Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.3.7.3-1 to subscribe the SS-UE2 (MCPTT client) to affiliation status changes of targeted user A? (see note)	-	-	1	P
3	Make the SS-UE2 (MCPTT client) request the establishment of a remotely initiated group call by user A in group A with verification of user A's affiliation status.	-	-	-	-
4	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to request the remote initiation of a group call for Group A by SS-UE1? (see note)	-	-	2	P
5	Make the SS-UE1 (MCPTT client) request the establishment of a group call for Group A to SS-UE2 with automatic commencement mode and implicit floor request.	-	-	-	-
6	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a pre-arranged group call with automatic commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.1.3-1?	-	-	3	P
7	Check: Does the MCPTT server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 for a remotely initiated group call response indicating successful status?	-	-	2	P
8	Make the SS-UE1 (MCPTT client) release the call.	-	-	-	-
9	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	4	P
NOTE: During this procedure the SS-UE1 performs the role of SS-UE2 and viceversa.					

6.1.5.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.1.5.1.3.3-1: SIP SUBSCRIBE from the SS-UE2 (Step 2, Table 6.1.5.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [15]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.5.1.3.3-2			

Table 6.1.5.1.3.3-2: MCPTT-Info in SIP SUBSCRIBE (Table 6.1.5.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.5.1.3.3-3: SIP NOTIFY from the MCPTT Server (Steps 2, Table 6.1.5.1.3.2-1; Step 3, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 6.1.5.1.3.3-4			

Table 6.1.5.1.3.3-4: PIDF in SIP NOTIFY (Table 6.1.5.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_A			
tuple				
Id attribute	Encrypted URI (see note) with value set to px_MCX_Client_A_ID			
status				
affiliation				
status	"affiliated"			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 6.1.5.1.3.3-5: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.1.5.1.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.5.1.3.3-6			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.1.5.1.3.3-7			

Table 6.1.5.1.3.3-6: MCPTT-Info in SIP MESSAGE (Table 6.1.5.1.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	The MCPTT group identity of the targeted MCPTT group for the remotely initiated call	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.1	
anyExt				
request-type	"remotely-initiated-group-call-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.1	
notify-remote-user	"true"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.5.1.3.3-7: Resource lists in SIP MESSAGE (Table 6.1.5.1.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_A	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.1	

Table 6.1.5.1.3.3-8: SIP MESSAGE from the MCPTT-Server (Step 4, Table 6.1.5.1.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity				
name-addr	px_MCX_SIP_PublicUserId_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.5.1.3.3-9			

Table 6.1.5.1.3.3-9: MCPTT-Info in SIP MESSAGE (Table 6.1.5.1.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.3.1	
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
anyExt				
request-type	"remotely-initiated-group-call-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
notify-remote-user	"true"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.5.1.3.3-10: SIP MESSAGE from the SS-UE1 (Step 7, Table 6.1.5.1.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.1.5.1.3.3-11			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.1.5.1.3.3-12			

Table 6.1.5.1.3.3-11: MCPTT-Info in SIP MESSAGE (Table 6.1.5.1.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID	The MCPTT group identity called by the remote MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.2	
anyExt				
response-type	"remotely-initiated-group-call-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.2	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
remotely-initiated-call-outcome	"success"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.1.5.1.3.3-12: Resource lists in SIP MESSAGE (Table 6.1.5.1.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_B	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.2	

Table 6.1.5.1.3.3-13: SIP MESSAGE from the MCPTT-Server (Step 7, Table 6.1.5.1.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.5.1.3.3-14			

Table 6.1.5.1.3.3-14: MCPTT-Info in SIP MESSAGE (Table 6.1.5.1.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
anyExt				
response-type	"remotely-initiated-group-call-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
remotely-initiated-call-outcome	"success"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

6.2 Private Calls

6.2.1 MCPTT Server - MCPTT Client / Private Call / On-demand / Automatic Commencement Mode / With Floor Control / Upgrade to Emergency Call / Cancellation of Emergency on User request

6.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a private call with automatic
commencement mode and implicit floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request and supporting MCPTT_FloorRequestQueueing }
ensure that {
  when { the SS-UE2 (MCPTT client) requests the floor}
    then { IUT (MCPTT Server) enforces Floor Queue Position Info }
            }
```

(3)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request and not supporting MCPTT_FloorRequestQueueing }
ensure that {
  when { the SS-UE2 (MCPTT client) requests the floor }
    then { IUT (MCPTT Server) enforces Floor Deny }
            }
```

(4)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request and supporting MCPTT_FloorRequestQueueing }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to release the floor }
    then { IUT (MCPTT Server) enforces floor release with Floor Ack, Floor Taken to SS-UE1 (MCPTT
client) and Floor Granted to SS-UE2 (MCPTT client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request and supporting MCPTT_FloorRequestQueueing }
ensure that {
  when { the SS-UE2 (MCPTT client) requests to release the floor }
    then { IUT (MCPTT Server) enforces floor release with Floor Ack, Floor Idle to SS-UE2 (MCPTT
client) and Floor Idle to SS-UE1 (MCPTT client) }
            }
```

(6)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request and not supporting MCPTT_FloorRequestQueueing }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to release the floor }
    then { IUT (MCPTT Server) enforces floor release with Floor Ack, Floor Idle to SS-UE1 (MCPTT
client) and Floor Idle to SS-UE2 (MCPTT client) }
            }
```

(7)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to upgrade the call to emergency private call with
implicit floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and sends Floor Taken message to
SS-UE2 (MCPTT Client) }
            }
```

(8)

```
with { IUT (MCPTT Server) having established an Emergency Private Call and implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to release the floor }
    then { IUT (MCPTT Server) enforces floor release with Floor Ack, Floor Idle to SS-UE1 (MCPTT
client) and Floor Idle to SS-UE2 (MCPTT client) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established an Emergency Private Call and implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) requests the floor }
    then { IUT (MCPTT Server) sends floor granted to the SS-UE1 (MCPTT client) and sends floor taken
to the SS-UE2 (MCPTT client) }
            }
```

(10)

```
with { IUT (MCPTT Server) having established an Emergency Private Call and implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to downgrade the call to private call without implicit
floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the downgraded
call with all affiliated users of the group without implicit floor request }
            }
```

(11)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to SS-UE2 (MCPTT Client) }
```

6.2.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.1.3.1.1, 11.1.1.3.1.3, 11.1.1.3.2, 11.1.1.3.3, 11.1.1.4.2, 11.1.1.4.3, 11.1.1.4.4, 11.1.1.4.5, 11.1.1.4.6, 11.1.3.2.1.1, 11.1.3.2.2.1, 11.1.3.3.1, 4.6.2, 6.3.2.1.3, 6.3.2.1.4, 6.3.2.1.6, 6.3.2.1.8.2, 6.3.2.1.8.3, 6.3.2.2.4, 6.3.2.2.4.1, 6.3.2.2.4.2, 6.3.2.2.5.1, 6.3.2.2.5.2, 6.3.3.2.1, 6.3.3.2.3.1, 6.3.3.2.3.2, 6.3.3.2.4, 6.6.3.3.2, ETSI TS 124 380 [5] (3GPP TS 24.380), clause 13, ETSI TS 133 180 [31] (3GPP TS 33.180), clause 5.4. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "private" or "first-to-answer", the participating MCPTT function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [29] and shall not continue with the rest of the steps;

…

3) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

…

5) shall:

a) if the <session-type> is set to "private", determine that the call is a private call; and

…

6) if the call is a:

a) private call, determine the public service identity of the controlling MCPTT function for the private call service associated with the originating user's MCPTT ID identity; or

…

11) if the call is a private call and:

a) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [18] with the value "Auto" and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCPTT user profile document on the participating MCPTT function or is present with the value "false" (see the MCPTT user profile document in 3GPP TS 24.484 [50]) indicating that the user identified by the MCPTT ID is not authorised to initiate private call with automatic commencement, shall reject the "SIP INVITE request for originating participating MCPTT function" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

…

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3…

16) shall set the Request-URI to the public service identity of the controlling MCPTT function hosting the private call service or first-to-answer call service as determined by step 6);

17) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

18) if the call is a private call and:

…

d) if a Priv-Answer-Mode header field containing the value of "Auto" has not been included in the outgoing SIP INVITE request as specified in step 17) above and the incoming "SIP INVITE request for originating participating MCPTT function" contained an Answer-Mode header field as specified in IETF RFC 5373 [18], then shall populate the Answer-Mode header field of the outgoing SIP INVITE request with the contents of the Answer-Mode header field from the incoming "SIP INVITE request for originating participating MCPTT function";

19) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCPTT function", as specified in clause 6.3.2.1.1.1;

…

20) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCPTT client;

…

22) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 180 (Ringing) response, the participating MCPTT function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall send the SIP 180 (Ringing) response to the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

5A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [7].

The participating MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.1.3]

"Upon receipt of a SIP re-INVITE request for an existing MCPTT private call session the participating MCPTT function:

1) may reject the SIP re-INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP re-INVITE request according to rules and procedures specified in IETF RFC 4412 [29] and skip the rest of the steps;

…

3) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

…

3a) if the participating MCPTT function cannot find a binding between the public user identity and an MCPTT ID or if the validity period of an existing binding has expired, then the participating MCPTT function shall reject the SIP re-INVITE request with a SIP 404 (Not Found) response …

4) shall validate the media parameters and if the MCPTT speech codec is not offered in the SIP re-INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

…

5) shall generate a SIP re-INVITE request as specified in clause 6.3.2.1.9;

6) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP re-INVITE request to the MCPTT ID of the calling user;

…

7) shall, if the SIP re-INVITE request was received within an on-demand session include in the SIP re-INVITE request an SDP containing the current media parameters used by the existing session;

…

9) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field if included in the SIP re-INVITE request from the MCPTT client; and

10) shall forward the SIP re-INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) if the SIP 200 (OK) response is to be sent within an on-demand session shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

…

4) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

5) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4]; and

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

The participating MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if one of the following conditions are met:

a) "SIP INVITE request for terminating participating MCPTT function" contains an Answer-Mode header field with the value "Auto";

b) "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) "SIP INVITE request for terminating participating MCPTT function" contains a Priv-Answer-Mode header field with the value "Auto"; and

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.3]

"Upon receipt of a SIP re-INVITE request for an existing MCPTT private call session the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP re-INVITE request to retrieve the binding between the MCPTT ID and public user identity;

3) if the binding between the MCPTT ID and public user identity does not exist, then the participating MCPTT function shall reject the SIP re-INVITE request with a SIP 404 (Not Found) response and skip the rest of the steps;

4) shall generate a SIP re-INVITE as specified in clause 6.3.2.2.10;

NOTE 2: As this is the modification of an in-progress MCPTT private call, this procedure does not attempt modification of the existing answer-mode of the call.

5) shall include in the SIP re-INVITE request an SDP offer containing the current media parameters used by the existing session; and

6) shall send the SIP re-INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving the SIP 200 (OK) response to the SIP re-INVITE request, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4].

The participating MCPTT function shall forward any other SIP response that does not contain SDP along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.2]

"Upon receipt of:

- a "SIP INVITE request for controlling MCPTT function of a private call"; or

…

the controlling MCPTT function:

…

9) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCPTT user is not in an in-progress emergency private call state with the targeted MCPTT user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCPTT user has initiated an MCPTT emergency private call to the targeted user; and

b) shall cache the information that the MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

10) shall perform actions as described in clause 6.3.3.2.2;

11) shall allocate an MCPTT session identity for the MCPTT session; and

…

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4]; and

…

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 11.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], if the <session-type> in the SIP INVITE request is set to "private", the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) shall include a P-Answer-State header field with the value "Unconfirmed";

…

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

6) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

…

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

…

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4].

Upon receiving a SIP BYE request from the originating MCPTT client containing an application/vnd.3gpp.mcptt-info+xml MIME body containing a <release-reason> element set to a value of "authentication of the MIKEY-SAKE I_MESSAGE failed", the controlling MCPTT function:

…

2) shall follow the procedures in clause 11.1.3.3.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.3]

"Upon receiving a SIP re-INVITE request with an emergency indication set to a value of "true", the controlling MCPTT function:

1) *shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCPTT function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

2) *shall validate the request as described in clause 6.3.3.1.17;*

   …

4) *if a Resource-Priority header field is included in the received SIP re-INVITE request and if the Resource-Priority header field is set to the value indicated for emergency calls, shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps if neither of the following conditions are true:*

   a) *the SIP re-INVITE request does contains an authorised request for an MCPTT emergency call as determined in step 2 above; or*

   b) *the originating MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;*

5) *if the SIP re-INVITE request contains an emergency indication set to a value of "true" and the originating MCPTT user is not in an in-progress emergency private call state with the targeted MCPTT user:*

   a) *shall cache the information that the MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user; and*

   …

6) *shall send a SIP re-INVITE invite towards the MCPTT user listed in the MIME resource-lists body of received SIP re-INVITE request as specified in clause 11.1.1.4.5.*

Upon receiving a SIP 200 (OK) response for the SIP re-INVITE request and if the SIP response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) *shall generate a SIP 200 (OK) response to the SIP re-INVITE request as specified in the clause 6.3.3.2.3 before continuing with the rest of the steps;*

2) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP re-INVITE request containing the current media parameters used by the existing session;*

…

4) *shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and*

5) *shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18:

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.4]

"Upon receiving a SIP re-INVITE request with an emergency indication set to a value of "false", the controlling MCPTT function:

1) *shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCPTT function …*

2) *shall validate the request as described in clause 6.3.3.1.17;*

3) if the SIP re-INVITE request contains an unauthorised request for an MCPTT emergency private call cancellation as determined by clause 6.3.3.1.13.4:

a) shall reject the SIP re-INVITE …

4) shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response if a Resource-Priority header field is included in the received SIP re-INVITE request set to the value configured for emergency calls, and skip the remaining steps; and

5 if the SIP re-INVITE request contains an authorised request for an MCPTT emergency private call cancellation as determined by clause 6.3.3.1.13.4:

a) shall clear the cache of the MCPTT ID of the originator of the MCPTT emergency private call that is no longer in an in-progress emergency private call state with the targeted MCPTT user; and

…

6) shall send a SIP re-INVITE request towards the MCPTT user listed in the MIME resource-lists body of received SIP re-INVITE request as specified in clause 11.1.1.4.6.

Upon receiving a SIP 200 (OK) response for the SIP re-INVITE request and if the SIP response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP re-INVITE request as specified in the clause 6.3.3.2.3 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP re-INVITE request as specified in the clause 6.3.3.2.2;

…

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.5]

"This clause describes the procedures for sending a re-INVITE request to an MCPTT user in an MCPTT private call for the purpose of upgrading the session to an emergency private call session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 11.1.1.4.3.

The controlling MCPTT function:

1) shall generate a SIP re-INVITE request as specified in clause 6.3.3.1.9;

2) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing re-INVITE request;

3) if the received SIP re-INVITE request contains an authorised request for an MCPTT emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

…

4) shall include a Resource-Priority header field populated with the values for an MCPTT emergency private call as specified in clause 6.3.3.1.19, if the received SIP re-INVITE request contains an authorised request for an MCPTT emergency private call as determined in step 2 above; and

5) shall send the SIP re-INVITE request towards the core network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP re-INVITE request the controlling MCPTT function:

1) shall cache the contact received in the Contact header field."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.6]

"This clause describes the procedures for sending a re-INVITE request to an MCPTT user in an MCPTT emergency private call for the purpose of downgrading the session to a normal priority private call session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 11.1.1.4.4.

The controlling MCPTT function:

1) shall generate a SIP re-INVITE request as specified in clause 6.3.3.1.9;

2) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing re-INVITE request.

3) if the received SIP re-INVITE request contains an authorised request for an MCPTT emergency private call cancellation as determined by clause 6.3.3.1.13.4:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false";

…

4) shall include a Resource-Priority header field populated with the values for a normal MCPTT private call as specified in clause 6.3.3.1.19, if the received SIP re-INVITE request contains an authorised request for an MCPTT emergency private call cancellation as determined in step 3 above; and

5) shall send the SIP re-INVITE request towards the core network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP re-INVITE request the controlling MCPTT function:

1) shall cache the contact received in the Contact header field."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.1.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.2.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), 11.1.3.3.1]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.6.2]

"Key aspects of MCPTT emergency private calls include:

- adjusted EPS bearer priority for both participants whether or not they are both in an emergency condition (i.e. both have their MCPTT emergency state set). This is achieved by using the Resource-Priority header field as specified in IETF RFC 4412 [29] with namespaces defined for use by MCPTT specified in IETF RFC 8101 [48];

- the initiator of the MCPTT emergency private call can override the other MCPTT user in the MCPTT emergency private call unless that user also has their MCPTT emergency state set;

…

- restoration of normal floor control priority participants when the emergency elevated priority is cancelled;

- *requires the MCPTT user to be authorised to either originate or cancel an MCPTT emergency private call;*
- *requires the targeted MCPTT user to be authorised to receive an MCPTT emergency private call;*

…

- *the originator of the MCPTT emergency private call can request that the call use either manaual or automatic commencement mode."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 13]

"Media plane security provides integrity and confidentiality protection of individual media streams and media plane control messages in MCPTT sessions.

The media plane security is based on 3GPP MCPTT security solution including key management and end-to-end media and floor control messages protection as defined in 3GPP TS 33.180 [18].

…

In an on-network private call:

…

2. *if protection of floor control messages sent using unicast between the MCPTT client and the participating MCPTT function serving the MCPTT client is negotiated, the CSK and the CSK-ID protect the floor control messages sent and received by the MCPTT client and by the participating MCPTT function;*

…

4. *if protection of media control messages sent using unicast between the MCPTT client and the participating MCPTT function serving the MCPTT client is negotiated, the CSK and the CSK-ID protect the media control messages sent and received using unicast by the MCPTT client and by a participating MCPTT function; and*

…"

[ETSI TS 133 180 [31] (3GPP TS 33.180), clause 5.4]

"The key (CSK) is distributed from the MCX client to the MCX Server(s) using the 'CSK upload' procedure. The procedure shall use the common key distribution mechanism described in clause 5.2.2, transported over the SIP bearer. Identity hiding may be supported as defined in clause 5.2.6. The MCX Server may respond with a KMS Redirect Response (KRR) as described in clause 5.2.8 (e.g. if the MC client has migrated or is roaming).

The initiating entity of the CSK upload procedure shall be the MCX UE and the receiving entity shall be the MCX Server. With respect to the common key distribution procedure, the initiating entity URI shall be the MCX Service user ID of the user andthe receiving entity URI shall be the MCX Server Domain Security Identifier (MDSI). The MDSI is added to the recipient field (IDRr) of the message. The distributed key, K, shall be the CSK and the distributed identifier K-ID shall be the CSK-ID.

Clause E.4 provides MIKEY message structure for CSK distribution.

Before the CSK upload procedure can be used by the client to securely share the encryption key, the MC user shall first be authorized by KMS for key management services. Once the MC user is authorized, the KMS distributes the user's key material to the client as specified in clause 5.3.3.

The server receives the SIP message with the protected CSK and retrieves it from the message. It associates the MC User's SIP Core identity (IMPU), MC Service user ID (e.g. MCPTT ID) and the received CSK. Identity binding is used to uniquely identify the CSK used in protection of the SIP payload in subsequent SIP messages sent by both the client and the servers within a MC domain."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.3]

"When generating an initial SIP INVITE request according to 3GPP TS 24.229 [4], on receipt of an incoming SIP INVITE request, the participating MCPTT function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcptt media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

…

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

…

shall copy the application/vnd.3gpp.mcptt-location-info+xml MIME body from the SIP INVITE request into the outgoing SIP INVITE."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.4]

"When sending SIP provisional responses other than the SIP 100 (Trying) response, the participating MCPTT function shall generate a SIP provisional response according to 3GPP TS 24.229 [4] and:

1) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

c) the isfocus media feature tag; and

d) an MCPTT session identity mapped to the MCPTT session identity if provided in the Contact header field of the incoming provisional response;

2) shall include the "norefersub" option tag in a Supported header field in accordance with 3GPP TS 24.229 [4];

3) may include a Resource-Share header field in accordance with clause 5.7.1.20.2 in 3GPP TS 24.229 [4]; and

4) if the incoming SIP provisional response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP provisional response."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.6]

"Upon receiving a SIP BYE request from the MCPTT client, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [4];

3) shall set the Request-URI to the MCPTT session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCPTT function, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCPTT function shall forward a SIP 200 (OK) response to the MCPTT client and shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the controlling MCPTT function."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.8.2]

"If the participating MCPTT function receives a SIP request from the MCPTT client including a request for an MCPTT emergency alert and:

1) if the <allow-activate-emergency-alert> element of the <actions> element of the <rule> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true"; and

…

then the MCPTT emergency alert request shall be considered to be an authorised request for an MCPTT emergency alert. In all other cases, it shall be considered to be an unauthorised request for an MCPTT emergency alert.

If the participating MCPTT function receives a SIP request from the MCPTT client including a request to cancel an MCPTT emergency alert to an MCPTT group, and if the <allow-cancel-emergency-alert> element of the <actions> element of a <rule> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true", then the MCPTT emergency alert cancellation request shall be considered to be an authorised request to cancel an MCPTT emergency alert. In all other cases, it shall be considered to be an unauthorised request to cancel an MCPTT emergency alert."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.8.3]

"To validate the combinations of <emergency-ind>, <imminentperil-ind> and <alert-ind> which are received in SIP requests, the participating MCPTT function shall follow the procedures specified in clause 6.3.3.1.17."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.4]

"The participating MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4] and:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

c) the isfocus media feature tag;

d) an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

5) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

6) shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];

7) may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4]; and

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.4.1]

"When sending a SIP provisional response other than the SIP 100 (Trying) response to the SIP INVITE request, the participating MCPTT function shall generate a SIP provisional response according to 3GPP TS 24.229 [4] and:

1) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag; and

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

2) if the outgoing SIP provisional response is to be sent in response to the receipt of a SIP provisional response and the response contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP provisional response; and

3) if the incoming SIP INVITE request included the Supported header field with the value "100rel" and according to local policy, may include the Require header field with the value "100rel"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.4.2]

"When sending SIP 200 (OK) responses, the participating MCPTT function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [4] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCPTT function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.5.1]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" that requires automatic commencement mode:

1) if:

a) the invited MCPTT client has one or more pre-established sessions without an associated MCPTT session;

b) the media-level section for the offered MCPTT speech media stream is the same as the media-level section for MCPTT speech media stream in an existing pre-established session; and

c) the media-level section of the offered media-floor control entity is the same as the media-level section for media-floor control entity in an existing pre-established session;

then the participating MCPTT function shall perform the actions specified in clause 6.3.2.2.5.3; or

2) otherwise the participating MCPTT function shall perform the actions specified in clause 6.3.2.2.5.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.5.2]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" for an on-demand session that requires automatic commencement mode the participating MCPTT function:

1) if:

a) the incoming SIP INVITE request contained a Priv-Answer-Mode header field set to the value of "Auto";

b) no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) the incoming SIP INVITE request contained an Answer-Mode header field set to "Auto" and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer";

then:

a) shall generate a SIP 183 (Session Progress) response to the "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.4.1;

…

NOTE: The SIP 183 (session Progress) response can be sent reliably or unreliably depending on the content of the received SIP INVITE request. Regardless of if the SIP 183 (Session Progress) response is sent reliably or unreliably, SDP is not included in the SIP 183 (Session Progress) response.

c) shall set the P-Answer-State header field to "Unconfirmed" in the SIP 183 (Session Progress) response;

2) shall copy the public user identity contained in the Request-URI of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the SIP 183 (Session Progress) response;

3) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

4) shall set the Request-URI to the public user identity associated to the MCPTT ID of the MCPTT user to be invited;

5) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request, into the outgoing SIP INVITE request;

6) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

7) if the Priv-Answer-Mode header field is present in the incoming SIP INVITE request with a value of "Auto", shall include a Priv-Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request. Otherwise, if the Answer-Mode header field is present in the incoming SIP INVITE request, the participating MCPTT function shall include an Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request;

8) if no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer", shall set the Answer-Mode header field to "Auto" in the outgoing SIP INVITE request;

9) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.1;

10) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

11) shall send the SIP INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].

If the SIP 183 (Session Progress) response was sent reliably, then upon receiving a SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the above SIP INVITE request sent to the MCPTT client, the participating MCPTT function:

1) if the SIP 183 (Session Progress) was sent unreliably, shall send the SIP 200 (OK) response immediately; and

2) if the SIP 183 (Session Progress) was sent reliably and,

a) if the SIP PRACK request to the SIP 183 (Session Progress) response has been received by the participating MCPTT function and the SIP 200 (OK) response to the SIP PRACK request has been sent, shall send the SIP 200 (OK) response immediately;

b) if the SIP PRACK request to the SIP 183 (Session Progress) response has not yet been received, then upon receipt of the SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4], before sending the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function".

When the participating MCPTT function sends the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

3A) if the SIP 183 (Session Progress) was sent with the P-Answer-State header field set to "Unconfirmed", shall set the P-Answer-State header field to "Confirmed" in the outgoing SIP 200 (OK) response;

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4].

The participating MCPTT function shall forward any other SIP response that does not contain SDP along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resource associated with the SIP session with the controlling MCPTT function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [4];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) *if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

5) *shall send the SIP BYE request to the MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCPTT function:

1) *shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCPTT function according to 3GPP TS 24.229 [4]; and*

2) *shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT client."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) *for the accepted media stream in the received SDP offer:*

   a) *shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function; and*

2) *for the accepted media channel control channel, if present in the received SDP offer:*

   a) *shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function, for the accepted media plane control channel, if present in the received SDP offer; and*

   b) *shall include 'fmtp' attributes as specified in 3GPP TS 24.380 clause 14."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.3.1]

"When sending SIP provisional responses with the exception of the SIP 100 (Trying) response to the SIP INVITE request the controlling MCPTT function:

1) *shall generate the SIP provisional response;*

2) *shall include a P-Asserted-Identity header field with the public service identity of the controlling MCPTT function;*

3) *shall include an MCPTT session identity in the Contact header field; and*

4) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcptt media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and*

   c) *the isfocus media feature tag."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.3.2]

"When sending a SIP 200 (OK) response to the initial SIP INVITE request, the controlling MCPTT function:

1) *shall generate the SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];*

2) *shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";*

3) *shall include the option tag "timer" in a Require header field;*

4) *shall include a P-Asserted-Identity header field with the public service identity of the controlling MCPTT function;*

5) shall include a SIP URI for the MCPTT session identity in the Contact header field identifying the MCPTT session at the controlling MCPTT function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

9) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

10) shall include the "explicitsub" and "nosub" option tags in a Supported header field according to IETF RFC 7614 [35]; and

11) void

12) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

NOTE: The non-controlling MCPTT function is also regarded as a MCPTT client in a temporary MCPTT group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];

3) shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCPTT session is not required, shall send a SIP NOTIFY request to all remaining MCPTT clients in the MCPTT session with a subscription to the conference event package as specified in clause 10.1.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.6.3.2]

"The procedures in clause 6.6.3.3 and clause 6.6.3.4 shall be used with a XPK equal to the CSK and a XPK-ID equal to the CSK-ID in the following circumstances as described in 3GPP TS 33.180 [78]:

1) MCPTT client sends integrity protected content to an MCPTT server; and

2) MCPTT server sends integrity protected content to an MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.6.3.3.2]

"If the <integrity-protection> element in the Service Configuration document as specified in 3GPP TS 24.484 [50] is set to "true", or no <integrity-protection> element is present in the Service Configuration document, then sending integrity protected content from the MCPTT server to the MCPTT client is enabled. If the <allow-signalling-protection> element of the <protection-between-mcptt-servers> element is set to "true" in the Service Configuration document as specified in 3GPP TS 24.484 [50] or no <allow-signalling-protection> element is present in the Service Configuration document, then sending integrity protected content between MCPTT servers is enabled.

When sending integrity protected content, the MCPTT server shall use the appropriate keying information specified in clause 6.6.3.2 and shall perform the procedures in clause 6.6.3.3.3 to integrity protect XML MIME bodies.

NOTE: Each XML MIME body is integrity protected separately.

If the <integrity-protection> element in the Service Configuration document as specified in 3GPP TS 24.484 [50] is set to "false", then sending integrity protected content from the MCPTT server to the MCPTT client is disabled, and all XML MIME bodies are sent without integrity protection."

6.2.1.3 Test description

6.2.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.1.3.2 Test procedure sequence

Table 6.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a private call to SS-UE2 with automatic commencement mode and implicit floor request	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a private call with automatic commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	The SS-UE2 (MCPTT client) requests the floor	-	-	-	-
4	EXCEPTION: Steps 4a1 - 4b1 describe behaviour that depends on pc_MCPTT_FloorRequestQueueing. The "lower case letter" identifies a step sequence that takes place when one or the other is the case	-	-	-	-
4a1	IF pc_MCPTT_FloorRequestQueueing THEN: Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Queue Position Info' as described in Annex A, Table A.3.12.3-1 for SS-UE2?	-	-	2	P
4b1	ELSE IF NOT pc_MCPTT_FloorRequestQueueing THEN: Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Deny' as described in Annex A, Table A.3.14.3-1 for SS-UE2?	-	-	3	P
5	The SS-UE1 (MCPTT client) releases the floor	-	-	-	-
6	EXCEPTION: Steps 6a1 - 6b2 describe behaviour that depends on pc_MCPTT_FloorRequestQueueing. The "lower case letter" identifies a step sequence that takes place when one or the other is the case	-	-	-	-
6a1	IF pc_MCPTT_FloorRequestQueueing THEN: Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Taken' as described in Annex A, Table A.3.16.3-1 for SS-UE1?	-	-	4	P
6a2	Check: Does the MCPTT server send a Floor Granted message to SS-UE2?	<-	Floor Granted	4	P
6a3	The SS-UE2 (MCPTT client) releases the floor	-	-	-	-
6a4	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE2?	-	-	5	P
6a5	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<-	Floor Idle	5	P
6b1	ELSE IF NOT pc_MCPTT_FloorRequestQueueing THEN: Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	6	P
6b2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	6	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7	The SS-UE1 (MCPTT client) requests upgrade of the call to an emergency private call with implicit floor request	-	-	-	-
8	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying ' as described in Annex A, Table A.3.1.3-1 to upgrade the call to emergency private call with implicit floor request, with option b.ii of note on Table A.3.1.3-1?	-	-	7	P
9	void	-	-	-	-
10	The SS-UE1 (MCPTT client) releases the floor	-	-	-	-
11	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	8	P
12	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	8	P
13	The SS-UE1 (MCPTT client) requests the floor	-	-	-	-
14	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	9	P
15	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	9	P
16	The SS-UE1 (MCPTT client) releases the floor	-	-	-	-
17	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	8	P
18	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	8	P
19	The SS-UE1 (MCPTT client) requests downgrade of the call to a normal private call	-	-	-	-
20	Check: Does the MCPTT server correctly perform procedure ' MCPTT E2E session establishment/modification without provisional responses other than 100 Trying ' as described in Annex A, Table A.3.1.3-1 to downgrade the call without implicit floor request, with option a of note on Table A.3.1.3-1?	-	-	10	P
21	The SS-UE1 (MCPTT client) requests the floor	-	-	-	-
22	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Request - Floor Granted' as described in Annex A, Table A.3.11.3-1 for SS-UE1?	-	-	9	P
23	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<-	Floor Taken	9	P
24	The SS-UE1 (MCPTT client) releases the floor	-	-	-	-
25	Check: Does the MCPTT server correctly perform procedure 'MCPTT Floor Release - Floor Idle' as described in Annex A, Table A.3.15.3-1 for SS-UE1?	-	-	8	P
26	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<-	Floor Idle	8	P
27	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
28	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	11	P

6.2.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.1.3.3-1: SIP INVITE from the SS-UE1
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-1B			

Table 6.2.1.3.3-1A: SDP in SIP INVITE from the SS-UE1
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions PRIVATE-CALL, IMPLICIT_GRANT_REQUESTED

Table 6.2.1.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions PRIVATE -CALL

Table 6.2.1.3.3-2: SIP INVITE from the MCPTT Server
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-2B			

Table 6.2.1.3.3-2A: SDP in SIP INVITE from the MCPTT Server
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions PRIVATE-CALL

Table 6.2.1.3.3-2B: MCPTT-Info in SIP INVITE from the MCPTT Server
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions PRIVATE-CALL

Table 6.2.1.3.3-3: SIP 200 (OK) from the SS-UE2
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-3A			

Table 6.2.1.3.3-3A: SDP in SIP 200 (OK) from the SS-UE2
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions PRIVATE -CALL

Table 6.2.1.3.3-4: SIP 200 (OK) from the MCPTT Server
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-4A			

Table 6.2.1.3.3-4A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 2, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions PRIVATE-CALL

Table 6.2.1.3.3-5: SIP re-INVITE from the SS-UE1
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions PRIVATE-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-5B			

Table 6.2.1.3.3-5A: SDP in SIP re-INVITE from the SS-UE1
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), PRIVATE-CALL, IMPLICIT_GRANT_REQUESTED
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.2.1.3.3-5B: MCPTT-Info in SIP re-INVITE from the SS-UE1
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions PRIVATE-CALL, EMERGENCY-CALL

Table 6.2.1.3.3-6: SIP re-INVITE from the MCPTT Server
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions PRIVATE-CALL, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-6B			

Table 6.2.1.3.3-6A: SDP in SIP re-INVITE from the MCPTT Server
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), PRIVATE-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.2.1.3.3-6B: MCPTT-Info in SIP re-INVITE from the MCPTT Server
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions PRIVATE-CALL, EMERGENCY-CALL

Table 6.2.1.3.3-7: SIP 200 (OK) from the SS-UE2
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-7A			

Table 6.2.1.3.3-7A: SDP in SIP 200 (OK) from the SS-UE2
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions PRIVATE-CALL

Table 6.2.1.3.3-8: SIP 200 (OK) from the MCPTT Server
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-8A			

Table 6.2.1.3.3-8A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions PRIVATE-CALL

Table 6.2.1.3.3-9: SIP re-INVITE from the SS-UE1
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, with conditions PRIVATE-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-9A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-9B			

Table 6.2.1.3.3-9A: SDP in SIP re-INVITE from the SS-UE1
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions SDP_OFFER (see note), PRIVATE-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.2.1.3.3-9B: MCPTT-Info in SIP re-INVITE from the SS-UE1
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions PRIVATE-CALL

Table 6.2.1.3.3-10: SIP re-INVITE from the MCPTT Server
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, with conditions PRIVATE-CALL, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.1.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.1.3.3-10B			

Table 6.2.1.3.3-10A: SDP in SIP re-INVITE from the MCPTT Server
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions SDP_OFFER (see note), PRIVATE-CALL
NOTE: INITIAL_SDP_OFFER condition is removed.

Table 6.2.1.3.3-10B: MCPTT-Info in SIP re-INVITE from the MCPTT Server
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions PRIVATE-CALL

Table 6.2.1.3.3-11: SIP 200 (OK) from the SS-UE2
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-11A			

Table 6.2.1.3.3-11A: SDP in SIP 200 (OK) from the SS-UE2
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions PRIVATE-CALL

Table 6.2.1.3.3-12: SIP 200 (OK) from the MCPTT Server
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.1.3.3-12A			

Table 6.2.1.3.3-12A: SDP in SIP 200 (OK) from the MCPTT Server
(Step 20, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions PRIVATE-CALL

Table 6.2.1.3.3-13: Floor Granted
(Step 8, Table 6.2.1.3.2-1; Step 2a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-14: Floor Taken
(Step 8, Table 6.2.1.3.2-1; Step 3a1, Annex A, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-15: Floor Release
(Steps 11 and 17, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-16: Floor Idle
(Steps 11 and 17, Table 6.2.1.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-17: Floor Idle (Steps 12 and 18, Table 6.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-18: Floor Request
(Step 14, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-19: Floor Granted (Step 14, Table 6.2.1.3.2-1; Steps 2a1 and 2b1, Annex A, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions EMERGENCY-CALL

Table 6.2.1.3.3-20: Floor Taken (Step 15, Table 6.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions EMERGENCY-CALL

6.2.2 MCPTT Server - MCPTT Client / Private Call / On-demand / Automatic Commencement Mode / Without Floor Control

6.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a private call with automatic
commencement mode and without floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
           }
```

(2)

```
with { IUT (MCPTT Server) having established a Private Call with Automatic Commencement Mode and
without implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
```

6.2.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.2.3.1, 11.1.1.3.1.1, 11.1.2.3.2, 11.1.1.3.2, 11.1.2.4.1, 11.1.1.4.1, 11.1.2.4.2, 11.1.1.4.2, 11.1.3.2.1.1, 11.1.3.2.2.1 and 11.1.3.3.1. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" for a private call or first-to-answer call with an SDP offer not including a media-level section for media plane control messages, the participating MCPTT function shall consider it as a request for a private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.1.1 for an on-demand session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "private" or "first-to-answer", the participating MCPTT function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [29] and shall not continue with the rest of the steps;

…

3) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

…

5) shall:

a) if the <session-type> is set to "private", determine that the call is a private call; and

…

6) if the call is a:

a) private call, determine the public service identity of the controlling MCPTT function for the private call service associated with the originating user's MCPTT ID identity; or

…

11) if the call is a private call and:

a) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [18] with the value "Auto" and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCPTT user profile document on the participating MCPTT function or is present with the value "false" (see the MCPTT user profile document in 3GPP TS 24.484 [50]) indicating that the user identified by the MCPTT ID is not authorised to initiate private call with automatic commencement, shall reject the "SIP INVITE request for originating participating MCPTT function" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

...

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3…

16) shall set the Request-URI to the public service identity of the controlling MCPTT function hosting the private call service or first-to-answer call service as determined by step 6);

17) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

18) if the call is a private call and:

…

d) if a Priv-Answer-Mode header field containing the value of "Auto" has not been included in the outgoing SIP INVITE request as specified in step 17) above and the incoming "SIP INVITE request for originating participating MCPTT function" contained an Answer-Mode header field as specified in IETF RFC 5373 [18], then shall populate the Answer-Mode header field of the outgoing SIP INVITE request with the contents of the Answer-Mode header field from the incoming "SIP INVITE request for originating participating MCPTT function";

19) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCPTT function", as specified in clause 6.3.2.1.1.1;

…

20) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCPTT client;

…

22) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 180 (Ringing) response, the participating MCPTT function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall send the SIP 180 (Ringing) response to the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

5A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [7].

The participating MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function" for the private call or first-to-answer call with SDP offer not including media-level section for media-floor control entity, the participating MCPTT shall consider it as the request for the private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if one of the following conditions are met:

a) "SIP INVITE request for terminating participating MCPTT function" contains an Answer-Mode header field with the value "Auto";

b) "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) "SIP INVITE request for terminating participating MCPTT function" contains a Priv-Answer-Mode header field with the value "Auto"; and

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.4.1]

"The controlling MCPTT function shall follow the procedures as specified in clause 11.1.1.4.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.1]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcptt-calling-user-id> containing the calling user's MCPTT ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

…

4) shall copy the MCPTT ID of the MCPTT user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

5) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited;

NOTE 2: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 3: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

6) shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

7) shall include a Resource-Priority header field populated with the values for an MCPTT emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined in step 2 above; or

b) the originating MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

8) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

9) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [4]; and

10) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request containing a Require header field with the option tag "100rel" and a P-Answer-State header field with the value "Unconfirmed" in response, the controlling MCPTT function:

1) shall send a SIP PRACK request towards the terminating network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

Upon expiry of the private call timer, the controlling MCPTT function shall follow the procedure for releasing private call session as specified in clause 11.1.4.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.4.2]

"Upon receiving of a "SIP INVITE request for controlling MCPTT function of a private call" or a "SIP INVITE request for controlling MCPTT function of a first-to-answer call", with SDP offer not including media-level section for media-floor control entity, the controlling MCPTT function shall consider it as the request for the private call without floor control or first-to-answer call without floor control, and shall follow the procedures as specified in clause 11.1.1.4.2"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.2]

"Upon receipt of:

- *a "SIP INVITE request for controlling MCPTT function of a private call"; or*

…

the controlling MCPTT function:

…

9) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCPTT user is not in an in-progress emergency private call state with the targeted MCPTT user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCPTT user has initiated an MCPTT emergency private call to the targeted user; and

b) shall cache the information that the MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

10) shall perform actions as described in clause 6.3.3.2.2;

11) shall allocate an MCPTT session identity for the MCPTT session; and

…

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4]; and

…

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 11.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], if the <session-type> in the SIP INVITE request is set to "private", the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) shall include a P-Answer-State header field with the value "Unconfirmed";

…

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

6) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

…

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

…

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4].

Upon receiving a SIP BYE request from the originating MCPTT client containing an application/vnd.3gpp.mcptt-info+xml MIME body containing a <release-reason> element set to a value of "authentication of the MIKEY-SAKE I_MESSAGE failed", the controlling MCPTT function:

…

2) shall follow the procedures in clause 11.1.3.3.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.1.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.2.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.3.1]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

6.2.2.3 Test description

6.2.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.2.3.2 Test procedure sequence

Table 6.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a private call to SS-UE2 (MCPTT client) with automatic commencement mode and without floor control.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a private call with automatic commencement mode and without floor control according to option c of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	The SS-UE1 (MCPTT client) releases the call.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	2	P

6.2.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.2.3.3-1: SIP INVITE from the SS-UE1 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.2.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.2.3.3-1B			

Table 6.2.2.3.3-1A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.2.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions PRIVATE -CALL

Table 6.2.2.3.3-2: SIP INVITE from the MCPTT Server (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.2.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.2.3.3-2B			

Table 6.2.2.3.3-2A: SDP in SIP INVITE from the MCPTT Server (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.2.3.3-2B: MCPTT-Info in SIP INVITE from the MCPTT Server (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions PRIVATE-CALL

Table 6.2.2.3.3-3: SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.2.3.3-3A			

Table 6.2.2.3.3-3A: SDP in SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.2.3.3-4: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.2.3.3-4A			

Table 6.2.2.3.3-4A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

6.2.3 MCPTT Server - MCPTT Client / Private Call / On-demand / Emergency Private Call / Automatic Commencement Mode / Force of automatic commencement mode / Without Floor Control

6.2.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of an emergency private call with
automatic commencement mode and without floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an Emergency Private Call with Automatic Commencement
Mode and without implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
```

6.2.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.2.3.1, 11.1.1.3.1.1, 11.1.2.3.2, 11.1.1.3.2, 11.1.2.4.1, 11.1.1.4.1, 11.1.2.4.2, 11.1.1.4.2, 11.1.3.3.1, 11.1.3.2.1.1, 11.1.3.2.2.1, 6.2.8.3.2 and 11.1.1.2.1.1. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" for a private call or first-to-answer call with an SDP offer not including a media-level section for media plane control messages, the participating MCPTT function shall consider it as a request for a private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.1.1 for an on-demand session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "private" or "first-to-answer", the participating MCPTT function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [29] and shall not continue with the rest of the steps;

…

3) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

…

5) shall:

a) if the <session-type> is set to "private", determine that the call is a private call; and

…

6) if the call is a:

a) private call, determine the public service identity of the controlling MCPTT function for the private call service associated with the originating user's MCPTT ID identity; or

…

11) if the call is a private call and:

a) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [18] with the value "Auto" and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCPTT user profile document on the participating MCPTT function or is present with the value "false" (see the MCPTT user profile document in 3GPP TS 24.484 [50]) indicating that the user identified by the MCPTT ID is not authorised to initiate private call with automatic commencement, shall reject the "SIP INVITE request for originating participating MCPTT function" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

...

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3…

16) shall set the Request-URI to the public service identity of the controlling MCPTT function hosting the private call service or first-to-answer call service as determined by step 6);

17) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

18) if the call is a private call and:

a) if a Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was received in the incoming SIP INVITE request with a value of "Manual", shall not include a Priv-Answer-Mode header field in the outgoing SIP INVITE request;

…

c) if the <allow-force-auto-answer> element of the <ruleset> element is present in the MCPTT user profile document with the value "true" (see the MCPTT user profile document in 3GPP TS 24.484 [50]) on the participating MCPTT function, and the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was received in the incoming SIP INVITE request with a value of "Auto", shall include the Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

…

19) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCPTT function", as specified in clause 6.3.2.1.1.1;

…

20) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCPTT client;

…

22) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 180 (Ringing) response, the participating MCPTT function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall send the SIP 180 (Ringing) response to the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

5A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [7].

The participating MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function" for the private call or first-to-answer call with SDP offer not including media-level section for media-floor control entity, the participating MCPTT shall consider it as the request for the private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if one of the following conditions are met:

a) "SIP INVITE request for terminating participating MCPTT function" contains an Answer-Mode header field with the value "Auto";

b) "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) "SIP INVITE request for terminating participating MCPTT function" contains a Priv-Answer-Mode header field with the value "Auto"; and

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.4.1]

"The controlling MCPTT function shall follow the procedures as specified in clause 11.1.1.4.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.1]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcptt-calling-user-id> containing the calling user's MCPTT ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

…

4) shall copy the MCPTT ID of the MCPTT user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

5) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited;

NOTE 2: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 3: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

6) shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

7) shall include a Resource-Priority header field populated with the values for an MCPTT emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined in step 2 above; or

b) the originating MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

8) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

9) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [4]; and

10) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request containing a Require header field with the option tag "100rel" and a P-Answer-State header field with the value "Unconfirmed" in response, the controlling MCPTT function:

1) shall send a SIP PRACK request towards the terminating network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

Upon expiry of the private call timer, the controlling MCPTT function shall follow the procedure for releasing private call session as specified in clause 11.1.4.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.4.2]

"Upon receiving of a "SIP INVITE request for controlling MCPTT function of a private call" or a "SIP INVITE request for controlling MCPTT function of a first-to-answer call", with SDP offer not including media-level section for media-floor control entity, the controlling MCPTT function shall consider it as the request for the private call without floor control or first-to-answer call without floor control, and shall follow the procedures as specified in clause 11.1.1.4.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.2]

"Upon receipt of:

- a "SIP INVITE request for controlling MCPTT function of a private call"; or

…

the controlling MCPTT function:

6) if received SIP INVITE request includes an <emergency-ind>, shall validate the request as described in clause 6.3.3.1.17;

…

9) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCPTT user is not in an in-progress emergency private call state with the targeted MCPTT user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCPTT user has initiated an MCPTT emergency private call to the targeted user; and

b) shall cache the information that the MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

10) shall perform actions as described in clause 6.3.3.2.2;

11) shall allocate an MCPTT session identity for the MCPTT session; and

…

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4]; and

…

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 11.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], if the <session-type> in the SIP INVITE request is set to "private", the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) shall include a P-Answer-State header field with the value "Unconfirmed";

…

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

6) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

…

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

…

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4].

Upon receiving a SIP BYE request from the originating MCPTT client containing an application/vnd.3gpp.mcptt-info+xml MIME body containing a <release-reason> element set to a value of "authentication of the MIKEY-SAKE I_MESSAGE failed", the controlling MCPTT function:

…

2) shall follow the procedures in clause 11.1.3.3.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.3.1]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.1.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.2.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.2.8.3.2]

"When the MCPTT emergency private call state is set to "MEPC 1: emergency-pc-capable" and this is an authorised request for an MCPTT emergency private call as determined by the procedures of clause 6.2.8.3.1.1, the MCPTT client:

1) shall set the MCPTT emergency state if not already set;

2) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP request an <emergency-ind> element set to "true" and set the MCPTT emergency private call state to "MEPC 2: emergency-pc-requested";

3) if the MCPTT user has also requested an MCPTT emergency alert to be sent and this is an authorised request for MCPTT emergency alert as determined by the procedures of clause 6.2.8.3.1.3, shall:

a) include in the application/vnd.3gpp.mcptt-info+xml MIME body the <alert-ind> element set to "true" and set the MCPTT private emergency alert state to "MPEA 2: emergency-alert-confirm-pending"; and

b) include in the SIP request the specific location information for MCPTT emergency alert as specified in clause 6.2.9.1;

4) if the MCPTT user has not requested an MCPTT emergency alert to be sent, shall set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "false"; and

5) if the MCPTT emergency private priority state of this private call is set to a value other than "MEPP 2: in-progress" shall set the MCPTT emergency private priority state to "MEPP 3: confirm-pending"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.2.1.1]

"…

3) if the MCPTT user has requested the origination of an MCPTT emergency private call or is originating an MCPTT private call and the MCPTT emergency state is already set, the MCPTT client:

a) shall, if this is an authorised request for an MCPTT emergency private call as determined by the procedures of clause 6.2.8.3.1.1, comply with the procedures in clause 6.2.8.3.2; and

b) should, if this is an unauthorised request for an MCPTT emergency private call as determined in step a) above, indicate to the MCPTT user that they are not authorised to initiate an MCPTT emergency private call;

…

14) if the MCPTT user is initiating a private call then:

a) if force of automatic commencement mode at the invited MCPTT client is requested by the MCPTT user, shall include in the SIP INVITE request a Priv-Answer-Mode header field with the value "Auto" according to the rules and procedures of IETF RFC 5373 [18];"

6.2.3.3 Test description

6.2.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.3.3.2 Test procedure sequence

Table 6.2.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of an emergency private call to SS-UE2 (MCPTT client) forcing automatic commencement mode and without floor control	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish an emergency private call and forcing automatic commencement mode, without floor control according to option c of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	2	P

6.2.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.3.3.3-1: SIP INVITE from the SS-UE1 (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition PRIVATE-CALL, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Priv-Answer-Mode				
answer-mode-value	"Auto"	if force of automatic commencement mode at the invited MCPTT client is requested by the MCPTT user,		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.3.3.3-1B			

Table 6.2.3.3.3-1A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.3.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions PRIVATE-CALL, EMERGENCY-CALL

Table 6.2.3.3.3-2: SIP INVITE from the MCPTT Server (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition PRIVATE-CALL, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.3.3.3-2B			

Table 6.2.3.3.3-2A: SDP in SIP INVITE from the MCPTT Server (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.3.3.3-2B: MCPTT-Info in SIP INVITE from the MCPTT Server (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions PRIVATE-CALL, EMERGENCY-CALL

Table 6.2.3.3.3-3: SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.3.3.3-3A			

Table 6.2.3.3.3-3A: SDP in SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

Table 6.2.3.3.3-4: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.2.3.3.3-4A			

Table 6.2.3.3.3-4A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions PRIVATE-CALL, WITHOUT_FLOORCONTROL

6.2.4 MCPTT Server - MCPTT Client / Private Call / Pre-established Session / Automatic Commencement Mode / Without Floor Control

6.2.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 and having established a private call using pre-
established session }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests to establish a private call with automatic commencement
mode and without floor control with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and establishes the call with SS-UE2
(MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having established a private call using pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT Client) requests the call release keeping the pre-established private
call with SIP REFERE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) and releases the SS-UE2 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having released the private call keeping pre-established session }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
            }
```

(4)

```
with { IUT (MCPTT Server) having released the private call keeping pre-established session }
ensure that {
  when { the SS-UE2 (MCPTT client) ends the pre-established session with SIP BYE message }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message }
           }
```

6.2.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.2.2, 11.1.2.3.1, 10.1.1.3.1.2, 11.1.2.3.2, 11.1.1.3.2, 6.3.2.2.5.3, 11.1.3.2.1.2, 11.1.3.2.2.2, ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 9.3.2.2.2, 9.3.2.3.2, 9.3.2.3.3, 9.3.2.4.5, 9.3.2.4.7, 9.3.2.4.10. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.2]

"…

When the MCPTT user wants to make a private call without floor control or first-to-answer call without floor control using a pre-established session, the MCPTT client shall follow the procedures in clause 11.1.1.2.2.1 with the following exceptions:

1) *step 8 c) i) is re-written as: if the SDP parameters of the pre-established session contain a media-level section of a media-floor control entity or if end-to-end security is required for the private call, an application/sdp MIME body containing the SDP parameters of the pre-established session according to 3GPP TS 24.229 [4] with the clarifications given in clause 6.2.1. If the pre-established session was established with implicit floor control, then the application/sdp MIME body shall not contain the implicit floor request as specified in clause 6.4; and*

2) *step 9a) is re-written as: if the SDP parameters of the pre-established session contain a media-level section of a media-floor control entity, an application/sdp MIME body containing the SDP parameters of the pre-established session according to 3GPP TS 24.229 [4] with the clarifications given in clause 6.2.1. If the pre-established session was established with implicit floor control, then the application/sdp MIME body shall not contain the implicit floor request as specified in clause 6.4.*

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.1]

"Upon receipt of a "SIP REFER request for a pre-established session" for a private call or first-to-answer call with an SDP offer including a media-level section for media plane control messages with the 'fmtp' attribute containing a value of "mc_no_floor_ctrl" as specified in 3GPP TS 24.380 [5] clause 14, the participating MCPTT function shall consider it as a request for a private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.1.2 for initiation using a pre-established session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.3.1.2]

"Upon receipt of a "SIP REFER request for a pre-established session", with:

1) *the Refer-To header field containing a Content-ID ("cid") Uniform Resource Locator (URL) as specified in IETF RFC 2392 [62] that points to an application/resource-lists MIME body as specified in IETF RFC 5366 [20] containing one or more <entry> element(s) with a "uri" attribute containing a SIP URI set to the MCPTT ID of the called user(s);*

2) *an hname body" parameter in the headers portion of the SIP URI specified above containing an application/vnd.3gpp.mcptt-info MIME body with the <session-type> element set to "private" or "first-to-answer"; and*

3) *a Content-ID header field set to the "cid" URL;*

the participating function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP REFER request;

…

6) if the received SIP REFER request contains an application/resource-lists MIME body referenced by a "cid" URL in the Refer-To header field with only one <entry> element with an application/vnd.3gpp.mcptt-info MIME body with the <session-type> element:

a) not set to "private", shall reject the "SIP REFER request for pre-established session" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps; or

b) set to "private", determine that the call is a private call;

7) if the call is a:

a) private call, shall determine the public service identity of the controlling MCPTT function for the private call service associated with the originating user's MCPTT ID; or

…

10) if the call is a private call:

a) if the received SIP REFER request includes an Answer-Mode header field as specified in IETF RFC 5373 [18] set to "Auto" contained in the headers portion of the SIP URI present in the application/resource-lists MIME body referenced by a "cid" URL in the Refer-To header field, and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) on the participating MCPTT function or is present with the value "false" (indicating that the user identified by the MCPTT ID is not authorised to initiate private call with automatic commencement), shall reject the "SIP REFER request for pre-established session" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

…

13) if the "SIP REFER request for a pre-established session" contained a Refer-Sub header field containing "false" value and a Supported header field containing "norefersub" value, shall handle the SIP REFER request as specified in 3GPP TS 24.229 [4], IETF RFC 3515 [25] as updated by IETF RFC 6665 [26], and IETF RFC 4488 [22] without establishing an implicit subscription;

14) shall generate a final SIP 200 (OK) response to the "SIP REFER request for a pre-established session" according to 3GPP TS 24.229 [4];

NOTE 3: In accordance with IETF RFC 4488 [22], the participating MCPTT function inserts the Refer-Sub header field containing the value "false" in the SIP 200 (OK) response to the SIP REFER request to indicate that it has not created an implicit subscription.

15) shall send the response to the "SIP REFER request for a pre-established session" towards the MCPTT client according to 3GPP TS 24.229 [4];

16) shall generate a SIP INVITE request as specified in clause 6.3.2.1.4 with the following clarifications:

…

17) shall set the Request-URI of the SIP INVITE request to the public service identity of the controlling MCPTT function hosting the private call service or first-to-answer call service for the calling MCPTT user as determined above in step 7);

18) if the call is a private call:

a) if the SIP REFER request contained a Priv-Answer-Mode header field as specified in IETF RFC 5373 [18] set to "Manual" in the headers portion of the SIP URI in the application/resource-lists MIME body referenced by a "cid" URL in the Refer-To header field, shall copy the Priv-Answer-Mode header field from the incoming SIP REFER request to the outgoing SIP INVITE request;

b) if the <allow-force-auto-answer> element of the <ruleset> element is present in the MCPTT user profile document with the value "true" (see the MCPTT user profile document in 3GPP TS 24.484 [50]) on the participating MCPTT function, and the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was received in the headers portion of the SIP URI in the application/resource-lists MIME body referenced by a "cid" URL in the Refer-To header field, with a value set to "Auto", shall copy the Priv-Answer-Mode header field to the outgoing SIP INVITE request; and

c) if a Priv-Answer-Mode header field containing the value of "Auto" has not been copied to the outgoing SIP INVITE request as specified in step 16) above, and the incoming SIP REFER request contained an Answer-Mode header field in the headers portion of the SIP URI in the application/resource-lists referenced by a "cid" URL in the Refer-To header field, then copy the Answer-Mode header field to the outgoing SIP INVITE request;

19) if the received SIP REFER request contained a Resource-Priority header field, shall include in the outgoing SIP INVITE request a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the received SIP REFER request;

NOTE 4: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

20) shall forward the SIP INVITE request according to 3GPP TS 24.229 [4].

Upon receiving SIP provisional responses for the SIP INVITE request the participating MCPTT function:

1) shall discard the received SIP responses without forwarding them.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the participating MCPTT function:

1) if:

a) the received SIP 2xx response was in response to a request for an MCPTT private call; or

b) the received SIP 2xx response was in response to a SIP INVITE request for a first-to-answer call which did not include an a=key-mgmt "mikey" attribute value containing a MIKEY-SAKKE I_MESSAGE in the SDP answer;

then:

a) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.2.3.2]

"Upon receipt of a "SIP INVITE request for terminating participating MCPTT function" for the private call or first-to-answer call with SDP offer not including media-level section for media-floor control entity, the participating MCPTT shall consider it as the request for the private call without floor control or first-to-answer call without floor control and shall follow the procedures as specified in clause 11.1.1.3.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.2]

"This clause covers both on demand session and pre-established session.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if one of the following conditions are met:

a) "SIP INVITE request for terminating participating MCPTT function" contains an Answer-Mode header field with the value "Auto";

b) "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) "SIP INVITE request for terminating participating MCPTT function" contains a Priv-Answer-Mode header field with the value "Auto"; and"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.2.5.3]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" for a pre-established session that requires automatic commencement mode the participating MCPTT function:

…

2) shall generate a SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function" as described in the clause 6.3.2.2.4.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.2.2.2 based on the SDP negotiated during the pre-established session establishment and SDP offer received in the "SIP INVITE request for terminating participating MCPTT function";

4) shall set the P-Answer-State header field to "Unconfirmed" in the SIP 200 (OK) response;

5) shall set the P-Asserted-Identity header field to the same value as included in the SIP INVITE request that was sent by the MCPTT client when the pre-established session was established, in the SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [4]; and

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 9.

NOTE 2: Received SIP INVITE requests handled by the present clause in steps 2) through 7) will result in the terminating client not receiving data normally contained in an application/ vnd.3gpp.mcptt-info+xml MIME body. This includes but is not limited to call type indicators such as XML elements <broadcast-ind> or <associated-group>."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.1.2]

"Upon receiving from the MCPTT client a SIP REFER request when using a pre-established session with the "method" SIP URI parameter set to value "BYE" in the URI in the Refer-To header field the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.7."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.2.2]

"Upon receiving a SIP BYE request from the controlling MCPTT function and if the MCPTT session id refers to an MCPTT user that has a pre-established session with the participating MCPTT function, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 9.3 in 3GPP TS 24.380 [5] for disconnecting the media plane resources towards the controlling MCPTT function;

2) shall send a SIP 200 (OK) response to the controlling MCPTT function;

3) shall interact with the media plane as specified in clause 9.3 in 3GPP TS 24.380 [5] for disconnecting media plane resources towards the MCPTT client from the media plane resources towards the controlling MCPTT function; and

4) shall maintain the pre-established session towards the MCPTT client."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.2.2]

"When a pre-established session is created between the participating MCPTT function and an MCPTT client, the participating MCPTT function:

1. *shall initiate any needed user plane resources for the pre-established session; and*
2. *shall enter the 'G: Pre-established session not in use' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.3.2]

"Upon receiving a SIP REFER request from the MCPTT client the participating MCPTT function:

1. *shall reserve user plane resources for the call session, if not yet reserved; and,*
2. *shall enter the 'G: Pre-established session in use' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.3.3]

"Upon receiving a SIP INVITE request from the controlling MCPTT function, if in automatic answer mode, the participating MCPTT function:

1. *shall send the Connect message to the invited MCPTT client using the media plane control transport channel defined for this call. The Connect message:*
    a. *shall include the SIP URI received in the Contact header field of the SIP INVITE request in the <MCPTT Session Identity> value in the MCPTT Session Identity field;*
    …
    c. *if the SIP INVITE request includes an "application/vnd.3gpp.mcptt-info+xml" MIME body with the <session-type> element set to a value of "private":*
        i. *shall include the content of the <mcptt-calling-user-id> element in the Inviting MCPTT User Identity field if available from the SIP INVITE message and privacy of the inviting MCPTT ID is not required;*
        ii. *shall include 'anonymous@anonymous.invalid' in the Inviting MCPTT User Identity field if the MCPTT ID of the inviting MCPTT user is not available from the SIP INVITE message or privacy of the inviting MCPTT ID is required;*
        iii. *shall include the value 'private' in the <Session Type> value in the MCPTT Session Identity field; and*
        iv. *if the MIKEY-SAKKE I_MESSAGE is included in the received SIP INVITE request, shall include the MIKEY-SAKKE I_MESSAGE as received, in the PCK I_MESSAGE field;*
    d. *shall include a Media Streams field, if the negotiated SDP in the pre-arranged session contains more media streams than required by the SDP in the SIP INVITE request; and*

    NOTE 1: For example if the call is a private call without floor control the Media Streams field needs to be included with the <Control Channel> value set to '0'.

    e. *shall set the first bit in the subtype of the Connect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;*
2. *shall start timer T55 (Connect);*
3. *shall initialize counter C55 (Connect) to 1;*
4. *shall reserve the media channel resources if not yet reserved; and*
5. *shall enter the 'G: Pre-established session in use' state.*

NOTE 2: The case of manual answer is described in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.5]

"Upon receiving an indication from the controlling MCPTT function that the call is released (see 3GPP TS 24.379 [2]), the participating MCPTT function:

1. *shall stop timer T55 (Connect)if running;*
2. *shall send Disconnect message to the MCPTT client. The Disconnect message:*
    a. *shall include the MCPTT Session Identity field with the same content as sent in the Connect message when the MCPTT call was established; and*
    b. *should set the first bit in the subtype of the Disconnect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;*
3. *shall start timer T56 (Disconnect);*
4. *shall initialise counter C56 (Disconnect) to 1; and*
5. *shall enter the 'G: Call releasing' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.7]

"Upon receiving an Acknowledge message from the MCPTT client, the participating MCPTT function:

1. *shall stop timer T55 (Connect);*
2. *if the Reason Code is not 'Accepted' the participating MCPTT function:*
    a. *shall send Disconnect message to the MCPTT client;*
        i. *shall include the MCPTT Session Identity field with the same content as sent in the Connect message when the MCPTT call was established; and*
        ii. *should set the first bit in the subtype of the Disconnect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;*
    b. *shall start timer T56 (Disconnect);*
    c. *shall initialise counter C56 (Disconnect) to 1;*
    d. *shall send a call release indication to the controlling MCPTT function (see 3GPP TS 24.379 [2]); and*
    e. *shall enter the 'G: Call releasing' state; and*
3. *otherwise the participating MCPTT function:*
    a. *shall send a SIP 200 (OK) response to the SIP INVITE request as specified in 3GPP TS 24.379 [2] if the SIP 200 (OK) response to the SIP INVITE request is not already sent; and*
    b. *shall remain in the 'G: Pre-established session in use' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 9.3.2.4.10]

"Upon receiving a SIP 200 (OK) response from the controlling MCPTT function, the participating MCPTT function:

1. *shall send the Connect message to the MCPTT client which initiated this call using the media plane control transport channel defined for this call. The Connect message:*
    a. *shall include the SIP URI received in the Contact header field of the SIP 200 (OK) response in the <MCPTT Session Identity> value in the MCPTT Session Identity field;*
    b. *if an "application/vnd.3gpp.mcptt-info+xml" MIME body with an <session-type> element is included in the SIP 200 (OK) response, shall include the <Session Type> value in the MCPTT Session Identity field with a value corresponding to the <session-type> element;*

c. if an "application/vnd.3gpp.mcptt-info+xml" MIME body with an <session-type> element is not included in the SIP 200 (OK) response, shall include the <Session Type> value set to 'no session type' in the MCPTT Session Identity field;

d. if a Warning header field with a SIP "warn-code" set to the value "399" is received in the SIP 200 (OK) response, shall include the content of the SIP "warn-text" parameter from the received Warning header field in the <Warning Text> value in the Warning Text field;

e. if a P-Answer-State header field is included in the SIP 200 (OK) response, shall include an Answer State field with an <Answer State> value corresponding to the "answer-type" parameter; and

f should set the first bit in the subtype of the Connect message to '1' (Acknowledgment is required) as described in subclause 8.3.2;

2. shall start timer T55 (Connect);

3. shall increment counter C55 (Connect) by 1; and,

4. shall remain in the 'G: Pre-established session in use' state."

6.2.4.3 Test description

6.2.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Pre-established Session Establishment as specified in step 1 on Table 5.4.3.2-1 in clause 5.4.

6.2.4.3.2 Test procedure sequence

Table 6.2.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a private call to SS-UE2 (MCPTT client), with automatic commencement mode and without floor control, reusing the pre-established session	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E call establishment using a pre-established session, automatic commencement' as described in Annex A, Table A.3.5.3-1 to establish a private call with automatic commencement mode and without floor control?	-	-	1	P
3	The SS-UE1 (MCPTT client) releases the call keeping the pre-established session.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E call release keeping the pre-established session' as described in Annex A, Table A.3.6.3-1?	-	-	2	P
5	The SS-UE1 (MCPTT client) releases the pre-established session.	-	-	-	-
6	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE1?	-	-	3	P
7	The SS-UE2 (MCPTT client) releases the pre-established session.	-	-	-	-
8	Check: Does the MCPTT Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session for SS-UE2?	-	-	4	P

6.2.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.4.3.3-1: SIP REFER from SS-UE1
(Step 2, Table 6.2.4.3.2-1; Step 1, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-1, with conditions PRIVATE-CALL and WITHOUT_FLOORCONTROL for all clauses and subclauses referenced on that Table.

Table 6.2.4.3.3-2: SIP 200 (OK)
(Step 2, Table 6.2.4.3.2-1; Step 2, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-3, with conditions PRIVATE-CALL and WITHOUT_FLOORCONTROL for all clauses and subclauses referenced on that Table.

Table 6.2.4.3.3-3: Connect
(Step 2, Table 6.2.4.3.2-1; Steps 3 and 5, Annex A, Table A.3.5.3-1)

Derivation Path: Annex A, Table A.3.5.4-4, with condition PRIVATE-CALL and WITHOUT_FLOORCONTROL.

Table 6.2.4.3.3-6: SIP BYE from SS-UE1 and SS-UE2 (Steps 6 and 8, Table 6.2.4.3.2-1; Step 1, Annex A, Table A.2.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

6.2.5 MCPTT Server - MCPTT Client / Private Call Call-Back Request / Private call call-back fulfilment / Private Call Call-Back Cancel Request

6.2.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { SS-UE2 (MCPTT Client) requests the private call call-back for targeted user by sending a
SIP MESSAGE message }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to SS-UE1 (MCPTT Client) and responds with
a SIP 200 (OK) to SS-UE2 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having requested the private call call-back for targeted user }
ensure that {
  when { SS-UE2 (MCPTT Client) requests to cancel the private call call-back for targeted user by
sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to SS-UE1 (MCPTT Client) and responds with
a SIP 200 (OK) to SS-UE2 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having requested the private call call-back for targeted user }
ensure that {
  when { the SS-UE1 (MCPTT client) initiates a private call with manual commencement mode and
implicit floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT private call with Manual Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the private group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to SS-UE2 (MCPTT Client) }
            }
```

6.2.5.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.5.2.1, 11.1.5.2.2, 11.1.5.2.3, 11.1.5.3.1, 11.1.5.3.1. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.5.2.1]

"…

The MCPTT client:

1) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

2) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

4) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the MCPTT user;

5) shall include in an application/resource-lists+xml MIME body, the MCPTT ID of the targeted MCPTT user, according to rules and procedures of IETF RFC 5366 [20];

6) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element with the <anyExt> element containing:

a) if the request is a private call call-back request:

i) the <request-type> set to a value of "private-call-call-back-request>;

ii) the <urgency-ind> set to a value of "low", "normal" or "high" to indicate the urgency of the call-back request; and

iii) the <time-of-request> set to the date and time of the request using the format specified in clause F.1.3; and

b) if the request is a private call call-back cancel request, the <request-type> set to a value of "private-call-call-back-cancel-request";

…

8) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.5.2.2]

"…

The MCPTT client:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33]:

2) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

3) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6] in the SIP MESSAGE request

4) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

5) shall set the Request-URI in the SIP MESSAGE request to the public service identity identifying the participating MCPTT function serving the MCPTT user;

6) shall include in an application/resource-lists+xml MIME body, the MCPTT ID contained in the <mcptt-calling-user-id> element in the application/ vnd.3gpp.mcptt-info+xml MIME body of the received SIP MESSAGE request;

7) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element with the <anyExt> element containing:

a) if the received SIP MESSAGE was a "SIP MESSAGE request for private call call-back request for terminating MCPTT client", the <response-type> element set to a value of "private-call-call-back-response"; and

b) if the received SIP MESSAGE was a "SIP MESSAGE request for private call call-back cancel request for terminating MCPTT client", the <response-type> element set to a value of "private-call-call-back-cancel-response"; and

8) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.5.2.3]

"When the target MCPTT user wants to make a private call call-back, the target MCPTT client shall initiate a private call in manual commencement mode towards the requesting MCPTT client using the MCPTT ID of the requesting MCPTT user as found in the "PCCB target client entry" stored on the UE, by following the procedures in:

1) clause 11.1.1.2.1.1 or clause 11.1.1.2.2.1 for private call with floor control; or

2) clause 11.1.2.2 for private call without floor control;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.5.3.1]

"Upon receiving a "SIP MESSAGE request for private call call-back for originating participating MCPTT function" the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;

NOTE: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

5) shall determine the public service identity of the controlling MCPTT function for the private call call-back service for the MCPTT user;

6) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

7) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCPTT function associated with the private call call-back service for the MCPTT user;

8) shall copy the contents of the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 included in the outgoing SIP MESSAGE request;

9) shall set the <mcptt-calling-user-id> element of the <mcpttinfo> element containing the <mcptt-Params> element to the MCPTT ID determined in step 2) above;

10) shall copy the contents of the application/resource-lists MIME body in the received SIP MESSAGE request into an application/resource-lists MIME body in the outgoing SIP MESSAGE request;

11) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request;

12) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

13) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

14) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

15) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response in response to the SIP MESSAGE request sent in step 13), the participating MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.5.3.1]

"Upon receiving a "SIP MESSAGE request for private call call-back for terminating participating MCPTT function" the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

…

4) shall generate an outgoing SIP MESSAGE request as specified in clause 6.3.2.2.11;

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request; and

6) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall forward the SIP 2xx response to the controlling MCPTT function."

6.2.5.3 Test description

6.2.5.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.5.3.2 Test procedure sequence

Table 6.2.7.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE2 (MCPTT client) sends a private call call-back request to SS-UE1 (MCPTT client)	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to request the private call call-back for targeted user A from user B?	-	-	1	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to respond the private call call-back request to user B? (see note)	-	-	1	P
4	The SS-UE2 (MCPTT client) cancels the private call call-back request sent to SS-UE1 (MCPTT client)	-	-		-
5	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to cancel the private call call-back for targeted user A from user B?	-	-	2	P
6	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to respond the private call call-back cancellation request to user B? (see note)	-	-	2	P
7	The SS-UE2 (MCPTT client) sends a private call call-back request to SS-UE1 (MCPTT client)	-	-	-	-
8	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to request the private call call-back for targeted user A from user B?	-	-	1	P
9	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to respond the private call call-back request to user B? (see note)	-	-	1	P
10	The SS-UE1 (MCPTT client) requests a private call to SS-UE2 (MCPTT client) with manual commencement mode and implicit floor request	-	-	-	-
11	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E private call establishment, manual commencement' as described in Annex A, Table A.3.3.3-1 to establish a private call with manual commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.3.3-1, from user A to user B?	-	-	3	P
12	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
13	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	4	P
NOTE: During this procedure the SS-UE1 performs the role of SS-UE2 and viceversa.					

6.2.5.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.7.3.3-1: SIP MESSAGE from the SS-UE2 (Steps 2 and 8, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-2			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.7.3.3-3			

Table 6.2.7.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
request-type	"private-call-call-back-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	
urgency-ind	"normal"	Arbitrarily choosen	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	
time-of-request	"YYYY-MM-DDThh:mm:ss"	set to the date and time of the request	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-3: Resource lists in SIP MESSAGE (Table 6.2.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_A	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	

Table 6.2.7.3.3-4: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Steps 2 and 8, Table 6.2.7.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-5			

Table 6.2.7.3.3-5: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
anyExt				
request-type	"private-call-call-back-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.5.2.1	
urgency-ind	"normal"	Arbitrarily choosen	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.5.2.1	
time-of-request	"YYYY-MM-DDThh:mm:ss"	set to the date and time of the request	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.5.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-6: SIP MESSAGE from the SS-UE1 (Steps 3 and 9, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-7			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.7.3.3-8			

Table 6.2.7.3.3-7: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
response-type	"private-call-call-back-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-8: Resource lists in SIP MESSAGE (Table 6.2.7.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_B	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	

Table 6.2.7.3.3-9: SIP MESSAGE from the MCPTT-Server to SS-UE2 (Steps 3 and 9, Table 6.2.7.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-10			

Table 6.2.7.3.3-10: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
anyExt				
response-type	"private-call-call-back-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-11: SIP MESSAGE from the SS-UE2 (Step 5, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-12			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.7.3.3-13			

Table 6.2.7.3.3-12: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
request-type	"private-call-call-back-cancel-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-13: Resource lists in SIP MESSAGE (Table 6.2.7.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_A	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.1	

Table 6.2.7.3.3-14: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Step 5, Table 6.2.7.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-15			

Table 6.2.7.3.3-15: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-14)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
anyExt				
request-type	"private-call-call-back-cancel-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 12.1.5.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-16: SIP MESSAGE from the SS-UE1 (Steps 6, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-17			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.7.3.3-18			

Table 6.2.7.3.3-17: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
response-type	"private-call-call-back-cancel-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.7.3.3-18: Resource lists in SIP MESSAGE (Table 6.2.7.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_B	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	

Table 6.2.7.3.3-19: SIP MESSAGE from the MCPTT-Server to SS-UE2 (Step 6, Table 6.2.7.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-20			

Table 6.2.7.3.3-20: MCPTT-Info in SIP MESSAGE (Table 6.2.7.3.3-19)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
anyExt				
response-type	"private-call-call-back-cancel-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.5.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

6.2.6 MCPTT Server - MCPTT Client / Ambient listening call / On-demand / Remotely initiated Ambient listening call / Remotely initiated ambient listening call release

6.2.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a private call with automatic
commencement mode and without floor control }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
           }
```

(2)

```
with { IUT (MCPTT Server) having established a private call with automatic commencement mode and
without floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) establishes a private call with automatic commencement mode }
    then { IUT (MCPTT Server) enforces floor granted to the SS-UE2 (MCPTT client) and floor taken
with the Permission to Request the Floor field to SS-UE1 (MCPTT client) }
           }
```

(3)

```
with { IUT (MCPTT Server) having established a private call with automatic commencement mode and
without floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the private call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
```

6.2.6.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.6.2.1.1, 11.1.6.3.1.1, 11.1.6.3.1.2, 11.1.6.3.2.1, 11.1.6.3.2.2, 11.1.6.4.1, 11.1.6.4.2, ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 6.3.4.2.2, 6.3.4.4.2, 6.3.5.2.2, 6.3.5.4.4. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.2.1.1]

"…

The MCPTT client:

1) *shall set the Request-URI of the SIP INVITE request to a public service identity of the participating MCPTT function serving the MCPTT user;*

2) *may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [4];*

3) *shall include the g.3gpp.mcptt media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [16];*

4) *shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

5) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP INVITE request;*

6) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref contain with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

7) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "ambient-listening";

8) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <ambient-listening-type> element set to a value of:

…

b) "remote-init", if the MCPTT user has requested a remotely initiated ambient listening call;

9) shall insert in the SIP INVITE request a MIME resource-lists body with the MCPTT ID of the targeted MCPTT user, according to rules and procedures of IETF RFC 5366 [20];

NOTE 1: the targeted MCPTT user is the listened-to MCPTT user in the case of a remotely initiated ambient listening call or the listening MCPTT user in the case of a locally initiated listening call.

10) if an end-to-end security context needs to be established then:

a) if necessary, shall instruct the key management client to request keying material from the key management server as described in 3GPP TS 33.180 [78];

b) shall use the keying material to generate a PCK as described in 3GPP TS 33.180 [78];

c) shall use the PCK to generate a PCK-ID with the four most significant bits set to "0001" to indicate that the purpose of the PCK is to protect private call communications and with the remaining twenty eight bits being randomly generated as described in 3GPP TS 33.180 [78];

d) shall encrypt the PCK to a UID associated to the MCPTT client using the MCPTT ID and KMS URI of the invited user and a time related parameter as described in 3GPP TS 33.180 [78];

e) shall generate a MIKEY-SAKKE I_MESSAGE using the encapsulated PCK and PCK-ID as specified in 3GPP TS 33.180 [78];

f) shall add the MCPTT ID of the originating MCPTT to the initiator field (IDRi) of the I_MESSAGE as described in 3GPP TS 33.180 [78]; and

g) shall sign the MIKEY-SAKKE I_MESSAGE using the originating MCPTT user's signing key provided in the keying material together with a time related parameter, and add this to the MIKEY-SAKKE payload, as described in 3GPP TS 33.180 [78];

11) shall include an SDP offer according to 3GPP TS 24.229 [4] with the clarification given in clause 6.2.1 and with a media stream of the offered media-floor control entity;

…

13) if this is a remotely initiated ambient listening call, shall comply with the conditions for an implicit request to grant the floor to the terminating MCPTT client as specified in clause 6.4;

14) shall include in the SIP INVITE request a Priv-Answer-Mode header field with the value "Auto" according to the rules and procedures of IETF RFC 5373 [18]; and

15) shall send the SIP INVITE request towards the participating MCPTT function according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "ambient-listening", the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 1: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

10) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

11) shall set the Request-URI to the public service identity of the controlling MCPTT function hosting the private call service;

12) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

13) if the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was received in the incoming SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the incoming SIP INVITE request or a Priv-Answer-Mode header field was received containing a value other than "Auto", shall include the Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

14) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCPTT function", as specified in clause 6.3.2.1.1.1;

…

16) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP INVITE request, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

6) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

7) shall send the SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4]; and

8) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.3.1.2]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function:

1) shall follow the procedures as specified in clause 11.1.3.2.1.1.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.3.2.1]

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

NOTE: The procedures in the present clause are applicable for both on-demand and pre-established sessions.

…

3) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

…

6) *shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18]."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.3.2.2]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 11.1.3.2.2.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.4.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT ambient listening session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 11.1.6.4.2.

The controlling MCPTT function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcptt-calling-user-id> containing the calling user's MCPTT ID is copied into the outgoing SIP INVITE.

2) *shall copy the MCPTT ID of the MCPTT user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;*

3) *shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited;*

NOTE 2: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 3: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

4) *shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;*

5) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;*

…

7) *if the <ambient-listening-type> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP INVITE request is set to a value of "remote-init", shall include in the outgoing SIP INVITE request an alert-info header field set to a value of "<file:///dev/null>" according to IETF RFC 3261 [24];*

8) *shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [4]; and*

9) *shall start a private call timer with a value set to the configured max private call duration for the user.*

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) *shall cache the contact received in the Contact header field; and*

2) *shall interact with the media plane as specified in 3GPP TS 24.380 [5].*

Upon expiry of the private call timer, the controlling MCPTT function shall follow the procedure for releasing the ambient listening call session as specified in clause 11.1.6.4.3 with the clarification that the <release-reason> element included in the SIP BYE request shall be set to "private-call-timer-expiry"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.6.4.2]

"Upon receiving of a "SIP INVITE request for controlling MCPTT function of an ambient listening call" the controlling MCPTT function:

1) shall check whether the public service identity contained in the Request-URI is allocated for ambient listening call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps;

2) shall perform actions to verify the MCPTT ID of the inviting MCPTT user in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request, and authorise the request according to local policy; and

…

5) shall perform actions as described in clause 6.3.3.2.2;

6) shall allocate an MCPTT session identity for the MCPTT ambient listening call session; and

7) shall invite the MCPTT user listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 11.1.6.4.1.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 11.1.6.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) shall include a P-Answer-State header field with the value "Unconfirmed";

4) shall include in the SIP 200 (OK) response the g.3gpp.mcptt.ambient-listening-call-release feature-capability indicator as described in clause D.3 in the Feature-Caps header field according to IETF RFC 6809 [60];

NOTE 1: The originator of the ambient listening call is either the initiator of a "remote-init" ambient listening type call or the originator of a "local-init" ambient listening type call. In either case, the originating user is allowed to release the ambient listening call.

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

6) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) if the <ambient-listening-type> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP INVITE request is set to a value of "remote-init":

a) shall cache the MCPTT ID contained in the <mcptt-calling-user-id> element of the received SIP INVITE request as the listening MCPTT user of the ambient listening call and cache the MCPTT ID contained in the MIME resource-lists body of the received SIP INVITE request as the listened-to MCPTT user; and

b) shall cache the ambient listening type of the ambient listening call as "remote-init"; and

…

If the procedures of clause 11.1.6.4.1 were successful in inviting the MCPTT user listed in the MIME resource-lists body of received SIP INVITE request and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

3) shall include in the SIP 200 (OK) response the g.3gpp.mcptt.ambient-listening-call-release feature-capability indicator as described in clause D.3 in the Feature-Caps header field according to IETF RFC 6809 [60];

NOTE 2: The originator of the ambient listening call is either the initiator of a "remote-init" ambient listening type call or the originator of a "local-init" ambient listening type call. In either case, the originating user is allowed to release the ambient listening call.

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

5) shall send a SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

6) if the <ambient-listening-type> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body in the received SIP INVITE request is set to a value of "remote-init":

a) shall cache the MCPTT ID contained in the <mcptt-calling-user-id> element of the received SIP INVITE request as the listening MCPTT user of the ambient listening call and cache the MCPTT ID contained in the MIME resource-lists body of the received SIP INVITE request as the listened-to MCPTT user; and

b) shall cache the ambient listening type of the ambient listening call as "remote-init"; and

…

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.2.2]

"…

3. when the 'basic floor control operation towards the floor participant' state machine is initialized and the initialised session is not a temporary group session:

a. if the "mc_granted" fmtp attribute is not negotiated as specified in clause 14 or this is an ambient listening call:

i. if the floor control server is granting an implicit floor request at MCPTT call establishment, shall act as if a Floor Request message was received and perform the actions specified in subclause 6.3.4.3.3; or

ii. if the floor control server is not granting an implicit floor request at MCPTT call establishment, shall enter the'G: Floor Idle' state as specified in subclause 6.3.4.3.2; or

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:

…

e. if the call is a remotely initiated ambient listening call, shall set the first bit in the subtype of the Floor Granted message to '1' (Acknowledgment is required) as described in subclause 8.2.2;

NOTE: If the call is an ambient listening call and the ambient listening call type is remote-initiated, then the floor participant to which the floor is granted is the terminating floor participant of the call. Otherwise the floor is granted to the participant which requested the floor.

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.2.2]

"…

When a SIP Session is established and if the session is an ambient listening call session then the floor control interface towards the MCPTT client:

1. *if the floor is granted to the associated floor participant*
   a. *shall forward the "Floor Granted" message to the associated floor participant; and*
   b. *shall enter the state 'U: permitted' as specified in the subclause 6.3.5.5.2; and*
2. *if the floor is not granted to the associated floor participant*
   a. *shall forward the "Floor Taken" message to the associated floor participant; and*
   b. *shall enter the state 'U: not permitted Floor Taken' as specified in the subclause 6.3.5.4.2."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.4.4]

"…

Upon receiving a Floor Request message from the associated floor participant and the session is a broadcast group call or an ambient listening call, the floor control interface towards the MCPTT client in the floor control server:

1. *shall send a Floor Deny message to the associated floor participant. The Floor Deny message:*
   a. *shall include in the Reject Cause field the <Reject Cause> value cause #5 (Receive only);*
   b. *may include in the Reject Cause field an additional text string explaining the reason for rejecting the floor request in the <Reject Phrase> value; and*
   c. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;*
2. *may set the first bit in the subtype of the Floor Deny message to '1' (Acknowledgment is required) as described in subclause 8.2.2; and*

NOTE 2: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

3. *shall remain in the 'U: not permitted and Floor Taken' state.*

…"

6.2.6.3 Test description

6.2.6.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)

- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1

- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.6.3.2 Test procedure sequence

Table 6.2.8.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a remotely initiated ambient listening call to SS-UE2 (MCPTT client)	-	-	-	-
2	Check: Does the MCPTT server correctly perform step 1 of procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a remotely initiated ambient listening call with automatic commencement mode and without implicit floor control according to option a of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	Check: Does the MCPTT server send a Floor Granted to SS-UE2?	<-	Floor Granted	2	P
4	Check: Does the MCPTT server send a Floor Taken to SS-UE1, with the Permission to Request the Floor field set to '0'?	<-	Floor Taken	2	P
5	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
6	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	3	P

6.2.6.3.2 Test procedure sequence

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.8.3.3-1: SIP INVITE from the SS-UE1 (Step 2, Table 6.2.6.3.2-1; Step 1, Annex A, Table A.3.1.3-1 Step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Answer-Mode	not present			
Priv-Answer-Mode				
answer-mode-value	"Auto"			
answer-mode-param	"require"			

Derivation Path: Annex A, Table A.3.1.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.6.3.3-2.			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.6.3.3-3.			

Table 6.2.6.3.3-2: SDP in SIP INVITE (Table 6.2.6.3.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media attribute		a=line attribute=recvonly		
recvonly				

Table 6.2.8.3.3-3: MCPTT-Info in SIP INVITE (Table 6.2.6.3.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
session-type	"ambient-listening"			
anyExt				
ambient-listening-type	"remote-init"	The anyExt field may contain other values - these need not be checked		

Table 6.2.6.3.3-4: SIP INVITE from MCPTT Server to SS-UE2 (Step 2, Table 6.2.6.3.2-1; Step 1, Annex A, Table A.3.1.3-1 Step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Answer-Mode	not present			
Priv-Answer-Mode				
answer-mode-value	"Auto"			
answer-mode-param	"require"			
Alert-Info				
value	"<file:///dev/null>"			
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.6.3.3-5.			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.6.3.3-6.			

Table 6.2.6.3.3-5: SDP in SIP INVITE (Table 6.2.6.3.3-4)

Derivation Path: Annex A, Table A.3.1.4-5, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media attribute		a=line attribute=recvonly		
recvonly				

Table 6.2.6.3.3-6: MCPTT-Info in SIP INVITE (Table 6.2.6.3.3-4)

Derivation Path: Annex A, Table A.3.1.4-6, condition PRIVATE-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
session-type	"ambient-listening"			
anyExt				
ambient-listening-type	"remote-init"	The anyExt field may contain other values - these need not be checked		

Table 6.2.6.3.3-6: SIP 200 (OK) from SS-UE2 (Step 2, Table 6.2.6.3.2-1; Step 1, Annex A, Table A.3.1.3-1 Step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.6.3.3-7			

Table 6.2.6.3.3-7: SDP in SIP 200 (OK) (Table 6.2.6.3.3-6)

Derivation Path: Annex A, Table A.3.1.4-8, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media attribute		a=line attribute=sendonly		
sendonly				

Table 6.2.6.3.3-9: SIP 200 (OK) from MCPTT Server to SS-UE1 (Step 2, Table 6.2.6.3.2-1; Step 1, Annex A, Table A.3.1.3-1 Step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Feature-Caps				
fcap-name	"g.3gpp.mcptt.ambient-listening-call-release"	Indicates that the MCPTT server is capable of receiving a SIP BYE from an MCPTT client to release an ambient-listening call		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.6.3.3-10			

Table 6.2.6.3.3-10: SDP in SIP 200 (OK) (Table 6.2.6.3.3-9)

Derivation Path: Annex A, Table A.3.1.4-10, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media attribute		a=line attribute=sendonly		
sendonly				

Table 6.2.6.3.3-11: Floor Taken (Step 4, Table 6.2.6.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Permission to Request the Floor			
Permission to Request the Floor	"0"	The receiver is NOT permitted to request floor	

6.2.7 MCPTT Server - MCPTT Client / First-to-answer call / On-demand / With Floor Control

6.2.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a first-to-answer call to targeting
users without implicit floor control with SIP INVITE message }
    then { IUT (MCPTT Server) accepts the request and sends SIP INVITE message to SS-UE2 (MCPTT
Client) and sends SIP INVITE message to SS-UE3 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having in process a first-to-answer call }
ensure that {
  when { the SS-UE2 (MCPTT Client) responds with SIP 180 Ringing message }
    then { IUT (MCPTT Server) receives the response and sends SIP 183 Session Progress message to
SS-UE1 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having in process a first-to-answer call }
ensure that {
  when { the SS-UE2 (MCPTT client) responds with SIP 200 OK message }
    then { IUT (MCPTT Server) receives the response and sends SIP 200 OK message to SS-UE1 (MCPTT
Client)  }
            }
```

(4)

```
with { IUT (MCPTT Server) having in process a first-to-answer call }
ensure that {
  when { the SS-UE2 (MCPTT client) responds with SIP 200 OK message }
    then { IUT (MCPTT Server) sends SIP CANCEL message to SS-UE3 (MCPTT Client) and sends SIP ACK
message to SS-UE3 (MCPTT Client) after received SIP 487 Request Terminated message }
            }
```

(5)

```
with { IUT (MCPTT Server) having established a first-to-answer call }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the a first-to-answer call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to SS-UE1 (MCPTT client)
and sends a SIP BYE message to SS-UE2 (MCPTT client) }
```

6.2.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.1.3.1.1, 11.1.1.4.1, 11.1.1.4.2, 11.1.3.2.1.1, 11.1.3.2.2.1, 11.1.3.3.1. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "private" or "first-to-answer", the participating MCPTT function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [29] and shall not continue with the rest of the steps;

…

3) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

…

5) shall:

..

b) if the <session-type> is set to "first-to-answer", determine that the call is a first-to-answer-call;

6) if the call is a:

…

b) first-to-answer, determine the public service identity of the controlling MCPTT function for the first-to-answer call service associated with the originating user's MCPTT ID identity;

…

15) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3 with the following clarifications:

a) if the call is a first-to-answer call and if the <PrivateCall> element exists in the MCPTT user profile document with one or more <entry> elements (see the MCPTT user profile document in 3GPP TS 24.484 [50]), then only the <entry> elements of the application/resource-lists MIME body that have a "uri" attribute that matched with an <entry> elements of the <PrivateCall> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]) are included in the application/resource-lists MIME body and the <session-type> is set to "first-to-answer" in the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request generated in clause 6.3.2.1.3;

16) shall set the Request-URI to the public service identity of the controlling MCPTT function hosting the private call service or first-to-answer call service as determined by step 6);

17) shall set the <mcptt-calling-user-id> element in an application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

…

19) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCPTT function", as specified in clause 6.3.2.1.1.1;

…

20) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCPTT client;

…

22) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 180 (Ringing) response, the participating MCPTT function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall send the SIP 180 (Ringing) response to the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response, the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

5A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [7].

The participating MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.1]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcptt-calling-user-id> containing the calling user's MCPTT ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

3) if the received SIP INVITE request contained a <session-type> element in an application/vnd.3gpp.mcptt-info+xml MIME body set to "first-to-answer" shall include in the SIP INVITE request a Priv-Answer-Mode header field with the value "Manual" according to the rules and procedures of IETF RFC 5373 [18];

4) shall copy the MCPTT ID of the MCPTT user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

5) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited;

NOTE 2: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 3: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

6) shall copy the public user identity of the calling MCPTT user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

7) shall include a Resource-Priority header field populated with the values for an MCPTT emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCPTT emergency private call as determined in step 2 above; or

b) the originating MCPTT user is in an in-progress emergency private call state with the targeted MCPTT user;

8) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

9) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [4]; and

10) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request containing a Require header field with the option tag "100rel" and a P-Answer-State header field with the value "Unconfirmed" in response, the controlling MCPTT function:

1) shall send a SIP PRACK request towards the terminating network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

Upon expiry of the private call timer, the controlling MCPTT function shall follow the procedure for releasing private call session as specified in clause 11.1.4.4.".

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.1.4.2]

"Upon receipt of:

- *a "SIP INVITE request for controlling MCPTT function of a private call"; or*
- *a "SIP INVITE request for controlling MCPTT function of a first-to-answer call";*

the controlling MCPTT function:

2) if the <session-type> in the SIP INVITE request is set to "first-to-answer" shall check whether the public service identity contained in the Request-URI is allocated for first-to-answer call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps;

…

10) shall perform actions as described in clause 6.3.3.2.2;

11) shall allocate an MCPTT session identity for the MCPTT session; and

12) if the <session-type> in the received SIP INVITE request is set to "first-to-answer" and if the SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <call-to-functional-alias-ind> element set to "true":

…

otherwise shall invite the MCPTT user(s) listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 11.1.1.4.1.

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCPTT client, the controlling MCPTT function:

…

2) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "first-to-answer", and no other SIP 180 (Ringing) responses have been received that are associated with a SIP INVITE that contained a <session-type> set to "first-to-answer", shall generate a SIP 183 (Session Progress) response to the SIP INVITE request and send the SIP 183 (Session Progress) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 11.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], if the <session-type> in the SIP INVITE request is set to "private", the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client, the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) shall include a P-Answer-State header field with the value "Unconfirmed";

…

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

6) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "first-to-answer" and the SIP final response has not yet been sent to the inviting MCPTT client the controlling MCPTT function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

…

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 4: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) for all other MCPTT clients that were invited due to the controlling MCPTT function receiving a SIP INVITE request with a <session-type> element set to the value of "first-to-answer":

a) shall send a SIP BYE request to release a SIP dialog that has been established since the SIP 200 (OK) response was sent in step 5) by following the procedures in clause 6.3.3.1.5 with the clarification that the SIP BYE request contain an application/vnd.3gpp.mcptt-info+xml MIME body including a <release-reason> element set to a value of "not selected for call";

b) shall generate and send a SIP CANCEL request according SIP IETF RFC 3261 [24], to cancel a SIP dialog that has not yet been established since the SIP 200 (OK) response was sent in step 5);

c) on receiving a SIP 200 (OK) to a SIP CANCEL request, shall wait to receive a SIP 487 (Request Terminated) to the original SIP INVITE request sent to the client; and

d) if a SIP 487 (Request Terminated) from the MCPTT client is not received within a time determined by the MCPTT server implementation, shall send a SIP BYE towards the MCPTT client by following the procedures in clause 6.3.3.1.5 with the clarification that the SIP BYE request contain an application/vnd.3gpp.mcptt-info+xml MIME body including a <release-reason> element set to a value of "not selected for call"; and

8) if not successful in cancelling or terminating SIP dialogs in step 6) above, may repeat the SIP CANCEL and SIP BYE requests.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCPTT function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4].

Upon receiving a SIP BYE request from the originating MCPTT client containing an application/vnd.3gpp.mcptt-info+xml MIME body containing a <release-reason> element set to a value of "authentication of the MIKEY-SAKE I_MESSAGE failed", the controlling MCPTT function:

…

2) shall follow the procedures in clause 11.1.3.3.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.1.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.2.2.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.3.3.1]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

6.2.7.3 Test description

6.2.7.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client), SS-UE2 (MCPTT Client) and SS-UE3 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) , the SS-UE3 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the SS-UE3 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the SS-UE3 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.7.3.2 Test procedure sequence

Table 6.2.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a first-to-answer call to targeting users B and C, in SS-UE2 and SS-UE3 respectively.	-	-	-	-
2	Check: Does the MCPTT server correctly perform steps 1-4 within step 1 of procedure 'MCPTT E2E private call establishment, manual commencement' as described in Annex A, Table A.3.3.3-1 to request the establishment of a first-to-answer call without implicit floor request with user B?	-	-	1	P
3	Check: Does the MCPTT server correctly perform steps 2-3 within step 1 of procedure 'MCPTT E2E private call establishment, manual commencement' as described in Annex A, Table A.3.3.3-1 to request the establishment of a first-to-answer call without implicit floor request with user C? (see note)	-	-	1	P
4	The SS-UE2 (MCPTT Client) sends a SIP 180 Ringing to the MCPTT Server	->	SIP 180 (Ringing)	-	-
5	Check: Does the MCPTT server send a SIP 183 (session Progress) to SS-UE1?	<-	SIP 183 (Session Progress)	2	P
6	The SS-UE2 (MCPTT Client) accepts the call.	-	-	-	-
7	Check: Does the MCPTT server correctly performs procedure 'MCPTT E2E private call establishment, manual commencement' except steps 7-8 within step 1, as described in Annex A, Table A.3.3.3-1 to complete the first-to-answer call establishment with user B?	-	-	3	P
8	Check: Does the MCPTT server send a SIP CANCEL to cancel the request (another Client has answered) towards user C.	<--	SIP CANCEL	4	P
9	The SS-UE3 (MCPTT client) responds with a SIP 200 (OK)?	-->	SIP 200 (OK)	-	-
10	The SS-UE3 (MCPTT client) sends a SIP 487 (Request Terminated) to confirm terminating of the dialog	-->	SIP 487 (Request Terminated)	-	-
10A	Check: Does the MCPTT server send a SIP ACK?	<--	SIP ACK	4	P
12	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
13	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	5	P
NOTE: During this procedure the SS-UE3 performs the role of SS-UE2.					

6.2.7.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.7.3.3-1: SIP INVITE from the SS-UE1 (Step 2, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.3.3.3-1; Step 1, Annex A, Table A.2.21.3-1)

Derivation Path: Annex A, Table A.3.3.4-1 with FIRST-TO-ANSWER instead of MANUAL condition				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Annex A, Table A.3.3.4-2 without PRIVATE-CALL condition			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Annex A, Table A.3.3.4-3 with FIRST-TO-ANSWER condition instead of PRIVATE-CALL.			
MIME body part		**Resource lists**		
MIME-part-body	Resource list as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.3.1-1 with FIRST-TO-ANSWER condition.			

Table 6.2.7.3.3-2: SIP INVITE from MCPTT Server to SS-UE2 (Step 2, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.3.3.3-1; Step 2, Annex A, Table A.2.21.3-1)

Derivation Path: Annex A, Table A.3.3.4-4 with FIRST-TO-ANSWER instead of MANUAL condition				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Annex A, Table A.3.3.4-6 with FIRST-TO-ANSWER condition instead of PRIVATE-CALL.			

Table 6.2.7.3.3-3: SIP INVITE from MCPTT Server to SS-UE3 (Step 3, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.3.3.3-1; Step 2, Annex A, Table A.2.21.3-1)

Derivation Path: Annex A, Table A.3.3.4-4 with FIRST-TO-ANSWER instead of MANUAL condition				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.7.3.3-4.			

Table 6.2.7.3.3-4: MCPTT-Info in SIP INVITE (Table 6.2.7.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition FIRST-TO-ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_C	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 6.2.7.3.3-5: SIP 200 (OK) from SS-UE2 (Step 6, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.3.3.3-1; Step 7, Annex A, Table A.2.21.3-1)

Derivation Path: Annex A, Table A.3.3.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Annex A, Table A.3.3.4-8 with WITH_SECURITY condition			

Table 6.2.7.3.3-5: SIP 200 (OK) from MCPTT Server to SS-UE1 (Step 6, Table 6.2.7.3.2-1; Step 1, Annex A, Table A.3.3.3-1; Step 8, Annex A, Table A.2.21.3-1)

Derivation Path: Annex A, Table A.3.3.4-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Annex A, Table A.3.3.4-10 with WITH_SECURITY condition			

Table 6.2.7.3.3-6: SIP ACK from the SS-UE3 (Step 10A, Table 6.2.21.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.2-1, condition NON-2XX

6.2.8 MCPTT Server - MCPTT Client / Remotely initiated private call / On-demand

6.2.8.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { SS-UE3 (MCPTT Client) requests the remote initiation of a private call by sending a SIP
MESSAGE message }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to SS-UE1 (MCPTT Client) and responds with
a SIP 200 (OK) to SS-UE3 (MCPTT Client) }
           }
```

(2)

```
with { IUT (MCPTT Server) having requested a remotely initiated private call }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a private call with automatic
commencement mode and implicit floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCPTT Client) }
           }
```

(3)

```
with { IUT (MCPTT Server) having established a remotely initiated private call }
ensure that {
  when { the SS-UE1 (MCPTT client) ends remotely initiated private call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to SS-UE1 (MCPTT client)
and sends a SIP BYE message to SS-UE2 (MCPTT client) }
```

6.2.8.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 11.1.7.2.1, 11.1.7.2.2, 11.1.7.3.1, 11.1.7.3.2, 11.1.7.4. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-15 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.7.2.1]

"Upon receiving a request from the MCPTT to send a remotely initiated private call request to the remote MCPTT user to originate a private call to an identified MCPTT user, the MCPTT client:

…

2) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

d) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-called-party-id> element set to the MCPTT ID of the identified MCPTT user of the remotely initiated private call; and

ii) an <anyExt> element containing:

A) the <request-type> element set to a value of "remotely-initiated-private-call-request";

B) the <notify-remote-user> element set to a value of "true" if the requesting MCPTT user has indicated that the remote MCPTT user be notified of the remotely initiated private call request; and

C) the <notify-remote-user> element set to a value of "false" if the requesting MCPTT user has indicated that the remote MCPTT user not be notified of the remotely initiated private call request;

e) shall insert in the SIP MESSAGE request a MIME resource-lists body with the MCPTT ID of the remote MCPTT user, according to rules and procedures of IETF RFC 5366 [20]; and

f) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the MCPTT user; and

3) shall send the SIP MESSAGE request towards the MCPTT server according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.7.2.2]

"…

Upon completion of the procedures of clause 11.1.1.2.1.1 or clause 11.1.1.2.2.1, the MCPTT client:

1) if:

a) the MCPTT ID of the identified MCPTT user is identical to the <mcptt-calling-user-id> element contained in the <mcptt-Params> element of the <mcpttinfo> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body contained in the received SIP MESSAGE request: and

b) the procedures of clause 11.1.1.2.1.1 or clause 11.1.1.2.2.1 were successful in originating an MCPTT private call to the identified MCPTT user;

then:

a) shall skip the remaining steps of the current clause;

NOTE: In this case, it is not necessary to send a response to the sender of the remotely initiated private call request as the sender and the terminating party of the successful private call origination are the same user and will be aware of the request's outcome.

2) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33] with the following clarifications:

a) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9] in the SIP MESSAGE request;

b) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

c) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [4];

d) shall include in an application/resource-lists+xml MIME body the MCPTT ID contained in the <mcptt-calling-user-id> element in the application/ vnd.3gpp.mcptt-info+xml MIME body of the received SIP MESSAGE request; and

e) shall include an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 with the <mcpttinfo> element containing the <mcptt-Params> element containing:

i) the <mcptt-called-party-id> set to the MCPTT ID of the identified MCPTT user called by the remote MCPTT user; and

ii) an <anyExt> element containing:

A) the <response-type> element set to a value of "remotely-initiated-private-call-response";

B) if the procedures of clause 11.1.1.2.1.1 or clause 11.1.1.2.2.1 were successful in originating an MCPTT private call to the identified MCPTT user, a <remotely-initiated-call-outcome> element set to a value of "success"; and

C) if the procedures of clause 11.1.1.2.1.1 or clause 11.1.1.2.2.1 were not successful in originating an MCPTT private call to the identified MCPTT user, a <remotely-initiated-call-outcome> element set to a value of "fail";

3) shall set the Request-URI to the public service identity identifying the participating MCPTT function serving the MCPTT user; and

4) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.7.3.1]

"Upon receiving a "SIP MESSAGE request for remotely initiated private call for originating participating MCPTT function" the participating MCPTT function:

…

2) shall determine the MCPTT ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;

NOTE: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3

…

4) shall determine the public service identity of the controlling MCPTT function for the remotely initiated private call service for the requesting MCPTT user;

5) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

6) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCPTT function associated with the remotely initiated private call service for the requesting MCPTT user;

7) shall copy the contents of the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body as specified in clause F.1 included in the outgoing SIP MESSAGE request;

8) shall set the <mcptt-calling-user-id> element of the <mcpttinfo> element containing the <mcptt-Params> element to the MCPTT ID determined in step 2) above;

9) shall copy the contents of the application/resource-lists MIME body in the received SIP MESSAGE request into an application/resource-lists MIME body in the outgoing SIP MESSAGE request;

10) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request;

11) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

12) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

14) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response in response to the SIP MESSAGE request sent in step 13), the participating MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.7.3.2]

"Upon receiving a "SIP MESSAGE request for remotely initiated private call for terminating participating MCPTT function" the participating MCPTT function:

…

2) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCPTT ID and public user identity;

…

4) shall generate an outgoing SIP MESSAGE request as specified in clause 6.3.2.2.11;

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request; and

6) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the participating MCPTT function shall forward the SIP 2xx response to the controlling MCPTT function.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 11.1.7.4]

Upon receiving a:

- *"SIP MESSAGE request for remotely initiated private call request for controlling MCPTT function"; or*
- *"SIP MESSAGE request for remotely initiated private call response for controlling MCPTT function";*

the controlling MCPTT function:

…

4) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

…

7) shall copy the contents of the application/vnd.3gpp. mcptt-info+xml MIME body in the received SIP MESSAGE request into an application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP MESSAGE request;

8) shall copy the MCPTT ID of the MCPTT user listed in the MIME resources body of the incoming SIP MESSAGE request, into the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP MESSAGE request;

…

12) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [4].

Upon receipt of SIP 2xx responses to the outgoing SIP MESSAGE requests, the controlling MCPTT function shall generate a SIP 200 (OK) response and forward the SIP 200 (OK) response to the originating participating MCPTT function."

6.2.8.3 Test description

6.2.8.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client), SS-UE2 (MCPTT Client) and SS-UE3 (MCPTT Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) , the SS-UE3 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

- The IUT (MCPTT Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the SS-UE3 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client) and the SS-UE3 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.8.3.2 Test procedure sequence

Table 6.2.8.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE3 (MCPTT client) requests the remote initiation of a private call by SS-UE1 (MCPTT client) to SS-UE2 (MCPTT client)	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to request the remote initiation of a private call from SS-UE1 to SS-UE2? (see note)	-	-	1	P
3	The SS-UE1 (MCPTT client) requests the establishment of a private call to SS-UE2 (MCPTT client) with manual commencement mode and implicit floor request.	-	-	-	-
4	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E private call establishment, manual commencement' as described in Annex A, Table A.3.3.3-1 to establish a private call with manual commencement mode and implicit floor control according to option b.ii of note in Annex A, Table A.3.3.3-1?	-	-	2	P
5	The SS-UE1 (MCPTT client) notifies the successful remote initiation of the private call to SS-UE3 (MCPTT client).	-	-	-	-
6	Check: Does the MCPTT Server correctly perform procedure 'MCX E2E SIP MESSAGE' as described in Annex A, Table A.2.10.3-1 to notify to SS-UE3 the successful remote initiation of a private call between SS-UE1 and SS-UE2?	-	-	1	P
7	The SS-UE1 (MCPTT client) releases the call	-	-	-	-
8	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	3	P
NOTE: During this procedure, SS-UE3 performs the role of SS-UE1, and SS-UE1 the role of SS-UE2.					

6.2.8.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.2.8.3.3-1: SIP MESSAGE from the SS-UE3 (Step 2, Table 6.2.8.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.8.3.3-2			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.8.3.3-3			

Table 6.2.8.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.2.8.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	not present			
mcptt-calling-user-id	not present			
mcptt-called-party-id	Encrypted (see note) <mcptt-called-party-id> with mcpttURI set to px_MCPTT_ID_User_B	User to be called	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	
anyExt				
request-type	"remotely-initiated-private-call-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	
notify-remote-user	"true"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.8.3.3-3: Resource lists in SIP MESSAGE (Table 6.2.8.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_A	The targeted MCPTT user initiating the call	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	

Table 6.2.8.3.3-4: SIP MESSAGE from the MCPTT-Server to SS-UE1 (Step 2, Table 6.2.8.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity				
name-addr	px_MCX_SIP_PublicUserId_C		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.4	
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.8.3.3-5			

Table 6.2.8.3.3-5: MCPTT-Info in SIP MESSAGE (Table 6.2.8.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A	The targeted MCPTT user initiating the call	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_C	The MCPTT user requesting a remote private call	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.3.1	
mcptt-called-party-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B	The MCPTT user to be called	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
anyExt				
request-type	"remotely-initiated-private-call-request"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	
notify-remote-user	"true"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.1	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.8.3.3-6: SIP MESSAGE from the SS-UE1 (Step 6, Table 6.2.8.3.2-1; Step 1, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1 with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1	
MIME-part-body	MCPTT-Info as described in Table 6.2.8.3.3-7			
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 6.2.8.3.3-8			

Table 6.2.8.3.3-7: MCPTT-Info in SIP MESSAGE (Table 6.2.8.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	not present			
mcptt-calling-user-id	not present			
mcptt-called-party-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
anyExt				
response-type	"remotely-initiated-private-call-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
remotely-initiated-call-outcome	"success"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.2.8.3.3-8: Resource lists in SIP MESSAGE (Table 6.2.8.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_C	The targeted MCPTT user	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.2.2	

Table 6.2.8.3.3-9: SIP MESSAGE from the MCPTT-Server to SS-UE3 (Step 6, Table 6.2.8.3.2-1; Step 3, Annex A, Table A.2.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCPTT and ACCEPT-CONTACT-WITH-MEDIA-FEATURE-TAG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.2.8.3.3-14			

Table 6.2.8.3.3-10: MCPTT-Info in SIP MESSAGE (Table 6.2.8.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_C		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.5.4	
mcptt-called-party-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
anyExt				
response-type	"remotely-initiated-private-call-response"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
remotely-initiated-call-outcome	"success"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 11.1.7.2.2	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

6.3 Location

6.3.1 MCPTT Server - MCPTT Client / Location / Event Triggered Location Information report

6.3.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { IUT (MCPTT Server) requests to send a location reporting configuration }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message containing the cell change trigger
criteria and the minimum report interval to the SS-UE1 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having received Location Information report }
ensure that {
  when { SS-UE1 (MCPTT Client) requests location report containing the cell change trigger
identification along with the location information report, the current location coordinates and the
current serving ECGI by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

6.3.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 13.2.2, 13.2.4, 13.3.4.1, 13.3.4.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.2.2]

"The participating MCPTT function may configure the location reporting in the MCPTT client by generating a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33]. The participating MCPTT function:

1) shall include a Request-URI set to the URI corresponding to the identity of the MCPTT client;

2) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref set to the value "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" in accordance with IETF RFC 3841 [6];

3) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with an <mcptt-request-uri> element containing the MCPTT ID of the MCPTT user to receive the configuration;

4) shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body with the <Configuration> element contained in the <location-info> root element set to the desired configuration;

5) shall include the TriggerId attribute where defined for the sub-elements defining the trigger criterion ;

6) shall include the public service identity of the participating MCPTT function in the P-Asserted-Identity header field;

7) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

8) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.2.4]

"If the participating MCPTT function receives a SIP request containing:

1) a Content-Type header field set to "application/vnd.3gpp.mcptt-location-info+xml"; and

2) an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element;

then the participating MCPTT function shall authorise the location report based on the MCPTT ID received. If the MCPTT user is authorised to send a location report the participating MCPTT function:

1) shall use the location information as needed; and

2) shall follow the procedure of clause 6.3.2.4.1, if the MCPTT client has entered into or exited from an emergency alert area ;and

3) shall follow the procedure of clause 6.3.2.4.2, if the MCPTT client has entered into or exited from a group geographic area."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.3.4.1]

"If a location reporting trigger fires the MCPTT client checks if the minimumReportInterval timer is running. If the timer is running the MCPTT client waits until the timer expires. When the minimumReportInterval timer fires, the MCPTT client:

1) shall, if any of the reporting triggers are still true, send a location information report as specified in clause 13.3.4.2.

If the MCPTT client receives a location information request as specified in clause 13.3.3, the MCPTT client shall send a location report as specified in clause 13.3.4.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.3.4.2]

"…

If the MCPTT client does not need to send a SIP request for other reasons, the MCPTT client shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33]. The MCPTT client;

1) shall include in the Request-URI, the SIP URI received in the P-Asserted-Identity header field in the received SIP MESSAGE request for location report configuration;

2) shall include a Content-Type header field set to "application/vnd.3gpp.mcptt-location-info+xml";

3) shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body and in the <location-info> root element include:

a) a <Report> element and if the Report was triggered by a location request include the <ReportID> attribute set to the value of the <RequestID> attribute in the received Request;

b) a <TriggerId> child element set to the value of each <Trigger-Id> value of the triggers that have fired; and

c) the location reporting elements corresponding to the triggers that have fired;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.mcptt along with parameters "require" and "explicit" in accordance with IETF RFC 3841 [6];

5) shall set the minimumReportInterval timer to the minimumReportInterval time and start the timer;

6) shall reset all triggers; and

7) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]."

6.3.1.3 Test description

6.3.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)

- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.3.1.3.2 Test procedure sequence

Table 6.3.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the MCPTT Server send a location reporting configuration to SS-UE1 (see note 1)	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to send a location reporting configuration message containing the cell change trigger criteria and the minimum report interval to SS-UE1?	-	-	1	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to send a location report from SS-UE1 containing the cell change trigger identification along with the location information report, the current location coordinates, and the current serving ECGI? (see note 2)	-	-	2	P
NOTE 1: This is expected to be done via a suitable implementation dependent MMI. NOTE 2: This action represents the UE changing from cell.					

6.3.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.3.1.3.3-1: SIP MESSAGE from the MCPTT Server (Step 2, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT and LOCATION-CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-info as described in Table 6.3.1.3.3-2			
MIME body part		**Location-info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.3	
MIME-part-body	Location-info as described in Table 6.3.1.3.3-3			

Table 6.3.1.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.3.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.3.1.3.3-3: Location-Info in SIP MESSAGE (Table 6.3.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
location-info				
Configuration				
NonEmergencyLocationInformation				
ServingEcgi	present	Provision of information in the report is requested.		
NeighbouringEcgi	present	Provision of information in the report is requested.		
MbmsSaId	not present	Provision of information in the report is not requested.		
MbsfnArea	not present	Provision of information in the report is not requested.		
GeographicalCoordinate	present			
minimumIntervalLength	"25"	the value in seconds		
EmergencyLocationInformation"				
ServingEcgi	present	Provision of information in the report is requested.		
NeighbouringEcgi	present	Provision of information in the report is requested.		
MbmsSaId	not present	Provision of information in the report is not requested.		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MbsfnArea	not present	Provision of information in the report is not requested.		
GeographicalCoordinate	present	Provision of information in the report is requested.		
minimumIntervalLength	"30"	the value in seconds		
TriggeringCriteria				
CellChange				
AnyCellChange				
TriggerID	any value			

Table 6.3.1.3.3-3: SIP MESSAGE
(Step 3, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body	MIME body not including MCPTT-Info and not including MCPTT-Affiliation-Command	not including any MIME body part with Content-Type being "application/vnd.3gpp.mcptt-info+xml" or "application/vnd.3gpp.mcptt-affiliation-command+xml"		
MIME body part		Location info	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.3	
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-location-info+xml"			
MIME-part-body	Location-info as described in Table 6.3.1.3.3-4			

Table 6.3.1.3.3-4: Location-Info in SIP MESSAGE (Table 6.3.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
location-info				
Report				
TriggerID	Same value as in the location configuration	Value included in Table 6.3.1.3.3-2		
CurrentLocation				
CurrentServingEcgi	The E-UTRAN Cell Global Identification (ECGI) of Cell 2			
NeighbouringEcgi	The ECGI of Cell 1			
CurrentCoordinate		The location simulated by the GNSS simulator at the time of transmission of the message.		
longitude	px_MCX_CoordinateLongitude_Client_A (see note)			
latitude	px_MCX_CoordinateLatitude_Client_A (see note)			
ReportType	"NonEmergency"			
NOTE: Shall be encrypted as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.4.1-1A.				

6.3.2 MCPTT Server - MCPTT Client / Location / On-demand Location Information Request

6.3.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { IUT (MCPTT Server) requests to send a location reporting configuration }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message containing the periodic trigger criteria
and the minimum report interval to the SS-UE1 (MCPTT Client) }
            }
```

(2)

```
with { IUT (MCPTT Server) having received Location Information report }
ensure that {
  when { IUT (MCPTT Server) requests to send an on-demand location report }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE1 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having received Location Information report }
ensure that {
  when { SS-UE1 (MCPTT Client) requests location report containing the location information report,
the current location coordinates and the current serving ECGI by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

6.3.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 13.2.3, 13.3.3, 13.3.4.1, 13.3.4.2. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.2.3]

"If the participating MCPTT function needs to request the MCPTT client to report its location, the participating MCPTT functions shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33]. The participating MCPTT function:

1) *shall include a Request-URI set to the URI corresponding to the identity of the MCPTT client;*

2) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref set to the value "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" in accordance with IETF RFC 3841 [6];*

3) *shall include an application/vnd.3gpp.mcptt-info+xml MIME body with an <mcptt-request-uri> element containing the MCPTT ID of the MCPTT user;*

4) *shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Request> element contained in the <location-info> root element;*

5) *shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and*

6) *shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]."*

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.3.3]

"Upon receiving a SIP MESSAGE request containing

1) *an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref set to the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";*

2) *a Content-Type header field set to "application/vnd.3gpp.mcptt-location-info+xml"; and*

3) an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Request> element included in the <location-info> root element;

then the MCPTT client:

1) shall send a location report as specified in clause 13.3.4; and

2) shall reset the minimumReportInterval timer."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.3.4.1]

"…

If the MCPTT client receives a location information request as specified in clause 13.3.3, the MCPTT client shall send a location report as specified in clause 13.3.4.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.3.4.2]

"…

If the MCPTT client does not need to send a SIP request for other reasons, the MCPTT client shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33]. The MCPTT client;

1) shall include in the Request-URI, the SIP URI received in the P-Asserted-Identity header field in the received SIP MESSAGE request for location report configuration;

2) shall include a Content-Type header field set to "application/vnd.3gpp.mcptt-location-info+xml";

3) shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body and in the <location-info> root element include:

a) a <Report> element and if the Report was triggered by a location request include the <ReportID> attribute set to the value of the <RequestID> attribute in the received Request;

b) a <TriggerId> child element set to the value of each <Trigger-Id> value of the triggers that have fired; and

c) the location reporting elements corresponding to the triggers that have fired;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.mcptt along with parameters "require" and "explicit" in accordance with IETF RFC 3841 [6];

5) shall set the minimumReportInterval timer to the minimumReportInterval time and start the timer;

6) shall reset all triggers; and

7) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]."

6.3.2.3 Test description

6.3.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)

- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

- The IUT (MCPTT Server) is the acting Participating Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1

- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

6.3.2.3.2 Test procedure sequence

Table 6.3.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the MCPTT Server send a location reporting configuration to SS-UE1? (see note)	-	-	-	-
2	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to send a location reporting configuration message containing the periodic trigger criteria and the minimum report interval to SS-UE1?	-	-	1	P
3	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to send an on-demand location report request to SS-UE1?	-	-	2	P
4	Check: Does the MCPTT Server correctly perform procedure 'MCX Participating Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to send a location report from SS-UE1 containing the location information report, the current location coordinates, and the current serving ECGI?	-	-	3	P
NOTE: This is expected to be done via a suitable implementation dependent MMI.					

6.3.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.3.1.3.3-1: SIP MESSAGE from the MCPTT Server (Step 2, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT and LOCATION-CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-info as described in Table 6.3.1.3.3-2			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT and LOCATION-CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**Location-info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.3	
MIME-part-body	Location-info as described in Table 6.3.1.3.3-3			

Table 6.3.1.3.3-2: MCPTT-Info in SIP MESSAGE (Table 6.3.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 13.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 6.3.1.3.3-3: Location-Info in SIP MESSAGE (Table 6.3.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
location-info				
Configuration				
NonEmergencyLocationInformation				
ServingEcgi	present	Provision of information in the report is requested.		
NeighbouringEcgi	present	Provision of information in the report is requested.		
MbmsSaId	not present	Provision of information in the report is not requested.		
MbsfnArea	not present	Provision of information in the report is not requested.		
GeographicalCoordinate	present			
minimumIntervalLength	"25"	the value in seconds		
EmergencyLocationInformation"				
ServingEcgi	present	Provision of information in the report is requested.		
NeighbouringEcgi	present	Provision of information in the report is requested.		
MbmsSaId	not present	Provision of information in the report is not requested.		
MbsfnArea	not present	Provision of information in the report is not requested.		
GeographicalCoordinate	present	Provision of information in the report is requested.		
minimumIntervalLength	"30"	the value in seconds		
TriggeringCriteria				
PeriodicReport				

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
extension base	"15"	periodicity of the reports (in seconds)		
TiggerID	any value			

Table 6.3.1.3.3-4: SIP MESSAGE from the MCPTT Server (Step 3, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT and LOCATION-CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-info as described in Table 6.3.1.3.3-2			
MIME body part		**Location-info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.3	
MIME-part-body	Location-info as described in Table 6.3.1.3.3-5			

Table 6.3.1.3.3-5: Location-Info in SIP MESSAGE (Table 6.3.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.3-1

Table 6.3.1.3.3-6: SIP MESSAGE (Step 4, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body	MIME body not including MCPTT-Info and not including MCPTT-Affiliation-Command	not including any MIME body part with Content-Type being "application/vnd.3gpp.mcptt-info+xml" or "application/vnd.3gpp.mcptt-affiliation-command+xml"		
MIME body part		Location info	ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.3	
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-location-info+xml"			
MIME-part-body	Location-info as described in Table 6.3.1.3.3-7			

Table 6.3.1.3.3-7: Location-Info in SIP MESSAGE (Table 6.3.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
location-info				
Report				
ReportID attribute	"1"	The same as the ID in the request message.		
ReportType attribute	"NonEmergency"			
CurrentLocation				
CurrentServingEcgi	The E-UTRAN Cell Global Identification (ECGI) of Cell 2			
NeighbouringEcgi	The ECGI of Cell 1			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.4.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
CurrentCoordinate		The location simulated by the GNSS simulator at the time of transmission of the message.		
longitude	px_MCX_CoordinateLongitude_Client_A (see note			
latitude	px_MCX_CoordinateLatitude_Client_A (see note)			
NOTE: Shall be encrypted as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.4.1-1A.				

6.4 MBMS

6.4.1 MCPTT Server - MCPTT Client / MBMS / MBMS Bearer Announcement / MBMS Bearer Listening Status / Transition to MBMS from Unicast / MBMS Floor Control / Transition to Unicast from MBMS

6.4.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a group call with automatic
commencement mode and without implicit floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes a pre-arranged
group call }
            }
```

(2)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send an MBMS bearer announcement to notify bearer
availability }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE1 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report by sending a SIP MESSAGE
message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send an MBMS bearer announcement to notify bearer
availability }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE2 (MCPTT Client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening
general-purpose subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(6)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to use MBMS subchannel for the conversation }
    then { IUT (MCPTT Server) sends a Map Group To Bearer message }
            }
```

(7)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening audio
and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(8)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening audio
and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send floor Granted }
    then { IUT (MCPTT Server) sends a Floor Granted message with request for acknowledgement using
unicast bearer }
            }
```

(10)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { IUT (MCPTT Server) requests to set the explicit MuSiK key }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message containing a MuSiK download message to the
SS-UE1 (MCPTT Client) and to the SS-UE2 (MCPTT Client) }
            }
```

(11)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1(MCPTT Client) performs floor control interactions }
    then { IUT (MCPTT Server) respondes floor control (Floor Ack, Floor Idle) }
            }
```

(12)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT client) stops using MBMS subchannel for the conversation }
    then { IUT (MCPTT Server) sends a Unmap Group To Bearer message }
            }
```

(13)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report to stop notify listening
audio and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(14)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to stop notify listening
audio and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(15)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { IUT (MCPTT Server) requests to send an MBMS bearer cancellation announcement to notify
bearer unavailability }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE1 (MCPTT Client) and to the
SS-UE2 (MCPTT Client) }
            }
```

(16)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT Client) sends MBMS Bearer listening status report message to MCPTT
Server, to notify not being listening to general-purpose subchannel }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(17)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE2 (MCPTT Client) sends MBMS Bearer listening status report message to MCPTT
Server, to notify not being listening to general-purpose subchannel }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(18)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and having stopped using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT client) sends Floor Request }
    then { IUT (MCPTT Server) replies with Floor Taken, Floor Idle, Floor Granted }
            }
```

(19)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and having stopped using a MBMS subchannel for the conversation }}
ensure that {
  when { the SS-UE1 (MCPTT client) ends the Pre-arranged group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

6.4.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 14.2.2.2, 14.2.3, 14.3.3.1, ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 10.2.2.2, 10.2.3.2, 10.2.3.3, 10.2.3.4, 10.2.3.5, 10.2.3.11, 10.2.3.12, 6.3.4.3.3, 6.3.4.4.2, 6.3.4.4.6. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.2.2.2]

"For each MCPTT client that the participating MCPTT function is sending an MBMS bearer announcement to, the participating MCPTT function:

1) shall generate an SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

2) shall set the Request-URI to the URI received in the To header field in a third-party SIP REGISTER request;

3) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media-feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

4) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";

5) shall include one application/sdp MIME body conforming to 3GPP TS 24.229 [4], where the application/sdp MIME body:

a) shall include the Content-Disposition header field with the value "render";

b) should include one or more "m=audio" media lines and media line attributes as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for audio and media control. Additionally, the participating MCPTT function:

i) shall set c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6;

ii) shall set port number of the media line to 9;

ii) shall include the "a=rtcp-mux" attribute as specified in IETF RFC 5761 [39]; and

iii) shall include the "a=rtcp:9" as specified in IETF RFC 5761 [39].

c) should include one or more "m=audio" media lines and media line attributes as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for audio only. Additionally, the participating MCPTT function:

i) shall set the c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6; and

ii) shall set the port number of the media line to 9;

NOTE 1: If an MBMS subchannel for audio only is included, the "a=rtcp-mux" and "a=rtcp:" attributes are not included in the media line.

d) shall include one "m=application" media line as defined in 3GPP TS 24.380 [5] to be used as the general purpose MBMS subchannel. The media line shall include a valid multicast IP address and a valid port number. If the protection of MBMS subchannel control messages sent over the general purpose MBMS subchannel of the MBMS bearer is required, the participating MCPTT function also includes an "a=key-mgmt" media-level attribute. The participating MCPTT function:

i) shall encrypt the MSCCK to a UID associated to the targeted MCPTT ID and a time related parameter as described in 3GPP TS 33.180 [78];

ii) shall generate a MIKEY-SAKKE I_MESSAGE using the encapsulated MSCCK and MSCCK-ID as specified in 3GPP TS 33.180 [78];

iii) shall add the public service identity of the participating MCPTT function to the initiator field (IDRi) of the I_MESSAGE as described in 3GPP TS 33.180 [78];

iv) shall sign the MIKEY-SAKKE I_MESSAGE using the public service identity of the participating MCPTT function signing key provided in the keying material together with a time related parameter, and add this to the MIKEY-SAKKE payload, as described in 3GPP TS 33.180 [78]; and

v) shall include the "mikey" key management and protocol identifier and the signed MIKEY-SAKKE I_MESSAGE in the value of the a=key-mgmt" media-level attribute according to IETF RFC 4567 [47]; and

NOTE 2: The media parameters to be used by the MBMS subchannel for media is included in the Map Group To Bearer message defined in 3GPP TS 24.380 [5] and not included in this application/sdp MIME body.

e) if "m=audio" media lines to be used in an MBMS subchannel for audio only are included above, shall include one or more "m=application" media line as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for floor control messages. The media line:

i) shall set c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6; and

ii) shall set the port number of the media line to 9;

NOTE 3: The use of a separate MBMS subchannel for floor control is optional. When a separate MBMS subchannel for floor control is not used, floor control messages are sent in the MBMS subchannel for media.

6) shall include an application/vnd.3gpp.mcptt-mbms-usage-info+xml MIME body defined in clause F.2 with the <version> element set to "1" and one or more <announcement> elements associated with the pre-activated MBMS bearers. Each set of an <announcement> element:

a) shall include a TMGI value in the <TMGI> element;

NOTE 4: The same TMGI value can only appear in one <announcement> element. The TMGI value is also used to identify the <announcement> when updating or cancelling the <announcement> element.

NOTE 5: The security key active for the general purpose MBMS subchannel on which the mapping (i.e. the Map Group To Bearer message) of media or media control to this MBMS bearer was indicated, is used for MBMS subchannels on this MBMS bearer, unless a different key or an indication of not using encryption is in place.

b) shall include the QCI value in the <QCI> element;

c) if multiple carriers are supported, shall include the frequency to be used in the <frequency> element;

NOTE 6: In the current release if the <frequency> element is included, the frequency in the <frequency> element is the same as the frequency used for unicast.

d) shall include one or more MBMS service area IDs in <mbms-service-area-id> elements in the <mbms-service-areas> element;

NOTE 7: Initial mappings of groups to MBMS subchannels on an MBMS bearer for the purpose of carrying media or media control can occur only where the MBMS service area for this bearer and the MBMS service area for the bearer carrying the general purpose MBMS subchannel on which the Map Group To Bearer message is sent intersect. However, once media or media control were successfully mapped to this bearer, the reception by the MCPTT client can continue (until Unmap Group To Bearer is received or until timeout) throughout the entire MBMS service area of this bearer.

e) may include the <report-suspension> element and set it to "true" value or the "false" value;

NOTE 8: The participating function can choose to direct some clients not to send an MBMS bearer suspension report when notified by RAN, by including the <report-suspension> element set to "false". The purpose is to prevent an avalanche of identical reports sent by clients roughly at the same time, to report the suspension of the same MBMS bearer. The way the participation function determines which clients are to send or not to send the report is outside the scope of the present document.

f) if the MBMS bearer is carrying the general purpose MBMS subchannel, shall include one <GPMS>element, giving the number of the "m=application" media line in the application/sdp MIME body generated in step 5 above to be used for the general purpose MBMS subchannel; and

g) if the packet headers are compressed with ROHC specified in RFC 5795 [80] in this MBMS bearer, the <anyExt> element in the <announcement> element in the <mcptt-mbms-usage-info> element shall include the <mcptt-mbms-rohc> element defined in clause F.2.3.

7) shall include the MBMS public service identity of the participating MCPTT function in the P-Asserted-Identity header field;

8) shall include in a MIME body with Content-Type header field set to "application/vnd.3gpp.mcptt-info+xml", the <mcptt-request-uri> element set to the MCPTT ID of the user; and

9) shall send the SIP MESSAGE request towards the MCPTT client according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.2.3]

"Upon receiving a "SIP MESSAGE request for an MBMS listening status update", the participating MCPTT function shall handle the request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33].

If the SIP MESSAGE request contains:

1) an application/vnd.3gpp.mcptt-mbms-usage-info+xml MIME body with an <mbms-listening-status> element; and

2) an application/vnd.3gpp.mcptt-info+xml MIME body containing an MCPTT ID in the <mcptt-request-uri> served by the participating MCPTT function;

then the participating MCPTT function:

1) shall verify that the public user identity in the P-Asserted-Identity header field is bound to the MCPTT ID in the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body, and if that is the case:

a) if the <mbms-listening-status> element is set to "listening":

i) if a <session-id> element is included, shall indicate to the media plane that the MCPTT client in the session identified by the <session-id> element is now listening to the MBMS subchannel; and

ii) if <general-purpose> element is included with the value "true", shall indicate to the media plane that the MCPTT client is now listening to the general purpose MBMS subchannel; and

b) if the <mbms-listening-status> element is set to "not-listening":

i) if a <session-id> element is included, shall indicate to the media plane that the MCPTT client in the sessions identified by the <session-id> elements is not listening to the MBMS subchannel;

ii) if <general-purpose> element is included with the value "false", shall indicate to the media plane that the MCPTT client is no longer listening to the general purpose MBMS bearer; and

iii) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

NOTE 1: If the MCPTT client reports that the MCPTT client is no longer listening to the general purpose MBMS subchannel it is implicitly understood that the MCPTT client no longer listens to any MBMS subchannel in ongoing conversations that the MCPTT client previously reported status "listening".

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.3.3.1]

"If one of the following conditions is fulfilled:

1) if the MCPTT client:

a) receives a Map Group To Bearer message over the general purpose MBMS channel;

b) participates in a group session identified by the Map Group To Bearer message; and

c) the status "listening" is not already reported; or

2) if the MCPTT client:

a) receives an announcement as described in clause 14.3.2;

b) enters an MBMS service area where a general purpose MBMS is available; and

c) experiences good MBMS bearer radio condition;

then the MCPTT client shall report that the MCPTT client is listening to the MBMS bearer as specified in clause 14.3.3.2.

If one of the following conditions is fulfilled:

1) if the MCPTT client:

a) receives an MBMS bearer announcement as described in the clause 14.3.2;

b) the MBMS bearer announcement contains a cancellation of an <announcement> element identified by the same TGMI value as received in a Map Group To Bearer message in an ongoing conversation; and

c) the status "not-listening" is not already reported;

2) if the MCPTT client:

a) receives an MBMS bearer announcement as described in the clause 14.3.2;

b) the MBMS bearer announcement contains a cancellation of an <announcement> element;

c) does not participate in an ongoing conversation;

d) the MCPTT client has reported the "listening" status due to the availability of the general purpose MBMS subchannel in the <announcement> element; and

e) the status "not-listening" is not already reported; or

3. if the MCPTT client:

a) suffers from bad MBMS bearer radio condition,

then the MCPTT client shall report that the MCPTT client is not listening to the MBMS subchannels as specified in clause 14.3.3.2.

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.2.2]

"Upon receiving a Floor Request message or a Floor Taken message and when the participating MCPTT function decides that an MBMS subchannel shall be used for a conversation in an ongoing group session, the participating MCPTT function needs to determine if the MBMS bearer has sufficient capacity for the new conversation. If the new MBMS bearer has sufficient capacity the participating MCPTT function:

NOTE: The participating MCPTT function can take that decision when receiving the first Floor Request from a floor participant or when receiving a Floor Taken message destined to one of the floor participants served by the participating MCPTT client.

1. shall create an instance of the 'Participating MCPTT function MBMS subchannel control' state machine;

2. shall send a Map Group To Bearer message over the general purpose MBMS subchannel. The Map Group To Bearer message:

a. shall include TMGI;

b. shall include the identifier of the media stream; and

c. shall include the MCPTT Group identifier field;

3. shall start timer T15 (Conversation);

4. shall start timer T16 (Unmap Group To Bearer);

5. shall enter the 'M: A conversation is active' state;

6. if the Floor Request was received, shall perform actions as described in subclause 6.4.2; and

7. if the Floor Taken was received, shall perform the actions described in subclause 10.2.3.3.

If the MBMS bearer does not have sufficient capacity for the new conversation the participating MCPTT function:

1. may free capacity for the new conversation by transfering an existing conversation over an MBMS bearer to unicast bearers following the procedure in subclause 10.2.3.12; or

2. may use the MBMS bearer for the signaling messages, while using unicast bearers for the media message."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.2]

"When a Floor Idle message destined to a floor participant listening to the MBMS subchannel is received, the participating MCPTT function:

1. shall for each received Floor Idle message, check if there is a stored message-sequence-number value associated with the conversation;

2. if a message-sequence-number value associated with the conversation is stored, compared the value of the stored message-sequence-number value with the value in the received Floor Idle message;

a. if the received message-sequence-number value is higher than the stored value:

i. shall replace the stored message-sequence-number value with the received message-sequence-number field;

ii. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

iii. shall send the received Floor Idle message over the MBMS subchannel; and

b. if the received message-sequence-number value is lower or the same as the stored value, shall discard the received Floor Idle message;

3. if a message-sequence-number value associated with the conversation is not stored (i.e. this is the first floor control message received in the conversation):

a. shall store the received message-sequence-number value and associate the value with the conversation;

b. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

c. shall send the received Floor Idle message over the MBMS subchannel;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.3]

"When a Floor Taken message destined to a floor participant listening to the MBMS subchannel is received, the participating MCPTT function:

1. shall for each received Floor Taken message, check if there is a stored message-sequence-number value associated with the conversation;

2. if a message-sequence-number value associated with the conversation is stored, compare the value of the stored message-sequence-number value with the value in the received Floor Taken message;

a. if the received message-sequence-number value is higher than the stored value:

i. shall replace the stored message-sequence-number value with the received message-sequence-number field;

ii. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

iii. shall send the received Floor Taken message over the MBMS subchannel; and

b. if the received message-sequence-number value is lower or the same as the stored value, shall discard the received Floor Taken message;

3. if a message-sequence-number value associated with the conversation is not stored (i.e. this is the first floor control message received in the conversation):

a. shall store the received message-sequence-number value and associate the value with the conversation;

b. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

c. shall send the received Floor Taken message over the MBMS subchannel;

4. if the received Floor Taken message indicates that a Floor Ack message is expected (i.e. the acknowledgment bit is set to '1' as specified in subclause 8.2.2), shall send a Floor Ack message towards the controlling MCPTT function The Floor Ack message:

a. shall include the Message Type field set to '2' (Floor Taken); and

b. shall include the Source field set to '1' (participating MCPTT function is the source);

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.4]

"When a floor control message other than the Floor Idle and Floor Taken message is received from a floor participant or received from the floor control server, the participating MCPTT function:

1. shall forward the floor control message as specified in subclause 6.4;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.5]

"When receiving a RTP media packet destined to one of the MCPTT client listening to the MBMS subchannel, the participating MCPTT function:

NOTE: An RTP media packet not destined to an MCPTT client listening to the MBMS subchannel is forwarded to the MCPTT client over the unicast bearer.

1. shall check if the media packet is already sent over the MBMS subchannel or not;

2. if the RTP media packet is already sent over the MBMS subchannel, shall discard the RTP media packet;

3. if the RTP media packet is not already sent over the MBMS sub channel, shall instruct the media distribution function to send the RTP media packet over the MBMS subchannel;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.11]

"If the control and signalling plane indicates that the group call session is released, the participating MCPTT function:

1. shall send the Unmap Group To Bearer message over the MBMS subchannel. The Unmap Group To Bearer message:

a. shall include the MCPTT Group ID field;

2. shall stop timer T15 (Conversation), timer T16 (Map Group To Bearer) and timer T17 (Unmap Group To Bearer), if running; and

3. shall release the instance of the 'Participating MCPTT function MBMS subchannel management' state machine used for the conversation."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.12]

"If the participating MCPTT server decides that an ongoing conversation over an MBMS bearer shall start using unicast bearers, the participating MCPTT function may send an Application Paging message over the MBMS subchannel associated with this conversation.

NOTE: The Application Paging message can be sent at the same time as the conversation is started using unicast bearers, this will improve the MCPTT access time since the Application Paging message in most cases will reach the client quicker than using normal paging procedures."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.3.3]

"Upon receiving a floor request message (from a floor participant that is permitted to make a floor request) the floor control arbitration logic in the floor control server:

…

3. if the floor request is granted the floor control server:

…

e. shall enter the 'G: Floor Taken' state as specified in the subclause 6.3.4.4.2."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:

a. shall include the value of timer T2 (Stop talking)in the Duration field;

b. shall include the granted priority in the Floor priority field;

…

3. shall send Floor Taken message to all other floor participants. The Floor Taken message:

a. if the floor is currently granted only to one particpant:

i shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field, if privacy is not requested;

ii. may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested; and

iii. shall include the Location field:

A. if the current location of the talker is not available or is not to be reported according to the MCPTT user profile, then the location type is set to '0' (Not provided); or

B. if the current location of the talker is available and may be reported according to the MCPTT user profile, then the location type and location value are set as specified in table 8.2.3.21-3;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.6]

"Upon receiving a Floor Release message the floor control arbitration logic in the floor control server:

1. shall request the media distributor in the MCPTT server to stop forwarding RTP media packets received from the participant that sent the Floor Release message;

…

4. if mutli-talker floor control is not supported shall enter the 'G: Floor Idle' state as specified in the subclause 6.3.4.3.2; and"

6.4.1.3 Test description

6.4.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.
- A pre-activated MBMS bearer exists.

6.4.1.3.2 Test procedure sequence

Table 6.4.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a group call to SS-UE2 (MCPTT client) with automatic commencement mode and without implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a pre-arranged group call with automatic commencement mode and without implicit floor control according to option a of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	Make the MCPTT Server activate an MBMS bearer and send an MBMS bearer announcement to notify bearer availability (see note)	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 4a-4d and steps 5a-5d might be performed in parallel	-	-	-	-
4a	Check: Does the MCPTT Server send an MBMS bearer announcement to notify MBMS bearer availability to SS-UE1?	<--	SIP MESSAGE	2	P
4b	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) message to MCPTT Server	-->	SIP 200 (OK)	-	-
4c	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to general-purpose subchannel	-->	SIP MESSAGE	-	-
4d	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	3	P
5a	Check: Does the MCPTT Server send an MBMS bearer announcement to notify MBMS bearer availability to SS-UE2?	<--	SIP MESSAGE	4	P
5b	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message to MCPTT Server	-->	SIP 200 (OK)	-	-
5c	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to general-purpose subchannel.	-->	SIP MESSAGE	-	-
5d	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	5	P
6	The SS-UE1 (MCPTT client) sends a Floor Request message to MCPTT Server	-->	Floor Request	-	-
7	Make the MCPTT Server use MBMS subchannel for the conversation (see note)	-	-	-	-
8	Check: Does the MCPTT Server send a Map Group To Bearer message over general purpose MBMS subchannel?	<--	Map Group To Bearer	6	P
9	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
10	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	7	P
11	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
12	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	8	P
13	Check: Does the MCPTT Server send a Floor Granted message with request for acknowledgement using unicast bearer to SS-UE1?	<--	Floor Granted	9	P
14	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 15a-15b and steps 16a-16b might be performed in parallel	-	-	-	-
15a	Check: Does the MCPTT Server send a message containing a MuSiK download message to set the explicit MuSiK key to be used for the floor control encryption over the MBMS subchannel to SS-UE1?	<--	SIP MESSAGE	10	P
15b	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
16a	Check: Does the MCPTT Server send a message containing a MuSiK download message to set the explicit MuSiK key to be used for the floor control encryption over the MBMS subchannel to SS-UE2?	<--	SIP MESSAGE	10	P
16b	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-
17	Check: Does the MCPTT Server send a Floor Taken message using multicast bearer?	<--	Floor Taken	11	P
18	The SS-UE1 (MCPTT client) sends a Floor Release message to MCPTT Server	-->	Floor Release	-	-
19	Check: Does the MCPTT Server send a Floor Idle message using multicast bearer?	<--	Floor Idle	11	P
20	Make the MCPTT Server stop using MBMS subchannel for the conversation (Note)	-	-	-	-
21	Check: Does the MCPTT Server send an Unmap Group To Bearer message over general purpose MBMS subchannel?	<--	Unmap Group To Bearer	12	P
22	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
23	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	13	P
24	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
25	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	14	P
26	Make the MCPTT Server stop using MBMS bearer	-	-	-	-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 27a-27b and steps 28a-28b might be performed in parallel	-	-	-	-
27a	Check: Does the MCPTT Server send a MESSAGE containing the MBMS bearer cancellation announcement for the SS-UE1 (MCPTT client)?	<--	SIP MESSAGE	15	P
27b	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-
28a	Check: Does the MCPTT Server send a MESSAGE containing the MBMS bearer cancellation announcement for the SS-UE2 (MCPTT client)?	<--	SIP MESSAGE	15	P
28b	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-
29	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to general-purpose subchannel	-->	SIP MESSAGE	-	-
30	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	16	P
31	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to general-purpose subchannel	-->	SIP MESSAGE	-	-
32	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	17	P
33	The SS-UE1 (MCPTT client) sends a Floor Request message to MCPTT Server	-->	Floor Request	-	-
34	Check: Does the MCPTT Server send a Floor Granted message with request for acknowledgement using unicast to SS-UE1?	<--	Floor Granted	18	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
35	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
36	Check: Does the MCPTT Server send a Floor Taken message with request for acknowledgement using unicast bearer to SS-UE2?	<--	Floor Taken	18	P
37	The SS-UE2 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
38	The SS-UE1 (MCPTT client) sends a Floor Release message to MCPTT Server	-->	Floor Release	-	-
39	Check: Does the MCPTT Server send a Floor Idle message with request for acknowledgement using unicast bearer to SS-UE1?	<--	Floor Idle	18	P
40	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
41	Check: Does the MCPTT Server send a Floor Idle message with request for acknowledgement using unicast bearer to SS-UE2?	<--	Floor Idle	18	P
42	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
43	The SS-UE1 (MCPTT client) releases the call.	-	-	-	-
44	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	19	P
NOTE: This is expected to be done via a suitable implementation dependent MMI.					

6.4.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.4.1.3.3-1: SIP MESSAGE from the MCPTT Server (Step 4a, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		
Message-body				
MIME body part		**SDP message**		
MIME-part-headers				
Content-Type	"application/sdp"			
Content-Disposition	"render"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
MIME-part-body	SDP message as described in Table 6.4.1.3.3-2			
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-3			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-4			

Table 6.4.1.3.3-2: SDP in SIP MESSAGE (Table 6.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media descriptions				
media description		m= line media = audio		
media	"audio"			
port	"9"			
proto	"RTP/AVP"			
fmt	"99"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4" or "IP6" depending on IP address			
connection-address	"0.0.0.0" (IPv4) or domain name within ".invalid" DNS top level domain (IPv6)			
media attribute		a= line attribute = rtcp-mux		
rtcp-mux	"rtcp-mux"			
media attribute		a= line attribute = rtcp		
rtcp	"rtcp"			
port	"9"			
media description		m= line media = application		
media	"application"			
port	"9"			
proto	"udp"			
fmt	"MCPTT"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4" or "IP6" depending on IP address			
connection-address	"0.0.0.0" (IPv4) or domain name within ".invalid" DNS top level domain (IPv6)			
media description		m= line media = application		
media	"application"			
port	port number assigned by the MCPTT Server			
proto	"udp"			
fmt	"MCPTT"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4"			
connection-address	multicast IP assigned by the MCPTT Server			
media attribute		a= line attribute = key-mgmt		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
key-mgmt			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mikey	MIKEY-SAKKE I_MESSAGE as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-4.	MSCCK distribution as specified in ETSI TS 133 180 [31] (3GPP TS 33.180), Annex H		

Table 6.4.1.3.3-3: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-1)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status	not present			
announcement				
TMGI				
MBMS Service ID	"0F0F0F"	The selected value is randomly chosen - a 6 digit hexadecimal number between 000000 and FFFFFF (see TS 23.003 [X] clause 15.2. The coding of the MBMS Service ID is the responsibility of each administration		
MCC	The same value as for PLMN1	Mobile Country Code		
MNC	The same value as for PLMN1	Mobile Network Code		
QCI	"65"	Mission Critical user plane Push To Talk voice		
mbms-service-areas	"0"	The selected value is randomly chosen. The value 0 has a special meaning; it shall denote the whole PLMN as the MBMS Service Area		
GPMS	"2"	The number of the "m=application" media line in the application/sdp MIME		
version	"1"			

Table 6.4.1.3.3-4: MCPTT-Info in SIP MESSAGE (Table 6.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.1.3.3-5: SIP MESSAGE from the MCPTT Server (Step 5a, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-6			

Table 6.4.1.3.3-6: MCPTT-Info in SIP MESSAGE (Table 6.4.1.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.1.3.3-7: SIP MESSAGE from the SS-UE1 (Step 4c, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	The value received in the P-Asserted-Identity header field of the announcement message (px_MBMS_Service_ID)			
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-8			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-9			

Table 6.4.1.3.3-8: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-7)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"listening"			
session-id	not present			
general-purpose	"true"			
TMGI	same value as the announcement in the SIP MESSAGE of Table 6.4.1.3.3-3			
announcement	not present			
version	"1"			

Table 6.4.1.3.3-9: MCPTT-Info in SIP MESSAGE (Table 6.4.1.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.3.3.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.1.3.3-10: SIP MESSAGE from the SS-UE2 (Step 5c, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	The value received in the P-Asserted-Identity header field of the announcement message (px_MBMS_Service_ID)			
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-8			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-11			

Table 6.4.1.3.3-11: MCPTT-Info in SIP MESSAGE (Table 6.4.1.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.3.3.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.1.3.3-12: SIP MESSAGE from the SS-UE1 (Step 9, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		MCPTT MBMS usage info		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-13			

Table 6.4.1.3.3-13: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-12)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"listening"			
session-id	px_session_A_ID			
general-purpose	not present			
TMGI	same value as the announcement in the SIP MESSAGE			
announcement	not present			
version	"1"			

Table 6.4.1.3.3-14: SIP MESSAGE from the SS-UE2 (Step 11, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		MCPTT MBMS usage info		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-13			

Table 6.4.1.3.3-15: SIP MESSAGE from the MCPTT Server (Steps 15a and 16a, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		
Message-body		not including any MIME body part other than the ones described below		
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-16			
MIME body part		**MIKEY message**		
MIME-part-headers				
Content-Type	"application/mikey"			
MIME-part-body	As described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-5.	MIKEY message containing the MuSiK		

Table 6.4.1.3.3-16: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-15)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status	not present			
announcement	not present			
version	"1"			
anyExt				
mbms-explicitMuSiK-download				
group	px_MCPTT_Group_A_ID			

Table 6.4.1.3.3-17: SIP MESSAGE from the SS-UE1 (Step 22, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-18			

Table 6.4.1.3.3-18: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-17)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"not-listening"			
session-id	px_sesson_A_ID			
general-purpose	not present			
TMGI	same value as the announcement in the SIP MESSAGE			
announcement	not present			
version	"1"			

Table 6.4.1.3.3-19: SIP MESSAGE from the SS-UE2 (Step 24, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-18			

Table 6.4.1.3.3-20: SIP MESSAGE from the MCPTT Server (Step 27a, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-21			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-4			

Table 6.4.1.3.3-21: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-20)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status	not present			
announcement				
TMGI				
MBMS Service ID	"0F0F0F"	The selected value is randomly chosen - a 6 digit hexadecimal number between 000000 and FFFFFF (see TS 23.003 [X] clause 15.2. The coding of the MBMS Service ID is the responsibility of each administration		
MCC	The same value as for PLMN1	Mobile Country Code		
MNC	The same value as for PLMN1	Mobile Network Code		
QCI	"65"	Mission Critical user plane Push To Talk voice		
mbms-service-areas	not present			
GPMS	not present			
version	"1"			

Table 6.4.1.3.3-22: SIP MESSAGE from the MCPTT Server (Step 28a, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-21			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-6			

Table 6.4.1.3.3-23: SIP MESSAGE from the SS-UE1 (Step 29, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-24			

Table 6.4.1.3.3-24: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.1.3.3-23)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"not-listening"			
session-id	not present			
general-purpose	"false"			
TMGI	same value as the announcement in the SIP MESSAGE			
announcement	not present			
version	"1"			

Table 6.4.1.3.3-25: SIP MESSAGE from the SS-UE2 (Step 31, Table 6.4.1.3.2-1)

Derivation Path: Table 6.4.1.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.1.3.3-24			

Table 6.4.1.3.3-26: SIP INVITE from the SS-UE1 (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.4.1.3.3-26A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-26B			

Table 6.4.1.3.3-26A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions GROUP-CALL

Table 6.4.1.3.3-26B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL

Table 6.4.1.3.3-27: SIP INVITE from the MCPTT Server (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.4.1.3.3-27A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.1.3.3-27B			

Table 6.4.1.3.3-27A: SDP in SIP INVITE from the MCPTT Server (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions GROUP-CALL

Table 6.4.1.3.3-27B: MCPTT-Info in SIP INVITE from the MCPTT Server (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL

Table 6.4.1.3.3-28: SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.4.1.3.3-28A			

Table 6.4.1.3.3-28A: SDP in SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.4.1.3.3-29: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.4.1.3.3-29A			

Table 6.4.1.3.3-29A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.4.1.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.4.1.3.3-30: Floor Granted (Steps 13 and 34, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions ACK

Table 6.4.1.3.3-31: Floor Taken (Step 36, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions ACK

Table 6.4.1.3.3-32: Floor Idle (Steps 39 and 41, Table 6.4.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions ACK

6.4.2 MCPTT Server - MCPTT Client / MBMS / Multi Talker

6.4.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCPTT Client) initiates establishment of a group call with automatic
commencement mode and without implicit floor request }
    then { IUT (MCPTT Server) accepts the call from the initiator and establishes a pre-arranged
group call }
            }
```

(2)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send an MBMS bearer announcement to notify bearer
availability }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE1 (MCPTT Client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report by sending a SIP MESSAGE
message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(4)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send an MBMS bearer announcement to notify bearer
availability }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message to the SS-UE2 (MCPTT Client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening
general-purpose subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(6)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) requests to use MBMS subchannel for the conversation }
    then { IUT (MCPTT Server) sends a Map Group To Bearer message }
            }
```

(7)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening audio
and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(8)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to notify listening audio
and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode }
ensure that {
  when { IUT (MCPTT Server) requests to send floor Granted }
    then { IUT (MCPTT Server) sends a Floor Granted message with request for acknowledgement using
unicast bearer }
            }
```

(10)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { IUT (MCPTT Server) requests to set the explicit MuSiK key }
    then { IUT (MCPTT Server) sends a SIP MESSAGE message containing a MuSiK download message to the
SS-UE1 (MCPTT Client) and the SS-U2 (MCPTT Client)}
            }
```

(11)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { IUT (MCPTT Server) initiates floor messages }
    then { IUT (MCPTT Server) enforces Flor Taken }
            }
```

(12)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE2 (MCPTT client) sends Floor Request }
    then { IUT (MCPTT Server) replies with Floor Taken to accept Multi-talker Floor Control request
}
            }
```

(13)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE2 (MCPTT client) sends Floor Release }
    then { IUT (MCPTT Server) replies with Floor Release Multi Talker }
            }
```

(14)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT client) sends Floor Release }
    then { IUT (MCPTT Server) replies with Floor Idle and Unmap Group To Bearer }
            }
```

(15)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { SS-UE1 (MCPTT Client) sends a MBMS Bearer listening status report to stop notify listening
audio and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(16)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and using a MBMS subchannel for the conversation }
ensure that {
  when { SS-UE2 (MCPTT Client) sends a MBMS Bearer listening status report to stop notify listening
audio and floor control subchannel by sending a SIP MESSAGE message }
    then { IUT (MCPTT Server) responds with a SIP 200 (OK) }
            }
```

(17)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and having stopped using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT client) sends Floor Request }
    then { IUT (MCPTT Server) replies with Floor Taken, Floor Idle, Floor Granted }
            }
```

(18)

```
with { IUT (MCPTT Server) having established pre-arranged group call with automatic commencement
mode and having stopped using a MBMS subchannel for the conversation }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the Pre-arranged group call }
    then { IUT (MCPTT Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to the other participants }
            }
```

6.4.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 14.2.2.2, 14.2.3, 14.3.3.1, ETSI TS 124 380 [5] (3GPP TS 24.380) clauses 10.2.2.2, 10.2.3.2, 10.2.3.3, 10.2.3.3a, 10.2.3.4, 10.2.3.5, 10.2.3.11, 6.3.4.4.2, 6.3.4.4.6, 6.3.4.4.7a. The following represents a copy/paste extraction of the requirements relevant to the test purpose; any references within the copy/paste text should be understood within the scope of the core specification they have been copied from. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.2.2.2]

"For each MCPTT client that the participating MCPTT function is sending an MBMS bearer announcement to, the participating MCPTT function:

1) shall generate an SIP MESSAGE request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33];

2) shall set the Request-URI to the URI received in the To header field in a third-party SIP REGISTER request;

3) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media-feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

4) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcptt";

5) shall include one application/sdp MIME body conforming to 3GPP TS 24.229 [4], where the application/sdp MIME body:

a) shall include the Content-Disposition header field with the value "render";

b) should include one or more "m=audio" media lines and media line attributes as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for audio and media control. Additionally, the participating MCPTT function:

i) shall set c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6;

ii) shall set port number of the media line to 9;

ii) shall include the "a=rtcp-mux" attribute as specified in IETF RFC 5761 [39]; and

iii) shall include the "a=rtcp:9" as specified in IETF RFC 5761 [39].

c) should include one or more "m=audio" media lines and media line attributes as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for audio only. Additionally, the participating MCPTT function:

i) shall set the c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6; and

ii) shall set the port number of the media line to 9;

NOTE 1: If an MBMS subchannel for audio only is included, the "a=rtcp-mux" and "a=rtcp:" attributes are not included in the media line.

d) shall include one "m=application" media line as defined in 3GPP TS 24.380 [5] to be used as the general purpose MBMS subchannel. The media line shall include a valid multicast IP address and a valid port number. If the protection of MBMS subchannel control messages sent over the general purpose MBMS subchannel of the MBMS bearer is required, the participating MCPTT function also includes an "a=key-mgmt" media-level attribute. The participating MCPTT function:

i) shall encrypt the MSCCK to a UID associated to the targeted MCPTT ID and a time related parameter as described in 3GPP TS 33.180 [78];

ii) shall generate a MIKEY-SAKKE I_MESSAGE using the encapsulated MSCCK and MSCCK-ID as specified in 3GPP TS 33.180 [78];

iii) shall add the public service identity of the participating MCPTT function to the initiator field (IDRi) of the I_MESSAGE as described in 3GPP TS 33.180 [78];

iv) shall sign the MIKEY-SAKKE I_MESSAGE using the public service identity of the participating MCPTT function signing key provided in the keying material together with a time related parameter, and add this to the MIKEY-SAKKE payload, as described in 3GPP TS 33.180 [78]; and

v) shall include the "mikey" key management and protocol identifier and the signed MIKEY-SAKKE I_MESSAGE in the value of the a=key-mgmt" media-level attribute according to IETF RFC 4567 [47]; and

NOTE 2: The media parameters to be used by the MBMS subchannel for media is included in the Map Group To Bearer message defined in 3GPP TS 24.380 [5] and not included in this application/sdp MIME body.

e) if "m=audio" media lines to be used in an MBMS subchannel for audio only are included above, shall include one or more "m=application" media line as defined in 3GPP TS 24.380 [5] to be used as the MBMS subchannel for floor control messages. The media line:

i) shall set c-line to the unspecified address (0.0.0.0), if IPv4, or to a domain name within the ".invalid" DNS top-level domain, if IPv6; and

ii) shall set the port number of the media line to 9;

NOTE 3: The use of a separate MBMS subchannel for floor control is optional. When a separate MBMS subchannel for floor control is not used, floor control messages are sent in the MBMS subchannel for media.

6) shall include an application/vnd.3gpp.mcptt-mbms-usage-info+xml MIME body defined in clause F.2 with the <version> element set to "1" and one or more <announcement> elements associated with the pre-activated MBMS bearers. Each set of an <announcement> element:

a) shall include a TMGI value in the <TMGI> element;

NOTE 4: The same TMGI value can only appear in one <announcement> element. The TMGI value is also used to identify the <announcement> when updating or cancelling the <announcement> element.

NOTE 5: The security key active for the general purpose MBMS subchannel on which the mapping (i.e. the Map Group To Bearer message) of media or media control to this MBMS bearer was indicated, is used for MBMS subchannels on this MBMS bearer, unless a different key or an indication of not using encryption is in place.

b) shall include the QCI value in the <QCI> element;

c) if multiple carriers are supported, shall include the frequency to be used in the <frequency> element;

NOTE 6: In the current release if the <frequency> element is included, the frequency in the <frequency> element is the same as the frequency used for unicast.

d) shall include one or more MBMS service area IDs in <mbms-service-area-id> elements in the <mbms-service-areas> element;

NOTE 7: Initial mappings of groups to MBMS subchannels on an MBMS bearer for the purpose of carrying media or media control can occur only where the MBMS service area for this bearer and the MBMS service area for the bearer carrying the general purpose MBMS subchannel on which the Map Group To Bearer message is sent intersect. However, once media or media control were successfully mapped to this bearer, the reception by the MCPTT client can continue (until Unmap Group To Bearer is received or until timeout) throughout the entire MBMS service area of this bearer.

e) may include the <report-suspension> element and set it to "true" value or the "false" value;

NOTE 8: The participating function can choose to direct some clients not to send an MBMS bearer suspension report when notified by RAN, by including the <report-suspension> element set to "false". The purpose is to prevent an avalanche of identical reports sent by clients roughly at the same time, to report the suspension of the same MBMS bearer. The way the participation function determines which clients are to send or not to send the report is outside the scope of the present document.

f) if the MBMS bearer is carrying the general purpose MBMS subchannel, shall include one <GPMS>element, giving the number of the "m=application" media line in the application/sdp MIME body generated in step 5 above to be used for the general purpose MBMS subchannel; and

g) if the packet headers are compressed with ROHC specified in RFC 5795 [80] in this MBMS bearer, the <anyExt> element in the <announcement> element in the <mcptt-mbms-usage-info> element shall include the <mcptt-mbms-rohc> element defined in clause F.2.3.

7) shall include the MBMS public service identity of the participating MCPTT function in the P-Asserted-Identity header field;

8) shall include in a MIME body with Content-Type header field set to "application/vnd.3gpp.mcptt-info+xml", the <mcptt-request-uri> element set to the MCPTT ID of the user; and

9) shall send the SIP MESSAGE request towards the MCPTT client according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.2.3]

"Upon receiving a "SIP MESSAGE request for an MBMS listening status update", the participating MCPTT function shall handle the request in accordance with 3GPP TS 24.229 [4] and IETF RFC 3428 [33].

If the SIP MESSAGE request contains:

1) an application/vnd.3gpp.mcptt-mbms-usage-info+xml MIME body with an <mbms-listening-status> element; and

2) an application/vnd.3gpp.mcptt-info+xml MIME body containing an MCPTT ID in the <mcptt-request-uri> served by the participating MCPTT function;

then the participating MCPTT function:

1) shall verify that the public user identity in the P-Asserted-Identity header field is bound to the MCPTT ID in the <mcptt-request-uri> element in the application/vnd.3gpp.mcptt-info+xml MIME body, and if that is the case:

a) if the <mbms-listening-status> element is set to "listening":

i) if a <session-id> element is included, shall indicate to the media plane that the MCPTT client in the session identified by the <session-id> element is now listening to the MBMS subchannel; and

ii) if <general-purpose> element is included with the value "true", shall indicate to the media plane that the MCPTT client is now listening to the general purpose MBMS subchannel; and

b) if the <mbms-listening-status> element is set to "not-listening":

i) if a <session-id> element is included, shall indicate to the media plane that the MCPTT client in the sessions identified by the <session-id> elements is not listening to the MBMS subchannel;

ii) if <general-purpose> element is included with the value "false", shall indicate to the media plane that the MCPTT client is no longer listening to the general purpose MBMS bearer; and

iii) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

NOTE 1: If the MCPTT client reports that the MCPTT client is no longer listening to the general purpose MBMS subchannel it is implicitly understood that the MCPTT client no longer listens to any MBMS subchannel in ongoing conversations that the MCPTT client previously reported status "listening".

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 14.3.3.1]

"If one of the following conditions is fulfilled:

1) if the MCPTT client:

a) receives a Map Group To Bearer message over the general purpose MBMS channel;

b) participates in a group session identified by the Map Group To Bearer message; and

c) the status "listening" is not already reported; or

2) if the MCPTT client:

a) receives an announcement as described in clause 14.3.2;

b) enters an MBMS service area where a general purpose MBMS is available; and

c) experiences good MBMS bearer radio condition;

then the MCPTT client shall report that the MCPTT client is listening to the MBMS bearer as specified in clause 14.3.3.2.

If one of the following conditions is fulfilled:

1) if the MCPTT client:

a) receives an MBMS bearer announcement as described in the clause 14.3.2;

b) the MBMS bearer announcement contains a cancellation of an <announcement> element identified by the same TGMI value as received in a Map Group To Bearer message in an ongoing conversation; and

c) the status "not-listening" is not already reported;

2) if the MCPTT client:

a) receives an MBMS bearer announcement as described in the clause 14.3.2;

b) the MBMS bearer announcement contains a cancellation of an <announcement> element;

c) does not participate in an ongoing conversation;

d) the MCPTT client has reported the "listening" status due to the availability of the general purpose MBMS subchannel in the <announcement> element; and

e) the status "not-listening" is not already reported; or

3. if the MCPTT client:

a) suffers from bad MBMS bearer radio condition,

then the MCPTT client shall report that the MCPTT client is not listening to the MBMS subchannels as specified in clause 14.3.3.2.

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.2.2]

"Upon receiving a Floor Request message or a Floor Taken message and when the participating MCPTT function decides that an MBMS subchannel shall be used for a conversation in an ongoing group session, the participating MCPTT function needs to determine if the MBMS bearer has sufficient capacity for the new conversation. If the new MBMS bearer has sufficient capacity the participating MCPTT function:

NOTE: The participating MCPTT function can take that decision when receiving the first Floor Request from a floor participant or when receiving a Floor Taken message destined to one of the floor participants served by the participating MCPTT client.

1. shall create an instance of the 'Participating MCPTT function MBMS subchannel control' state machine;

2. shall send a Map Group To Bearer message over the general purpose MBMS subchannel. The Map Group To Bearer message:

a. shall include TMGI;

b. shall include the identifier of the media stream; and

c. shall include the MCPTT Group identifier field;

3. shall start timer T15 (Conversation);

4. shall start timer T16 (Unmap Group To Bearer);

5. shall enter the 'M: A conversation is active' state;

6. if the Floor Request was received, shall perform actions as described in subclause 6.4.2; and

7. if the Floor Taken was received, shall perform the actions described in subclause 10.2.3.3.

If the MBMS bearer does not have sufficient capacity for the new conversation the participating MCPTT function:

1. may free capacity for the new conversation by transfering an existing conversation over an MBMS bearer to unicast bearers following the procedure in subclause 10.2.3.12; or

2. may use the MBMS bearer for the signaling messages, while using unicast bearers for the media message."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.2]

"When a Floor Idle message destined to a floor participant listening to the MBMS subchannel is received, the participating MCPTT function:

1. shall for each received Floor Idle message, check if there is a stored message-sequence-number value associated with the conversation;

2. if a message-sequence-number value associated with the conversation is stored, compared the value of the stored message-sequence-number value with the value in the received Floor Idle message;

a. if the received message-sequence-number value is higher than the stored value:

i. shall replace the stored message-sequence-number value with the received message-sequence-number field;

ii. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

iii. shall send the received Floor Idle message over the MBMS subchannel; and

b. if the received message-sequence-number value is lower or the same as the stored value, shall discard the received Floor Idle message;

3. if a message-sequence-number value associated with the conversation is not stored (i.e. this is the first floor control message received in the conversation):

a. shall store the received message-sequence-number value and associate the value with the conversation;

b. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

c. shall send the received Floor Idle message over the MBMS subchannel;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.3]

"When a Floor Taken message destined to a floor participant listening to the MBMS subchannel is received, the participating MCPTT function:

1. shall for each received Floor Taken message, check if there is a stored message-sequence-number value associated with the conversation;

2. if a message-sequence-number value associated with the conversation is stored, compare the value of the stored message-sequence-number value with the value in the received Floor Taken message;

a. if the received message-sequence-number value is higher than the stored value:

i. shall replace the stored message-sequence-number value with the received message-sequence-number field;

ii. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

iii. shall send the received Floor Taken message over the MBMS subchannel; and

b. if the received message-sequence-number value is lower or the same as the stored value, shall discard the received Floor Taken message;

3. if a message-sequence-number value associated with the conversation is not stored (i.e. this is the first floor control message received in the conversation):

a. shall store the received message-sequence-number value and associate the value with the conversation;

b. shall set the acknowledgment bit to '0' as specified in subclause 8.2.2, if not already set; and

c. shall send the received Floor Taken message over the MBMS subchannel;

4. if the received Floor Taken message indicates that a Floor Ack message is expected (i.e. the acknowledgment bit is set to '1' as specified in subclause 8.2.2), shall send a Floor Ack message towards the controlling MCPTT function The Floor Ack message:

a. shall include the Message Type field set to '2' (Floor Taken); and

b. shall include the Source field set to '1' (participating MCPTT function is the source);

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.3a]

"When a Floor Release Multi Talker message destined to a floor participant listening to the MBMS subchannel is received, the participating MCPTT function:

1. shall for each received Floor Release Multi Talker message, check if there is a stored message-sequence-number value associated with the conversation;

2. if a message-sequence-number value associated with the conversation is stored, compare the value of the stored message-sequence-number value with the value in the received Floor Release Multi Talker message;

a. if the received message-sequence-number value is higher than the stored value:

i. shall replace the stored message-sequence-number value with the received message-sequence-number field; and

ii. shall send the received Floor Release Multi Talker message over the MBMS subchannel; and

b. if the received message-sequence-number value is lower or the same as the stored value, shall discard the received Floor Release Multi Talker message;

3. if a message-sequence-number value associated with the conversation is not stored (i.e. this is the first floor control message received in the conversation):

a. shall store the received message-sequence-number value and associate the value with the conversation; and

b. shall send the received Floor Release Multi Talker message over the MBMS subchannel;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.4]

"When a floor control message other than the Floor Idle and Floor Taken message is received from a floor participant or received from the floor control server, the participating MCPTT function:

1. shall forward the floor control message as specified in subclause 6.4;

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.5]

"When receiving a RTP media packet destined to one of the MCPTT client listening to the MBMS subchannel, the participating MCPTT function:

NOTE: An RTP media packet not destined to an MCPTT client listening to the MBMS subchannel is forwarded to the MCPTT client over the unicast bearer.

1. *shall check if the media packet is already sent over the MBMS subchannel or not;*
2. *if the RTP media packet is already sent over the MBMS subchannel, shall discard the RTP media packet;*
3. *if the RTP media packet is not already sent over the MBMS sub channel, shall instruct the media distribution function to send the RTP media packet over the MBMS subchannel;*

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 10.2.3.11]

"If the control and signalling plane indicates that the group call session is released, the participating MCPTT function:

1. *shall send the Unmap Group To Bearer message over the MBMS subchannel. The Unmap Group To Bearer message:*
   a. *shall include the MCPTT Group ID field;*
2. *shall stop timer T15 (Conversation), timer T16 (Map Group To Bearer) and timer T17 (Unmap Group To Bearer), if running; and*
3. *shall release the instance of the 'Participating MCPTT function MBMS subchannel management' state machine used for the conversation."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. *shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:*
   a. *shall include the value of timer T2 (Stop talking)in the Duration field;*
   b. *shall include the granted priority in the Floor priority field;*

   …

3. *shall send Floor Taken message to all other floor participants. The Floor Taken message:*

   …

   b. *if multi-talker is supported and the floor is currently granted to multiple participants:*
      i. *shall include the Floor Indicator field with the I-bit set to '1' (Multi-talker);*
      ii. *shall include the list of granted users in the multi-talker group in List of Granted Users field, including a new granted talker;*
      iii. *shall include the list of SSRCs of granted floor participants;*
      iv) *may include the list of functional aliases of the granted floor participants in the List of Functional Aliases field; and*
      v. *shall include the List of Locations of granted floor participants;"*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.6]

"Upon receiving a Floor Release message the floor control arbitration logic in the floor control server:

1. *shall request the media distributor in the MCPTT server to stop forwarding RTP media packets received from the participant that sent the Floor Release message;*

…

5. *if multi-talker floor control is supported, then:*

   a. *shall send a Floor Release Multi Talker message. The Floor Release Multi Talker message:*

      i. *shall include SSRC field set to the SSRC of the floor participant that has released the floor; and*

      ii. *shall include the User ID set to the MCPTT ID of the floor participant releasing the floor;*

   b. *shall remove the participant that sent the Floor Release message from the list of currently granted talkers; and*

   c. *if the list of currently granted talkers is empty, shall enter the 'G: Floor Idle' state as specified in the subclause 6.3.4.3.2; or if the list of currently granted talkers is not empty shall stay in state the 'G: Floor Taken'."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.7a]

"On receipt of a floor request message and if the group is configured as multi-talker group the floor control arbitration logic in the floor control server:

1. *shall select one of the following options:*

   a. *if the maximum number of simultaneous talkers applicable for multi-talker control is reached and if the floor request message has effective priority indicating pre-emptive priority, determine from all participants having permission to send media, the one with the lowest priority and revoke the floor from the participant with the lowest priority; or*

   b. *if the maximum number of simultaneous talkers applicable for multi-talker control is not reached, allow media from both the current speaker(s) and from the participant now requesting floor;*

…

3. *if allow media from both the current speaker(s) and from the participant now requesting floor is selected:*

   a. *shall send a Floor Granted message to the requesting floor participant. The Floor Granted message:*

      i. *shall include the value of the T2 (Stop talking) timer in the Duration field;*

      ii. *shall include the granted priority in the Floor priority field;*

      iii. *if a Track Info field associated with the floor control server state transition diagram for 'multe-talker floor control operation' is stored, shall include the stored Track Info field;*

      iv. *shall include the Floor Indicator field with the I-bit set to '1' (Multi-talker); and*

      v. *shall include the SSRC of granted floor participant;*

   b. *shall add the MCPTT ID of the user to which the floor is granted to the list of currently granted talkers;*

   c. *shall send a Floor Taken message to any non-controlling MCPTT functions involved and to floor participants controlled by the controlling MCPTT function that will listen to the RTP media from the multi-talker MCPTT client according to local policy. The Floor Taken message:*

      i. *shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the associated functional alias in the Functional Alias field, if privacy is not requested;*

ii. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1;

iii. shall include the Floor Indicator field with the I-bit set to '1' (Multi-talker);

iv. shall include the list of granted users in the multi-talker group in List of Granted Users field;

v. shall include the list of SSRCs of granted floor participants; and

vi) may include the list of functional aliases of the granted floor participants in the List of Functional Aliases field.

d. shall start the T1 (End of RTP media) timer for the particpant to which the floor is granted;

e. shall start timer T20 (Floor Granted) for the particpant to which the floor is granted, if the floor request was queued and initialise the counter C20 (Floor Granted) to 1;

f. shall stay in the state to 'G: Floor Taken' state."

6.4.2.3 Test description

6.4.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client)
- For the underlying "transport bearer" over which the SS-UE1 (MCPTT client) and the MCPTT Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [23] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCPTT operation in the MCPTT configuration document).

IUT:

- IUT (MCPTT Server)
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCPTT Server) is the acting Participating Server.

Preamble:

- The IUT (MCPTT Server) is connected to PLMN1
- The IUT (MCPTT Server) is connected to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) performs the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) are affiliated to the Prearranged group.
- The IUT (MCPTT Server) is the controlling server of the Prearranged group.
- A pre-activated MBMS bearer exists.

6.4.2.3.2 Test procedure sequence

Table 6.4.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) requests the establishment of a group call to SS-UE2 (MCPTT client) with automatic commencement mode and without implicit floor request.	-	-	-	-
2	Check: Does the MCPTT server correctly perform procedure 'MCPTT E2E session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.3.1.3-1 to establish a pre-arranged group call with automatic commencement mode and without implicit floor control according to option a of note in Annex A, Table A.3.1.3-1?	-	-	1	P
3	Make the MCPTT Server activate an MBMS bearer and send an MBMS bearer announcement to notify bearer availability (Note)	-	-	-	-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 4a-4d and steps 5a-5d might be performed in parallel	-	-	-	-
4a	Check: Does the MCPTT Server send an MBMS bearer announcement to notify MBMS bearer availability to SS-UE1?	<--	SIP MESSAGE	2	P
4b	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) message to MCPTT Server	-->	SIP 200 (OK)	-	-
4c	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to general-purpose subchannel	-->	SIP MESSAGE	-	-
4d	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	3	P
5a	Check: Does the MCPTT Server send an MBMS bearer announcement to notify MBMS bearer availability to SS-UE2?	<--	SIP MESSAGE	4	P
5b	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message to MCPTT Server	-->	SIP 200 (OK)	-	-
5c	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to general-purpose subchannel.	-->	SIP MESSAGE	-	-
5d	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	5	P
6	The SS-UE1 (MCPTT client) sends a Floor Request message to MCPTT Server	-->	Floor Request	-	-
7	Make the MCPTT Server use MBMS subchannel for the conversation (see note)	-	-	-	-
8	Check: Does the MCPTT Server send a Map Group To Bearer message over general purpose MBMS subchannel?	<--	Map Group To Bearer	6	P
9	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
10	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	7	P
11	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
12	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	8	P
13	Check: Does the MCPTT Server send a Floor Granted message with request for acknowledgement using unicast bearer to SS-UE1?	<--	Floor Granted	9	P
14	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 15a-15b and steps 16a-16b might be performed in parallel	-	-	-	-
15a	Check: Does the MCPTT Server send a message containing a MuSiK download message to set the explicit MuSiK key to be used for the floor control encryption over the MBMS subchannel to SS-UE1?	<--	SIP MESSAGE	10	P
15b	The SS-UE1 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-
16a	Check: Does the MCPTT Server send a message containing a MuSiK download message to set the explicit MuSiK key to be used for the floor control encryption over the MBMS subchannel to SS-UE2?	<--	SIP MESSAGE	10	P
16b	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) to MCPTT Server	-->	SIP 200 (OK)	-	-
17	Check: Does the MCPTT Server send a Floor Taken message using multicast bearer?	<--	Floor Taken	11	P
18	The SS-UE2 (MCPTT client) sends a Floor Request message to MCPTT Server	-->	Floor Request	-	-
19	Make the MCPTT Server accept Multi-talker Floor Control request (Note)	-	-	-	-
20	Check: Does the MCPTT Server send a Floor Taken message with l-bit in the Floor Indicator set to '1' (Multi-talker) using MBMS subchannel?	<--	Floor Taken	12	P
21	The SS-UE2 (MCPTT client) sends a Floor Release message to MCPTT Server	-->	Floor Release	-	-
22	Make the MCPTT Server use accept Multi-talker Floor Control release (Note)	-	-	-	-
23	Check: Does the MCPTT Server send a Floor Release Multi Talker using MBMS subchannel?	<--	Floor Release Multi Talker	13	P
24	The SS-UE1 (MCPTT client) sends a Floor Release message to MCPTT Server	-->	Floor Release	-	-
25	Check: Does the MCPTT Server send a Floor Idle message using multicast bearer?	<--	Floor Idle	14	P
26	Make the MCPTT Server stop using MBMS subchannel for the conversation (see note)	-	-	-	-
27	Check: Does the MCPTT Server send an Unmap Group To Bearer message over general purpose MBMS subchannel?	<--	Unmap Group To Bearer	14	P
28	The SS-UE1 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
29	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	15	P
30	The SS-UE2 (MCPTT client) sends MBMS Bearer listening status report message to MCPTT Server, to notify not being listening to audio and floor control over MBMS subchannel	-->	SIP MESSAGE	-	-
31	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE2?	<--	SIP 200 (OK)	16	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
32	The SS-UE1 (MCPTT client) sends a Floor Request message to MCPTT Server	-->	Floor Request	-	-
33	Check: Does the MCPTT Server send a Floor Granted message with request for acknowledgement using unicast to SS-UE1?	<--	Floor Granted	17	P
34	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
35	Check: Does the MCPTT Server send a Floor Taken message with request for acknowledgement using unicast bearer to SS-UE2?	<--	Floor Taken	17	P
36	The SS-UE2 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
37	The SS-UE1 (MCPTT client) sends a Floor Release message to MCPTT Server	-->	Floor Release	-	-
38	Check: Does the MCPTT Server send a Floor Idle message with request for acknowledgement using unicast bearer to SS-UE1?	<--	Floor Idle	17	P
39	The SS-UE1 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
40	Check: Does the MCPTT Server send a Floor Idle message with request for acknowledgement using unicast bearer to SS-UE2?	<--	Floor Idle	17	P
41	The SS-UE2 (MCPTT client) responds with a Floor Ack message to MCPTT Server	-->	Floor Ack	-	-
42	The SS-UE1 (MCPTT client) releases the call.	-	-	-	-
43	Check: Does the MCPTT server correctly perform procedure 'MCX E2E call release' as described in Annex A, Table A.2.19.3-1?	-	-	18	P
NOTE: This is expected to be done via a suitable implementation dependent MMI.					

6.4.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 6.4.2.3.3-1: SIP MESSAGE from the MCPTT Server (Step 4a, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		
Message-body				
MIME body part		**SDP message**		
MIME-part-headers				
Content-Type	"application/sdp"			
Content-Disposition	"render"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
MIME-part-body	SDP message as described in Table 6.4.2.3.3-2			
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-3			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-4			

Table 6.4.2.3.3-2: SDP in SIP MESSAGE (Table 6.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media descriptions				
media description		m= line media = audio		
media	"audio"			
port	"9"			
proto	"RTP/AVP"			
fmt	"99"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4" or "IP6" depending on IP address			
connection-address	"0.0.0.0" (IPv4) or domain name within ".invalid" DNS top level domain (IPv6)			
media attribute		a= line attribute = rtcp-mux		
rtcp-mux	"rtcp-mux"			
media attribute		a= line attribute = rtcp		
rtcp	"rtcp"			
port	"9"			
media description		m= line media = application		
media	"application"			
port	"9"			
proto	"udp"			
fmt	"MCPTT"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4" or "IP6" depending on IP address			
connection-address	"0.0.0.0" (IPv4) or domain name within ".invalid" DNS top level domain (IPv6)			
media description		m= line media = application		
media	"application"			
port	port number assigned by the MCPTT Server			
proto	"udp"			
fmt	"MCPTT"			
Connection Data		c= line		
nettype	"IN"			
addrtype	"IP4"			
connection-address	multicast IP assigned by the MCPTT Server			
media attribute		a= line attribute = key-mgmt		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
key-mgmt			ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mikey	MIKEY-SAKKE I_MESSAGE as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-4.	MSCCK distribution as specified in ETSI TS 133 180 [31] (3GPP TS 33.180), Annex H		

Table 6.4.2.3.3-3: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.2.3.3-1)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status	not present			
announcement				
TMGI				
MBMS Service ID	"0F0F0F"	The selected value is randomly chosen - a 6 digit hexadecimal number between 000000 and FFFFFF (see TS 23.003 [X] clause 15.2. The coding of the MBMS Service ID is the responsibility of each administration		
MCC	The same value as for PLMN1	Mobile Country Code		
MNC	The same value as for PLMN1	Mobile Network Code		
QCI	"65"	Mission Critical user plane Push To Talk voice		
mbms-service-areas	"0"	The selected value is randomly chosen. The value 0 has a special meaning; it shall denote the whole PLMN as the MBMS Service Area		
GPMS	"2"	The number of the "m=application" media line in the application/sdp MIME		
version	"1"			

Table 6.4.2.3.3-4: MCPTT-Info in SIP MESSAGE (Table 6.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.2.3.3-5: SIP MESSAGE from the MCPTT Server (Step 5a, Table 6.4.2.3.2-1)

Derivation Path: Table 6.4.2.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-6			

Table 6.4.2.3.3-6: MCPTT-Info in SIP MESSAGE (Table 6.4.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.2.2.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.2.3.3-7: SIP MESSAGE from the SS-UE1 (Step 4c, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	The value received in the P-Asserted-Identity header field of the announcement message (px_MBMS_Service_ID)			
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-8			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-9			

Table 6.4.2.3.3-8: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.2.3.3-7)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"listening"			
session-id	not present			
general-purpose	"true"			
TMGI	same value as the announcement in the SIP MESSAGE of Table 6.4.2.3.3-3			
announcement	not present			
version	"1"			

Table 6.4.2.3.3-9: MCPTT-Info in SIP MESSAGE (Table 6.4.2.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.3.3.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.2.3.3-10: SIP MESSAGE from the SS-UE2 (Step 5c, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	The value received in the P-Asserted-Identity header field of the announcement message (px_MBMS_Service_ID)			
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-8			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-11			

Table 6.4.2.3.3-11: MCPTT-Info in SIP MESSAGE (Table 6.4.2.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 14.3.3.2	
mcptt-calling-user-id	not present			
NOTE: Encrypted element as described in Table 5.5.3.2.1-1A.				

Table 6.4.2.3.3-12: SIP MESSAGE from the SS-UE1 (Step 9, Table 6.4.2.3.2-1)

Derivation Path: Table 6.4.2.3.3-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		MCPTT MBMS usage info		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-13			

Table 6.4.2.3.3-13: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.2.3.3-12)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"listening"			
session-id	px_session_A_ID			
general-purpose	not present			
TMGI	same value as the announcement in the SIP MESSAGE			
announcement	not present			
version	"1"			

Table 6.4.2.3.3-14: SIP MESSAGE from the SS-UE2 (Step 11, Table 6.4.2.3.2-1)

Derivation Path: Table 6.4.2.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		MCPTT MBMS usage info		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-13			

Table 6.4.2.3.3-15: SIP MESSAGE from the MCPTT Server (Steps 15a and 16a, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity				
addr-spec	px_MBMS_Service_ID	URI of the participating MCPTT function		
Message-body		not including any MIME body part other than the ones described below		
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-16			
MIME body part		**MIKEY message**		
MIME-part-headers				
Content-Type	"application/mikey"			
MIME-part-body	As described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-5.	MIKEY message containing the MuSiK		

Table 6.4.2.3.3-16: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.2.3.3-15)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status	not present			
announcement	not present			
version	"1"			
anyExt				
mbms-explicitMuSiK-download				
group	px_MCPTT_Group_A_ID			

Table 6.4.2.3.3-17: SIP MESSAGE from the SS-UE1 (Step 28, Table 6.4.2.3.2-1)

Derivation Path: Table 6.4.2.3.3-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-18			

Table 6.4.2.3.3-18: MCPTT-MBMS-Usage-Info in SIP MESSAGE (Table 6.4.2.3.3-17)

Derivation Path: ETSI ETSI TS 124 379 [4] (3GPP TS 24.379), Clause F.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcptt-mbms-usage-info				
mbms-listening-status				
mbms-listening-status	"not-listening"			
session-id	px_sesson_A_ID			
general-purpose	not present			
TMGI	same value as the announcement in the SIP MESSAGE			
announcement	not present			
version	"1"			

Table 6.4.2.3.3-19: SIP MESSAGE from the SS-UE2 (Step 30, Table 6.4.2.3.2-1)

Derivation Path: Table 6.4.2.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT MBMS usage info**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-mbms-usage-info+xml"			
MIME-part-body	MCPTT-MBMS-Usage-Info as described in Table 6.4.2.3.3-18			

Table 6.4.2.3.3-20: SIP INVITE from the SS-UE1 (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-1, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.4.2.3.3-20A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-20B			

Table 6.4.2.3.3-20A: SDP in SIP INVITE from the SS-UE1 (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-2, with conditions GROUP-CALL

Table 6.4.2.3.3-20B: MCPTT-Info in SIP INVITE from the SS-UE1 (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 1, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-3, with conditions GROUP-CALL

Table 6.4.2.3.3-21: SIP INVITE from the MCPTT Server (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1; step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-4, condition GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.4.2.3.3-21A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.4.2.3.3-21B			

Table 6.4.2.3.3-21A: SDP in SIP INVITE from the MCPTT Server (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-5, with conditions GROUP-CALL

Table 6.4.2.3.3-21B: MCPTT-Info in SIP INVITE from the MCPTT Server (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 2, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-6, with conditions GROUP-CALL

Table 6.4.2.3.3-22: SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-7, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.4.2.3.3-22A			

Table 6.4.2.3.3-22A: SDP in SIP 200 (OK) from the SS-UE2 (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 5, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-8, with conditions GROUP-CALL

Table 6.4.2.3.3-23: SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-9, with conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP Message	As described in Table 6.4.2.3.3-23A			

Table 6.4.2.3.3-23A: SDP in SIP 200 (OK) from the MCPTT Server (Step 2, Table 6.4.2.3.2-1; Step 1, Annex A, Table A.3.1.3-1, step 6, Annex A, Table A.2.18.3-1)

Derivation Path: Annex A, Table A.3.1.4-10, with conditions GROUP-CALL

Table 6.4.2.3.3-24: Floor Granted (Steps 13 and 33, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1, with conditions ACK

Table 6.4.2.3.3-25: Floor Taken (Step 20, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions Multi-Talker

Table 6.4.2.3.3-26: Floor Taken (Step 35, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, with conditions ACK

Table 6.4.2.3.3-27: Floor Idle (Steps 38 and 40, Table 6.4.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, with conditions ACK

Annex A (normative): Common Test Environment

A.1 Introduction

This annex specifies additional generic test procedures to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), to be used by the test cases wherever applicable.

A.2 Generic test procedures for Server-Client MCS operation

A.2.1 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing procedure sequences which can be referred from the relevant server-to-client TCs specified in the present document.

The procedures specified are required to ensure that any MC service can take place between a MC client and a MC server or specific MC relevant pre-conditions are met before a test case can be executed.

A.2.2 Void

A.2.3 MCX Server Originated notifications regarding temporary group creation or tear down

A.2.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.3.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.3.3 Procedure

Table A.2.3.3-1: MCX Server originated notifications regarding temporary group creation or tear down

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP NOTIFY to the SS-UE informing about change of group A's configuration document?	<--	SIP NOTIFY	-	P
2	The SS-UE (MCX Client) sends a SIP 200 (OK) message to the MCX Server.	-->	SIP 200 (OK)	-	-
3	The SS-UE (MCX Client) sends an HTTP GET Request message to the MCX Server that contains the access token and the XCAP-URI of the Group Configuration document.	-->	HTTP GET	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the MCX Server send a HTTP 200 (OK) message including the updated Group Document?	<--	HTTP 200 (OK)	-	P
5	Check: Does the MCX Server send a SIP NOTIFY message to the SS-UE (MCX Client) containing the group key transport payloads (GKTP) document including the group keys?	<-	SIP NOTIFY	-	P
6	The SS-UE (MCX Client) sends a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.3.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).
- Clarifications specified in clause 5.3.22.4 shall also be followed.

Table A.2.3.4-1: SIP NOTIFY (Step 1, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-1

Table A.2.3.4-2: SIP 200 (OK) (Steps 2, 6, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-2

Table A.2.3.4-3: HTTP GET (Step 3, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-3

Table A.2.3.4-4: HTTP 200 (OK) (Step 4, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-4

Table A.2.3.4-5: SIP NOTIFY (Step 5, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-6

A.2.4 MCX Server Terminated Group Creation

A.2.4.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.4.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.4.3 Procedure

Table A.2.4.3-1: MCX Server Terminated Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP PUT to the Group Management Server for requesting the creation of the new group. (see note)	-->	HTTP PUT	-	-
2	Check: Does the MCX Server respond with a HTTP 201 (Created) response message?	<--	HTTP 201 (Created)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.4.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.4.4-1: HTTP PUT (Step 1, Table A.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.4-1, condition GROUPCREATE

A.2.5 MCX Server Terminated Temporary Group Creation

A.2.5.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.5.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.5.3 Procedure

Table A.2.5.3-1: MCX Participating Terminated Temporary Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP POST to the MCX Server to request for creation of a temporary group. (see note)	-->	HTTP POST	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) containing the GMOP group-regroup-creation-response?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.5.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.5.4-1: HTTP POST (Step 1, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TEMPGROUP

Table A.2.5.4-2: HTTP 200 (OK) (Step 2, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TEMPGROUP

A.2.6 MCX Server Terminated Temporary Group Tear Down

A.2.6.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.6.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.6.3 Procedure

Table A.2.6.3-1: MCX Server Terminated Temporary Group Tear Down procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP DELETE to the MCX Server to request for tear down of a temporary group (see note).	-->	HTTP DELETE	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) response message?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.6.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.6.4-1: HTTP DELETE (Step 1, Table A.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, condition TEMPGROUP

A.2.7 MCX Subscription and Notification

A.2.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.7.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.7.3 Procedure

Table A.2.7.3-1: MCX Subscription and Notification

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP SUBSCRIBE message request.	-->	SIP SUBSCRIBE	-	-
2	Check: Does the MCX Server respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX Server send a SIP NOTIFY message?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)		
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.7.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.7.4-1: SIP SUBSCRIBE (Step 1, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"4294967295"	to receive the current status and later notification	ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3; ETSI TS 124 281 [10] (3GPP TS 24.281) clause 8.2.1.3; ETSI TS 124 282 [9] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.1.3 and 9A.2.1.3	MCPTT
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCVIDEO Info**	ETSI TS 124 281 [10] (3GPP TS 24.281) clause 8.2.1.3	MCVIDEO
MIME-part-body	MCVIDEO-Info as described in Table A.2.7.4-3			
MIME body part		**MCDATA Info**	ETSI TS 124 282 [9] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	MCDATA
MIME-part-body	MCDATA-Info as described in Table A.2.7.4-4			

Table A.2.7.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.7.4-3: MCVideo-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (see note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [10] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.7.4-4: MCData-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (see note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [9] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.7.4-5: SIP 200 (OK) (Step 2, Table A.2A.7.3-1)

Derivation Path: Table ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP

A.2.8 MCX Participating Terminated SIP MESSAGE

A.2.8.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.8.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.8.3 Procedure

Table A.2.8.3-1: MCX Participating Terminated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) send a SIP MESSAGE message (see note)	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server responds with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.8.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.9 MCX Participating Originated SIP MESSAGE

A.2.9.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.9.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.9.3 Procedure

Table A.2.9.3-1: MCX Participating Originated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP MESSAGE message?	<--	SIP MESSAGE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (see note)	→	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.9.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.10 MCX E2E SIP MESSAGE

A.2.10.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.10.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.10.3 Procedure

Table A.2.10.3-1: MCX E2E SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server send a SIP MESSAGE to SS-UE2?	<--	SIP MESSAGE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.10.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.11 MCX E2E SIP MESSAGE Request - Accept

A.2.11.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.11.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.11.3 Procedure

Table A.2.11.3-1: MCX E2E SIP MESSAGE Request - Accept Participating Originated

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message.	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX server send a SIP MESSAGE accepting the request to SS-UE1?	<--	SIP MESSAGE	-	P
4	The SS-UE1 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
-	EXCEPTION: Depending on the MCX Server implementation, the following steps may be performed in parallel to steps 3-4.	-	-	-	-
5	Check: Does the MCX Server send a SIP MESSAGE message to SS-UE2?	<--	SIP MESSAGE	-	P
6	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-

A.2.11.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.12 MCX Presence Event Status Change

A.2.12.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.12.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.12.3 Procedure

Table A.2.12.3-1: MCX Presence Event Status Change

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP PUBLISH message.	-->	SIP PUBLISH	-	P
2	Check: Does the MCX Server respond to the SIP PUBLISH message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	-
3	Check: Does the MCX Server send a SIP NOTIFY message informing about the status change progress?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	Check: Does the MCX Server send a SIP NOTIFY informing about the final status?	<--	SIP NOTIFY	-	P
6	The SS-UE responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.12.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.12.4-1: SIP PUBLISH (Step 1, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT

Table A.2.12.4-2: MCPTT-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.12.4-3: MCVideo-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (see note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [10] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.12.4-4: MCData-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (see note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [9] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.12.4-5: SIP 200 (OK) (Step 2, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, condition PUBLISH-RSP

Table A.2.12.4-3: SIP NOTIFY (Steps 3 and 5, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition PRESENCE-EVENT

Table A.2.12.4-4: SIP 200 (OK) (Steps 4 and 6, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1

A.2.13 MCX Participating Terminated pre-established session establishment

A.2.13.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.13.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.13.3 Procedure

Table A.2.13.3-1: MCX Participating Terminated pre-established session establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE message in order to create a pre-established session	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 (Trying) message?	<--	SIP 100 Trying	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) responds with a SIP ACK message.	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.13.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.13.4-1: SIP INVITE (Step 1, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-1

Table A.2.13.4-2: SIP 200 (OK) (Step 3, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-2

A.2.14 MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

A.2.14.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.14.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.14.3 Procedure

Table A.2.14.3-1: MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call (see note)	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 Trying?	<--	SIP 100 (Trying)	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK)?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification (Note)	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.14.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.14.4-1: SIP INVITE (Step 1, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.14.4-1: SIP 200 (OK) (Step 3, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.15 MCX Participating Terminated call release

A.2.15.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.15.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.15.3 Procedure

Table A.2.15.3-1: MCX Participating Terminated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP BYE request to terminate the MCX session (see note)	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.15.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.16 MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

A.2.16.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.16.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.16.3 Procedure

Table A.2.16.3-1: MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP INVITE requesting the establishment/modification of a call?	<--	SIP INVITE	-	P
2	The SS-UE (MCX Client) sends a SIP 100 Trying (see note)	-->	SIP 100 (Trying)	-	-
3	The SS-UE (MCX Client) responds with a SIP 200 (OK) (see note)	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification?	<--	SIP ACK	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.16.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.16.4-1: SIP INVITE (Step 1, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note).
NOTE: With condition re_INVITE for session modifications**.**

Table A.2.16.4-2: SIP 200 (OK) (Step 3, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

A.2.17 MCX Participating Originated call release

A.2.17.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.17.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.17.3 Procedure

Table A.2.17.3-1: MCX Participating Originated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP BYE request to terminate the MCX session?	<--	SIP BYE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (see note).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.17.4 Specific message contents

All message contents are as specified in clause 5.5. and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.18 MCX E2E session establishment/modification without provisional responses other than 100 Trying

A.2.18.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.18.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.18.3 Procedure

Table A.2.18.3-1: MCX E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
7	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.18.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.18.4-1: SIP INVITE (Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.18.4-2: SIP INVITE (Step 2, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.18.4-1: SIP 200 (OK) (Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [**1**] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.18.4-1: SIP 200 (OK) (Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.19 MCX E2E call release

A.2.19.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.19.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.19.3 Procedure

Table A.2.19.3-1: MCX E2E call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP BYE request to terminate the MCX session	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond a SIP BYE request to terminate the MCX session to SS-UE2 (MCX Client)?	<--	SIP BYE	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.19.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.19.4-1: SIP BYE from the SS-UE1 (Step 1, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

Table A.2.19.4-2: SIP BYE from the MCPTT Server (Step 2, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1

A.2.20 MCX E2E group call establishment, manual commencement

A.2.20.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.20.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.20.3 Procedure

Table A.2.20.3-1: MCX E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 183 (Session Progress) reliably.	-->	SIP 183 (Session Progress)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6	Check: Does the MCX Server acknowledge the receipt of SIP 183 (Session Progress) from SS-UE2?	<--	PRACK	-	P
7	The SS-UE2 (MCX Client) responds a SIP 200 (OK) to PRACK message	-->	SIP 200 (OK)	-	-
8	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
9	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
10	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
11	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.20.4 Specific message contents

All message contents are as specified in clause 5.5 with conditions GROUP-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

For chat group calls, CHAT-GROUP-CALL condition applies instead of GROUP-CALL.

Table A.2.20.4-1: SIP 200 (OK) (Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.20.4-2: SIP 200 (OK) (Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.21 MCX E2E private call establishment, manual commencement

A.2.21.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.21.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.21.3 Procedure

Table A.2.21.3-1: MCX E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 180 (Ringing) unreliably.	-->	SIP 180 (Ringing)	-	-
6	Check: Does the MCX Server respond with a unreliable SIP 180 (Ringing) to SS-UE1 (MCX Client)?	<--	SIP 180 (Ringing)	-	P
7	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
8	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
9	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
10	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.21.4 Specific message contents

All message contents are as specified in clause 5.5 with conditions PRIVATE-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.21.4-1: SIP 200 (OK) (Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.21.4-1: SIP 200 (OK) (Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.3 Generic test procedures for MCPTT Server-Client operation

A.3.0 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing MCPTT procedure sequences which can be referred from the relevant TCs specified in the present document.

The procedures specified are required to ensure that MCPTT service can take place between a MCPTT Client and a MCPTT server or specific MCPTT relevant pre-conditions are met before a test case can be executed.

A.3.1 MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

A.3.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.1.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.1.3 Procedure

Table A.3.1.3-1: MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E session establishment/modification without provisional responses other than 100 Trying' for as described in Table A.2.18.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (see note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly (see note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1 (see note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).
c. SS-UE1 sends SDP offer without media description for floor control.

A.3.1.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.1.4-1: SIP INVITE (Step 1, Table A.3.1.3-1; Step 1, Table A.2.18.3-1)

Derivation Path: **ETSI TS 136 579-1 [1]** (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note), with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [22]	
generic-param	"1800"	The recommended initial value is 1 800 in IETF RFC 4028 [22].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-2			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-3			
NOTE: With condition re_INVITE for session modifications.				

Table A.3.1.4-2: SDP in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 3	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.1.3-1), apply condition IMPLICIT_GRANT_REQUESTED.				
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL.				
NOTE 3: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.1.4-3: MCPTT-Info in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE_REFER

Table A.3.1.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.1.3-1; Step 2 Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note), with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [20] IETF RFC 7134 [16] IETF RFC 8101 [18]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note), with condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [22]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-6			
NOTE: With condition re_INVITE for session modifications.				

Table A.3.1.4-5: SDP in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE1 requests a call without media floor control (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL.				
NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.1.4-6: MCPTT-Info in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A				

Table A.3.1.4-7: SIP 200 (OK)
(Step 1, Table A.3.1.3-1; Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [14]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-8			

Table A.3.1.4-8: SDP in SIP 200 (OK) (Table A.3.1.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.1.4-9: SIP 200 (OK)
(Step 1, Table A.3.1.3-1; Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-10			

Table A.3.1.4-10: SDP in SIP 200 (OK) (Table A.3.1.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_ FLOORCO NTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
NOTE 1: If the SS-UE requests an implicit floor (Option b of note on Table A.3.1.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.1.3-1) apply condition WITHOUT_FLOORCONTROL NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

A.3.2 MCPTT E2E group call establishment, manual commencement

A.3.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.2.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.2.3 Procedure

Table A.3.2.3-1: MCPTT E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E group call establishment, manual commencement' as described in Table A.2.20.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (see note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly (see note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1. (see note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).

A.3.2.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.2.4-1: SIP INVITE from the SS-UE1
(Step 1, Table A.3.2.3-1;Step 1, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [22]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [22].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-3			

Table A.3.2.4-2: SDP in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.2.4-3: MCPTT-Info in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition GROUP-CALL, INVITE_REFER

Table A.3.2.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.2.3-1; Step 2, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [20] IETF RFC 7134 [16] IETF RFC 8101 [18]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [22]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-6			

Table A.3.2.4-5: SDP in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.2.4-6: MCPTT-Info in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition GROUP-CALL (note 1)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note 2) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (ote 2) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE 1: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL. NOTE 2: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.2.4-7: SIP 200 (OK) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT, INVITE-RSP and GROUP-CALL (note)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [14]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-8			
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.				

Table A.3.2.4-8: SDP in SIP 200 (OK) (Table A.3.2.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.2.4-9: SIP 200 (OK) from the MCPTT Server (Step 1, Table A.3.2.3-1; Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-10			

Table A.3.2.4-10: SDP in SIP 200 (OK) (Table A.3.2.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description [2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_ssrc	any value		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
NOTE 1: If the SS-UE requests an implicit floor (Option b of note on Table A.3.2.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

Table A.3.2.4-11: SIP 183 (Session Progress) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 5, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with condition MCPTT and 100rel

A.3.3 MCPTT E2E private call establishment, manual commencement

A.3.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.3.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.3.3 Procedure

Table A.3.3.3-1: MCPTT E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E private call establishment, manual commencement' as described in Table A.2.21.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (Note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly. (Note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1 (Note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P
NOTE: Possibilities in SDP-offer/answer depend on the test case requirements: a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request. b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request: i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted). ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1). iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor). c. SS-UE1 sends SDP offer without media description for floor control.					

A.3.3.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.3.4-1: SIP INVITE from the SS-UE1 (Step 1, Table A.3.3.3-1; Step 1, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT, PRIVATE-CALL and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [22]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [22].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-3			

Table A.3.3.4-2: SDP in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.3.3-1), apply condition IMPLICIT_GRANT_REQUESTED. NOTE 2: If the SS-UE requests a call without media floor (Option c of NOTE1 on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.				

Table A.3.3.4-3: MCPTT-Info in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition PRIVATE-CALL, INVITE_REFER

Table A.3.3.4-4: SIP INVITE from the MCPTT-Server to SS-UE2 (Step 1, Table A.3.3.3-1; Step 2, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [20] IETF RFC 7134 [16] IETF RFC 8101 [18]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [22]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-6			

Table A.3.3.4-5: SDP in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE: If the SS-UE1 requests a call without media floor control (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.				

Table A.3.3.4-6: MCPTT-Info in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.3.4-7: SIP 200 (OK) from the SS-UE2 to MCPTT Server (Step 1, Table A.3.3.3-1; Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [14]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-8			

Table A.3.3.4-8: SDP in SIP 200 (OK) (Table A.3.3.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.3.4-9: SIP 200 (OK) from the MCPTT Server to SS-UE1 (Step 1, Table A.3.3.3-1; Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-10			

Table A.3.3.4-10: SDP in SIP 200 (OK) (Table A.3.3.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_ FLOORCO NTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
NOTE 1: If the SS-UE1 request an implicit floor (Option b of note on Table A.3.3.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS-UE1 request a call without media floor (Option c of note on Table A.3.3.3-1) apply condition WITHOUT_FLOORCONTROL. NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

Table A.3.3.4-11: SIP 180 (Ringing) response
(Step 1, Table A.3.3.3-1; Steps 5, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.1-1, with condition MCPTT

Table A.3.3.4-12: SIP 180 (Ringing) response
(Step 1, Table A.3.3.3-1; Steps 6, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with condition MCPTT

A.3.4 MCPTT E2E call establishment using a pre-established session, manual commencement

A.3.4.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.4.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.4.3 Procedure

Table A.3.4.3-1: MCPTT E2E call establishment using a pre-established session, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server send a SIP re-INVITE requesting the establishment of a call to SS-UE2 (MCPTT Client)?	<--	SIP re-INVITE	-	P
4	The SS-UE2 (MCPTT Client) responds a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
5	The SS-UE2 (MCPTT Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCPTT server send a SIP ACK to acknowledge the session establishment to SS-UE2?	<--	SIP ACK	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
7	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
8	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
9	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
10	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.4.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.4.4-1: SIP REFER (Step 1, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [15]	
MIME body part		**Resource list**	IETF RFC 5366 [24]	
MIME-part-body	Resource-lists as described in Table A.3.4.4-2			

Table A.3.4.4-2: Resource-lists in SIP REFER (Table A.3.4.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.4.4-3A

Table A.3.4.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.4.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME-part-body	SDP Message as described in Table A.3.4.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-2C			

Table A.3.4.4-2B: SDP in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
NOTE 1: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.4.4-2C: MCPTT-Info in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.4.4-3: SIP 200 (OK) (Step 2, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.4.4-3A			

Table A.3.4.4-3A: SDP in SIP 200 (OK) (Table A.3.4.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Table A.3.4.4-4: SIP INVITE (Step 3, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 with conditions MCPTT, MANUAL and re_INVITE

Information Element	Value/remark	Comment	Reference	Condition
Resource-Priority			IETF RFC 4412 [20] IETF RFC 7134 [16] IETF RFC 8101 [18]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 with conditions MCPTT, MANUAL and re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [22]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-6			

Table A.3.4.4-5: SDP in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE requests a call without media floor control condition WITHOUT_FLOORCONTROL shall be applied. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.4.4-6: MCPTT-Info in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.4.4-7: SIP 200 (OK) (Step 4, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with conditions MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [14]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-8			

Table A.3.4.4-8: SDP in SIP 200 (OK) (Table A.3.4.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.4.4-9: Connect (Step 7, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.4.4-10: Connect (Step 9, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	

A.3.5 MCPTT E2E call establishment using a pre-established session, automatic commencement

A.3.5.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.5.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.5.3 Procedure

Table A.3.5.3-1: MCPTT E2E call establishment using a pre-established session, automatic commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
3	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
4	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
5	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
6	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.5.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.5.4-1: SIP REFER (Step 1, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [15]	
MIME body part		**Resource list**	IETF RFC 5366 [24]	
MIME-part-body	Resource-lists as described in Table A.3.5.4-2			

Table A.3.5.4-2: Resource-lists in SIP REFER (Table A.3.5.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.5.4-3A

Table A.3.5.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.5.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table A.3.5.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.5.4-2C			

Table A.3.5.4-2B: SDP in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [12] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [11] (3GPP TS 26.114) Table K.6	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
"RS"	0		IETF RFC 3556 [8]	
"RR"	2000		IETF RFC 3556 [8]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
NOTE 1: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.5.4-2C: MCPTT-Info in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.5.4-3: SIP 200 (OK) (Step 2, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.5.4-3A			

Table A.3.5.4-3A: SDP in SIP 200 (OK) (Table A.3.5.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [13]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	any value			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
			TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_ FLOORCO NTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
NOTE 1: If the SS-UE requests an implicit floor more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied. NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

Table A.3.5.4-4: Connect (Step 3, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.5.4-5: Connect (Step 5, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
PCK I_MESSAGE field	MIKEY-SAKKE I_MESSAGE as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-2A		PRIVATE-CALL

A.3.6 MCPTT E2E call release keeping the pre-established session

A.3.6.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.6.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.6.3 Procedure

Table A.3.6.3-1: MCPTT E2E call release keeping the pre-established session

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message with method "BYE" to release the MCPTT session and keep the pre-established session.	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) to SS-UE1?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server release the call by sending a Disconnect message to SS-UE2 (MCPTT Client)?	<--	Disconnect	-	P
4	The SS-UE2 (MCPTT Client) sends an Acknowledgement to accept the release of the call	-->	Acknowledge	-	-

A.3.6.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.6.4-1: SIP REFER (Step 1, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and METHOD-BYE

Table A.3.6.4-2: SIP 200 (OK) (Step 2, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP

Table A.3.6.4-3: Disconnect (Steps 3, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.13-1 with condition ACK

A.3.7 MCPTT Group Affiliation Status Determination and Subscription

A.3.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.7.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.7.3 Procedure

Table A.3.7.3-1: MCPTT Group affiliation status determination and subscription

Same procedure as defined in: Table A.2.7.3-1

A.3.7.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.7.4-1: SIP SUBSCRIBE (Step 1, Table A.3.7.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT and PRESENCE-EVENT

Table A.3.7.4-2: SIP NOTIFY (Step 3, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.7.4-3			

Table A.3.7.4-3: PIDF in SIP NOTIFY (Table A.3.7.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (note)
NOTE: PIDF document contains tuple with empty <status> element.

A.3.8 MCPTT Group Affiliation Status Change

A.3.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.8.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.8.3 Procedure

Table A.3.8.3-1: MCPTT Group Affiliation Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.8.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.8.4-1: SIP PUBLISH (Step 1, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.8.4-2: PIDF in SIP PUBLISH (Table A.3.8.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition AFFILIATION

Table A.3.8.4-3: SIP 200 (OK) (Step 2, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [15] IETF RFC 3903 [19]	
value	same value as in the request or lower			

Table A.3.8.4-4: SIP NOTIFY (Step 3, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-5			

Table A.3.8.4-5: PIDF in SIP NOTIFY (Table A.3.8.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

Table A.3.8.4-6: SIP NOTIFY (Step 5, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-7			

Table A.3.8.4-7: PIDF in SIP NOTIFY (Table A.3.8.4-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

A.3.9 MCPTT Functional Alias Status Determination and Subscription

A.3.9.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.9.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.9.3 Procedure

Table A.3.9.3-1: MCPTT Functional Alias Status Determination and Subscription

Same procedure as defined in Table A.2.7.3-1

A.3.9.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.9.4-1: SIP SUBSCRIBE (Step 1, Table A.3.9.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.1.3	
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			

Table A.3.9.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.3.9.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
request-type	functional-alias-status-determination		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.1.3	

Table A.3.9.4-3: SIP NOTIFY (Step 3, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1 with conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.9.4-4			

Table A.3.9.4-4: PIDF in SIP NOTIFY (Table A.3.9.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (note)
NOTE: PIDF document contains tuple with empty <status> element.

A.3.10 MCPTT Functional Alias Status Change

A.3.10.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.10.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.10.3 Procedure

Table A.3.10.3-1: MCPTT Functional Alias Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.10.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.10.4-1: SIP PUBLISH (Step 1, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.10.4-2: PIDF in SIP PUBLISH (Table A.3.10.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition FUNCTIONAL_ALIAS_STATUS_CHANGE

Table A.3.10.4-3: SIP 200 (OK) (Step 2, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [15] IETF RFC 3903 [19]	
value	same value as in the request or lower			

Table A.3.10.4-4: SIP NOTIFY (Step 3, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-5			

Table A.3.10.4-5: PIDF in SIP NOTIFY (Table A.3.10.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	

Table A.3.10.4-6: SIP NOTIFY (Step 5, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-7			

Table A.3.10.4-7: PIDF in SIP NOTIFY (Table A.3.10.4-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	

A.3.11 MCPTT Floor Request - Floor Granted

A.3.11.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.11.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.11.3 Procedure

Table A.3.11.3-1: MCPTT Floor Request - Floor Granted

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Granted message with acknowledgement required?	<--	Floor Granted	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Granted message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Granted message with no acknowledgement required?	<--	Floor Granted	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.11.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).

- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.11.4-1: Floor Granted (Step 2a1, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition ACK

A.3.12 MCPTT Floor Request - Floor Queue Position Info

A.3.12.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.12.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.12.3 Procedure

Table A.3.12.3-1: MCPTT Floor Request - Floor Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.12.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.12.4-1: Floor Queue Position Info (Step 2a1, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.13 MCPTT Queuing Position Request

A.3.13.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.13.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.13.3 Procedure

Table A.3.13.3-1: MCPTT Queuing Position Request

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Queue Position Request message. (Note)	-->	Floor Queue Position Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.13.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.13.4-1: Floor Queue Position Info (Step 2a1, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.14 MCPTT Floor Request - Floor Deny

A.3.14.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.14.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.14.3 Procedure

Table A.3.14.3-1: MCPTT Floor Request - Floor Deny

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with acknowledgement required?	<--	Floor Deny	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Deny message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with no acknowledgement required?	<--	Floor Deny	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.14.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.14.4-1: Floor Deny (Step 2a1, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1 condition ACK

A.3.15 MCPTT Floor Release - Floor Idle

A.3.15.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.15.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.15.3 Procedure

Table A.3.15.3-1: MCPTT Floor Release - Floor Idle

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (Note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Idle message with acknowledgement required?	<--	Floor Idle	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Idle message. (Note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Idle message with no acknowledgement required?	<--	Floor Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.15.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.15.4-1: Floor Release (Step 1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.15.4-2: Floor Idle (Step 3a1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition ACK

A.3.16 MCPTT Floor Release - Floor Taken

A.3.16.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.16.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.16.3 Procedure

Table A.3.16.3-1: MCPTT Floor Release - Floor Taken

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (Note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Taken message with acknowledgement required?	<--	Floor Taken	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Taken message. (Note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Taken message with no acknowledgement required?	<--	Floor Taken	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.16.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.16.4-1: Floor Release (Step 1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.16.4-2: Floor Taken (Step 3a1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition ACK

A.4 MCPTT Conformance testing test point models

Based on the test points models shown in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 4.1, the following section extends the test environment implementation examples provided in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 4.2 for Server-Client and Server-Server conformance testing.

Figure A.4.1 shows test configuration where the IUT and the system simulator, simulating MCPTT Clients, communicate, one with the other, over the LTE radio interface (test points (1)).

Figures A.4.2, A.4.3, A.4.4, A.4.5 and A.4.6 show test configuration where the IUT and the System Simulator communicate, one with the other, over the MCPTT-3 or IWF-1 interfaces, as defined by ETSI TS 123 179 [3] (3GPP TS 23.179), clause 7.5.2.4, and ETSI TS 123 283 [25] (3GPP TS 23.283), clause 7.4.1, respectively (test points (4)).

Figures A.4.7 shows test configuration where the IUT and the System Simulator communicate, one with the other, over the CSC-16 or IWF-3 interface, as defined by ETSI TS 123 280 [26] (3GPP TS 23.280), clause 7.5.2.16, and ETSI TS 123 283 [25] (3GPP TS 23.283), clause 7.4.3, respectively (test points (4)).

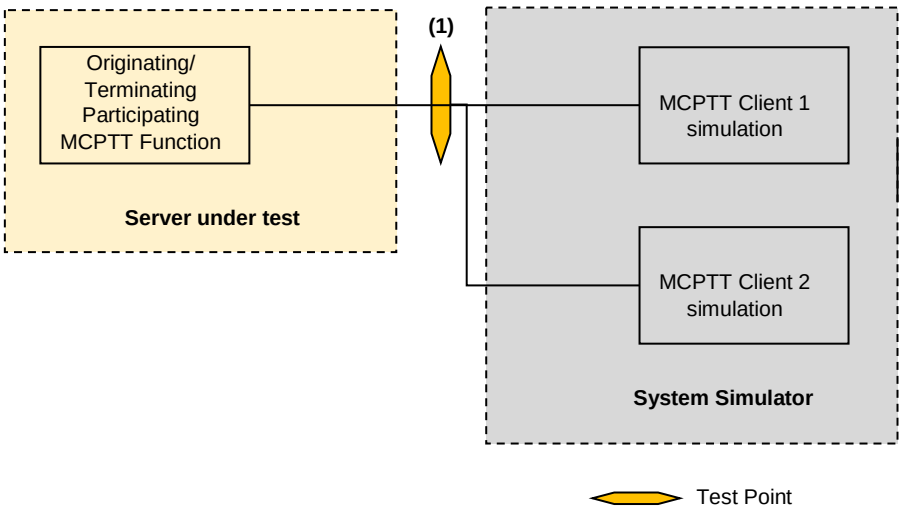


Figure A.4.1: Testing the MCPTT Server (server-to-client)

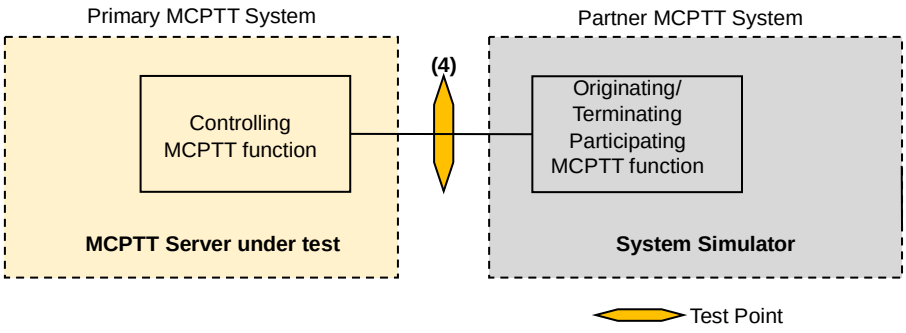


Figure A.4.2: Testing the MCPTT Server (server-to-server), Controlling function

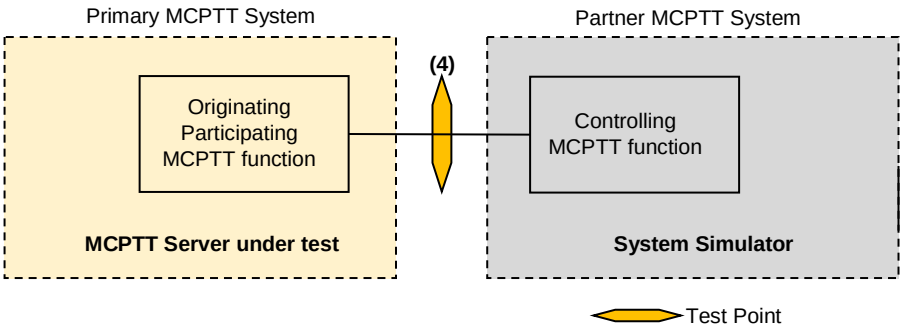


Figure A.4.3: Testing the MCPTT Server (server-to-server), Originating function

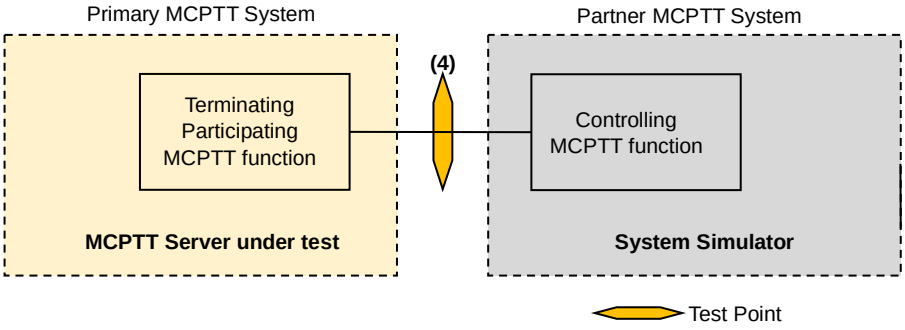


Figure A.4.4: Testing the MCPTT Server (server-to-server), Terminating function

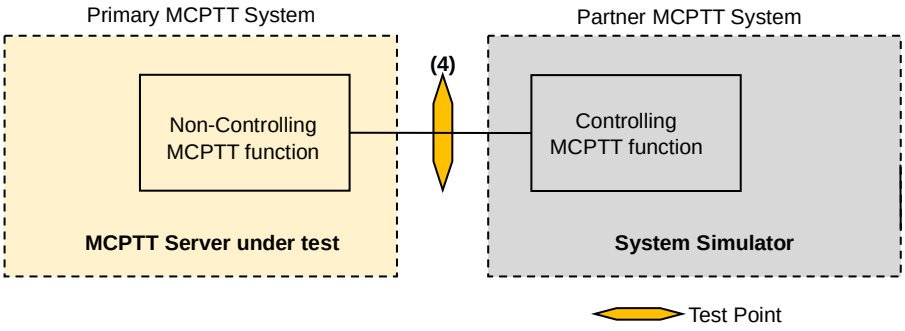


Figure A.4.5: Testing the MCPTT Server (server-to-server), Non-controlling function

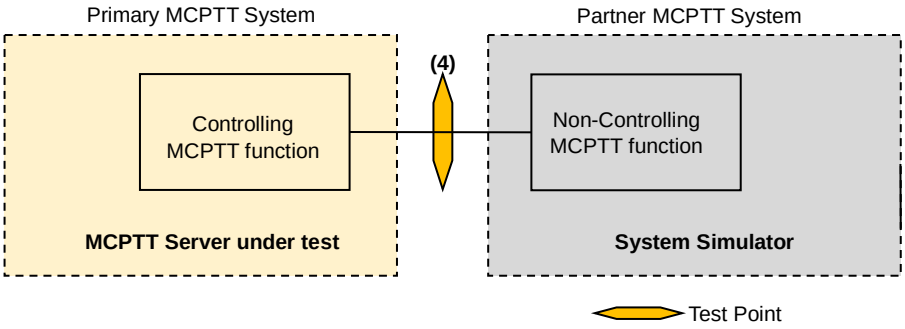


Figure A.4.6: Testing the MCPTT Server (server-to-server), Controlling function; trusted and untrusted modes

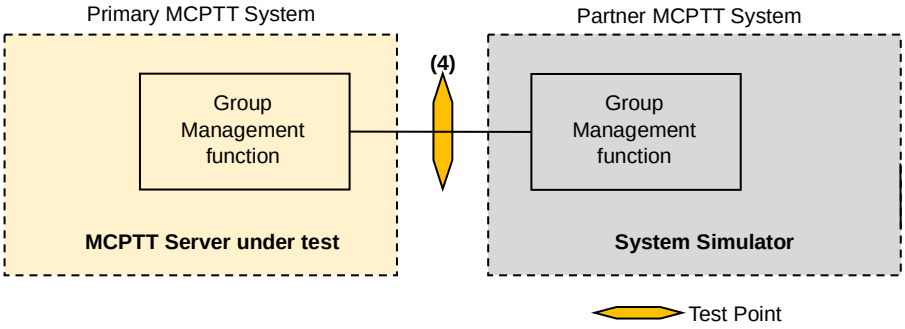


Figure A.4.7: Testing the MCPTT Server (server-to-server), Group Management function

Annex B (informative): Bibliography

- 3GPP TS 22.280: "Mission Critical Services Common Requirements (MCCoRe); Stage 1".
- 3GPP TS 23.379: "Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2".
- 3GPP TS 24.109: "Bootstrapping interface (Ub) and network application function interface (Ua); Protocol details".
- 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".
- 3GPP TS 24.382: "Mission Critical Push To Talk (MCPTT) identity management; Protocol specification".
- 3GPP TS 29.283: "Diameter data management applications".
- IETF RFC 2392: "Content-ID and Message-ID Uniform Resource Locators".
- IETF RFC 2616: "Hypertext Transfer Protocol -- HTTP/1.1".
- IETF RFC 3515: "The Session Initiation Protocol (SIP) Refer Method".
- IETF RFC 3711: "The Secure Real-time Transport Protocol (SRTP)".
- IETF RFC 3840: "Indicating User Agent Capabilities in the Session Initiation Protocol (SIP)".
- IETF RFC 3841: "Caller Preferences for the Session Initiation Protocol (SIP)".
- IETF RFC 4488: "Suppression of Session Initiation Protocol (SIP) REFER Method Implicit Subscription".
- IETF RFC 4538: "Request Authorization through Dialog Identification in the Session Initiation Protocol (SIP)".
- IETF RFC 4825: "The Extensible Markup Language (XML) Configuration Access Protocol (XCAP)".
- IETF RFC 4964: "The P-Answer-State Header Extension to the Session Initiation Protocol for the Open Mobile Alliance Push to Talk over Cellular".
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- IETF RFC 5875: "An Extensible Markup Language (XML) Configuration Access Protocol (XCAP) Diff Event Package".
- IETF RFC 6050: "A Session Initiation Protocol (SIP) Extension for the Identification of Services".
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- IETF RFC 6750: "The OAuth 2.0 Authorization Framework: Bearer Token Usage".
- IETF RFC 6809: "Mechanism to Indicate Support of Features and Capabilities in the Session Initiation Protocol (SIP)".
- IETF RFC 7647: "Clarifications for the Use of REFER with RFC 6665".
- IETF RFC 8445: "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal".

History

Version	Date	Status
V1.1.1	January 2026	Publication